T&T Industrial, Inc.

Health, Safety, and Environmental Manual

Rev. 1/7/2026

T&T Industrial, Inc.

Mission Statement

1. MISSION STATEMENT

The HSE Department at T&T Industrial, Inc. develops and implements comprehensive environmental health and safety policies throughout the organization to support its mission and goals. Their focus is on protecting public health, preventing personal injury, and ensuring regulatory compliance in areas such as chemical, biological, and radiation safety, occupational health and safety, and environmental stewardship. HSE is committed to reducing injuries, accidents, and environmental impact through high-quality training, workplace evaluation, emergency response, hazardous materials management, and regulatory compliance.

1.1 Responsibilities

At T&T Industrial, Inc., safety is everyone's responsibility, and all employees participate in ensuring a safe work environment. New Employee Orientation covers all Company safety policies and procedures, and the Safety Coordinator and Safety Committee lead the overall safety effort by providing necessary resources for accident prevention. The Safety Committee regularly reviews and updates HSE procedures through a quarterly review process to maintain the integrity of the Company safety management system. Supervisors are responsible for maintaining safe work conditions, and employees are responsible for following established safety procedures, reporting potential hazards promptly, and promoting a proactive culture of safe and responsible facility use. Maintaining a safe work environment is a top priority for T&T Industrial, Inc. and a personal goal for each employee.

Endorsed By: (Title)			
Signature:		Date:	

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1. ABRASIVE BLASTING PROGRAM

1.2 Purpose and Scope

The purpose of this program is to outline safe work practices to protect workers from hazardous dust levels and toxic metals that may be generated from blasting material and the underlying substrate and coatings being blasted.

This procedure applies to all T&T Industrial, Inc. employees conducting abrasive blasting.

1.3 References

Number	Title
29 CFR 1926 Subpart D	Occupational Health and Environmental Controls - Ventilation
29 CFR 1926 Subpart D	Occupational Health and Environmental Controls - Gases, Vapors, Fumes, Dusts, and Mists
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders – Control of Hazardous Substances
ANSI Z87.1	American National Standards Institute-Standard Practice for Occupational and Educational Eye and Face Protection
29 CFR 1910 Subpart G	Occupational Safety and Health Standards - Ventilation
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances – Air Contaminants
42 CFR 84	Approval of Respiratory Protection Devices
CMS-FM-0005	Abrasive Blasting Equipment Checklist
CMS-FM-0006	Carbon Monoxide Testing and Monitoring Checklist

1.4 Definitions

Acronym/Term	Definition
Abrasive	Solid particles made from various man-made or natural materials, (e.g., black beauty, steel shot, walnuts, etc.).
Abrasive Blast Hood	A respirator that uses compressed air and consists of a helmet that covers the entire head, a cape that covers the shoulders, and an apron that hangs from the bottom of the cape to the crotch.
Abrasive Blast Nozzle	The device through which the abrasive is accelerated and directed from the hose to the surface being cleaned.
Abrasive Blasting	The forcible application of an abrasive to a surface by air pressure, hydraulic pressure, or centrifugal force, to clean and/or remove surface contaminants, or to improve the surface profile prior to application of a coating.
Breathing Air	Air that meets or exceeds respiratory specifications and does not present a health

Acronym/Term	Definition
	hazard to anyone breathing the air.
Competent Employee / Person	A person who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them

1.5 Abrasive Blasting Requirements

Abrasive blasting involves the use of an abrasive media introduced into a high-pressure air stream. The potential hazard requires specific precautions throughout the surface preparation operation to prevent injury or damage to personnel and equipment.

1.6 Exposure

Exposure of employees to inhalation, ingestion, skin absorption, or contact with any material or substance at a concentration above those specified in the "Threshold Limit Values (TLVs) of Airborne Contaminants for 1970" of the American Conference of Governmental Industrial Hygienists, also provided in 29 CFR 1910.1000, shall be avoided.

Abrasives and surface coatings on the materials blasted are shattered and pulverized during blasting operations and the dust formed will contain particles of respirable size. The composition and toxicity of the dust from these sources shall be considered in making an evaluation of the potential health hazards.

1.7 Safe Work Practices

Dust shall not be permitted to accumulate on the floor or on ledges outside of an abrasive-blasting enclosure, and dust spills shall be cleaned up promptly. Aisles and walkways shall be kept clear of steel shot or similar abrasive which may create a slipping hazard.

The blast nozzle shall be bonded and grounded to prevent the buildup of static charges.

The blast cleaning nozzles shall be equipped with an operating valve which must be held open manually or must be equipped with a “Deadman” valve. A support shall be provided on which the nozzle may be mounted when it is not in use.

Compressed air shall not be used for cleaning purposes except where the pressure is reduced to less than 30 p.s.i.

1.8 Respiratory Protection

A respiratory protection program shall be established wherever it is necessary to use respiratory protective equipment including worksite-specific procedures and elements for required respirator use.

All abrasive-blasting operators shall use abrasive-blasting respirators:

- When working inside of blast-cleaning rooms.
- When using silica sand in manual blasting operations where the nozzle and blast are not physically separate from the operator in an exhaust ventilated enclosure.
- Where concentrations of toxic dust dispersed by the abrasive blasting may exceed the limits set in 29 CFR 1910.1000 and the nozzle and blast are not physically separated from the operator in an exhaust-ventilated enclosure.

Only respirators approved by the National Institute for Occupational Safety and Health (NIOSH) under 42 CFR part 84 shall be used to protect employees from dust produced during abrasive blasting operations.

Air for abrasive-blasting respirators shall be free of harmful quantities of dusts, mists, or noxious gases, and shall meet the requirements for supplied-air quality and use specified in 29 CFR 1910.134(i).

1.9 Personal Protective Equipment (PPE)

Equipment for protection of the eyes and face shall be supplied to the operator when the respirator design does not provide such protection.

Equipment for protection of the eyes and face shall be supplied to any other personnel working in the vicinity of abrasive blasting operations.

1.10 Ventilation

The construction, installation, inspection, and maintenance of exhaust systems shall conform to the principles and requirements set forth in American National Standard Fundamentals Governing the Design and Operation of Local Exhaust Systems, Z9.2-1960, and ANSI Z33.1-1961, which are incorporated by reference as specified in 1910.6.

Blast-cleaning enclosures shall be exhaust ventilated in such a way that a continuous inward flow of air will be maintained at all openings in the enclosure during the blasting operation.

2. Access to Medical Records Program

1.11 Purpose and Scope

This section establishes the minimum requirements for providing information to employees in accordance with medical and exposure records.

This procedure applies to all T&T Industrial, Inc. employees.

1.12 References

Number	Title
29 CFR 1910 Subchapter Z	Toxic and Hazardous Substance - Access to Employee Exposure and Medical Records
CMS-FM-0007	Notice to Employees Form

1.13 Definitions

Acronym/Term	Definition
Designated Representative	Any individual or organization to whom an employee gives written authorization to exercise a right of access. For the purposes of access to employee exposure records and analyses using exposure or medical records, a recognized or certified collective bargaining agent shall be treated automatically as a designated representative without regard to written employee authorization.
Employee Medical Records	Records that concern the health status of an employee, and are made or maintained by a physician, nurse, or other health care personnel or technician.
Employee Exposure Records	A record which measures or monitors the amount of a toxic substance or harmful physical agent to which the employee is or has been exposed.
Exposure	An employee is subjected to a toxic substance or harmful physical agent in the course of employment through any route of entry (inhalation, ingestion, skin contact or absorption, etc.), and includes past exposure and potential (e.g., accidental or possible) exposure.

1.14 Medical and Exposure Records

The purpose for exposure and medical records access is to improve the detection, treatment, and prevention of occupational illness and disease.

1.15 Access

Employees shall be provided the right of access to relevant exposure and medical records. Upon employment and at least annually thereafter, current employees shall be informed of the existence, location, and availability of any records, the person responsible for maintaining and providing access to records, and each employee's rights of access to these records.

Whenever an employee or designated representative requests access to a record, access shall be provided in a reasonable time, place, and manner. If access to the record cannot be reasonably provided within 15 working days, the employee shall be notified of the reason for the delay within the 15 working days.

Whenever an employee or designated representative requests a copy of a record, either a copy of the record shall be provided without cost to the employee or representative, the necessary mechanical copying facilities (e.g., photocopying) made available without cost to the employee or representative for copying the record, or the record shall be loaned to the employee or representative for a reasonable time to enable a copy to be made.

Whenever access is requested to an analysis which reports the contents of employee medical records by either direct identifier (name, address, social security number, payroll number, etc.) or by information which could reasonably be used under the circumstances indirectly to identify specific employees (e.g., exact age, height, weight, race, sex, date of initial employment, job title, etc.), personal identifiers shall be removed before access is provided.

1.16 Retention

Employee medical and exposure records shall be preserved and retained for the duration of employment plus 30 years.

An employee medical record is a record concerning the health status of an employee that is made or maintained by a physician, nurse, or other health care personnel or technician including:

- Medical and employment questionnaires or histories, including job description and occupational exposure.
- Results of medical examinations and laboratory tests.
- Medical opinions, diagnoses, progress notes, and recommendations.
- First aid records.
- Descriptions of treatments and prescriptions.

- Employee medical complaints.

An employee exposure record is any of the following kinds of information:

- Environmental (workplace) monitoring or measuring of a toxic substance or harmful physical agent, including personal, area, grab, wipe, or other form of sampling, as well as related collection and analytical methodologies, calculations, and other background data relevant to interpretation of the results obtained.
- Biological monitoring results which directly assess the absorption of a toxic substance or harmful physical agent by body systems (e.g., the level of a chemical in the blood, urine, breath, hair, fingernails, etc.) but not including results which assess the biological effect of a substance or agent or which assess an employee's use of alcohol or drugs.
- Safety data sheets indicating that the material may pose a hazard to human health.
- A chemical inventory or any other record which reveals where and when used and the identity (e.g., chemical, common, or trade name) of a toxic substance or harmful physical agent.

1.17 Transfer of Records

If the Company ceases to do business, all exposure and medical records shall be transferred to the successor employer. The successor employer shall receive and maintain these records. If the Company ceases to do business and there is no successor employer, affected current employees shall be notified of their rights of access to records.

3. Aerial Lifts (Elevated and Rotating Work Platforms) Program

1.18 Purpose and Scope

The purpose of this program is to provide minimum safe work practices for the operation of aerial lifts (elevated and rotating work platforms) to ensure the safety of employees.

This document applies to all T&T Industrial, Inc. employees working with or on aerial lifts.

1.19 References

Number	Title
29 CFR 1910 Subpart F	Powered Platforms, Manlifts, and Vehicle-Mounted Work Platforms - Vehicle Mounted Elevating and Rotating Work Platforms
29 CFR 1910 Subpart S	Electrical - Selection and Use of Work Practices
29 CFR 1926 Subpart L	Scaffolds - Aerial Lifts
29 CFR 1926 Subpart V	Electric Power Transmission and Distribution - Job Briefing
29 CFR 1926 Subpart O	Motor Vehicles, Mechanized Equipment, and Marine Operations - Motor Vehicles
29 CFR 1926 Subpart M	Fall Protection - Fall Protection Systems Criteria and Practices
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Order - Operating Instructions (Aerial Devices)
Cal/OSHA T8 CCR Subchapter 4	Construction Safety Orders - Warning Methods
Cal/OSHA T8 CCR Subchapter 5	Electrical Safety Orders - Warning Signs Required

1.20 Definitions

Acronym/Term	Definition
Aerial Lift	Any vehicle-mounted device used to elevate personnel, including extendable boom platforms, aerial ladders, articulating (jointed) boom platforms, vertical towers, and any combination of these.
Insulated Aerial Lifts	Offer protection from electric shock and electrocution by insulating from electrical ground.

1.21 Aerial Lifts (Elevated and Rotating Work Platforms) Program

Aerial lifts are commonly used in construction, inspection, maintenance, and repair services to lift employees to an elevated work position. Proper operation and use of aerial lifts can make the completion of tasks at elevation safer and more efficient. However, with unsafe use, work with aerial lifts can result in serious injury.

1.22 General

The brakes shall be set, and outriggers, when used, shall be positioned on pads or a solid surface. Wheel chocks shall be installed before using an aerial lift on an incline.

An aerial lift truck may not be moved when the boom is elevated in a working position with employees in the basket, except for equipment that is specifically designed for this type of operation.

Articulating boom and extensible boom platforms, primarily designed as personnel carriers, shall have both platform (upper) and lower controls. Lower-level controls shall not be operated unless permission has been obtained from the employee in the lift, except in case of emergency.

Before moving an aerial lift for travel, the boom(s) shall be inspected to see that it is properly cradled and outriggers are in stowed position.

Field modifications for use other than intended by the manufacturer are prohibited.

1.23 Training

Only trained persons shall operate an aerial lift.

Training shall include:

- Explanations of electrical, fall, and falling object hazards.
- Procedures for handling hazards.
- Recognizing and avoiding unsafe conditions.
- Instructions for correct operation of the lift, including maximum load and load capacity.
- Demonstrations of the skills and knowledge needed to operate an aerial lift prior to operation.
- When and how to perform inspections.

- Manufacturer’s requirements.

1.24 Qualifications

Anyone operating aerial lifts (JLG’s, scissor lifts, genie lifts, spiders, lulls, zoon boom, etc.) must be competent and qualified as follows:

- Training and Certification: Aerial lift operators need to complete formal training and obtain certification from a recognized training organization. This training covers both theoretical knowledge and practical skills related to operating aerial lifts safely.
- Knowledge of Regulations: Operators must be familiar with relevant safety regulations and standards, such as those set by OSHA (Occupational Safety and Health Administration). This includes understanding load capacities, safe operating procedures, and inspection requirements.
- Technical Skills: Aerial lift operators must be skilled in operating the specific type of aerial lift they will be using. This includes knowledge of the lift's controls, functions, and safety features.
- Safety Procedures: Operators must have a strong understanding of safety procedures, including pre-operation inspections, proper use of personal protective equipment (PPE), and emergency protocols.
- Risk Assessment: Competent operators must assess job sites for potential hazards and make decisions to ensure safe operation. This includes identifying obstacles, unstable surfaces, overhead obstructions, and more.
- Communication Skills: Aerial lift operators must communicate with ground personnel or other operators. Effective communication is essential for coordinating movements and ensuring safety.
- Emergency Response: Operators must be trained in emergency response procedures, including how to safely evacuate the aerial lift in case of malfunction or other emergencies.
- Continuous Learning: Industries and equipment standards evolve over time. Aerial lift operators must be open to ongoing training and staying up-to-date with the latest safety regulations and best practices.

1.25 Testing and Inspection

A pre-start inspection shall be conducted to determine safe working conditions including the following:

- Manufacturer’s recommendations,

- Vehicle components,
- Proper fluid level (oil, hydraulic, fuel, coolant, etc.),
- Leaks of fluids,
- Wheels and tires,
- Battery and charger,
- Lower-level controls,
- Horn, gauges, lights,
- Backup alarms,
- Steering and brakes,
- Lift components and controls,
- Operating and emergency controls,
- Personal protective devices,
- Hydraulic, air, pneumatic, fuel, and electrical systems,
- Fiberglass and other insulating components,
- Missing or unreadable placards, warnings, or operational, instructional, and control markings,
- Mechanical fasteners and locking pins,
- Cable and warning harnesses,
- Outriggers, stabilizers, and other structures,
- Loose or missing parts, and
- Guardrail systems.

Lift controls shall be tested each day prior to use to determine that such controls are in safe working condition.

Defective aerial lifts shall be removed from service until repairs are made.

Work zone inspections shall be conducted, including the following:

- Drop-offs, holes, or unstable surfaces such as loose dirt,
- Inadequate ceiling heights,
- Slopes, ditches, or bumps,
- Debris and floor obstructions,
- Overhead electrical power lines and communication cables,
- Other overhead obstructions,
- Other hazardous locations and atmospheres,
- High wind and other severe weather conditions, and
- The presence of others near the work.

Corrective action shall be taken to eliminate any hazards found.

1.26 Load Limits

Boom and basket load limits specified by the manufacturer shall not be exceeded.

1.27 Ladder and Tower Trucks

Before ladder or tower trucks are moved for highway travel, aerial ladders shall be secured in the lower traveling position by the locking device above the truck cab and the manually operated device at the base of the ladder or by other equally effective means.

1.28 Spotter / Backup Alarm

Aerial lifts having an obstructed view shall not be moved backward unless the vehicle has a reverse signal alarm audible above the surrounding noise level and/or an observer signals it is safe to do so.

1.29 Overhead Power Lines

For overhead power lines rated 50 kV or below, minimum clearance between the lines and any part of the equipment or load shall be at least 10 feet.

If the aerial lift is insulated for the voltage involved, and if the work is performed by a qualified person, the clearance distance (between the uninsulated portion of the aerial lift and the power line) may be referenced to the distance provided in 1910.333(c)(3)(ii)(C) Table 5-5.

1.30 Fall Protection

A personal fall arrest or travel restraint system shall be worn and attached to the boom or basket when working from an aerial lift. Employees are prohibited from attaching to adjacent poles or structures.

1.31 Field Modifications

Aerial work platforms shall not be field modified for uses other than those intended by the manufacturer unless the modification has been certified in writing by the manufacturer or by any other equivalent entity, such as a nationally recognized testing laboratory, to be in compliance with the applicable ANSI standard and this rule, and to be at least as safe as the equipment was before modification.

4. Arsenic Awareness Program

1.32 Purpose and Scope

The purpose of this program is to provide information about arsenic, the potential health effects associated with exposure, and safety procedures that should be followed to reduce exposure and protect the health of employees.

This program applies to all T&T Industrial, Inc. employees.

1.33 Resources

Number	Title
OSHA 29 CFR 1910 Subpart Z	Toxic and Hazardous Substances - Asbestos

1.34 Arsenic Awareness Program

Arsenic occurs naturally in the environment as an element of the earth's crust. Arsenic is combined with other elements such as oxygen, chlorine, and sulfur to form inorganic arsenic compounds. Exposure to higher-than-average levels of arsenic occurs mainly in workplaces, near or in hazardous waste sites, and areas with high levels naturally occurring in soil, rocks, and water. Exposure to high levels of arsenic can cause death. Exposure to arsenic at low levels for extended periods of time can cause a discoloration of the skin and the appearance of small corns or warts.

Arsenic is a silver-gray or white metallic, odorless, brittle solid.

1.35 Hazard Recognition

Arsenic exposure in the workplace occurs through inhalation, ingestion, dermal, or eye contact. Chronic exposure to arsenic leads to distinct skin diseases, such as arsenical keratinosis, which is characterized by excessive formation of scaly skin on the palms and soles; darkened patches of skin; wart formation; skin lesions; acne; and increased risk of skin cancers. Chronic arsenic poisoning can also cause sudden constriction in arteries or veins, reducing blood flow; decreased nerve function; lung, liver, kidney and bladder, and other cancers. Acute exposures can cause lung distress and death.

1.36 Training

Employees that are exposed to airborne concentrations of arsenic must complete training in accordance with local legislation prior to exposure.

1.37 Exposure Evaluation

Industries that use inorganic arsenic and its compounds, where sampling may be necessary, include wood preservation, glass production, nonferrous metal alloys, and electronic semiconductor manufacturing. Inorganic arsenic is also found in coke oven emissions associated with the smelter industry. Arsenic and its compounds occur in crystalline, powder, amorphous, or vitreous forms.

1.38 Engineering Controls and Work Practices

Engineering controls and safe work practices must be used to reduce/maintain arsenic exposure below applicable limits. Engineering controls and safe work practices include:

- Container labeling
- Providing employees with hazard information and training
- Monitoring airborne chemical concentrations.
- Providing eye wash stations and emergency showers.
- Wash or shower if skin comes in contact with a hazardous material.
- Always wash at the end of a work shift.
- Change into clean clothing if clothing becomes contaminated.
- Do not take contaminated clothing home.
- Require special training for employees that was contaminated clothing.
- Do not eat, smoke, or drink in areas where chemicals are being handled, processed, or stored.
- Wash hands carefully before eating, smoking, drinking, applying cosmetics, or using the toilet.
- Use a vacuum or a wet method to reduce dust during clean up.
- Do not dry sweep.
- Use a high efficiency particulate air (HEPA) filter when vacuuming. Do not use a standard shop vacuum.

1.39 First Aid

Eye Contact – Immediately flush with large amounts of water for at least 15 minutes, lifting upper and lower lids. Remove contact lenses, if worn, while rinsing. Seek medical attention.

Skin Contact – Quickly remove contaminated clothing. Immediately wash contaminated skin with large amounts of soap and water.

Inhalation – Remove the person from exposure. Begin rescue breathing (using universal precautions) if breathing has stopped and CPR if heart action has stopped. Transfer promptly to a medical facility.

5. Asbestos Awareness Program

1.40 Purpose and Scope

The purpose of this program is to provide information about asbestos, the potential health effects associated with exposure, and safety procedures that should be followed to reduce exposure and protect the health of employees.

This program applies to all T&T Industrial, Inc. employees.

1.41 Resources

Number	Title
OSHA 29 CFR 1910 Subpart Z	Toxic and Hazardous Substances - Asbestos
NIH Pub. No. 89-1647	OSHA Asbestos Training
CMS-FM-0008	Asbestos Exposure Control Checklist

1.42 Definitions

Acronym/Term	Definition
Asbestos	Asbestos is a generic term describing a family of naturally occurring fibrous silicate minerals. As a group, the minerals are noncombustible, do not conduct heat or electricity, and are resistant to many chemicals. Although there are several other varieties that have been used commercially, the most common asbestos mineral types likely to be encountered in buildings are chrysotile (white asbestos), amosite (brown asbestos), and crocidolite (blue asbestos). Among these, white asbestos is by far the most common asbestos mineral present in buildings.
Friable Asbestos	Friable asbestos material means finely divided asbestos or asbestos-containing material or any asbestos-containing material that can be crumbled, pulverized, or powdered by hand pressure. Individual fibers in friable asbestos-containing material can potentially become airborne and can then present a health hazard. Three types of friable material commonly used in buildings are: sprayed fibrous fireproofing; decorative or acoustic texture coatings; and thermal insulation.
Non-friable Asbestos	Non-friable asbestos includes a range of products in which asbestos fiber is effectively bound in a solid matrix from which asbestos fiber cannot normally escape. Non-friable asbestos includes a variety of products including asbestos cement tiles and boards and asbestos reinforced vinyl floor tiles. Cutting, braking, sanding, and drilling or similar activities can release asbestos fibers from even non-friable asbestos materials.

1.43 Asbestos Awareness Program

Asbestos is a common, naturally occurring group of fibrous minerals. Asbestos fibers have been used in a variety of building materials. Asbestos materials are used in the manufacture of heat-resistant clothing, automotive brake and clutch linings, and a variety of building materials including insulation, soundproofing, floor tiles, roofing felts, ceiling tiles, asbestos-cement pipe and sheet, and fire-resistant drywall. Asbestos is also present in pipe and boiler insulation materials, pipeline wrap and in sprayed-on materials located on beams, in crawlspaces, and between walls.

1.44 Hazards

Friable asbestos (that is, material that contains more than 0.1% asbestos by weight and can be crumbled by hand) is a potential hazard because it can release fibers into the air if damaged. Long-term exposure to airborne asbestos is necessary for chronic lung disease. Asbestos fibers are significantly more hazardous than dust or other materials even of the same composition because the shape of the fibers allows them to be inhaled but not eliminated by the respiratory system as with other particles.

Significant and long-term exposure to asbestos from activities that directly disturb asbestos-containing materials (such as asbestos mining) can lead to a variety of respiratory diseases, including asbestosis and mesothelioma (cancer of the lung lining), and cancer of the stomach and colon. Asbestosis is a non-malignant, irreversible disease resulting in fibrosis of the lung. Asbestos-related cancers tend also to result from substantial long-term exposure; however, mesothelioma may result from much smaller exposures to asbestos.

1.45 Hazard Control

- Engineering Controls - Engineering controls include the use of enclosures such as monitoring equipment, glove bags, tenting, negative pressure work areas, HEPA filters, controlled vacuums, water misters, and other equipment to ensure containment and cleanup of asbestos work areas.
- Administrative Controls - All qualified employees shall be issued proper personal protective equipment (PPE), such as respirators, disposable coveralls, gloves, etc. Written procedures and management authorizations are required for all work involving asbestos containing material.
- Training Controls – Training shall be provided to all employees with the potential to come into contact with asbestos before work begins and yearly thereafter. Training shall include identification of the locations for potential exposure, the health hazards as previously listed,

and that employees must not disturb asbestos containing materials. Examples of where asbestos maybe found include manufacturing of heat resistant clothing, automotive brake and clutch linings, certain building materials such as insulation, sound proofing, floor tiles, roofing felts, ceiling tiles, asbestos-cement pipe and sheet, fire resistant drywall, pipe and boiler insulating materials, pipeline wrap, and in sprayed on materials located on beams, in crawlspaces, and between walls. Training must include:

  - Prohibiting unauthorized employees from disturbing asbestos containing materials
  - Signs and labels will be used to identify areas where asbestos is present
  - Appropriate work practices will be used to ensure ACM and presumed ACM are not inadvertently disturbed

- Signs and labels shall identify the material which is present, where it is located, and appropriate work practices which will ensure that asbestos containing material (ACM) and/or presumed asbestos containing material (PACM) will not be disturbed.

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Asbestos Awareness Program

Appendix 1 Asbestos Exposure Control Checklist

General Information				
Competent Person:			Date:	
Location:				
Description of Job:				

Competent Person				
		Yes	**No**	**N/A**
1.	On site			
2.	Adequately trained			
3.	Authority to correct hazards			
4.	Conducts regularly scheduled inspections			
5.	Develops asbestos exposure control plan			
6.	Ensures exposure assessment and monitoring program			
7.	Ensures PPE is used and employees are trained			
8.	Hygiene facilities and engineering controls in use			
Method(s) of Asbestos Determination				
		Yes	**No**	**N/A**
1.	Owner/client documentation			
2.	Project specifications			
3.	Prior maintenance records			
4.	Other documentation			
5.	Bulk samples			
6.	Other sample methods			
Employee Asbestos Exposure Assessment & Monitoring				
		Yes	**No**	**N/A**
1.	Competent person to conduct exposure assessment			
2.	Initial exposure assessment			
3.	Negative exposure assessment			
4.	Daily / periodic monitoring			

Asbestos Exposure Control Plan Content				
		Yes	No	N/A
1.	Tasks / activities disturbing asbestos			
2.	Engineering controls			
3.	Work practices			
4.	Personal protective equipment			
5.	Respiratory protection			
6.	Medical surveillance			
7.	Exposure monitoring data			
8.	Multi-contractor communication			
9.	Employee training			
Regulated Work Area				
		Yes	No	N/A
1.	Established / supervised by competent person			
2.	Signs posted			
3.	Access controlled			
4.	Separated from other parts of facility			
5.	Respiratory protection used			
6.	No eating, drinking, smoking, etc.			
7.	Work supervised by competent person			
Negative Pressure Enclosure				
		Yes	No	N/A
1.	Negative pressure enclosure practical			
2.	Integrity of Enclosure			
3.	Established access control			
4.	Employee monitoring			
5.	PPE in use			
6.	Entry / exit procedures training			
7.	Engineering controls			
8.	Containment leak check each shift			
9.	Electric circuits deactivated unless GFCI			
10.	Work practices			

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Engineering Controls				
		Yes	**No**	**N/A**
1.	Local exhaust system with HEPA filters			
2.	General ventilation with HEPA filters			
3.	Vacuum cleaners with HEPA filters			
4.	Enclosure or isolation			
5.	Wet methods used for cleanup			
6.	Leak-tight labeled containers used for disposal			
Work Practices				
		Yes	**No**	**N/A**
1.	Regulated work area established			
2.	Work plan developed			
3.	Housekeeping plan			
4.	Proper tools used for work			
Personal Protective Equipment (General)				
		Yes	**No**	**N/A**
1.	Head protection / covering			
2.	Eye protection			
3.	Hand protection (gloves)			
4.	Foot protection / covering			
5.	Protective clothing (coveralls)			
6.	Disposal / decontamination procedures			
Respiratory Protection				
		Yes	**No**	**N/A**
1.	Written program established			
2.	Designated program administrator (competent person)			
3.	Employee medical qualification			
4.	Employee fit test (negative pressure respirator)			
5.	NIOSH / MSHA approved respirators			
6.	Cleaning, maintenance, and storage procedures			
7.	Grade D air supply for supplied air respirators			
8.	Program audits			

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Personal Hygiene				
		Yes	**No**	**N/A**
1.	Hand washing area (regardless of exposure level)			
2.	Shower and change room (separate for male and female)			
3.	Hot and cold water (water filter 5 micrometers)			
4.	Soap and towels			
5.	Lunchroom / area wet wipe / mop daily			
6.	No food, smoking devices, chew, or cosmetics in work area			
7.	Non-work area protected from contamination			
8.	Personnel decontamination techniques established			
Medical Surveillance				
		Yes	**No**	**N/A**
1.	Medical advisor supervising program (provided with required information)			
2.	Medical surveillance program			
3.	Employee(s) provided monitoring results within 30 days			
Employee Information and Training				
		Yes	**No**	**N/A**
1.	HazCom training provided all employees			
2.	Asbestos training – all employees in accordance with EPA model accreditation plan			
Signs				
		Yes	**No**	**N/A**
1.	Containers of contaminated clothing labeled			
2.	Work areas posted			
Recordkeeping				
		Yes	**No**	**N/A**
1.	Asbestos exposure control plan			
2.	Exposure assessment			
3.	Medical surveillance records			
4.	Employee training records			

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Asbestos Awareness Program

Environmental				
		Yes	No	N/A
1.	RCRA generator permit required			
2.	RCRA disposal permit required			
3.	RCRA licensed transporter			
Additional Requirements				
		Yes	No	N/A
1.	Heat stress monitoring			
2.	Heat stress training			
Comments				

6. Assured Grounding Conductors

1.46 Purpose and Scope

The purpose of this program is to provide guidelines to eliminate injuries resulting from possible malfunctions, improper grounding, and/or defective electrical tools.

This program applies to all T&T Industrial, Inc. employees working with electrical equipment.

1.47 Resources

Number	Title
29 CFR 1926 Subpart K	Electrical – Wiring Design and Protection

1.48 Assured Equipment Grounding Conductors Program

The assured equipment grounding conductor program covers all cord sets, receptacles which are not a part of the permanent wiring of the building or structure, and equipment connected by cord and plug which are available for use or used by employees.

Conductors used as ground conductors or equipment grounding conductors shall be identifiable and distinguishable from all other conductors.

Any equipment that has not met the requirements of this program shall not be made available or permitted to be used by employees.

All 120-volt, single-phase 15- and 20-ampere receptacle outlets on construction sites, which are not a part of the permanent wiring of the building or structure, and which are in use by employees, shall have approved ground-fault circuit interrupters for personnel protection. Receptacles on a two-wire, single-phase portable or vehicle-mounted generator rated not more than 5kW, where the circuit conductors of the generator are insulated from the generator frame and all other grounded surfaces, need not be protected with ground-fault circuit interrupters.

1.49 Inspections

Each cord set, attachment cap, plug and receptacle of cord sets, and any equipment connected by cord and plug, except cord sets and receptacles which are fixed and not exposed to damage, shall be visually inspected before each day's use for external defects, such as deformed or missing pins or insulation damage, and for indications of possible internal damage. Equipment found damaged or defective shall not be used until repaired. Damaged items shall be tagged 'DO NOT USE' and be removed from service until repaired and tested.

1.50 Testing

The following tests shall be performed on all cord sets, receptacles which are not a part of the permanent wiring of the building or structure, and cord- and plug-connected equipment required to be grounded:

- All equipment grounding conductors shall be tested for continuity and shall be electrically continuous.
- Each receptacle and attachment cap or plug shall be tested for correct attachment of the equipment grounding conductor. The equipment grounding conductor shall be connected to its proper terminal.

All required tests shall be performed:

- Before first use.
- Before equipment is returned to service following any repairs.
- Before equipment is used after any incident which can be reasonably suspected to have caused damage (e.g., when a cord set is run over).
- At intervals not to exceed 3 months, except that cord sets and receptacles which are fixed and not exposed to damage shall be tested at intervals not exceeding 6 months.

1.51 Recordkeeping

All tests conducted shall be recorded, identifying each:

- receptacle,
- cord set, and
- cord- and plug-connected equipment that passed the test.

The record shall indicate the last date it was tested or the interval for which it was tested. This record shall be kept by means of logs and shall be maintained until replaced by a more current record. The record shall be made available on the jobsite for inspection by the OSHA Assistant Secretary and any affected employee.

7. Behavior Based Safety Program

1.52 Purpose and Scope

The purpose of this program is to create a total safety culture in the workplace using Behavior Based Safety (BBS). The program addresses how to observe and correct behaviors that are critical to safety.

The program applies to all T&T Industrial, Inc. employees.

1.53 Resources

Number	Title
CMS-FM-0011	Behavior Observation Checklist
CMS-FM-0012	Safety Observation Checklist

1.54 Behavior Based Safety Program

Observation and feedback techniques of behavior-based safety (BBS) may be used to predict whether safety problems are increasing. Intensifying the BBS observation cycle will often prevent an injury or incident.

Employees will conduct direct observations of affected company employee's work behavior documenting both safe and unsafe acts in a manner that provides direct and measurable data.

1.55 Training

Training provided shall include:

- Review of the program objectives and incident metrics.
- How to conduct observations.
- How to complete the observation form.
- What the behaviors mean.
- Mentoring and coaching by role play for feedback training.
- Employee awareness of being observed at any time.

1.56 Design Team

The design team shall be comprised of hourly employees, supervisors, managers, and safety personnel.

Responsibilities of the team include designing forms, establishing training protocol, collecting data, setting goals, and identifying roles and responsibilities for the BBS process.

1.57 Identifying Critical Behaviors

1.57.1 Steps to Identify Critical Behaviors

- Analyze incident trends to determine the areas that have the greatest risk potential for incidents.
- Conduct a hazard evaluation of the site to determine the areas that have the greatest risk potential for incidents.
- Analyze tasks that have the greatest potential for serious injury or death.

1.57.2 Implement Controls

Implement effective engineering and/or administrative controls. Eliminate the hazard if possible.

1.58 Process Steps

Break down each step of the behavior process using enough detail so that independent observers evaluating the same employee will come to the same result.

1.59 Critical Behaviors

Break down the tasks into the following four critical behaviors:

- Personal Protective Equipment (PPE) – Determine what PPE is required to perform the task. Be specific so the observer knows what to look for.
- Housekeeping – The work area is to be evaluated by the observer with the condition documented.
- Use of Tools and Equipment – Determine what tools and equipment are used for the task and how they are used safely.

- Body Positioning / Protection – The observer will need to determine if the employee is performing the task in a manner that will protect them from strains, falling objects, exposure to a sudden release of chemicals, etc.

1.60 Behavior Observation Checklist

Use the critical behaviors and practices identified previously to develop the observation checklist. This information can later be used to provide measurable information on employee work practices and measure progress towards goals.

The checklist is used to document safe and unsafe behaviors.

For simplification and ease of use, limit the checklist to no more than 10 critical behaviors.

1.61 Observation and Feedback

Observation and feedback are the most important components of the BBS process. Observations provide direct, measurable information on employees' work practices, identifying both safe and unsafe behaviors.

Frequent, objective feedback is essential in maintaining safe behavior.

For safe behaviors, provide positive feedback. For unsafe behaviors, provide non-threatening, instructive feedback.

1.62 Observation and Feedback Plan

- Determine who will conduct the observations.
- Determine the frequency of the observations.
- Develop the observation procedures.
- Determine who will provide feedback and when.
- Train employees on how to conduct observations and provide feedback.

1.62.1 Observers

Observers may be any employee that is trained on the process. One observer should be assigned per shift or per department. Management shall allow employee time to be allocated to this process.

Management shall also be included as observers for quality control and leadership engagement purposes.

1.62.2 Frequency of Observations

Frequency (daily, weekly, monthly) shall be determined by the risk associated with the task. Higher risk warrants more frequent observations.

1.62.3 Observation Procedure

- The observer will observe the employee while working and will use the Safety Observation Checklist to record the number of safe and unsafe behaviors performed along with the number of appreciative and constructive feedback given, if any.
- Positive feedback should be given immediately when safe behaviors are exhibited.
- Constructive, non-threatening feedback should be given for unsafe behaviors.
- Stop work authority may be used as needed.
- Observations should last no longer than 20 minutes.

1.62.4 Providing Feedback

Employees providing feedback shall be trained as coaches.

Frequency of feedback shall be determined by the risk associated with the task, the number of targeted employees, the separate work areas, or level of employees.

Typically, feedback is given immediately following the observation, reviewing which critical behaviors were performed safely or unsafely.

If the feedback should include disciplinary action, it should only be given by the employees' supervisor.

For overall corrective action, regular department meetings should be held to review the behaviors.

The observer will:

- Review the observation with the observed employee.
- Start with positive comments.
- Reinforce safe behaviors observed first.
- Describe and discuss unsafe behaviors observed.
- Solicit from the observed employee explanations of their unsafe behaviors with open-ended questions.
- Re-emphasize no consequence to observed employee.

1.63 Improvement Goals

Setting improvement goals increases the effectiveness of feedback and the success of the BBS process.

Goals should be developed based on employee work practices and how they can be improved.

Action plans should then be developed to help employees achieve their goals.

Options for goals include:

- Percent Safe Goals – based on safe work practices observed; based on an evaluation of the areas level of safety; set for a short period (1-3 months).
- Process Goals – focused on improving a specific work practice such as using proper lifting techniques, the goal of which would be to reduce the number of times unsafe techniques are used.
- Implementation Goals – focused on maintaining the BBS process, such as setting a goal to increase the number of observations conducted in a specified period.

1.64 Developing Goals

A baseline must first be developed such as conducting observations for 4 weeks. Future observations can then be compared to the baseline and tracked for improvements. Improvements can then be placed in graphs and displayed for positive reinforcement.

1.64.1 Providing Positive Reinforcement

Providing positive reinforcement when employees improve or attain goals is the key to a successful BBS program.

Positive reinforcement can be immediate verbal feedback, graphical feedback placed in strategic locations in the workplace, regular meetings providing analysis and detailed feedback about specific behaviors.

1.65 Action Plans

Once trend analysis is complete, appropriate action plans shall be developed to address unsafe behaviors.

Action planning shall include:

- Evaluated and prioritized unsafe behaviors from trend analysis.
- Comments and feedback from data sheets.
- Designated responsible employees and timeframes.

- Defined responsible employees for action planning.
- Assurance of management support.

1.66 Success Measurement

Individual departments, as well as the Company as a whole, will compare measurements and track these results by an acceptable method so that numerical and statistical comparisons can be made over time.

If employees are performing their tasks with a higher percentage of safe behaviors, injuries are less likely to occur.

Calculate the incident rate and evaluate at set intervals.

Incident Rate = (number of incidents x 200,000) / total man hours.

Tracking these measurements allows for statistical comparison over time for continuous improvement.

T&T Industrial, Inc.

Behavior Based Safety Program

Appendix 2 Behavior Observation Checklist

General Information			
Observer:		Date:	
Department:		Time:	

Operating Procedures

	Safe	At Risk	Comments	% Safe
PPE: Using required personal protective equipment. Face shield, safety glasses, gloves and hearing protection.				
Housekeeping: Work area maintained safely (e.g., trash and scrap picked up, no spills, walkways clear, materials and tools organized).				
Using Tools and Equipment: Guards are in place, tool rest adjusted to within 1/8," grinding wheel in good condition, grinder secured.				
Body Positioning/Protecting: Hand positioned to avoid pinch point.				

*To determine percent safe, divide number of safe observations by the total number of observations for each task.

Appendix 3 Safety Observation Checklist

Observer:		Date:				
Department:		Time:				
Conditions	**Safe**	**At-Risk**	**Apprec. Feedback**	**Constr. Feedback**	**% Safe**	
1. Forklift warning devices operational						
2. Forklift driver's compartment free of debris						
3. Forklift propane tank clamps locked in place						
Behaviors	**Safe**	**At-Risk**	**Apprec. Feedback**	**Constr. Feedback**	**% Safe**	
1. Operator's driver's license displayed above the waist						
2. Forks 6" or less from ground when traveling						
3. Seat belts worn during forklift operation						
4. Sets parking brake, puts forks to floor, puts gear in neutral, and shuts off when leaving forklift unattended						
5. Sounds horn when exiting trailer						

6. Wears authorized safety footwear, gloves, and eye protection					
7. Uses approved lift cage when transporting or elevating people					
8. Removes freight from side of forks					
Comments:					

*To determine percent safe, divide number of safe observations by the total number of observations for each task.

8. Benzene Exposure Control and Awareness Program

1.67 Purpose and Scope

The purpose of this program is to provide information on an awareness level basis about the properties and hazards of benzene along with general guidelines.

This program applies to all T&T Industrial, Inc. employees with the potential for exposure to benzene.

1.68 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances - Benzene
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders – Control of Hazardous Substances
CMS-GL-0001	Benzene Data Sheet
CMS-GL-0002	Respiratory Protection for Benzene

1.69 Benzene Exposure Control and Awareness Program

Benzene may be encountered at refineries and laboratories, during refueling and tank gauging, and when completing oil field and pipeline maintenance operations.

Benzene is a clear, colorless liquid with a pleasant, sweet odor. The odor of benzene does not provide adequate warning of its hazard.

1.70 Compliance

In the event that benzene becomes a health hazard or potential health hazard at any site, implementation of engineering controls, work practice controls, respiratory protection, or any combination of these controls shall be implemented and established as a written program to reduce employee exposure to benzene to or below the Permissible Exposure Limit (PEL).

It is the responsibility of management and local health and safety representatives to see that these written compliance programs are made available to the OSHA Assistant Secretary, the OSHA Director, affected employees, and designated employee representatives.

The written program and plans shall be furnished upon request for examination and copying by the OSHA Assistant Secretary, the OSHA Director, affected employees, and designated employee representatives.

1.71 Engineering and Work Practice Controls

Developing and implementing engineering and work practice controls for benzene exposure safety is crucial to protect workers and comply with safety regulations. The company must institute work practice and engineering controls to reduce and maintain employee exposures at or below the permissible exposure limit (PEL) of 1 part per million (ppm) wherever feasible. When engineering and work practice controls are not feasible to reduce employee exposures to or below the PEL, the company will utilize these controls to reduce company employee exposures to the lowest levels achievable. The following schedule is implemented to ensure controls are in place:

- Phase 1: Assessment and Planning (Duration: 2-3 months)
  Month 1: Preliminary Assessment
  Month 2: Hazard Analysis
  Month 3: Planning
- Phase 2: Engineering Controls (Duration: 6-12 months)
  Months 4-6: Feasibility Study
  Months 7-9: Design and Procurement
  Months 10-12: Installation and Testing
- Phase 3: Work Practice Controls (Duration: Ongoing)
  Month 13: Work Practice Assessment
  Months 14-16: Procedure Development
  Months 17-18: Training and Implementation
- Phase 4: Monitoring and Continuous Improvement (Duration: Ongoing)
  Months 19+: Monitoring and Evaluation
- Ongoing: Improvement Initiatives
  Annual Reviews

1.72 Hazard Data

Benzene can affect the body through inhalation, skin / eye contact, or accidental ingestion. Benzene has a pleasant, sweet odor, but the odor does not provide adequate warning of its hazard.

- Short-Term (Acute) Health Effects:

  Exposure to high concentrations of benzene, well above the levels where its odor is first recognizable, may result in breathlessness, feelings of irritability, euphoric, or giddy; irritation in eyes, nose, and respiratory tract; headache, feel dizzy, nauseated, or intoxicated. Severe exposures may lead to convulsions and loss of consciousness.

- Long-Term (Chronic) Health Effects:

  Repeated or prolonged exposure to benzene, even at relatively low concentrations, may result in various blood disorders, ranging from anemia to leukemia, an irreversible, fatal disease. Benzene has been shown to cause cancer in humans. Benzene exposure has been associated with cancers such as myeloid leukemia, Hodgkin's disease, and lymphomas. Many blood disorders associated with benzene exposure may occur without symptoms.

- Physical Hazards:

  Benzene poses a serious fire and explosion hazard when exposed to heat or flame. Benzene vapor is heavier than air and may collect in low areas. Vapors can also travel for some distance and may come into contact with ignition sources. Smoking is prohibited in areas where benzene is used or stored. Since Benzene is highly flammable and vapors may form explosive mixtures in air, fire extinguishers must be readily available in areas where benzene is used or stored.

1.73 Permissible Exposure Limits

OSHA has issued the following guidelines for employee exposures to reduce the potential for adverse health effects:

- Action Level - The concentration of a chemical in air, calculated as an eight-hour time-weighted average (TWA), which initiates certain required activities such as exposure monitoring and medical surveillance. *The action level for benzene is 0.5 parts per million (0.5 ppm).*

- Permissible Exposure Limit (PEL) - The greatest concentration, calculated as an eight-hour TWA, to which nearly all employees may be repeatedly exposed during their eight-hour work schedule without experiencing adverse health effects. *The PEL for benzene is 1 part per million (1 ppm).*

- Short-Term Exposure Limit (STEL) - The greatest concentration that nearly all employees may be exposed during any one 15-minute period without experiencing adverse health effects. *The STEL for benzene is 5 parts per million (5 ppm).*

1.74 Personal Protective Equipment (PPE)

- General

  Contact with the eyes or skin with liquids containing benzene shall be prevented using protective garments and equipment which are impervious to benzene over any parts of the body that could be exposed to liquid benzene. The type of PPE necessary will vary depending on the concentration, amount used, and the potential for splashing. It may include goggles, boots, face shields, gloves, gowns, lab coats, aprons, and arm sleeves.

If there is potential for liquid benzene to get into the eyes, splash-proof safety goggles shall be worn. If there is potential for liquid benzene to splash onto the face, a face shield shall be worn.

PPE contaminated with benzene shall not be brought home by employees.

All PPE shall be inspected by employees prior to each use. PPE shall be stored in a clean and sanitary manner. Respirators shall be inspected monthly to ensure they are being used, stored, and cleaned properly. The Company shall provide protective clothing and equipment at no cost to the employee and ensure its use where appropriate. Proper eye and face protection equipment shall meet the requirements of 29 CFR 1910.133.

- Respiratory Protection

  If employee exposures are found to exceed the PEL or STEL, proper respiratory protection shall be provided until feasible engineering or administrative controls can be implemented. Employees working in areas with exposure to benzene shall receive appropriate respiratory protection provided by the Company. Respirators shall be provided that comply with the following requirements:

  - Periods necessary to install or implement feasible engineering and work-practice controls.
  - Work operations for which the job site establishes compliance with either the TWA or STEL through the use of engineering and work-practice controls is not feasible including but not limited to maintenance and repair activities, vessel repair / cleaning, or other operations for which engineering and work-practice controls are infeasible because exposures are intermittent and limited in duration.
  - Emergencies involving benzene discharge / spill.
  - PPE shall be provided and worn when appropriate to prevent eye contact and limit dermal exposure to liquid benzene.
  - For employees who are or may be exposed to benzene at or above the action level, 30 or more days per year above the PEL, or 20 or more days for employees who have been exposed to more than 10 ppm of benzene for 30 or more days in a year prior to effective date.
  - Respiratory protection shall be selected according to airborne concentrations of benzene. Reference local regulation for additional information.

1.75 Storage

Since benzene liquid is highly flammable, it shall be stored in tightly closed containers in a cool, well ventilated area. Benzene vapor may form explosive mixtures in air. All sources of ignition shall be controlled. Smoking is prohibited in areas where benzene is used or stored.

1.76 Signage and Labeling

Regulated Areas:

Areas where the airborne levels of benzene are found to exceed or can reasonably be expected to exceed the PELs, either the 8-hour TWA exposure of 1 ppm or the short-term exposure limit of 5 ppm for 15 minutes shall be designated as regulated areas. Access to these areas shall be limited to persons trained to recognize the hazards of benzene. All entrances and access ways shall be posted with signs bearing the following information:

DANGER
Benzene
Cancer Hazard
Flammable – No Smoking
Authorized Personnel Only
Respirator Required

Container Labels:

If a chemical product containing benzene is transferred into a container other than the original, it shall be labeled with the following information:

DANGER
Contains Benzene
Cancer Hazard

Labels shall be provided upon request.

When labeling containers, use the following hazard ratings:

- Health - 3
- Flammability - 3
- Reactivity - 0
- Personal Protective Equipment - This rating will vary based upon the use, but shall be at least a B.

1.77 Monitoring

In areas that employees are subject to benzene exposure, monitoring shall be conducted and recorded to determine the employees' exposure level.

- Measurements shall be taken that represent an average 8-hour exposure from one 8-hour sample or two separate 4-hour samples.
- Air samples shall also be taken in the employee's area where they will be working to represent air that is inhaled by the employee.
- Sampling and analysis of benzene samples may be conducted by a compilation of benzene vapor or charcoal absorption tubes, with chemical analysis by gas chromatography. Portable monitors, continuous monitoring, and passive dosimeters are also methods of sampling and monitoring.
- Monitoring shall have an accuracy to a 95% confidence level of not less than plus or minus 25% for concentrations of benzene greater than or equal to 0.5 ppm.
- For more specific monitoring requirements and air sample methods refer to local regulation.

1.78 Employee Information and Training

Every employee working with benzene shall receive training on its hazards. The following information shall be reviewed with employees prior to and annually when working with or around benzene:

- Requirements of the local regulation standard.
- The physical characteristics of benzene being a colorless liquid with a sweet smell and that smell is not an adequate warning of the hazards of benzene.
- Contents of the Safety Data Sheet.
- Description of the medical surveillance program.
- Description of the health hazards associated with exposure.
- Signs and symptoms of exposure.
- Instructions to report any signs or symptoms that may be attributable to benzene exposure.
- Description of the operations in the work area where benzene is present.
- Work practices to reduce exposure, including engineering and administrative controls and PPE required.
- Instructions for handling spills and emergency procedures.

- Contingency plans and provisions.

This training shall be conducted whenever a new hazard is introduced into the work area, when the employee transfers to another job, and whenever the employee demonstrates behavior that indicates a lack of understanding of the safe handling of chemicals.

Supervisors are responsible for ensuring that employees with potential exposure to benzene receive the appropriate training prior to working with the substance.

The individual presenting the training session shall document all training.

Benzene awareness training conducted for affected company employees must include the potential health effects from exposure. Exposure to benzene, even at low concentrations, can lead to inflammation of the nasal airways and throat. High level exposures can severely damage the lungs causing fluid accumulation and bleeding, which is often fatal. The major long-term effects of benzene on the blood of those exposed which include harmful effects on the bone marrow and can cause a decrease in red blood cells, leading to anemia. Benzene exposure can lead to excessive bleeding and negatively affect the immune system, increasing the chance for infection. Females that have breathed high levels of benzene for many months experienced irregular menstrual periods and decrease in the size of their ovaries. The Department of Health and Human Services (DHHS) has determined that benzene causes cancer in humans and that long-term exposure to high levels of benzene can cause leukemia.

Benzene is an extremely flammable substance with a flash point of 12 degrees Fahrenheit (-11 degrees Celsius), an autoignition temperature of 1,076 degrees Fahrenheit (580 degrees Celsius), and a lower explosion limit (LEL) of 1.3% and an upper flammable limit (UFL) of 7.5%. All ignition sources must be controlled when benzene is used, handled, or stored. Benzene vapors are heavier than air; thus, the vapors may travel along the ground and be ignited by open flames or sparks at locations remote from where the benzene is used, handled, or stored.

1.79 Medical Surveillance

Employees who are or may be exposed to benzene and meet the following requirements shall be involved in a medical surveillance program at no cost to the employee.

- Exposed at or above the action level for 30 or more days per year.
- Exposure or has the potential for exposure at or above the PELs 10 or more days a year.
- Exposed to more than 10 ppm of benzene for 30 or more days in a year prior to effective date.

These employees shall complete a medical questionnaire annually and receive a physical examination. The physical shall include blood tests to determine if any blood disorders exist.

- Employees exposed to benzene shall receive medical attention under the following circumstances:
    - When an employee has developed signs or symptoms associated with exposure to benzene.
    - When an employee is involved in a spill, leak, or other occurrence resulting in a possible exposure to benzene.
- Medical surveillance shall be recorded.

The written program shall be reviewed and revised as appropriate based on the most recent exposure monitoring data to reflect the current status of the program.

Appendix 4 Benzene Data Sheet

PERMISSIBLE EXPOSURE

Except as to the use of gasoline, motor fuels and other fuels subsequent to discharge from bulk terminals and other exemptions specified in 1910.1028(a)(2):

- Airborne: The maximum time-weighted average (TWA) exposure limit is 1 part of benzene vapor per million parts of air (1 ppm) for an 8-hour workday and the maximum short-term exposure limit (STEL) is 5 ppm for any 15-minute period.
- Dermal: Eye contact shall be prevented and skin contact with liquid benzene shall be limited.

APPEARANCE AND ODOR

Benzene is a clear, colorless liquid with a pleasant, sweet odor. The odor of benzene does not provide adequate warning of its hazard. It is toxic, not soluble in water, and is flammable.

HEALTH HAZARD DATA

Benzene can affect your health if you inhale it, or if it comes in contact with your skin or eyes. Benzene is also harmful if you happen to swallow it.

Effects of overexposure.

- Short-term (acute) overexposure: If you are overexposed to high concentrations of benzene, well above the levels where its odor is first recognizable, you may feel breathless, irritable, euphoric, or giddy; you may experience irritation in eyes, nose, and respiratory tract. You may develop a headache, feel dizzy, nauseated, or intoxicated. Severe exposures may lead to convulsions and loss of consciousness.
- Long-term (chronic) exposure. Repeated or prolonged exposure to benzene, even at relatively low concentrations, may result in various blood disorders, ranging from anemia to leukemia, an irreversible, fatal disease. Many blood disorders associated with benzene exposure may occur without symptoms.

PROTECTIVE CLOTHING AND EQUIPMENT

- Respirators. Respirators are required for those operations in which engineering controls or work practice controls are not feasible to reduce exposure to the permissible level. However, where employers can document that benzene is present in the workplace less than 30 days a year, respirators may be used in lieu of engineering controls. If respirators are worn, they must have joint Mine Safety and Health Administration and the National Institute for Occupational Safety and Health (NIOSH) seal of approval, and cartridge or canisters must be replaced before the end of their service life, or the end of the shift, whichever occurs first. If you experience difficulty breathing while wearing a respirator, you may request a positive pressure respirator from your employer. You must be thoroughly trained to use the assigned respirator, and the training will be provided by your employer.
- Protective Clothing. You must wear appropriate protective clothing (such as boots, gloves, sleeves, aprons, etc.) over any parts of your body that could be exposed to liquid benzene.
- Eye and Face Protection. You must wear splash-proof safety goggles if it is possible that benzene may get into your eyes. In addition, you must wear a face shield if your face could be splashed with benzene liquid.

EMERGENCY AND FIRST AID PROCEDURES

- Eye and face exposure. If benzene is splashed in your eyes, wash it out immediately with large amounts of water. If irritation persists or vision appears to be affected, see a doctor as soon as possible.
- Skin exposure. If benzene is spilled on your clothing or skin, remove the contaminated clothing and wash the exposed skin with large amounts of water and soap immediately. Wash contaminated clothing before you wear it again.
- Breathing. If you or any other person breathes in large amounts of benzene, get the exposed person to fresh air at once. Apply artificial respiration if breathing has stopped. Call for medical assistance or a doctor as soon as possible. Never enter any vessel or confined space where the benzene concentration might be high without proper safety equipment and at least one other person present who will stay outside. A lifeline should be used.
- Swallowing. If benzene has been swallowed and the patient is conscious, do not induce vomiting. Call for medical assistance or a doctor immediately.

MEDICAL REQUIREMENTS

If you are exposed to benzene at a concentration at or above 0.5 ppm as an 8-hour time-weighted average, or have been exposed at or above 10 ppm in the past while employed by your current employer, your employer is required to provide a medical examination and history and laboratory tests within 60 days of the effective date of this standard and annually thereafter. These tests shall be provided without cost to you. In addition, if you are accidentally exposed to benzene (either by ingestion, inhalation, or skin/eye contact) under emergency conditions known or suspected to constitute toxic exposure to benzene, your employer is required to make special laboratory tests available to you.

OBSERVATION OF MONITORING

Your employer is required to perform measurements that are representative of your exposure to benzene and you or your designated representative are entitled to observe the monitoring procedure. You are entitled to observe the steps taken in the measurement procedure, and to record the results obtained. When the monitoring procedure is taking place in an area where respirators or personal protective clothing and equipment are required to be worn, you or your representative must also be provided with, and must wear the protective clothing and equipment.

ACCESS TO RECORDS

You or your representative are entitled to see the records of measurements of your exposure to benzene upon written request to your employer. Your medical examination records can be furnished to yourself, your physician or designated representative upon request by you to your employer.

PRECAUTIONS FOR SAFE USE, HANDLING AND STORAGE

Benzene liquid is highly flammable. It should be stored in tightly closed containers in a cool, well ventilated area. Benzene vapor may form explosive mixtures in air. All sources of ignition must be controlled. Use non-sparking tools when opening or closing benzene containers. Fire extinguishers, where provided, must be readily available. Know where they are located and how to operate them. Smoking is prohibited in areas where benzene is used or stored. Ask your supervisor where benzene is used in your area and for additional plant safety rules.

Appendix 5 Respiratory Protection for Benzene

*Any employee who cannot use a negative-pressure respirator must be allowed to use a respirator with less breathing resistance, such as a powered air-purifying respirator or supplied-air respirator

Airborne Concentration of Benzene or Condition of Use	Required Respirator
Less than or equal to 10 ppm	A half mask, air-purifying respirator equipped with organic vapor cartridges
Less than or equal to 50 ppm	A full-face piece with organic vapor cartridges full-face gas mask with chin style canister
Less than or equal to 100 ppm	A full-face piece powered air-purifying respirator with organic vapor canister
Less than or equal to 1,000 ppm	A supplied-air respirator with full-face piece, operated in positive pressure mode.
Greater than 1,000 ppm or unknown concentration	• A self-contained breathing apparatus with full-face piece operated in positive pressure mode • Full-face piece positive-pressure supplied air respirator with auxiliary self-contained air supply
Escape	• Any organic vapor gas mask; or • Any self-contained breathing apparatus with full face-piece.
Fire Fighting	Full face-piece self-contained breathing apparatus in positive pressure mode

NOTE: Canisters must have a minimum service life of 4 hours when tested at 150 ppm benzene, at a flow rate of 64 LPM, 25°C and 84% relative humidity for non-powered air purifying respirators. The flow rate shall be 115 LPM and 170 LPM, respectively for light fitting and tight fitting powered air-purifying respirators.

9. Bloodborne Pathogens Program

1.80 Purpose and Scope

The purpose of this section is to provide general principles to be followed when working with potentially infectious material.

This procedure applies to all T&T Industrial, Inc. employees with the potential for exposure to infectious materials.

1.81 References

Number	Title
29 CFR1910 Subpart Z	Toxic and Hazardous Substances - Blood Borne Pathogens
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances - Access to Employee Exposure and Medical Records Standard
29 CFR 1910 Subpart I	Personal Protective Equipment - Eye and face protection
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Order – Control of Hazardous Substances
CMS-FM-0013	Hepatitis B Vaccination Consent Form
CMS-FM-0014	Hepatitis B Vaccination Declination Form
CMS-FM-0015	Employee Report of Occupational Exposure
CMS-FM-0016	Bloodborne Pathogens Exposure Control Plan

1.82 Definitions

Acronym/Term	Definition
Bloodborne Pathogens	Pathogenic microorganisms that are present in human blood and can cause disease in humans. These pathogens include, but are not limited to, Hepatitis B Virus (HBV) and Human Immunodeficiency Virus (HIV).
Contaminated Laundry	Laundry that has been soiled with blood or other potentially infectious materials or that may contain sharps.
Contaminated Sharps	Any contaminated object that can penetrate the skin including but not limited to needles, scalpels, broken glass, broken capillary tubes, and exposed ends of dental wires.
Decontamination	The use of physical or chemical means to remove, inactivate, or destroy bloodborne pathogens on a surface or item to the point where they are no longer capable of transmitting infectious particles and the surface or item is rendered safe for handling, use, or disposal.
Exposure Incident	A specific eye, mouth, other mucous membrane, non-intact skin, or parenteral contact with blood or other potentially infectious material that results from the

Acronym/Term	Definition
	performance of an employee's duties.
Infectious Waste	Also called biomedical waste, this includes human waste, animal waste, and objects and materials contaminated with blood and body fluids containing disease-causing micro-organisms or viruses.
Occupational Exposure	Reasonably anticipated skin, eye, mucous membrane, or parenteral contact with blood or other potentially infectious materials that may result from the performance of an employee's duties.
Other Potentially Infectious Materials	Semen, vaginal secretions, cerebrospinal fluid, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid, amniotic fluid, saliva in dental procedures, any body fluid that is visibly contaminated with blood, and all body fluids in situations where it is difficult or impossible to differentiate between body fluids. Any unfixed tissue or organ (other than intact skin) from a human (living or dead). Also, HIV-containing cell or tissue cultures, organ cultures, and HIV- or HBV-containing culture medium or other solutions, and blood, organs, or other tissues from experimental animals infected with HIV or HBV.
Parenteral	Piercing mucous membranes or the skin barrier through such events as needle sticks, human bites, cuts, and abrasions.
Sharp	Medical slang for a needle or similar pointed object.
Source Individual	Any individual, living or dead, whose blood or other potentially infectious materials may be a source of occupational exposure to the employee.
Sterilization	The use of a physical or chemical procedure to destroy all microbial life, including highly resistant bacterial endospores.
Universal Precautions	An approach to infection control. According to the concept of Universal Precautions, all human blood and certain human body fluids are treated as if known to be infectious for HIV, HBV, or other bloodborne pathogens.
Work Practice Controls	Controls that reduce the likelihood of exposure by altering the way a task is performed (e.g., prohibiting recapping of needles by a two-handed technique).

1.83 Bloodborne Pathogens Program

Approximately 5.6 million American employees are at risk of developing various types of illnesses due to their exposure to bloodborne pathogens such as the human immunodeficiency (HIV) and hepatitis B (HBV) viruses and other potentially infectious materials in the workplace. In recent years there has been a significant increase in the number of cases reported. This poses a serious problem for exposed employees and their employer.

The Exposure Control Plan must be readily available to employees and employees must be informed of its location.

1.84 General

Good general principles shall be followed when working with potentially infectious material. For example:

- It is prudent to minimize all exposure to bloodborne pathogens / potentially infectious material.
- Risk of exposure to bloodborne pathogens should never be underestimated.
- Institute as many work practices and engineering controls as possible to eliminate or minimize employee exposure to bloodborne pathogens / potentially infectious materials.
- Any site working with potentially infectious materials shall have an Exposure Control Plan that meets the letter and intent of the local regulatory Pathogens Standard. The plan shall be easily accessible to all employees and they shall be informed of where to find it. The objective of this Plan is twofold:
  - To protect employees from the health hazards associated with bloodborne pathogens by eliminating or minimizing employee exposure.
  - To provide appropriate treatment and counseling should an employee be exposed to bloodborne pathogens / potentially infectious materials.

It is important to keep the Exposure Control Plan up to date. To ensure currency, the Plan shall be revised and updated as follows:

- Bi-annually.
- Whenever new or modified tasks and procedures are implemented that affect occupational exposure of employees.
- Whenever employees' jobs are revised such that new instances of occupational exposure may occur.
- Whenever new functional positions are established that may include exposure to bloodborne pathogens.

1.85 Exposure Determination

The Exposure Control Plan shall identify job classifications in which employees have regular exposure, some exposure, or no occupational exposure to bloodborne pathogens and shall identify tasks and procedures in which occupational exposure to bloodborne pathogens occur.

The exposure determination shall be made without regard to the use of personal protective equipment (PPE).

1.86 Methods of Compliance

Universal precautions shall be observed by personnel to prevent contact with blood or other potentially infectious materials. In accordance with the concept of universal precautions, personnel shall treat blood and other potentially infectious materials as though potentially infected with HBV, HIV, or other bloodborne pathogens, particularly when differentiation between body fluid types is difficult or impossible. Proper PPE must be utilized.

1.87 Engineering and Work Practice Controls

Engineering and work practice controls shall be used to minimize employee exposure. These controls and a copy of the site-specific exposure control plan shall be provided to employees in a reasonable time, place, and manner.

The Exposure Control Plan must include the various types of bodily fluids that the effected employees can reasonably be exposed to in the workplace to include, but is not limited to blood, mucus, and saliva.

Where occupational exposure remains after institution of these controls, PPE shall also be used. Engineering controls may include installation of mechanical pipe-fitting devices, biosafety cabinets, and safety equipment for centrifuges; suitable facilities and mechanisms for quick drenching or flushing of the eyes and mucous membranes shall be provided in the work area for immediate emergency use; and handwashing facilities shall have hot and cold running water.

Engineering controls shall be examined and maintained or replaced on a regular schedule to ensure their effectiveness.

Training and monitoring of proper work practices shall be provided with employees instructed in performing all tasks in the use of appropriate precautions, engineering and work practice controls, and PPE.

1.87.1 Handwashing

Handwashing facilities shall be readily accessible to employees (e.g., no further than what would be considered reasonable for location of restrooms).

When provision of handwashing facilities is not feasible, an appropriate antiseptic hand cleanser in conjunction with clean cloth / paper towels or antiseptic towelettes shall be provided to employees.

When antiseptic hand cleansers or towelettes are used, hands shall be washed with soap and running water as soon as feasible following exposure incidents whether or not exposure is apparent.

Employees shall wash their hands immediately, or as soon as feasible, after removal of gloves or other PPE. Employees shall wash hands and any other skin with soap or germicidal agents and water, or flush mucous membranes with water, immediately or as soon as feasible following contact of such body areas with blood or other potentially infectious materials, whether or not contact is apparent.

1.87.2 Skin Surfaces

Eating, drinking, smoking, applying cosmetics or lip balm, handling contact lenses, and touching skin surfaces with contaminated hands is prohibited in work areas where there is the reasonable likelihood of occupational exposure to bloodborne pathogens and other potentially infectious materials. Personnel shall thoroughly wash hands with soap and running water prior to engaging in these types of activities.

Food and drink shall not be kept in refrigerators, freezers, shelves, or cabinets or on countertops or benchtops where blood or other potentially infectious materials are present.

1.88 Personal Protective Equipment (PPE)

Where there is potential for occupational exposure, the Company shall provide, at no cost to the employee, appropriate PPE such as, but not limited to, gloves, gowns, replaceable coveralls, laboratory coats, face shields or masks, eye protection, mouthpieces, resuscitation bags, pocket masks, or other ventilation devices.

PPE will be considered appropriate only if it does not permit blood or other potentially infectious materials to pass through to or reach the employee's work clothes, street clothes, undergarments, skin, eyes, mouth, or other mucous membranes under normal conditions of use and for the duration of time that the protective equipment will be used.

The Company shall ensure that employees use appropriate PPE unless it can be shown that an employee temporarily and briefly declined to use PPE when, under rare and extraordinary circumstances, it was the employee's professional judgment that in the specific instance its use would have prevented the delivery of healthcare or public safety services or would have posed an increased hazard to the safety of the employee or coworker. When the employee makes this judgment, the circumstances shall be investigated and documented to determine whether changes can be instituted to prevent such occurrences in the future.

Training and instruction on the proper use and limitations of PPE shall be provided with proper use enforced.

Appropriate PPE in the proper sizes shall be readily accessible at the worksite, on response vehicles, or issued to employees, and shall be maintained to ensure and promote its use. Employees shall know where to obtain protective equipment.

Hypoallergenic gloves, glove liners, powderless gloves, or other similar alternatives shall be readily accessible to those employees who are allergic to the gloves normally provided.

1.88.1 Cleaning, Repair, and Replacement

The cleaning, laundering, and disposal of PPE shall be provided in accordance with this procedure at no cost to the employee.

The repair or replacement of PPE shall be provided as needed to maintain its effectiveness at no cost to the employee.

Personnel shall be responsible for notifying their supervisor of the need to repair or replace PPE or clothing.

1.88.2 Contaminated PPE

If a garment(s) is penetrated by blood or other potentially infectious materials, the garment(s) shall be removed immediately or as soon as feasible. In such cases where there is reasonable expectation that an employee's garments may become contaminated by blood or other potentially infectious materials during the performance of duties, the employee shall take steps to ensure that clean garments are available.

Clothing, disposable PPE, and other items visibly contaminated with blood and not known to be contaminated with blood shall be treated as biohazardous waste regardless of the amount of blood present and shall be subject to regulated waste handling requirements. Contaminated items shall be placed in red bags or containers that meet the requirements of this procedure and shall be transported to a designated central collection point for biohazardous waste. Biohazardous waste shall be handled in accordance with local, state, and federal regulations.

1.88.3 Gloves

Disposable (single use) gloves, such as surgical or examination gloves, shall be replaced as soon as practical when contaminated, or as soon as feasible, if they are torn, punctured, or when their ability to function as a barrier is compromised.

Disposable (single use) gloves shall not be washed or decontaminated for re-use but shall be placed in an appropriately designated area or container for disposal in accordance with this procedure.

Utility gloves may be decontaminated for re-use if the integrity of the glove is not compromised. However, they shall be discarded if they are cracked, peeling, torn, punctured, or exhibit other signs of deterioration or when their ability to function as a barrier is compromised.

Disposable (single use) gloves, such as surgical or examination gloves, shall be worn under utility gloves to protect against penetration of fluids.

Gloves visibly contaminated with blood shall be considered infectious waste and shall be disposed of in red biohazard waste containers in the work area.

1.88.4 Eye and Face Protection

Masks in combination with eye protection devices, such as goggles / glasses with solid side shields or chin length face shields, shall be worn whenever splashes, spray, spatter, or droplets of blood or other potentially infectious materials may be generated and eye, nose, or mouth contamination can be reasonably anticipated.

Eye protection and face shields issued for the exclusive use of one employee shall be cleaned and disinfected after each day's use, or more often if necessary and following contamination. Those used by more than one employee shall be thoroughly cleaned and disinfected after each use.

Personal protective clothing shall be considered the same as PPE and shall be treated as such for the purposes of this procedure.

1.89 Housekeeping

Work areas shall be maintained in a clean and sanitary condition. Appropriate written schedules for cleaning and methods of decontamination based upon the location within the facility, type of surface to be cleaned, type of soil present, and tasks or procedures being performed in the area shall be developed and implemented.

All equipment and environmental and working surfaces shall be cleaned and decontaminated after contact with blood or other potentially infectious materials using any sterilization or disinfection procedures or sterilizing agent or high-level disinfectant that will kill viruses if used as directed.

All personnel covered under this procedure shall observe universal precautions in the performance of housekeeping duties and shall adhere to the requirements of this procedure for use of PPE to protect themselves and their fellow employees against unnecessary exposure.

1.90 Regulated Waste

Contaminated sharps shall be discarded immediately or as soon as feasible in containers that are:

- Closable.
- Puncture resistant.
- Constructed to contain all contents and prevent leakage of fluids during handling, storage, transport, or shipping.
- Labeled or color-coded (in accordance with this procedure). During use, containers for contaminated sharps shall be:
  - Easily accessible to personnel and located as close as feasible to the immediate area where sharps are to be used or can be reasonably anticipated to be found.
  - Maintained upright throughout use.
  - Replaced routinely and not be allowed to overfill.

When moving containers of contaminated sharps from the area of use, the container shall be:

- Closed immediately prior to removal or replacement to prevent spillage or protrusion of contents during handling, storage, transport, or shipping.
- Placed in a secondary container if outside contamination of the regulated waste container occurs or leakage is possible. The secondary container shall meet the criteria of this procedure.

Regulated waste shall be placed in containers that meet the criteria identified for contaminated sharps in this procedure.

Disposal of all regulated waste shall be handled in accordance with applicable Company procedures governing waste removal and regulations of the States and Territories.

Contaminated or potentially contaminated laundry shall be handled as little as possible with a minimum of agitation.

Contaminated or potentially contaminated laundry shall be bagged or containerized at the location where it was used and shall not be sorted or rinsed in the location of use.

Contaminated or potentially contaminated laundry shall be placed and transported in bags or containers labeled or color-coded in accordance with this procedure. However, when a facility utilizes universal precautions in the handling of all soiled laundry, alternative labeling or color-coding is sufficient if it permits all employees to recognize the containers as requiring compliance with universal precautions.

Whenever contaminated or potentially contaminated laundry is wet and presents a reasonable likelihood of soak-through or leakage of fluids from the bag or container, the laundry shall be placed and transported in bags or containers which prevent soak-through or leakage of fluids to the exterior.

Employees who have contact with contaminated or potentially contaminated laundry shall wear protective gloves and other appropriate PPE.

When a facility ships contaminated or potentially contaminated laundry offsite to a second facility that does not utilize universal precautions in the handling of all laundry, the facility generating the contaminated or potentially contaminated laundry shall place such laundry in bags or containers that are labeled or color-coded in accordance with this procedure.

1.91 Hepatitis B Virus (HBV) and Post-Exposure Evaluation and Follow Up

The Company shall make available the HBV vaccine, vaccination series, and post-exposure evaluation and follow up to all employees who have had an exposure incident.

All medical evaluations and procedures, including the HBV vaccine, vaccination series, and post-exposure evaluation and follow up shall be:

- Made available at no cost to the employee.
- Made available to the employee at a reasonable time and place.
- Performed by or under the supervision of a licensed physician or under the supervision of another licensed healthcare professional.
- Provided according to recommendations of the governing Public Health Service that are current at the time these evaluations and procedures take place, except as specified herein.

The Company shall ensure that employees covered under this procedure have been offered the Hepatitis B vaccination and have signed either consent or a declination form.

All laboratory tests shall be conducted by an accredited laboratory at no cost to the employee.

1.91.1 Hepatitis B Virus (HBV) Vaccination Unit

Persons at substantial risk for HBV who are demonstrated or judged likely to be susceptible should be vaccinated. The HBV vaccination is recommended for any previously unvaccinated healthcare worker who has a needle stick or other percutaneous accident with a sharp instrument or per mucosal (ocular or mucous membrane) exposure to blood.

An HBV vaccination shall be made available to employees after employees have received training required by local regulation and within 10 working days of initial assignment to tasks with occupational exposure, unless:

- The employee has previously received the complete HBV vaccination series.
- Antibody testing has revealed that the employee is immune.
- The vaccine is contraindicated for medical reasons.

Participation in a prescreening program shall not make a prerequisite for receiving the HBV vaccination.

If the employee initially declines the HBV vaccination, but at a later date while still covered under this procedure, decides to accept the vaccination, the HBV vaccination shall be made available to the employee at that time at no cost to the employee.

If a routine booster dose(s) of the HBV vaccine is recommended by the Public Health Service at a future date, such booster dose(s) shall be made available in accordance with local regulation and this procedure.

Copies of consent and declination forms shall be retained and shall be maintained in employee medical files in accordance with this procedure.

1.92 Reporting Occupational Exposure

Employees shall report all occurrences of occupational exposure as soon as feasible after the exposure. The Company will initiate the post-exposure evaluation and follow up process in response to reports of occupational exposure.

The following steps shall be taken in reporting occupational exposure to bloodborne pathogens or other potentially infectious materials:

- Employees shall notify their immediate supervisor as soon as feasible following an exposure incident.
- Employees shall complete an occupational exposure report.
- Employees shall sign the occupational exposure report and give the signed and completed form to the immediate supervisor for review and signoff.
- The supervisor shall immediately forward a copy of the report to management to initiate post-exposure evaluation and follow up.

Employees and supervisors may refer to the instructions contained in the Exposure Control Plan for reporting occupational exposure incidents to ensure that proper notifications and paperwork have been completed.

1.92.1 Post-Exposure Evaluation and Follow Up

Following a report of an exposure incident, the exposed employee shall immediately be offered confidential medical evaluation and testing.

Post-exposure evaluation and follow up shall consist of at least the following elements:

- Documentation of the route(s) of exposure.
- Identification, documentation, and testing of the source individual, unless it can be established that identification is infeasible or prohibited by state or local law.
- Collection and testing of blood for HBV and HIV serological status.
- Post-exposure prophylaxis as recommended by the Public Health Service when medically indicated.
- Counseling.
- Evaluation of reported illnesses.

If the employee consents to baseline blood collection but does not give consent at that time for HIV serological testing, the sample shall be preserved for at least 90 days. If, within 90 days of the exposure incident, the employee elects to have the baseline sample tested, such testing shall be done as soon as feasible.

After obtaining the exposed employee's consent for follow up testing, a sample of blood shall be collected and tested for HBV and/or HIV as soon as feasible following the exposure incident. The sample shall be collected and tested within 30 days of the exposure incident.

Post-exposure evaluation and follow up shall also include:

- Counseling.
- Evaluation of reported illnesses.

Following post-exposure evaluation and follow up, the exposed employee shall be provided with a copy of the evaluating healthcare professional's written opinion.

1.92.2 Post-Exposure Testing of the Source Individual

A good faith effort shall be made to both identify and obtain consent for HBV and HIV testing of the source individual.

The source individual's blood shall be collected and tested as soon as feasible and after consent is obtained to determine HBV and HIV infectivity.

If consent is not obtained, the Company shall establish that legally required consent cannot be obtained, and the source individual shall not be tested.

When the source individual's consent is not required by law, the source individual's blood, if available, shall be collected, tested, and the results documented. The condition, "if available," applies to blood samples that have been drawn from the source individual for other testing.

When the source individual is already known to be infected with HBV or HIV, testing for the source individual's known HBV or HIV status need not be repeated.

Results of the source individual's testing shall be made available to the exposed employee, and the exposed employee shall be informed of applicable laws and regulations concerning disclosure of the identity and infectious status of the source individual.

1.92.3 Post-Exposure Evaluation and Follow Up Results

Healthcare professionals who are responsible for evaluating employees following exposure incidents shall receive and review:

- A copy of the regulation.
- A description of the exposed employee's duties as they related to the exposure incident.
- Documentation of the route(s) of exposure and circumstances under which exposure occurred.
- Results of the source individual's blood testing, if available.

Maintenance of all medical records relevant to the treatment of the exposed employee, including vaccination status, is the responsibility of the Company.

Following evaluation of the exposed employee and testing of the source individual (if testing is done), the exposed employee shall be provided with a report, at minimum, containing:

- Documentation of the route(s) of exposure and the circumstances under which the exposure incident occurred.

- Identification and documentation of the source individual unless it can be established that identification is infeasible or prohibited by state or local law.

The exposed employee shall be provided with a copy of the evaluating healthcare professional's written opinion within 15 days of the completion of the evaluation.

The healthcare professional's written opinion for an HBV vaccination shall be limited to whether an HBV vaccination is indicated for an employee and whether the employee has received such vaccination.

The healthcare professional's written opinion for post-exposure evaluation and follow up shall be limited to:

- A statement that the employee has been informed of the results of the evaluation.
- A statement that the employee has been told about medical conditions resulting from exposure to blood or other potentially infectious materials which require further evaluation or treatment.

The healthcare professional's other findings or diagnoses shall remain confidential and shall not be included in the written report.

Following all evaluations and testing, an evaluation shall be made of the “failures of control” at the time of the incident using the data compiled to identify and make recommendations for correction of problems in order to prevent recurrence of similar incidents.

1.93 Communications of Hazards to Employees

Warning labels shall be affixed to warn employees of items containing blood or other potentially infectious material:

- Containers of regulated waste, refrigerators, and freezers containing blood or other potentially infectious materials which require further evaluation or treatment.
- Other containers used to store, transport, or ship blood or other potentially infectious materials, except where red bags or red containers have been substituted for labels.
- Labels required by this section shall include a picture of the biohazard symbol shown in the Exposure Control Plan.
- Labels shall be fluorescent orange, orange-red, or predominantly so, with lettering or the symbols in a contrasting color.
- Labels required by this section shall be affixed as close as feasible to the container by adhesive, string, wire, or other method that prevents their loss or unintentional removal.

- Red bags or red containers may be substituted for labels.
- Individual containers of blood or other potentially infectious materials that are placed in a labeled container during storage, transport, shipment, or disposal are exempted from the labeling requirement.
- Contaminated equipment shall be labeled in accordance with this procedure and shall also state those portions of the equipment that remain contaminated.

1.94 Information and Training

Employees who have the potential to be exposed to bodily fluids must be trained on the subject of bloodborne pathogens upon initial hire and annually thereafter. The training must be documented and retained for a minimum of 3 years.

Communicating hazards to employees and providing training and information are paramount in the implementation of this procedure since protective measures such as PPE and proper work practices will not be effective unless employees are instructed in their correct use. Training is also an important factor in risk reduction because not all employees are aware of the risks they may face in the workplace. Information programs can increase employee acceptance of the HBV vaccine and employee compliance with policies regarding PPE.

The Company shall be responsible for ensuring that employees covered under this procedure participate in the Bloodborne Pathogens Awareness Training Program, which shall be provided during working hours at no cost to the employee.

Employees that are exposed to bloodborne pathogens must undergo training applicable in accordance with local jurisdiction.

Training shall be provided:

- At the time of initial assignment.
- At least annually thereafter.

Annual training for all employees shall be provided within one year of their previous training.

Additional training shall be provided when changes such as modification of tasks or procedures or institution of new tasks or procedures affect the employee's occupational exposure. The additional training may be limited to addressing the new exposures created.

Material appropriate in content and vocabulary to educational level, literacy, and language of employees shall be used.

The training program shall contain at a minimum the following elements:

- An accessible copy of the regulatory text and an explanation of its contents.
- A general explanation of the epidemiology, symptoms, and modes of transmission of bloodborne pathogens and diseases.
- An explanation of an Exposure Control Plan and how the employee can obtain a copy of the written plan.
- An explanation of the appropriate methods for recognizing tasks and other activities that may involve exposure to blood and other potentially infectious materials.
- An explanation of the use and limitations of methods that will prevent or reduce exposure, including appropriate engineering controls, work practices, PPE, and clothing.
- Information on the types, proper use, location, removal, handling, decontamination, and disposal of PPE and clothing.
- An explanation of the basis for selection of PPE and clothing.
- Information on the HBV vaccine, including information on its efficacy, safety, method of administration, the benefits of being vaccinated, and that the vaccine and vaccination will be offered free of charge.
- Information on the appropriate actions to take and persons to contact in an emergency involving blood or other potentially infectious materials.
- An explanation of the procedure to follow if an exposure incident occurs, including the method of reporting the incident and the medical follow up that will be made available.
- Information on the post-evaluation and follow up that the Company is required to provide for the employee following an exposure incident.
- An explanation of the signs, labels, and color-coding system being utilized by the Company as defined in this procedure.
- An opportunity for interactive questions and answers with the trainer.

The person conducting the training shall be knowledgeable in the subject matter covered by the elements contained in the training program as it relates to the workplace that the training will address.

The person conducting the training shall be required to ensure that the training program and training records meet the requirements of and are maintained in accordance with regulation and this procedure.

1.95 Recordkeeping

1.95.1 Medical Records

Accurate medical records for each employee with occupational exposure shall be maintained for at least the duration of employment plus 30 years. These records shall be kept by the Company. The medical provider shall not be relied upon to keep the records for this timeframe.

The employee record shall include:

- The name and social security number of the employee.
- A copy of the employee's vaccination status including the date of all vaccinations and any medical records relative to the employee's ability to receive the vaccination as defined in this procedure.
- A copy of all results of examinations, medical testing, and follow up procedures as defined in this procedure.
- A copy of the healthcare professional's written report as required by this procedure.
- A copy of the information provided to the healthcare professional.

Employee medical records shall be kept confidential and not disclosed or reported without the employee's express written consent to any person within or outside the workplace, except as required by this procedure or as may be required by law.

All records required to be maintained by this procedure shall be made available upon request for examination and copying, to the subject employee or anyone having written consent of the subject employee, to the responsible party. Records shall be made available in accordance with the regulatory requirements.

Access to personal information shall be controlled in accordance with applicable legal, regulatory, and Company requirements (e.g., FFD Rule, Privacy Act, 29 CFR1910.1020).

1.95.2 Training Records

Training records shall include:

- The dates of the training sessions.
- The contents or a summary of the training sessions.
- The names and qualifications of the person(s) conducting the training.

- The names and job titles of all persons attending the training sessions.

Training records shall be maintained for 3 years from the date on which the training occurred in the Company designated electronic learning management system.

1.96 Regulatory Reporting Requirements

As a basic requirement, all work-related needle stick injuries and cuts from sharp objects that are contaminated with another person's blood or other potentially infectious material shall be recorded. The case shall be entered on the OSHA 300 Log as an injury. To protect the employee's privacy, the employee's name shall not be entered on the OSHA 300 Log.

- If the date of the event or exposure is known, the original injury shall be recorded with the date of the event or exposure.
- If there are multiple events or exposures, the most recent injury shall be recorded with the date that seroconversion is determined.

10. Business Continuity Program

1.97 Purpose and Scope

The purpose of this program is to set out our approach at a corporate level to responding to an incident and the key activities both in recovering critical service delivery and our community role in assisting in an emergency.

This is at a high-level, detailed plans for individual services are retained separately.

This program applies to all T&T Industrial, Inc. employees.

1.98 Objectives

The business continuity objectives shall be measurable, be monitored, be communicated, and be updated as appropriate. Objectives and targets can be measured qualitatively and/or quantitatively. Written documentation should be maintained of these objectives. When planning business continuity objectives, what will be done, what resources will be required, who will be responsible, when it will be completed, and how the results will be evaluated shall be determined. Goals and objectives include:

- Risk Prevention, Reduction, and Mitigation
- Organizational Resilience Enhancement
- Financial, Operational, and business continuity requirements
- Compliance with legal requirements
- Continual Improvement

1.99 Training

Employees who play a role in the business continuity / organizational resilience plan must be trained and prepared to fulfill their role. Training and documentation should be maintained in writing. Employees who are expected to play a role in the business continuity / organizational resilience program shall be trained.

1.100 Roles and Responsibilities

The individual department business continuity / resilience plans should clearly state the roles and/or responsibilities of those who are responsible for implementing continuity strategies.

1.101 Communication

The types of communications relevant to the business continuity management system shall be determined, taking into consideration what will be communicated, when to communicate, with whom to communicate, how to communicate and who at the company will be responsible for communication. This communication system should be designed to keep internal employees informed on issues related to business continuity.

External clients shall be notified in the event of a business disruption.

1.102 Risk Assessment

Analysis of the potential risks and impacts that could occur from disruptions to normal business operations must be performed. These analyses identify the potential areas where the Company will be most severely disrupted, the legal obligations, the key activities that help the Company provide goods and services, the persons, and other entities dependent on the Company, the impact of the being unable to deliver on its own obligations, and anything else that may be affected in a significant disruption.

1.103 Review

The effectiveness of the Company's response to incidents after an exercise or incident occurs must be evaluated. Any opportunities for improvement to the business continuity management system overall or its components and the roles of the various stakeholders in ensuring the business continuity management system is in its best form possible ahead of a disruptive incident occurring must be identified.

The business continuity plan must be reviewed annually to ensure that the procedures put in place to maintain business operations are appropriate and effective.

T&T Industrial, Inc.
Cadmium Program

11. CADMIUM PROGRAM

1.104 Purpose and Scope

The purpose of this program is to protect employees against potential exposure to cadmium in accordance with regulation on cadmium standards. The program defines the requirements to manage potential exposure of employees to cadmium where it may be encountered.

This program applies to all T&T Industrial, Inc. employees with potential exposure to cadmium.

1.105 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances - Cadmium

1.106 Cadmium Program

Cadmium and its compounds are highly toxic and exposure to this metal is known to cause cancer. It targets the body's cardiovascular, renal, gastrointestinal, neurological, reproductive, and respiratory systems.

Sources of cadmium exposure include smelting and refining of metals, manufacturing batteries, plastics, coatings, and solar panels, electroplating, metal machining, welding, painting, landfill operations, the recycling of electronic parts, the recycling of plastics, composting and waste collection, and incineration of municipal waste.

1.107 Permissible Exposure Limit (PEL)

No employee shall be exposed to an airborne concentration of cadmium in excess of 2.5 micrograms per cubic meter of air (2.5 $\mu g/m^3$), calculated as an 8-hour time-weighted average (TWA).

1.108 Roles and Responsibilities

The competent person plays a crucial role in ensuring the health and safety of workers who may be exposed to cadmium in the workplace. The competent person must have the necessary knowledge, training, and experience to effectively fulfill their responsibilities in accordance with local jurisdiction. Typical roles and responsibilities of a competent person include:

- Risk Assessment: Conducting thorough risk assessments to identify potential sources of cadmium exposure and evaluating the associated risks to workers.

- Exposure Monitoring: Implementing a comprehensive monitoring program to measure and assess workers' exposure levels to cadmium. This may involve air sampling, biological monitoring, or other appropriate methods.
- Engineering Controls: Recommending and overseeing the implementation of engineering controls to minimize or eliminate cadmium exposure. This may include ventilation systems, enclosure or isolation of processes, and substitution of less hazardous materials.
- Work Practices: Developing and enforcing safe work practices and procedures for handling cadmium, including proper storage, handling, use, and disposal methods. This may involve establishing restricted areas and implementing decontamination procedures.
- Training and Education: Providing training and education programs to workers on the hazards of cadmium exposure, safe work practices, and the proper use of personal protective equipment (PPE). Ensuring that workers are informed about the risks associated with cadmium and understand the necessary precautions.
- Medical Surveillance: Implementing a medical surveillance program to monitor the health of workers exposed to cadmium. This may include pre-placement and periodic medical examinations, biological monitoring, and maintaining medical records.
- Personal Protective Equipment (PPE): Assessing the need for and recommending appropriate PPE, such as respirators, gloves, protective clothing, and eyewear. Ensuring that workers are trained in the proper use, maintenance, and limitations of PPE.
- Recordkeeping and Documentation: Maintaining accurate records of exposure monitoring results, medical surveillance records, training documentation, and any incidents or accidents related to cadmium exposure. This includes documenting control measures implemented and reviewing the effectiveness of the program.
- Regulatory Compliance: Staying up to date with relevant regulations and standards regarding cadmium exposure and ensuring compliance with applicable laws, regulations, and guidelines.
- Program Evaluation and Improvement: Regularly reviewing and evaluating the effectiveness of the cadmium exposure program, identifying areas for improvement, and implementing corrective actions as necessary.

1.109 Training

A training program shall be instituted for all employees who are potentially exposed to cadmium. Employees shall participate in the training and records of content shall be maintained. Training shall be provided prior to initial assignment and at least annually.

1.110 Exposure Monitoring

Initial monitoring shall be performed to determine the 8-hour TWA exposure for each employee to characterize full shift exposure on each shift, for each job classification, in each work area.

If monitoring reveals employee exposures to be at or above the action level, periodic monitoring shall be performed at least every 6 months.

Additional monitoring shall be performed when there has been any change in the production process, raw materials, equipment, personnel, work practices, or control methods that may result in new or additional exposures to cadmium, or when there is any reason to believe that new or additional exposures have occurred.

1.111 Regulated Areas

Regulated areas shall be established wherever an employee's exposure to airborne concentrations of cadmium is, or can reasonably be expected to be, in excess of the PEL.

1.112 Controls

Engineering and work practice controls shall be used to reduce and maintain employee exposure to cadmium to or below the PEL unless it can be demonstrated that such controls are not feasible.

Procedures shall be developed and implemented to minimize employee exposure to cadmium during the maintenance of ventilation systems and the changing of filters. Examples include pre-work assessments, personal protective equipment (PPE), engineering controls, specific work practices, housekeeping and decontamination, training, and medical surveillance.

1.113 Compliance Program

Where the PEL is exceeded, a written compliance program shall be established and implemented to reduce employee exposure to or below the PEL by means of engineering and work practice controls. To the extent that engineering and work practice controls cannot reduce exposures to or below the PEL, the use of appropriate respiratory protection shall be included to achieve compliance with the PEL.

The following are OSHA requirements for the written program:

- Description of each operation where cadmium is emitted.
- A description of the specific means that will be employed to meet compliance including engineering plans.
- A report of technology considered in meeting the PEL.

- Air monitoring data.
- A detailed schedule for implementation.
- A work practice program.
- A written plan for emergency situations.
- Other relevant information.

The written program shall be reviewed and updated annually or more often to reflect significant changes in compliance status.

The program shall be provided for examination and copying upon request of affected employees, their representatives, the OSHA Assistant Secretary, and the OSHA Director.

1.114 Respiratory Protection

Appropriate respirators shall be selected and provided to employees.

1.115 Emergency Planning

A written plan for dealing with emergency situations involving substantial releases of airborne cadmium shall be developed and implemented.

1.116 Personal Protective Equipment (PPE)

Where a hazard is present or is likely to be present from skin or eye contact with cadmium, appropriate personal protective clothing and equipment shall be provided at no cost to employees. Employees shall use such clothing and equipment.

1.117 Medical Surveillance

A medical surveillance program shall be instituted for all employees who are or may be exposed to cadmium at or above the action level unless the employee is not exposed 30 or more days per year.

12. Cold Weather Safety Program

1.118 Purpose and Scope

The purpose of this program is to provide information on cold weather safety and precautions for working in cold weather to prevent cold stress.

This program applies to all T&T Industrial, Inc. employees with the potential to work in cold weather.

1.119 Resources

Number	Title
30 CFR 56.11016	Snow and Ice on Walkways and Travelways

1.120 Cold Stress Prevention

Cold weather conditions can be hazardous to the health and safety of employees, endanger the stability of the body system, and cause problems such as hypothermia and frostbite. It is of vital importance that adequate precautions are taken to alleviate the effect of cold environments and to ensure that personnel can work safely and efficiently.

Before work in cold environments can begin, an assessment shall be conducted to assess jobs and tasks, wind chill and environmental conditions. Employees physical condition shall be monitored to assess risk of cold stress.

1.121 Wind Chill Index

- Air temperature alone is not sufficient to assess the cold hazard in certain environments. Therefore, the Wind Chill Index, along with the air temperature, shall be used. Heat loss from convection is the greatest and most deceptive factor in loss of body heat.
- The Wind Chill Index is the cooling effect of any combination of temperature and wind velocity or air movement.
- The Wind Chill Index considers the wind velocity. If there is no anemometer (to measure wind speed), the following is a suggested guide for estimating wind speed:
    - 5 mph: light flag moves
    - 10 mph: light flag fully extended
    - 15 mph: raises newspaper sheet

  - 20 mph: causes blowing and drifting snow.

- The Wind Chill Index should be used to evaluate the cold hazard.
- The Wind Chill Index does not consider the body part(s) exposed to cold, the level of activity effect on the body's heat production, or the amount of clothing worn.

1.122 Control Measures

1.122.1 Engineering Controls

Cold stress can be reduced by the following controls:

- General or spot heating should be used to increase temperature.
- If work is being performed with bare hands for 10 or more minutes, special provisions shall be made to keep hands warm. Warm air jets, radiant heaters, or contact warm heaters shall be supplied.
- The work area shall be shielded if the air velocity is increased by the wind, draft, or ventilation equipment.
- At temperatures below 40 °F (4 °C), metal handles of tools and control bars shall be covered with thermal insulation. This can include insulation from gloves provided that the gloves are not wet.
- When necessary, equipment and processes shall be substituted, isolated, relocated, or redesigned to reduce the cold stress.
- Power tools, hoists, cranes, and lifting aids shall be used to reduce the metabolic workload.
- Heated warming shelters such as tents, cabins, automobiles, or trucks shall be made available if work is performed continuously in an equivalent chill temperature of 30 °F (-1 °C) or below. Employees shall be encouraged to use them.
- Regularly used walkways and travel ways shall be sanded, salted, or cleared of snow and ice as soon as practicable.
- Regular inspections on cold weather supplies (e.g., hand warmers, jackets, shovels, etc.) shall be carried out to ensure that supplies are always in stock.

1.122.2 Administrative Controls

These controls include the following work practices and rules designed to reduce total cold stress burden on the body. See the Appendix – Work / Warm-up Schedule for a 4-Hour Shift.

- Scheduling a work rest to reduce the peak of cold stress, enforcing scheduled work breaks out of the cold.
- Proper hydration must be made available to employees exposed to cold weather temperature extremes. Urging frequent intake of warm, sweet, caffeine-free, nonalcoholic drinks or soup provided at regular intervals.
- Consume warm, high calorie food to maintain energy reserves.
- Scheduling the coldest, heavy work for the warmest part of the day.
- Moving work to warmer areas whenever possible; pre-planning the activities prior to entering the cold environment.
- Assigning extra employees to highly demanding jobs.
- Allowing employees to pace themselves and take extra work breaks when needed to avoid fatigue.
- Providing relief employees for break times.
- Teaching employees the basic principles of preventing cold stress and emergency response to cold stress.
- Informing employees of the dangers associated with working around unstable snow buildup, sharp icicles, and ice dams and know how to prevent accidents caused by them.
- Maintaining protective supervision or a buddy system for those who work at 20 °F (-6 °C) or below.
- Allowing new employees time to adjust to conditions before they work fulltime in cold environments.
- Arranging work to minimize sitting still or standing for long periods at a time.
- Reorganizing work procedures to ensure as much of a job as possible is performed in a warm environment.
- Including the weight and bulkiness of clothing when estimating work performance criteria.

1.122.3 Personal Protective Equipment

It is the responsibility of the employee to dress in the clothing appropriate to the expected work conditions. The correct clothing shall be addressed in the following manner:

- It is important to preserve the air space between the body and the outer layer of clothing to retain body heat.
- It is most important to protect the feet, hands, head, and face. The hands and feet are the farthest from the heart and become cooled most easily. Keeping the head covered is important because as much as 40% of heat is lost when the head is exposed to the elements.
- All clothing and equipment shall be fitted properly and not interfere with circulation.
- Clothing should be loose fitting. Tight clothing of synthetic fabrics interferes with evaporation. At least three layers of clothing shall be worn.
  - Wear at least three layers of clothing. An inner layer of wool, silk or synthetic to wick moisture away from the body. A middle layer of wool or synthetic to provide insulation even when wet. An outer wind and rain protection layer that allows some ventilation to prevent overheating.
- A change of dry clothing should be kept in case clothes become wet.
- Do not underestimate the wetting effects of perspiration. Oftentimes wicking and venting of the body's sweat and heat are more important than protecting from rain or snow.
- Socks with high wool content are best. When two pairs are worn, the inside sock should be smaller, and made of cotton. If needed, wool socks can also double for mittens.
- Wool or thermal trousers are preferred. The best kind is either quilted or specially lined.
- Tight belts are not recommended because they cut off the circulation at the waist. Suspenders are encouraged where practical.
- Trousers should fit over the top of the boot to prevent snow and ice from entering.
- Boots should be felt-lined (insulated), rubber-bottomed, and leather-topped with a removable felt insole. Boots should be waterproofed and socks should be changed whenever the sock is sweat soaked.
- A wool sweater over a cotton shirt should be worn. Tops should be worn in a layering effect to ensure proper insulation.

- An anorak or snorkel coat or arctic parka should fit loosely and have a drawstring at the waist. The sleeves should fit snugly. A hood prevents the escape of warm air from the neck and tunnels the warm air past the face to give a slightly warmer breathing air. A wool cap should also be worn under the hood.
- Flame Resistant (FR) outer layers shall be worn where required.
- When wearing a hard hat, liners should be worn.
- A face mask or scarf is vital when working in cold wind. A ski mask gives better visibility than a snorkel hood. Face protectors should be removed periodically to check for frostbite.
- Safety glasses with side shields should be worn when outside. Special safety goggles to protect against ultraviolet light and glare are required when there is snow coverage that could cause a potential eye exposure hazard.

1.123 Cold Disorder Symptoms

1.123.1 Frostnip

This occurs when the face or extremities are exposed to cold wind, causing the skin to turn white.

1.123.2 Frostbite

Frostbite occurs when the skin actually freezes and loses water. In severe cases, amputation of the frostbitten area may be required. While frostbite usually occurs when the temperatures are 30° F or lower, wind chill factors can allow frostbite to occur in above freezing temperatures. Frostbite typically affects the extremities, particularly the feet and hands. The affected body part will be cold, tingling, stinging, or aching followed by numbness. Skin color turns red, then purple, then white, and is cold to the touch. There may be blisters in severe cases.

1.123.3 Trenchfoot or Immersion Foot

Trench Foot or immersion foot is caused by having feet immersed in cold water at temperatures above freezing for long periods of time. It is similar to frostbite but considered less severe. Symptoms usually consist of tingling, itching, or burning sensation. Blisters may be present.

1.123.4 Hypothermia

Hypothermia occurs when body heat is lost faster than it can be replaced. When the core body temperature drops below the normal 98.6° F to around 95° F, the onset of symptoms normally begins. The person may begin to shiver and stomp their feet to generate heat. Employees may lose coordination, have slurred speech, and fumble with items in the hand. The skin will likely be pale and cold. Pain in extremities can be the first warning of overexposure. The symptoms include uncontrolled shivering fits, sense of cold, slow heartbeat, vague or slow speech, glassy stare, apathy, memory lapses, incoherence, drowsiness, cool skin, slow irregular breathing, sometimes irregular pulse, weakened pulse, apparent exhaustion, and fatigue after rest. In an industrial environment, decreased mental acuity due to hypothermia can present additional risk, as the affected worker(s) are not focused on the hazards of the job and may put themselves or other personnel at risk.

Because many of the symptoms are behavioral and are not often noticed by the affected person, personnel should always watch for these symptoms in their co-workers. It is especially important to watch out for employees who are not dressed appropriately for the weather conditions (e.g., fewer layers, no hat or gloves, wet clothing, or an outer layer which is inadequate for the conditions). Personnel are responsible for stopping the work if these symptoms are noticed so that the affected person can be brought into a warmer environment for evaluation.

1.124 First Aid

All employees who are required to perform work in cold conditions must be trained on how to administer first aid treatment on cold induced injuries or illnesses.

1.124.1 Frostbite

- Never rub affected area. Rubbing may cause further damage to soft tissue.
- Warm area gently by soaking in water. The water should start out cold and be warmed up every five minutes by adding water that is 5 ºF (-15 ºC) warmer. Do not immerse affected part in water that is more than 105 ºF (40 ºC). If a thermometer is not available, test the water temperature with your hand. If the water temperature is uncomfortable, it is too hot.
- Keep the affected area under water until it looks red and feels warm.
- Loosely bandage the area with dry, sterile dressing. If fingers and toes are frostbitten, place cotton or gauze between them before applying the loose bandage.
- Do not break blisters.
- Get professional help immediately.

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1.124.2 Hypothermia

- Remove any wet clothing and dry the injured person.
- Warm the body gradually by wrapping the injured person in blankets or putting on dry clothing and moving the individual to a warmer place. Do not warm body quickly by immersing the person in hot water. Rapid warming can cause dangerous heart problems. If available, apply heating pads or other heating source to the body. Keep a protective barrier, such as towel, blanket, or clothing between heat source and injured person to avoid burning the individual.
    - If the injured person is alert, give warm liquid to drink. Never give liquids to an individual who is unconscious or semi-conscious.
- Handle the patient gently.
- Get professional help immediately.

1.125 Training

Personnel shall be instructed in cold stress prevention. Training shall be conducted before initial exposure and annually thereafter.

Training shall include the following instructions:

- Environmental and workplace conditions that can lead to cold stress
- Proper re-warming procedures and appropriate first aid treatment specifically for cold induced injuries or illnesses.
- Proper clothing practice.
- Proper use of warming shelters.
- Recognition of signs and symptoms of impending cold conditions such as hypothermia or excessive cooling of the body (even when shivering does not occur), frostnip, or frostbite.
- Safe work procedures such as the buddy system, vehicle breakdown procedures, and proper eating and drinking habits for working in the cold.

Appendix 6 Work / Warm-Up Schedule for a 4-Hour Shift

Work/Warm-up Schedule for a 4-Hour Shift

Air Temperature--Sunny Sky		No Noticeable Wind		5 mph Wind		10 mph Wind		15 mph Wind		20 mph Wind	
°C (approximate)	°F (approximate)	Maximum Work Period	Number of Breaks	Maximum Work Period	Number of Breaks	Maximum Work Period	Number of Breaks	Maximum Work Period	Number of Breaks	Maximum Work Period	Number of Breaks
-26 to -28	-15 to -19	(Normal Breaks) 1		(Normal Breaks) 1		75 min	2	55 min	3	40 min	4
-29 to -31	-20 to -24	(Normal Breaks) 1		75 min	2	55 min	3	40 min	4	30 min	5
-32 to -34	-25 to -29	75 min	2	55 min	3	40 min	4	30 min	5	Non-emergency work should cease	
-35 to -37	-30 to -34	55 min	3	40 min	4	30 min	5	Non-emergency work should cease			
-38 to -39	-35 to -39	40 min	4	30 min	5	Non-emergency work should cease					
-40 to -42	-40 to -44	30 min	5	Non-emergency work should cease							
-43 & below	-45 & below	Non-emergency work should cease									

Schedule applies to any 4-hour work period with moderate to heavy work activity; with warm-up periods of ten (10) minutes in a warm location and with an extended break (e.g. lunch) at the end of the 4-hour work period in a warm location.

Adapted from ACGIH 2012 TLVs

13. Compressed Air Safety Program

1.126 Purpose and Scope

The purpose of this program is to provide information on the properties and hazards of working with compressed air.

This program applies to all T&T Industrial, Inc. employees who work with or around compressed air.

1.127 Resources

Number	Title
29 CFR 1910 Subpart P	Hand and Portable Powered Tools and Equipment
29 CFR 1926 Subpart I	Power-Operated Hand Tools
29 CFR 1910 Subpart M	Compressed Gas and Compressed Air Equipment-Air Receivers

1.128 Compressed Air Safety Program

Because working with compressed air is so commonplace, people fail to recognize it as the potential hazard that it is. As little as 12 p.s.i. can be deadly in the wrong circumstances and a small slip up can lead to big problems.

People shall not be cleaned off with compressed air. The regulations relate to the cleaning of objects or items, (e.g., blow drying parts, etc.).

It is prohibited to use compressed air to clean clothing while worn or any part of the body under any circumstances.

Compressed air shall not be used for cleaning or blow down activities unless air pressure is regulated to below 30 p.s.i. and areas have been isolated from pedestrian traffic and effective chip guarding and personal protective equipment is implemented.

1.129 Guarding

The use of protective cone air nozzles is generally acceptable for protection of the operator; however, barriers, baffles or screens may be required to protect other employees near the operator if they are exposed to flying chips or particles.

1.130 Safety Valves

All safety valves shall be tested frequently and at regular intervals to ensure they are in proper operating condition and that they cannot be made inoperable by any means.

Safety valves, indicating / controlling devices, and other safety appliances shall be constructed, located, and installed so they cannot be rendered inoperative by any means.

1.131 Inspections

Compressed air cylinders shall be inspected regularly according to manufacturers recommended methods and frequency.

The outside of the cylinder shall be inspected for:

- Cracks
- Leaks
- Bulging
- Defective valves
- Signs of physical abuse
- Fire or heat damage
- Detrimental rusting or corrosion

Any cylinder that has a crack or leak, is bulged, has a defective valve, or a leaking or defective pressure relief device, or bears evidence of physical abuse, fire or heat damage, or detrimental rusting or corrosion, shall not be used, filled, or offered for transportation.

1.132 Air Receivers

All air receivers shall be equipped with a readily visible indicating pressure gauge that is equipped with one or more spring-loaded safety valves.

The total relieving capacity of these safety valves should be such as to prevent pressure in the receiver from exceeding the maximum allowable working pressure of the receiver by more than 10%.

The drain valve on air receivers shall be opened and the receiver completely drained frequently and at such intervals as to prevent the accumulation of excessive amounts of liquid in the receiver.

1.133 Training

Employees handling compressed air shall be adequately trained in the inherent hazards as well as proper handling, storage, and use.

14. Concrete and Masonry Program

1.134 Purpose and Scope

The purpose of this program is to provide specific requirements and safety principles to ensure that concrete and masonry operations are conducted safely and effectively.

This program applies to all T&T Industrial, Inc. employees that work in concrete and masonry operations.

1.135 Resources

Number	Title
29 CFR 1926 Subpart Q	Concrete and Masonry Construction

1.136 Concrete and Masonry

This document sets forth requirements with which construction sites must comply to protect construction employees from accidents and injuries resulting from the premature removal of formwork, the failure to brace masonry walls, the failure to support precast panels, the inadvertent operation of equipment, and the failure to guard reinforcing steel.

1.137 Construction Loads

No construction loads shall be placed on a concrete structure or portion of a concrete structure unless it is determined, based on information received from a person who is qualified in structural design, that the structure or portion of the structure is capable of supporting the loads.

1.138 Reinforcing Steel

All protruding reinforcing steel, onto and into which employees could fall, shall be guarded to eliminate the hazard of impalement.

1.139 Post-Tensioning Operations

No employee (except those essential to the post-tensioning operations) shall be permitted to be behind the jack during tensioning operations.

Signs and barriers shall be erected to limit employee access to the post-tensioning area during tensioning operations.

1.140 Concrete Buckets

No employee shall be permitted to ride concrete buckets.

Concrete buckets equipped with hydraulic or pneumatic gates shall have positive safety latches or similar safety devices installed to prevent premature or accidental dumping.

Concrete buckets shall be designed to prevent concrete from hanging up on top and the sides.

1.141 Working Under Loads

No employee shall be permitted to work under concrete buckets while buckets are being elevated or lowered into position.

To the extent practical, elevated concrete buckets shall be routed so that no employee, or the fewest number of employees, are exposed to the hazards associated with falling concrete buckets.

1.142 Personal Protective Equipment (PPE)

No employee shall be permitted to apply a cement, sand, and water mixture through a pneumatic hose unless the employee is wearing protective head and face equipment.

1.143 Requirements for Equipment and Tools

1.143.1 Bulk Cement Storage

Bulk storage bins, containers, and silos shall be equipped with the following:

- Conical or tapered bottoms
- Mechanical or pneumatic means of starting the flow of material.

No employee shall be permitted to enter storage facilities unless the ejection system has been shut down, locked out, and tagged to indicate that the ejection system is not to be operated.

1.143.2 Concrete Mixers

Concrete mixers with one cubic yard (-8 m3) or larger loading skips shall be equipped with the following:

- A mechanical device to clear the skip of materials.
- Guardrails installed on each side of the skip.

1.143.3 Power Concrete Trowels

Powered and rotating type concrete troweling machines that are manually guided shall be equipped with a control switch that will automatically shut off the power whenever the hands of the operator are removed from the equipment handles.

1.143.4 Concrete Buggies

Concrete buggy handles shall not extend beyond the wheels on either side of the buggy.

1.143.5 Concrete Pumping Systems

Concrete pumping systems using discharge pipes shall be provided with pipe supports designed for 100% overload.

Compressed air hoses used on concrete pumping system shall be provided with positive fail-safe joint connectors to prevent separation of sections when pressurized.

1.143.6 Tremies

Sections of tremies and similar concrete conveyances shall be secured with wire rope (or equivalent materials) in addition to the regular couplings or connections.

1.143.7 Bull Floats

Bull float handles used where they might contact energized electrical conductors, shall be constructed of nonconductive material, or insulated with a nonconductive sheath whose electrical and mechanical characteristics provide the equivalent protection of a handle constructed of nonconductive material.

1.143.8 Masonry Saws

Masonry saws shall be guarded with a semicircular enclosure over the blade.

A method for retaining blade fragments shall be incorporated in the design of the semicircular enclosure.

1.143.9 Lockout / Tagout Procedures

No employee shall be permitted to perform maintenance or repair activity on equipment (such as compressors mixers, screens, or pumps used for concrete and masonry construction activities) where the inadvertent operation of the equipment could occur and cause injury, unless all potentially hazardous energy sources have been locked out and tagged.

Tags shall read "Do Not Start" or similar language to indicate that the equipment is not to be operated.

1.144 Requirements for Cast-in-Place Concrete

1.144.1 General Requirements for Formwork

Formwork shall be designed, fabricated, erected, supported, braced, and maintained so that it will be capable of supporting without failure all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork. Formwork, which is designed, fabricated, erected, supported, braced, and maintained in conformance with ANSI A10.9-1983 will be deemed to meet the requirements of this paragraph.

Drawings or plans, including all revisions, for the jack layout, formwork (including shoring equipment), working decks, and scaffolds, shall be available at the jobsite.

1.144.2 Shoring and Reshoring

All Shoring equipment (including equipment used in reshoring operations) shall be inspected prior to erection to determine that the equipment meets the requirements specified in the formwork drawings.

Shoring equipment found to be damaged such that its strength is reduced to less than that required shall not be used for shoring.

Erected shoring equipment shall be inspected immediately prior to, during, and immediately after concrete placement.

Shoring equipment that is found to be damaged or weakened after erection, such that its strength is reduced to less than that required, shall be immediately reinforced.

The sills for shoring shall be sound, rigid, and capable of carrying the maximum intended load.

All base plates, shore heads, extension devices, and adjustment screws shall be in firm contact, and secured when necessary, with the foundation and the form.

Eccentric loads on shore heads and similar members shall be prohibited unless these members have been designed for such loading.

Whenever single post shores are used one on top of another (tiered), the work shall comply with the following specific requirements in addition to the general requirements for formwork:

- The design of the shoring shall be prepared by a qualified designer and the erected shoring shall be inspected by an engineer qualified in structural design.
- The single post shores shall be vertically aligned.
- The single post shores shall be spliced to prevent misalignment.
- The single post shores shall be adequately braced in two mutually perpendicular directions at the splice level. Each tier shall also be diagonally braced in the same two directions.
- Adjustment of single post shores to raise formwork shall not be made after the placement of concrete.
- Reshoring shall be erected, as the original forms and shores are removed, whenever the concrete is required to support loads in excess of its capacity.

1.144.3 Vertical Slip Forms

The steel rods or pipes on which jacks climb or by which the forms are lifted shall be:

- Specifically designed for that purpose.
- Adequately braced where not encased in concrete.

Forms shall be designed to prevent excessive distortion of the structure during the jacking operation.

All vertical slip forms shall be provided with scaffolds or work platforms where employees are required to work or pass.

Jacks and vertical supports shall be positioned in such a manner that the loads do not exceed the rated capacity of the jacks.

The jacks or other lifting devices shall be provided with mechanical dogs or other automatic holding devices to support the slip forms whenever failure of the power supply or lifting mechanism occurs.

The form structure shall be maintained within all design tolerances specified for plumbness during the jacking operation.

The predetermined safe rate of lift shall not be exceeded.

1.144.4 Reinforcing Steel

Reinforcing steel for walls, piers, columns, and similar vertical structures shall be adequately supported to prevent overturning and to prevent collapse.

Measures shall be taken to prevent unrolled wire mesh from recoiling. Such measures may include, but are not limited to, securing each end of the roll, or turning over the roll.

1.144.5 Removal of Formwork

Forms and shores (except those used for slabs on grade and slip forms) shall not be removed until it is determined that the concrete has gained sufficient strength to support its weight and superimposed loads. Such determination shall be based on compliance with one of the following:

- The plans and specifications stipulate conditions for removal of forms and shores, and such conditions have been followed, or
- The concrete has been properly tested with an appropriate ASTM standard test method designed to indicate the concrete compressive strength, and the test results indicate that the concrete has gained sufficient strength to support its weight and superimposed loads.

Reshoring shall not be removed until the concrete being supported has attained adequate strength to support its weight and all loads in place upon it.

1.145 Requirements for Precast Concrete

Precast concrete wall units, structural framing, and tilt-up wall panels shall be adequately supported to prevent overturning and to prevent collapse until permanent connections are completed.

Lifting inserts which are embedded or otherwise attached to tilt-up precast concrete members shall be capable of supporting at least two times the maximum intended load applied or transmitted to them.

Lifting inserts which are embedded or otherwise attached to precast concrete members, other than the tilt-up members, shall be capable of supporting at least four times the maximum intended load applied or transmitted to them.

Lifting hardware shall be capable of supporting at least five times the maximum intended load applied transmitted to the lifting hardware.

No employee shall be permitted under precast concrete members being lifted or tilted into position except those employees required for the erection of those members.

1.146 Requirements for Lift-Slab Operations

Lift-slab operations shall be designed and planned by a registered professional engineer who has experience in lift-slab construction. Such plans and designs shall be implemented and shall include detailed instructions and sketches indicating the prescribed method of erection. These plans and designs shall also include provisions for ensuring lateral stability of the building / structure during construction.

Jacks / lifting units shall be marked to indicate their rated capacity as established by the manufacturer.

Jacks / lifting units shall not be loaded beyond their rated capacity as established by the manufacturer.

Jacking equipment shall be capable of supporting at least two and one-half times the load being lifted during jacking operations and the equipment shall not be overloaded. For the purpose of this provision, jacking equipment includes any load bearing component which is used to carry out the lifting operation(s). Such equipment includes, but is not limited, to the following: threaded rods, lifting attachments, lifting nuts, hook-up collars, T-caps, shear heads, columns, and footings.

Jacks / lifting units shall be designed and installed so that they will neither lift nor continue to lift when they are loaded in excess of their rated capacity.

Jacks / lifting units shall have a safety device installed which will cause the jacks / lifting units to support the load in any position in the event any jack lifting unit malfunctions or loses its lifting ability.

Jacking operations shall be synchronized in such a manner to ensure even and uniform lifting of the slab. During lifting, all points at which the slab is supported shall be kept within 1/2 inch of that needed to maintain the slab in a level position.

If leveling is automatically controlled, a device shall be installed that will stop the operation when the 1/2-inch tolerance set forth is exceeded or where there is a malfunction in the jacking (lifting) system.

If leveling is maintained by manual controls, such controls shall be located in a central location and attended by a competent person while lifting is in progress. The competent person must be experienced in the lifting operation and with the lifting equipment being used.

The maximum number of manually controlled jacks / lifting units on one slab shall be limited to a number that will permit the operator to maintain the slab level within specified tolerances, but in no case shall that number exceed 14.

No employee, except those essential to the jacking operation, shall be permitted in the building / structure while any jacking operation is taking place unless the building / structure has been reinforced sufficiently to ensure its integrity during erection. The phrase "reinforced sufficiently to ensure its integrity" used in this paragraph means that a registered professional engineer, independent of the engineer who designed and planned the lifting operation, has determined from the plans that if there is a loss of support at any jack location, that loss will be confined to that location, and the structure as a whole will remain stable.

Under no circumstances, shall any employee who is not essential to the jacking operation be permitted immediately beneath a slab while it is being lifted.

A jacking operation begins when a slab or group of slabs is lifted and ends when such slabs are secured (with either temporary connections or permanent connections).

When making temporary connections to support slabs, wedges shall be secured by tack welding, or an equivalent method of securing the wedges to prevent them from falling out of position. Lifting rods may not be released until the wedges at that column have been secured.

All welding on temporary and permanent connections shall be performed by a certified welder, familiar with the welding requirements specified in the plans and specifications for the lift-slab operation.

Load transfer from jacks / lifting units to building columns shall not be executed until the welds on the column shear plates (weld blocks) are cooled to air temperature.

Jacks / lifting units shall be positively secured to building columns so that they do not become dislodged or dislocated.

Equipment shall be designed and installed so that the lifting rods cannot slip out of position or the employer shall institute other measures, such as the use of locking or blocking devices, which will provide positive connection between the lifting rods and attachments and will prevent components from disengaging during lifting operations.

1.147 Requirements for Masonry Construction

A limited access zone shall be established whenever a masonry wall is being constructed. The limited access zone shall conform to the following:

- The limited access zone shall be established prior to the start of construction of the wall.

- The limited access zone shall be equal to the height of the wall to be constructed plus 4 feet and shall run the entire length of the wall.
- The limited access zone shall be established on the side of the wall which will be unscaffolded.
- The limited access zone shall be restricted to entry by employees actively engaged in constructing the wall. No other employees shall be permitted to enter the zone.
- The limited access zone shall remain in place until the wall is adequately supported to prevent overturning and to prevent collapse unless the height of wall is over 8 feet, in which case, the limited access zone shall remain in place until the following requirements have been met.

All masonry walls over 8 feet in height shall be adequately braced to prevent overturning and to prevent collapse unless the wall is adequately supported so that it will not overturn or collapse. The bracing shall remain in place until permanent supporting elements of the structure are in place.

15. Confined Space Safety Program

1.148 Purpose and Scope

The purpose of this program is to ensure a safe working environment when work is performed in a confined space and ensure that hazards of those confined spaces have been evaluated.

This program applies to all T&T Industrial, Inc. employees who work in confined spaces.

1.149 Resources

Number	Title
29 CFR 1910 Subpart J	General Environmental Controls - Permit Required Confined Spaces
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders - Permit Required Confined Spaces
29 CFR 1926 Subpart AA	Confined Spaces in Construction
CMS-FM-0017	Confined Space Entry Annual Review Log
CMS-FM-0018	Confined Space Entry Permit
CMS-FM-0019	Confined Space Entrant Log
CMS-FM-0020	Contractor Hazard Information Identification

1.150 Confined Spaces

A safe and successful confined space entry requires preplanning. Preplanning includes identifying anticipated hazards inside the confined spaces, selecting proper equipment to control the hazards, providing good documentation on the necessary controls via the written permit system, providing appropriate training for employees involved in the entry, and rapidly responding to emergency situations.

1.151 Confined Space

A confined space involves the following:

- Adequate size and configuration for an employee to enter and to perform assigned work.
- Limited means of entry or outlet.
- Not designed for continuous employee occupancy.

1.152 Permit-Required Confined Space

A permit-required confined space is a confined space that needs a permit to be entered. A permit is required if the confined space includes, or potentially includes the following:

- Atmospheric Hazard - Hazards related to atmospheric conditions, such as:
  - oxygen deficiency
  - flammable conditions
  - toxic conditions
- Engulfment Hazard - Contains a material like bulk grains, soil, liquid, or dry cement, which has the potential for engulfing an entrant.
- Entrapment Design - Has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor, which slopes downward and tapers to a smaller cross-section; or makes escape or rescue difficult.
- Other recognized serious hazard, such as:
  - inadequate ventilation
  - burns from high temperatures
  - internal electrical or rotating equipment, lockout / tagout
  - emergencies or hazards outside the confined space
  - physical injury from slips and falls
  - high noise levels inside the confined space
  - energy hazards from steam or electrical equipment inside the confined space

If it is necessary to enter the permit space to eliminate hazards, such entry shall be performed as required by regulation. If testing and inspection during that entry demonstrate that the hazards within the permit space have been eliminated, the permit space may be reclassified as a non-permit confined space for as long as the hazards remain eliminated. If hazards arise within a permit space that has been declassified to a non-permit space, each employee in the space shall exit the space. The space shall be re-evaluated to determine whether it must be reclassified as a permit space.

1.153 Non-Permit Confined Space

A non-permit confined space is any confined space that is known not to have contained any hazardous material. In addition, all hazards are controlled, and air monitoring indicates that the atmosphere is safe for entry. Entry into these areas for inspections or minor maintenance (i.e., adjustments, tightening of fittings, etc.) may be made without the use of retrieval systems or standby personnel.

1.154 Roles and Responsibilities

All affected employees shall understand the hazards of going into a confined space.

1.155 Authorized Entrants

Those persons who have completed the training and are authorized to enter permit spaces (authorized entrants) are assigned specific duties and responsibilities which they must perform when they work in the permit space. Their duties and responsibilities, which are covered in the training program, include:

- Know the hazards they may face during entry including mode, signs, symptoms of exposure, and understand the consequences of exposure to hazards.
- Understand the proper use of any needed equipment and compliance with the provisions on the entry form and or permit.
- Communicate with the attendant as necessary if the entry involves a hazardous atmosphere confined space entry.
- Communicate with the attendant to check in or ask for rescue.
- Alert attendant when a warning symptom or other hazardous condition exists, or a prohibited condition is detected.
- Exit as quickly as possible whenever ordered or alerted by an evacuation alarm, warning sign, a symptom of exposure is detected, or prohibited condition.
- Keep lifelines orderly and untangled within the confined space.

1.156 Attendants

Attendants are individuals stationed outside a permit space who monitor authorized entrants. An attendant must be on duty outside of a confined space for the duration of the operation. Those persons who have completed the training and have been designated as permit space attendants are assigned specific duties and responsibilities, which they must perform in permit space job duties. Their duties and responsibilities, which are covered in the training program, include:

- Attendants must know the hazards of a confined space including information on the mode, signs, symptoms, and consequences of exposure.
- Keep lifelines orderly, untangled, and the end secured outside of the confined space.
- Know the possible behavioral effects of exposure, which include the following:

% of Oxygen in Air	Effects of Oxygen Deficiency
16 to 12%	Deep breathing, accelerated heartbeat, impaired attention, impaired thinking, impaired coordination.
14 to 10%	Very faulty judgement, very poor coordination, rapid fatigue from exertion that may cause permanent heart damage, intermittent breathing.
10% or below	Nausea, vomiting, inability to perform vigorous movement or loss of all movement, unconsciousness followed by death.
Less than 6%	Spasmodic breathing, convulsive movements, death in minutes.

- Check permits of authorized entrants.
- Prevent entry by those without a permit and take the following actions when unauthorized persons approach or enter a permit space while entry is underway:
  - Warn the unauthorized persons that they must stay away from the permit space.
  - Advise the unauthorized persons that they must exit immediately if they have entered the permit space.
  - Inform the authorized entrants and the entry supervisor if unauthorized persons have entered the permit space.

- Maintain a continuous count of those in a confined space.
- Monitor activity in the confined space and alert entrants of the need to evacuate.
- Summon rescue and emergency services when entrants may need assistance to escape a permit space hazard.
- Perform non-entry rescues. The attendant may only enter a confined space for rescue if they are trained and part of a rescue team and are then relieved by another qualified and trained attendant.
- Perform no duties that may interfere with attendant's primary duty to monitor and protect authorized entrants.
- Remain outside the confined space until relieved by another attendant.

1.157 Entry Supervisors

Those persons who have completed the training and have been designated as permit space entry supervisors are assigned specific duties and responsibilities, which they must perform in permit space job duties. Their duties and responsibilities, which are covered in the training program include:

- Know the hazards they may face during entry including the mode, signs, and symptoms of exposure. Also, they must understand the consequences of exposure to hazards.
- Verify all tests specified by the permit have been conducted, all hazards have been identified and mitigated, and all procedures and equipment are in place before endorsing the permit and allowing work to begin. Also, all procedures and equipment specified in the permit shall be in place before endorsing the permit and allowing entry to begin. The supervisor must sign the entry permit to authorize entry.
- Terminate the entry and cancel the permit when necessary.
- Verify rescue services are available and a means of communication are in place and operable if needed.
- Remove unauthorized individuals who enter or attempt to enter the permit space during entry operations.
- Determine that entry operations remain consistent with terms of the entry permit whenever responsibility is transferred or at intervals dictated by the hazards or operations performed within the space.
- Post-Operations Procedures: The entry supervisor shall terminate the entry and cancel the permit upon job completion or if conditions change within the confined space. The permit

will be reviewed, verification will occur that equipment is operational, and any employee concerns addressed. Procedural revisions may be required if the entry supervisor identifies that an unauthorized entry occurred, a hazard was identified that was not covered by the permit, the occurrence of an injury or near miss, or there are employee complaints. Management shall be informed if these referenced items are identified.

1.158 Confined Space Classification

The Safety Coordinator shall be the qualified person in confined spaces. An evaluation shall be conducted to assess the hazards. This evaluation will provide on-site personnel with the information necessary to identify the existence and location of permit-required confined spaces that must be covered by the Permit-Required Confined Space Entry Program and the hazards associated with them. All confined spaces will either be classified as a permit-required confined space (PRCS) or non-permit confined space (NPCS). All confined spaces shall be treated as permit spaces until determined to be otherwise.

The Company will provide the following to its clients and exposed sub-contractors, if applicable, prior to job tasks involving confined spaces:

- A list of confined spaces to be entered on the project in question for review by the qualified person and the client's / sub-contractor's representative, if applicable. This will provide all parties the opportunity to review the hazards and any specific equipment and personnel needs.
- The names of employees or sub-contracted employees, if applicable, working on the specific project who may serve as the Entry Supervisor, the Authorized Attendant(s), and the Authorized Entrant(s). The qualified person will act as the Entry Supervisor. The Authorized Attendants and Authorized Entrants will be other craft members working on site who have been properly trained as Entrants and/or Attendants. The individuals involved in these roles will be identified and named on each Entry Permit.

1.159 General Requirements – Non-Permit Confined Spaces

A qualified person that has received documented training in confined spaces shall determine what condition and precautions must be in place to allow for safe entry and what would constitute a change in conditions, which would require a re-evaluation of the confined space. Please refer to the definition of permit-required confined spaces within this document as a guide to determine whether a space is a PRCS or NPCS.

1.160 General Requirements – Permit-Required Confined Spaces

The following requirements apply to entry into Permit-Required Confined Spaces:

- Any condition making it unsafe to remove a cover shall be eliminated before the cover is removed.
- Covers shall be removed and replaced using tools designed for that purpose. When entrance covers are removed, the entrance shall be properly guarded to prevent an accidental fall of employees from foreign objects entering the space.
- Atmospheric Testing
    - Before entry into a confined space, necessary testing shall be conducted for hazardous atmospheres by a qualified person. If there is no potential for a hazardous atmosphere, the atmospheric testing may be waived. A qualified person shall possess the knowledge and skill to understand the test instrument's use, calibration procedures, limitations, and have the ability to interpret results.
    - Before any employee enters the space, the internal atmosphere shall be tested with a calibrated, direct reading instrument for the following conditions in the following order:
        1. Oxygen content (between 19.5% and 23.5%)
        2. Flammable gases and vapors (not over 10% of the Lower Flammable Limit)
        3. Potential toxic air contaminants (not over Permissible Exposure Limit)
    - The atmosphere within the space shall be periodically tested as necessary to ensure that the continuous forced air ventilation is preventing the accumulation of a hazardous atmosphere. Any employee who enters the space, or that employee's authorized representative shall be provided with an opportunity to observe the periodic testing.
    - Reevaluate the permit space in the presence of any authorized entrant or that employee's authorized representative who requests that the employer conduct such reevaluation because the entrant or representative has reason to believe that the evaluation of that space may not have been adequate
    - Immediately provide each authorized entrant or that employee's authorized representative with the results of any testing conducted.

  - It is recommended on a vertical entry that remote probes be used to test at various levels of the confined space as vapors and gases have different density's and could accumulate at the bottom, middle, or top of a confined space.
  - Atmospheric testing for the confined space should be based on the configuration and design of the space, and physical, and chemical characteristics of suspected contaminants that may be encountered. Continuous monitoring should be considered in situations when an employee is present in a space where atmospheric conditions have the potential to change. Examples include broken or leaking pipe or vessels, disturbance of exiting materials, the potential for adjacent work activities that can create a hazardous environment or any space that is not capable of being isolated.
  - All tests must be completed, accurate, and documented before entry.
- The space must be free of any hazardous atmosphere whenever an employee is inside.
- Smoking and open flames shall not be permitted within 20 feet of the confined space opening unless air samples indicate an atmosphere safe for hot work. No welding or burning will be permitted unless a special hot work permit is obtained.
- All ladders shall be inspected prior to entry.
- Only approved low voltage light and extension cords, or electrical apparatus provided with a ground fault circuit interrupter shall be used in a confined space.
- When entering a permit required confined space, there shall always be an attendant present to ensure the authorized entrant is following procedures and to monitor activity in the confined space.
- Continuous air ventilation shall be used, as follows:
  - An employee shall not enter the space until the ventilation has eliminated the hazard.
  - The ventilation shall be directed as to ventilate the immediate areas when an employee is working and shall continue until the employee leaves that area.
  - The air supply for the forced air ventilation shall come from a clean source and may not increase the hazards in the space.
- If oxygen consuming equipment is going to be used or if an oxygen consuming evolution will take place, continuous oxygen monitoring is required no matter what the classification of the space or the ventilation required.

1.161 Permits

All entry to a permit-space is restricted to those with permits. The written permit controls entry into a confined space. The use of a written permit will ensure that the following controls are in place:

1. The current classification reflects existing conditions in the space.
2. The checklist of entry requirements is appropriate for that classification.
3. Recordkeeping requirements are met for each entry.

Permits shall be available to all employees requiring entry to a confined space that requires a permit (permit-space). The permit shall extend only for the duration of the task. All permits shall be retained for a year to facilitate review of the Confined Space Program.

Permits shall include the following:

- Identification of the space
- Purpose of entry
- Date
- Time of issue and expiration
- List of authorized entrants with method to determine which authorized entrants are inside the permit space (sign-in sheet)
- Names of current attendants and entry supervisor
- List of hazards in the permit-space
- List of measures to isolate the permit-space and eliminate or control hazards before entry
- The acceptable entry conditions
- Results of initial and periodic tests initialed by the persons performing test and the time tests were performed
- Rescue and emergency services including equipment and phone numbers
- Communication procedures for attendants and entrants to maintain contact during entry
- Required PPE (respirators, communications devices, alarm, and rescue equipment)
- Any other information which is necessary

- Any additional permits issued to authorized work in the permit space (such as for hot work under the Fire Safety Program).

1.162 Safe Permit Space Entry Procedures

When confined space entry becomes necessary, the person in charge of the work to be performed shall make the need for the entry known to their supervisor. The Qualified Person will serve as the Entry Supervisor responsible for authorizing entry and issuing entry permits for work in confined spaces.

The qualified person in charge of the entry will be responsible for ensuring compliance with the entry requirements specified on the checklist and will ensure the permit is properly filled out. Upon conclusion of the entry, the qualified person will review, close off the permit space, and cancel the permit. A copy shall be filed in the job folder. The permit shall be posted outside of the confined space while the authorized entrant is inside.

The duration of the permit may not exceed the time required to complete the identified assigned task on the permit and will never exceed 12 hours. One permit is required for each specific confined space. If circumstances cause an interruption in the work or a change in the alarm condition for which entry was approved, a new Confined Space Entry Permit shall be completed.

The permit expires when any of the following conditions are met:

- The entry operations covered by the permit have been completed.
- A change in work conditions introduces a new hazard.
- The time period of the permit has elapsed.
- When personnel (entrants, attendants, entry supervisor) who were not originally identified on the permit are assigned to the confined space work operation.
- Any other changes in the existing conditions occur that may cause a new hazard or casts doubt upon the ability to continue safely in the same fashion.

Permits shall be readily available to all employees before entering a confined space, and the permits shall remain at the work site as long as the work is being performed there.

The Permit Program and completed permit checklists shall be reviewed at least annually.

When atmospheric testing shows a dangerous air contamination, oxygen deficiency, or oxygen enrichment, the written permit form or record showing the results of the atmospheric testing shall be retained for a minimum of one year. Canceled entry permits shall be kept for one year to facilitate the review of problems encountered and the appropriate changes made during the review.

1.163 Preventing Unauthorized Entry

To provide a safe work environment and to prevent exposed employees from accidentally entering a permit space, procedures shall be implemented to inform all employees of the existence, location, and danger posed by permit spaces at work sites. To inform employees of the existence of a permit space, warning signs and/or verbal communication by the qualified person shall be used.

If the space is found to be a permit-required confined space, it shall be labeled by posting a sign reading:

"DANGER -- PERMIT REQUIRED CONFINED SPACE, DO NOT ENTER"

This sign shall be permanently posted at the potential entry or access point to the space. If the space will not be entered, effective measures shall be taken to see that entrances are adequately marked and blocked. To ensure that unauthorized employees do not enter and work in permit spaces, confined space entry is restricted to employees that have been specifically trained as authorized entrants. The qualified person shall have the overall on-site safety responsibility.

1.164 Pre-Entry Procedures

Measures such as barriers or barricades necessary to prevent unauthorized entry shall be implemented. Pedestrian, vehicle, and other barriers necessary to protect entrants from external hazards shall be provided and verified throughout the duration of the authorized entry.

Each authorized entrant shall have the opportunity to observe any pre-entry monitoring or testing.

To ensure the safety and health of our employees, before allowing authorized employees to enter a permit space, conditions shall be evaluated in that space to determine if the conditions are safe for entry. The following steps shall be taken before entry into a confined space is permitted:

1. Disconnection of lines – Lines that may convey flammable, explosive, toxic, or otherwise injurious or incapacitating substances into the space shall be disconnected, blinded, locked out, or blocked off by other positive means to prevent the development of dangerous air contamination, oxygen deficiency, or oxygen enrichment within the space. The disconnection or blind shall be so located or done in such a manner that inadvertent reconnection of the line or removal of the blind is effectively prevented. If the jobsite is a Process Safety Management site, additional instruction is needed before proceeding.

2. Calibration of testing and monitoring equipment – Air testing and monitoring equipment shall be maintained and calibrated according to manufacturers' instructions. This equipment shall be periodically calibrated with an appropriate test gas to assure proper operation. Records of calibration shall be maintained for a minimum of one year.

   a) Calibration Check – Multigas and Single gas detectors:

      The calibration check is simple and should only take about one minute. Perform this calibration check before each day's use for each installed sensor. Daily calibration checks are performed to ensure the monitor and alarms are working correctly.

      i. Turn ON the Multigas / Single gas detector in clean, fresh air.
      ii. Verify that readings indicate no gas is present.
      iii. Attach regulator (supplied with calibration kit) to the cylinder.
      iv. Connect tubing (supplied with calibration kit) to the regulator.
      v. Attach other end of tubing to the instrument.
      vi. Open the valve on the regulator, if so supplied.

      The reading of the Multigas / Single gas detector display should be within the limits stated on the calibration cylinder.

      If readings are not within these limits, the Multigas / Single gas detector requires recalibration.

3. Air Monitoring – Prior to entry in a confined space, the space shall first be tested and ventilated. The atmosphere shall be tested using a three-gas (O_2, LEL, H_2S) or 4-gas (O_2, LEL, CO, H_2S) monitor. Ensure that the detector is fully charged, properly calibrated, and in operation for at least 10 minutes before the testing begins. For vertical entry confined spaces, the detector shall be lowered into the space to test each quarter section for 4 minutes in each position. The internal atmosphere shall be tested for the following conditions in the order given:

   a) oxygen content,
   b) flammable gas and vapors, and
   c) potential toxic air contaminants.

   All areas of the confined space shall be monitored to ensure the following conditions are met:

   Oxygen: 19.5% to 23.5%

Combustibility: < 10% LEL

CO: < 35 PPM

H_2S: <5 PPM

Other Toxins: <= PEL/TLV

- If the monitor alarms at any time during testing, immediately remove the monitor and note the readings indicated (oxygen, combustible, CO, H_2S) to determine which reading(s) caused the alarm. Ventilation is now required to bring the readings within the acceptable parameters. At this point, the confined space shall remain, as a permit required confined space.
- If no alarms occur during the testing, proceed with purging. The space may be classified as a non-permit required confined space if all the hazards have been controlled and ventilation maintains a safe air environment.

Air monitoring shall continue during all entries. Record the findings on the Entry Permit. Do not enter the confined space at any time during the initial testing. Physical entry shall not be made into an unknown atmosphere.

4. Ventilation – Where the existence of dangerous air contamination, oxygen deficiency, or oxygen enrichment is demonstrated by air testing, existing ventilation shall be augmented by appropriate means. Purging shall be performed using an approved air blower. The minimum purge time shall be 10 minutes. When the blower is set up, the intake shall be kept away from any source of contamination (i.e., vehicle exhaust, traffic, etc.). When the blower hose is lowered into the manhole, it shall reach to the lower one-third of the manhole and shall be angled toward a wall to ensure complete ventilation.

   Once purging of the atmosphere has been completed, the atmosphere shall be re-tested using the following procedures:

   I. If no alarm occurs during the re-testing, then entry may be made in accordance with the entry procedures discussed in the next section of this program.

   II. If an alarm occurs during the re-testing, check for the following:

      a) Placement of blower intake - It may be necessary to hook a straight section of hose to intake in order to obtain an air supply away from any contaminants.

      b) Any vehicles or engines near the manhole that are running may be contaminating the manhole with exhaust. Shut the vehicles and/or engines off or move them away from the manhole.

c) If applicable, check placement of blower hose into a manhole to ensure it reaches the lower one-third and is directed toward a wall.

d) If any of these factors (or others) are found to be potential sources of problems, then correct them and re-ventilate and recheck.

e) If, after re-ventilating, the detector does not alarm, then complete the Confined Space Entry Permit and proceed with entry operations.

f) If, after re-ventilating, the detector still alarms, call for a second detector and notify the qualified person. The second detector may be used to re-check the manhole. If the second detector does not alarm upon re-check, then inspect the first detector for low battery or possible malfunctions. If no malfunctions are found, then contact the Entry Supervisor before proceeding with entry. If the second detector alarms during recheck, no entry shall be made. Under no circumstances is entry to be made until the confined space has been established to be safe of atmospheric hazards.

5. Injurious or Corrosive Substances - Employees in confined spaces that have last contained injurious or corrosive substances to the eyes or body shall be provided with, and shall be required to wear, appropriate personal protective equipment. In addition, an eyewash and safety shower shall be provided within the work area outside of the confined space for immediate emergency use.
6. Ignition Sources – No sources of ignition shall be introduced into the space until implementation of appropriate provisions of this section has ensured that dangerous air contamination due to flammable or explosive substances does not exist.
7. Oxygen Consuming Equipment – Whenever oxygen-consuming equipment is to be used, measures shall be taken to ensure adequate combustion air and exhaust gas venting.
8. Oxygen Enrichment Condition – Whenever oxygen enrichment is possible due to conditions within the space, measures shall be taken to ensure that the oxygen level does not exceed 23% in the confined space. If tests indicate the oxygen levels to be higher than 23%, hot work is prohibited until the ventilating techniques have reduced the oxygen level to less than 23%.
9. Smoking – Smoking shall not be allowed in confined spaces or within 20 feet of a confined space opening.
10. Automatic Fire Protection Systems – Where there is no ready exit from spaces equipped with automatic fire suppression systems employing harmful design concentrations of

toxic or oxygen displacing gases, or total foam flooding, such systems shall be deactivated.

1.165 Entry Safeguard Requirement

A confined space entry is defined as an action, which results in any part of an employee's body breaking the plane of any opening of a confined space. This section specifies the appropriate actions and equipment required to safely access the space in question.

Once the confined space atmosphere has been tested, purged, and re-tested, and determined to be safe, then entry may be made provided the following guidelines are followed:

- The person entering the confined space shall carry the gas detector with them and it shall remain in operation at all times while the occupant remains in the confined space. If, at any time, the gas detector alarms, the confined space shall be evacuated immediately. If it is found that the detector is alarming due to a low battery, entry may be made using another detector provided the atmosphere is re-tested before re-entry.
- Ventilation (natural or mechanical as applicable) shall continue while the authorized entrant is in the confined space. For entry into manholes, the ventilation supply hose may be temporarily removed from the opening to allow for entry, but the hose shall be reinserted immediately after occupant has entered manhole.
- A standby person (attendant) shall be stationed outside of the manhole or confined space and shall remain in visual, or voice contact with the occupant at all times.
  - The standby person shall be specially trained in Confined Space Entry.
  - The standby person shall have the authority to order immediate evacuation of the space if there is any sign of intoxication from an undetected hazardous atmosphere or any other danger that warrants immediate evacuation.
  - If injury occurs or occupant becomes unconscious while in confined space, refer to the section titled "Rescue and Retrieval".
  - At least one attendant shall be stationed outside a confined space for the duration of entry operations. If multiple spaces are to be monitored by a single attendant, they shall have the means to respond to an emergency in one space while continuing oversight of the others, or they shall be relieved by another individual.

1.166 Rescue and Retrieval

Rescue and retrieval procedures and a means of communication to summon rescue shall be developed and in place prior to entry for summoning rescue and emergency services shall be developed and implemented to perform rescue, to provide necessary emergency services to rescued employees, and to prevent unauthorized personnel from attempting a rescue.

Affected employees will be provided the necessary PPE to protect them from the hazards posed by performing rescue in a permit-required confined space, at no cost to those employees.

Employees shall not enter permit spaces to perform rescue operations. Employees shall use the provided retrieval systems to remove personnel in confined spaces. Affected employees shall be trained to perform assigned rescue duties. At least one employee on the rescue team shall maintain current certification in basic first aid and cardiopulmonary resuscitation (CPR) skills. They shall also practice making permit space rescues at least once every 12 months by means of simulated rescue operations in which they remove dummies, mannequins, or personnel through representative openings and portals whose size, configuration, and accessibility closely approximate those of the permit spaces on-site.

Local emergency services shall be contacted immediately to assist in confined space emergencies. Prior to confined space work operation commencing, the local responders shall be informed of the site work tasks, confined space configuration, and the specific hazards. Rescue service shall have access to all permit spaces from which rescue may be necessary so they can develop appropriate rescue plans and practice rescue operations. Rescue service shall be on-site for immediately dangerous to life and health (IDLH) conditions while work is being performed.

Local emergency rescue teams shall be trained to perform the assigned rescue functions and must have received the training required of authorized entrants. They shall also practice making permit space rescues at least once every 12 months by means of simulated rescue operations in which they remove dummies, mannequins, or personnel through representative openings and portals whose size, configuration, and accessibility closely approximate those of the permit spaces on-site. At least one member of the rescue team shall maintain current certification in basic first aid and cardiopulmonary resuscitation (CPR) skills.

To facilitate non-entry rescue, retrieval systems (body harness or wristlets, and lifelines) shall be used, at no cost to the employee, whenever an authorized entrant enters a confined space. Retrieval systems shall meet the following requirements.

1. Each entrant shall use a chest or full body harness, with a retrieval line attached at the center of the entrants back near shoulder level or above the entrant's head. Wristlets may be used when the use of a body harness would create an additional hazard or interfere with the retrieval through a small man way entrance.
2. The other end of the retrieval line shall be attached to a mechanical device or fixed point outside the permit space in such a manner that rescue can begin as soon as the attendant becomes aware that rescue is necessary. A mechanical device shall be made available to retrieve personnel from a vertical entry permit space more than 5 feet in depth.
3. The use of retrieval lines that have a reasonable probability of becoming entangled with the retrieval lines used by other authorized entrants, or due to the internal configuration of the PRCS, are prohibited.

1.167 Multiple Employer Entry Procedures

All contractor or sub-contractor personnel acting under Company control shall comply with all applicable provisions of these procedures or show proof that their procedures and employee training are at least as effective as these procedures.

Contractors are responsible for the permit space entry of their own personnel. They shall be informed of the permit space hazards by giving them a copy of the form titled "Contractor Hazard Information Identification" for the permit space(s). Also, they shall be informed of Company safety procedures. They shall not be permitted to enter space(s) until the qualified person acting as the entry supervisor has determined that they have a permit space entry program and that the contractor's program does not endanger Company employees.

When a contractor's personnel and Company personnel perform permit space entry operations in the same permit space at the same time, both the contractor and Company shall provide an entry supervisor as a check and balance system. One or both parties may provide the attendant(s). The entry supervisors shall coordinate the work so neither crew endangers the other.

Entrants shall be instructed to comply with each other's evacuation orders and evacuation alarms. Attendants shall be instructed to immediately inform the other attendant if an evacuation order is issued or if an evacuation alarm is activated. This means the entrants of all employers shall evacuate the permit space if any attendant, any entry supervisor, or any entrant issues an evacuation order. If there is a dispute over the necessity to evacuate, all entrants shall evacuate and remain outside of the permit space until the dispute is settled.

When contractors perform work that involves permit space entry, then:

1. Inform the contractor that the workplace contains permit spaces and that permit space entry is allowed only through compliance with a permit space program meeting the requirements of the OSHA Standard. Supply the contractor with the form titled “Contractor Hazard Information Identification”.
2. Apprise the contractor of the hazards identified for the confined space.
3. Apprise the contractor of any precautions or procedures that the Company has implemented for the protection of employees in or near the permit spaces where contractor personnel will be working.
4. Coordinate entry operations with the contractor(s) when both Company personnel and contractor(s) personnel will be working in or near permit spaces.
5. The Company entry supervisor shall debrief the contractor at the conclusion of the entry operations regarding the procedures followed and regarding any hazards confronted.

1.168 Safety Equipment and Clothing

The entry permit shall include a list of necessary protective equipment to be used in the confined space as determined by the qualified person. The Company is responsible for the proper use of the safety equipment and the inspection and maintenance procedures performed on the safety equipment. The qualified person shall determine the type of protective equipment required.

- Eye and Face Protection - Employees shall use appropriate eye or face protection when exposed to hazards from: flying particles, molten metal, liquid chemicals, acids or caustic liquids, and potentially injurious radiation. Requirements for side protection, prescription lenses, filter lenses, and identification of the manufacturer of safety equipment shall be specified.
- Head Protection - Employees shall wear protective helmets when working in areas where there is a potential for a head injury from falling objects and protection from any other hazard identified by the qualified person. Employees who are near exposed electrical conductors shall wear protective helmets designed to reduce electrical shock hazards. Protective helmets purchased on or after July 5, 1994, shall comply with ANSI Z89.1-1986 or be equally effective.
- Foot Protection - Employees shall wear protective footwear when working in areas where there is a danger of foot injuries from falling or rolling objects, objects piercing the sole, or exposure of employees’ feet to electrical hazards. Protective footwear purchased on or after July 5, 1994, must comply with ANSI Z41-1991 or be equally effective.

- Hand Protection - Employees shall use the appropriate hand protection whenever employees' hands are exposed to hazards from skin absorption of harmful substances, severe cuts or lacerations, severe abrasions, punctures, chemical burns, thermal burns, and harmful temperature extremes.

  The Company will select appropriate hand protection based on an evaluation of:

  - The performance of the hand protection relative to the task to be performed,
  - The conditions present,
  - The duration of use; and
  - The actual and potential hazards identified.
- Hearing Protection - Employees shall wear hearing protection when engineering technology is insufficient to control the noise level and the ambient exposure limit exceeds those allowed in Table G-16 of 29 CFR 1910.95.
- Respiratory Protection - The need for respiratory protection shall be determined by a qualified person based upon conditions and test results of the confined space and the work activity to be performed. The respirators used shall be National Institute for Occupational Safety and Health (NIOSH) and or Mine Safety and Health Administration (MSHA) approved devices and shall be fitted and maintained in accordance with 29 CFR 1910.134.
- Body Protection - All employees entering a confined space shall wear full coverage work clothing as specified by the qualified person. Gloves and clothing made of impervious rubber or similar material are to be worn to protect against toxic or irritating materials. If the hazards of heat or cold stress exist in the confined space, clothing which has been tested to provide protection from over-exposure to these hazards shall be worn. Other body protection required in specific operations such as welding (flame resistant), riveting (heat resistant), and abrasive blasting (abrasion resistant) shall be provided to insure employee's safety

1.169 Training

Training shall be provided to all employees involved in confined space entry so that they acquire the understanding, knowledge, and skills necessary to perform their job safely. Training must establish proficiency in assigned duties in accordance with jurisdictional requirements. All training related materials, documents, and signed certificates shall be maintained.

The training programs shall consist of classroom (lecture and group activities), hands-on (equipment, monitors, PPE, entries), and audiovisuals. All new employees shall be trained to an awareness level as to the recognition of confined spaces.

Only specific employees shall be trained in confined spaces.

All employees shall be trained to be able to recognize hazardous conditions; properly use monitoring equipment; space preparation; entry and work procedures; the permit system; and emergency response actions.

Confined space entry training shall include:

- The written program and its requirements.
- The proper use of air monitoring equipment.
- The proper use and limitations of body harnesses, lifelines, retrieval systems, and other PPE.
- The proper use of all respiratory equipment.
- The typical hazards that may be encountered and the consequences of exposure to hazards.
- Recognizing the signs and symptoms of exposure to hazards.
- Understanding the duties specific to their role in confined space entry work as well as the duties of others who are involved as provided in this plan.
- Evaluating and preparing a confined space for entry.
- The proper use of the permit system.
- The importance and methods of maintaining communications between entrants and attendants.
- Conditions that require evacuation of the confined space.
- The importance and requirements for maintaining site control.

- The requirements for concluding an entry and terminating the permit.
- Proper confined space non-entry rescue procedures and assigned rescue duties.
- Welding, cutting, and brazing in a confined space.
- The proper use and calibration of monitoring equipment.

Certification of the training shall be documented and made available to employees and their authorized representatives.

1.170 Training Intervals

Training shall be provided to each affected employee:

- Prior to initial assignment.
- Prior to any changes in assigned duties.
- If a new hazard has been created or special deviations have occurred.
- Whenever there is reason to believe that there are deviations from entry procedures that could pose hazardous to employees.
- When inadequacies in the employee's knowledge or use of these procedures is found.
- At least every two years

1.171 Lockout / Tagout (LOTO)

Safety equipment required during this procedure shall be designated by the qualified person and dependent upon the potential hazards involved. A confined space shall be completely isolated from all other systems by physical disconnection, double block and bleed, or blanking off all lines. Blanks used to seal off lines shall be capable of withstanding the maximum working pressure or load of the line (with a minimum safety factor of four), be provided with a gasket on the pressure side to ensure a leak-proof seal and be made of chemically non-reactive material. Shutoff valves serving the confined space shall be locked in the closed position and tagged for identification. In addition to blanking, pumps and compressors serving these lines entering the confined space shall be locked out to prevent accidental activation.

If a drain line is located within the confined space, provision shall be made when necessary to tag it and leave it open. This shall also be recorded on the entry permit.

Additional procedures, which are necessary when the confined space is of a double wall construction, (e.g., water-jacketed) or similar type shall be determined by the qualified person and noted on the entry permit.

Electrical isolation of the confined space to prevent accidental activation of moving parts that would be hazardous to the employee shall be achieved by locking circuit breakers and/or disconnects in the open (off) position with a key-type padlock.

The only key is to remain with the person working inside the confined space. If more than one person is inside the confined space, each person shall place their own lock on the circuit breaker. In addition to the Lockout system, there shall be an accompanying tag that identifies the operation and prohibits use.

Disconnecting linkages or removing drive belts or chains can achieve mechanical isolation of moving parts. Equipment with moving mechanical parts shall also be blocked in such a manner that there can be no accidental rotation.

1.172 Review

Entry operations shall be reviewed when there is reason to believe that the measures taken under the permit space program may not protect employees. Circumstances requiring the review of the permit space program are as follows:

- Any unauthorized entry of a permit space.
- The detection of a permit space hazard not covered by the permit.
- The detection of a condition prohibited by the permit.
- The occurrence of an injury or near-miss during entry.
- A change in the use or configuration of a permit space.
- Employee complaints about effectiveness of the program.

The program shall be revised to correct deficiencies found to exist before subsequent entries are authorized.

This written program shall be reviewed annually. It shall be revised as necessary to protect employees from confined space hazards.

1.173 Recordkeeping

A written record of training shall be maintained including safety drills, inspections, tests, and maintenance. The records shall be retained for 3 years after the last date of training, inspection, test, or maintenance. In the event of separation of the employee, disposal of equipment or appliance, records may be disposed of after 1 year.

Where atmospheric testing indicates the presence of a toxic substance, records shall be maintained in accordance with the existing Federal regulation(s). These records shall include the dates and times of measurements, duties, and location of the employees within the confined space, samples taken, and PEL concentrations estimated from these samples. Records shall be available to the designated representatives of the Secretary of Health, Education, and Welfare, to the employer, and to the employee or former employee.

Appendix 7 **Confined Space Entry Annual Review Log**

Entry Location	Purpose of Entry	Permit Issued			Permit Canceled			Annual Review		
		Date	Time	Initials	Date	Time	Initials	Date	Time	Initials

Appendix 8 **Confined Space Entry Permit**

General Information			
Unit:		Entry Date	
Entry Supervisor		Permit Expires:	
Location:		Job No.:	
Purpose of Entry:			
Attendant(s):			

Rescue Information	
Telephone Number(s):	

Hazard Control Checklist			
		Yes	N/A
1.	Has the confined space been drained and purged?	☐	☐
2.	Has the confined space been cleaned?	☐	☐
3.	Has the confined space been ventilated?	☐	☐
4.	Has the confined space been blinded or isolated?	☐	☐
5.	Have all energy sources been locked out/tagged out and in a zero-energy state?	☐	☐
6.	Have all radiation sources been locked into their shielded containers?	☐	☐
7.	Do each open man way or entrance to the confined space have a posted notice?	☐	☐
8.	Is rescue equipment required?	☐	☐
9.	Will entry involve any of the following:	☐	☐
	-Oxygen deficiency (less than 19.5%)?	☐	☐
	-Flammable gases or vapors greater than 10% of the Lower Flammable Limit or greater than 23.5% oxygen?	☐	☐
	-Toxic gases or vapors greater than the Permissible Exposure Limit?	☐	☐
	-Configuration hazards?	☐	☐
	-Electrical shock?	☐	☐
	-Engulfment?	☐	☐
	-Materials harmful to the skin?	☐	☐
	-Mechanical hazards?	☐	☐

10.	Have all employees on this permit been trained in confined spaces?	☐	☐

If **YES** to any of the items in #9 above, contact the Safety Department.

Required PPE		
☐ **Communication Equipment**	☐ **Ventilation**	☐ **Respiratory Protection: HF/FF SCBA**
☐ **Electrical Equipment**	☐ **Safety Glasses, Goggles, Face shield**	☐ **Fall Protection**
☐ **Protective Clothing (FRC/Acid/Clicker)**	☐ **Gloves-Chemical/Thermal**	☐ **Foot Protection**
☐ **Rescue Equipment**	☐ **Hard Hat**	☐ **Other (Special Precautions/Restrictions)**

Air Monitoring			
Air Monitoring Sampling Required	☐ **Initial**	☐ **Periodic**	☐ **Continuous**
Multigas Detector Calibration Date:	**Month:**	**Day:**	**Year:**
Multigas Detector Calibration Daily Check	☐ **Pass**	☐ **Fail-Recalibrate**	☐ **Recalibrated**

T&T Industrial, Inc.

Confined Space Safety Program

Tests Required	Initial					Safe Limit	
1. Oxygen						Min 19.5%	Max 23.5%
2. Combustible Gases (%LEL)						Less than 10%	
3. Carbon Monoxide (CO)						25 ppm	
4. Hydrogen Sulfide (H_2S)						5ppm	
5. Sulfur Dioxide (SO_2)						2 ppm	
6. Total Hydrocarbons						300 ppm	
7. Benzene						1 ppm	
8. Other							

Calibration Unit Serial Number:	
Multigas Detector Serial Number:	

Authorization of Entry Supervisor

I, ______________________________________, attest that all entrants and attendants of this confined space have received training per the Confined Space Procedure.

Signature:		Date:	

Appendix 9 Confined Space Entrant Log

General Information			
Unit:		Entry Date	
Entry Supervisor		Permit Expires:	
Location:		Job No.:	
Purpose of Entry:			
Attendant(s):			

Rescue Information	
Telephone Number(s):	

Entrant Log						
Entrant Name	**Time In**	**Time Out**	**Time In**	**Time Out**	**Time In**	**Time Out**

Appendix 10 Contractor Hazard Information Identification

This is a list of permit spaces and the potential hazards of these spaces that the contractor will be working in or in close proximity to. The hazards of some spaces may change with use. Note this possibility as "change with use" in the Potential Hazards column then list the anticipated hazards.

NO WELDING OR BURNING PERMITTED UNLESS SPECIAL HOT WORK PERMIT IS OBTAINED

Permit Space	Potential Hazards

Signatures			
Contractor Employee Signature:		Date:	
Contractor Employee Signature:		Date:	
Contractor Employee Signature:		Date:	
Contractor Employee Signature:		Date:	

16. Cranes Program – US

1.174 Purpose and Scope

The purpose of this program is to ensure a safe working environment when work is performed using cranes.

This program applies to all T&T Industrial, Inc. employees who work with or near cranes.

1.175 Resources

Number	Title
29 CFR 1926 Subpart CC	Cranes and Derricks in Construction
29 CFR 1926 Subpart C	General Safety and Health Provisions
CMS-PR-0105	Signaling for Cranes

1.176 Cranes Program - US

Preventing incidents from the use of cranes requires employees to recognize the hazards involved.

1.177 Operator Qualification

Only those employees qualified by training are allowed to operate equipment and machinery. Operators shall be qualified / certified by one of the following methods:

- Certification by an accredited crane operator testing organization.
- Qualification by an audited Company program.
- Qualification by the US military.
- Licensing by a government entity.

Refresher training is required according to the certification body.

1.178 Operator Responsibilities

The procedures applicable to the operation of the equipment, including rated capacities (load charts), recommended operating speeds, special hazard warnings, instructions, and operator's manual, shall be readily available in the cab at all times for use by the operator.

Whenever there is a concern as to safety, the operator has the authority to stop and refuse to handle loads until a qualified person has determined that safety has been assured.

1.179 Employee Competency

Only those employees qualified and deemed competent through training may be the designated signal person and maintenance and repair employees.

1.180 Safe Work Practices

1.180.1 Assembly / Disassembly

Equipment shall not be assembled or used unless ground conditions are firm, drained, and graded to a sufficient extent so that, in conjunction (if necessary) with the use of supporting materials, the equipment manufacturer's specifications for adequate support and degree of level of the equipment are met.

The manufacturer's procedures and prohibitions shall be complied with when assembling and disassembling equipment.

Assembly / disassembly shall be directed by an individual who qualifies as both a competent person and a qualified person, or by a competent person who is assisted by one or more qualified persons. The competent person is referred to as the A/D director. The A/D director shall understand the applicable assembly / disassembly procedures. The A/D director shall review the applicable assembly / disassembly procedures immediately prior to the commencement of assembly / disassembly unless the A/D director understands the procedures and has applied them to the same type and configuration of equipment.

The A/D director is responsible for ensuring crew members are knowledgeable of their tasks prior to commencing work.

Prior to starting work, the A/D director is responsible for ensuring crew members are knowledgeable of the hazards associated with their tasks.

The A/D director is responsible for ensuring crew members are knowledgeable before commencing work of the hazardous positions/locations they need to avoid.

The A/D director shall be responsible for managing specific hazards associated with assembly/disassembly operations. These include the following hazards:

- Site and ground bearing conditions
- Blocking material
- Proper location of blocking
- Verifying assist crane loads
- Boom and jib pick points
- Center of gravity
- Stability upon pin removal
- Snagging
- Struck by counterweights
- Boom hoist brake failure
- Loss of backward stability
- Wind speed and weather

The boundaries of the swing radius of the crane must be marked or barricaded by barriers, lines, railings, or comparable warnings.

Rated capacity limits for loads imposed on the equipment, equipment components (including rigging), lifting lugs, and equipment accessories must not be exceeded for the equipment being assembled/disassembled during all phases of assembly/disassembly.

1.180.2 Work Zone

The work zone shall be identified by either:

a) Demarcating boundaries (such as with flags, or a device such as a range limit device or range control warning device) and prohibiting the operator from operating the equipment past those boundaries, or

b) Defining the work zone as the area 360 degrees around the equipment, up to the equipment's maximum working radius.

c) Erect and maintain control lines, warning lines, railings, or similar barriers to mark the boundaries of the hazard areas.

1.180.3 Work Near Power Lines

Determine if any part of the equipment, load line, or load (including rigging and lifting accessories), if operated up to the equipment's maximum working radius in the work zone, could get closer than 20 feet to a power line. If so, one of the following options shall be performed:

a) Deenergize and ground. Confirm from the utility owner / operator that the power line has been deenergized and visibly grounded at the worksite.

b) Ensure 20-foot clearance. Ensure that no part of the equipment, load line, or load gets closer than 20 feet to the power line.

c) Determine the line's voltage and the minimum approach distance permitted. Determine if any part of the equipment, load line, or load, while operating up to the equipment's maximum working radius in the work zone, could get closer than the minimum approach distance of the power line permitted. If so, then ensure that no part of the equipment, load line, or load (including rigging and lifting accessories), gets closer to the line than the minimum approach distance.

1.180.4 Critical and Engineered Lifts

Critical and/or engineered lifts must follow jurisdictional permitting requirements.

1.181 Signaling

A signal person shall be provided in each of the following situations:

- The point of operation, meaning the load travel or the area near or at load placement, is not in full view of the operator.
- When the equipment is traveling, the view in the direction of travel is obstructed.
- Due to site specific safety concerns, either the operator or the person handling the load determines that it is necessary.

Signals to operators shall use the hand, voice, audible method. Means of transmitting the signals (direct line of sight, radio, etc.) shall be suitable and appropriate for the site conditions. Hand signals shall follow the Standard Method in Appendix A of Subpart CC included in the Appendix of this document.

The device used to transmit signals shall be tested on site before beginning operations to ensure that the signal transmission is effective, clear, and reliable.

The ability to transmit signals between the operator and signal person shall be maintained. If the ability to transmit signals is interrupted at any time, the operator shall safely stop operations requiring signals until communication is reestablished and a proper signal is given and understood.

Only one person shall give signals to a crane at a time unless the emergency stop signal is given due to safety issues.

Each signal person shall:

- Know and understand the type(s) of signals used,
- Be competent in the application of the type of signals used,
- Have a basic understanding of equipment operation and limitations, including the crane dynamics involved in swinging and stopping loads and boom deflection from hoisting loads,
- Demonstrate that they meet the qualification requirements through an oral or written test, and through a practical test.

1.182 Inspection

All inspections shall be conducted by a competent person.

A visual inspection of equipment shall be conducted prior to each shift. The inspection shall consist of observation for apparent deficiencies.

Inspection items shall include:

- Pre-erection inspections
- Control mechanisms
- Pressurized lines
- Hooks and latches
- Wire rope
- Electrical apparatus
- Tires (when used)
- Ground conditions

Monthly inspections shall be conducted and documented. Documentation shall include:

- Items checked
- Results of inspection

- Name and signature of the inspector

Inspection records and preventative maintenance record documentation shall be maintained and retained for 3 months. Documented monthly inspection is not required if the daily inspection is documented and records are retained for 3 months.

The post assembly inspection shall ensure the selection of components, and the configuration of equipment that affect the safe operation of the crane are in accordance with manufacturer instructions, prohibitions, limitations, and specifications.

1.183 Critical Lifts

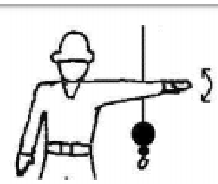
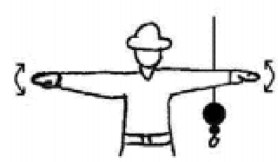
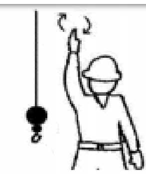
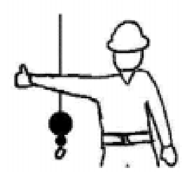
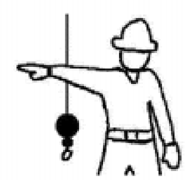
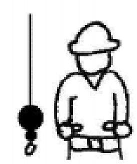
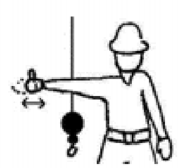
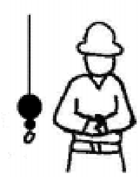
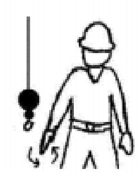
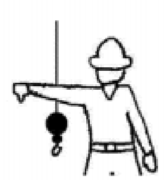
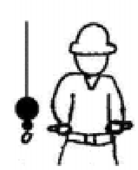
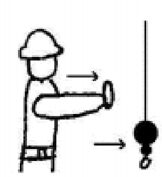
A critical lift is a lift where:

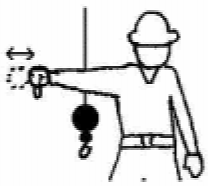
- The load exceeds 80% of the crane's capacity.
- Weight of the lift exceeds 50% of the load chart rating of the equipment being used and the lift is over power lines, process equipment, piping, or personnel are being lifted.
- Two booms are required.
- Poles or derricks have been erected.
- Personnel are being lifted.
- Crane is traveling with load.
- Any lift in a Critical Lift Area.

A Crane Safe Work Permit must be issued for the following:

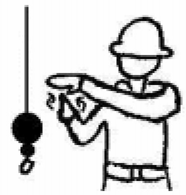
- All lifts with cranes having a capacity greater than 10 tons.
- All critical lifts.

Appendix 11 **Hand Signals**

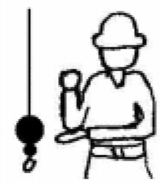
 **STOP** – With arm extended horizontally to the side, palm down, arm is swung back and forth.	 **EMERGENCY STOP** – With both arms extended horizontally to the side, palms down, arms are swung back and forth.	 **HOIST** – With upper arm extended to the side, forearm and index finger pointing straight up, hand and finger make small circles.
 **RAISE BOOM** – With arm extended horizontally to the side, thumb points up with other fingers closed.	 **SWING** – With arm extended horizontally, index finger points in direction that boom is to swing.	 **RETRACT TELESCOPING BOOM** – With hands to the front at waist level, thumbs point at each other with other fingers closed.
 **RAISE THE BOOM AND LOWER THE LOAD** – With arm extended horizontally to the side and thumb pointing up, fingers open and close while load movement is desired.	 **DOG EVERYTHING** – Hands held together at waist level.	 **LOWER** – With arm and index finger pointing down, hand and finger make small circles.
 **LOWER BOOM** – With arm extended horizontally to the side, thumb points down with other fingers closed.	 **EXTEND TELESCOPING BOOM** – With hands to the front at waist level, thumbs point outward with other fingers closed.	 **TRAVEL/TOWER TRAVEL** – With all fingers pointing up, arm is extended horizontally out and back to make a pushing motion in the direction of travel.



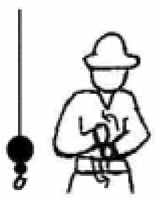
LOWER THE BOOM AND RAISE THE LOAD – With arm extended horizontally to the side and thumb pointing down, fingers open and close while load movement is desired.



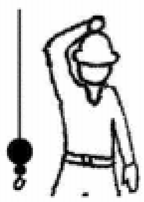
MOVE SLOWLY – A hand is placed in front of the hand that is giving the action signal.



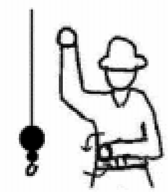
USE AUXILIARY HOIST (whipline) – With arm bent at elbow and forearm vertical, elbow is tapped with other hand. Then regular signal is used to indicate desired action.



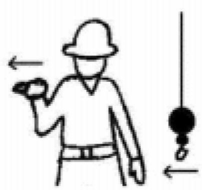
CRAWLER CRANE TRAVEL, BOTH TRACKS – Rotate fists around each other in front of body; direction of rotation away from body indicates travel forward; rotation towards body indicates travel backward.



USE MAIN HOIST – A hand taps on top of the head. Then regular signal is given to indicate desired action.



CRAWLER CRANE TRAVEL, ONE TRACK – Indicate track to be locked by raising fist on that side. Rotate other fist in front of body in direction that other track is to travel.



TROLLEY TRAVEL – With palm up, fingers closed and thumb pointing in direction of motion, hand is jerked horizontally in direction trolley is to travel.

17. Demolition and Blasting Safe Work Practices Program

1.184 Purpose and Scope

The purpose of this program is to identify and mitigate potential hazards associated with demolition and blasting activities, implement proper safety protocols and procedures, conduct regular inspections, and continuously improve safety measures to create a secure working environment and promote responsible and effective demolition and blasting practices.

This program applies to all T&T Industrial, Inc. employees involved with demolition and blasting.

1.185 Resources

Number	Title
29 CFR 1926 Subpart T	Demolition
29 CFR 1926 Subpart U	Blasting and the Use of Explosives

1.186 Demolition and Blasting Safe Work Practices

Before the start of every demolition job, the Company shall take steps to safeguard the health and safety of employees at the job site. These preparatory operations involve the overall planning of the demolition job, including the methods to be used to bring the structure down, the equipment necessary to do the job, and the measures to be taken to perform the work safely. Planning for a demolition job is as important as actually doing the work. Therefore, a competent / qualified person experienced in all phases of the demolition work to be performed shall perform all planning work.

1.187 Pre-Demolition / Engineering Survey

Prior to starting a demolition operation, a written pre-demolition survey or engineering survey of the structure shall be conducted. The purpose of this survey is to determine the condition of the framing, floors, and walls so that measures can be taken, if necessary, to prevent the premature collapse of any portion of the structure. In addition, the survey will identify designated / hazardous substances, physical hazards, and health hazards, etc. When indicated as advisable, any adjacent structure(s) or improvements shall also be similarly checked. The Company will maintain a written copy of this survey. Photographing existing damage in neighboring structures will also take place by the competent / qualified person designated.

The pre-demolition survey or engineering survey provides the Company with the opportunity to evaluate the job in its entirety. The Company shall plan for the wrecking of the structure, the equipment to do the work, manpower requirements, and the protection of the public. The safety of all employees on the job site will be a prime consideration. During the preparation of the pre-demolition survey or engineering survey, the Company shall plan for potential hazards such as fires, cave-ins, and injuries. If the structure to be demolished has been damaged by fire, flood, explosion, or some other cause, appropriate measures, including bracing and shoring of walls and floors, shall be taken to protect employees and any adjacent structures. It shall also be determined if any type of hazardous chemicals, gases, explosives, flammable material, or similar dangerous substances have been used or stored on the site. If the nature of a substance cannot be easily determined, samples shall be taken and analyzed by a qualified person prior to demolition.

1.188 Utility Location

One of the most important elements of the pre-job planning is the location of all utility services. All electric, gas, water, steam, sewer, and other services lines shall be shut off, capped, or otherwise controlled, at or outside the building before demolition work is started. In each case, any utility company that is involved shall be notified in advance, and its approval or services, if necessary, shall be obtained.

If it is necessary to maintain any power, water, or other utilities during demolition, such lines shall be temporarily relocated as necessary and/or protected. The location of all overhead power sources shall also be determined, as they can prove especially hazardous during any machine demolition. All employees shall be informed of the location of any existing or relocated utility service. The telephone numbers of the local police, ambulance, and fire departments shall be available at each job site. This information can prove useful to the job supervisor in the event of any traffic problems, such as the movement of equipment to or from the job site.

1.189 Medical Services and First Aid

Prior to starting work, provisions shall be made for prompt medical attention in case of serious injury. The nearest hospital, infirmary, clinic, or physician shall be located as part of the pre-demolition survey or engineering survey. The job supervisor shall be provided with instructions for the most direct route to these facilities. Proper equipment for prompt transportation of an injured employee, as well as a communication system to contact any necessary ambulance service, will be available at the job site. The telephone numbers of the hospitals, physicians, or ambulances shall be conspicuously posted.

A properly stocked first aid kit as determined by an occupational physician will be available at the job site. The first aid kit shall contain approved supplies in a weatherproof container with individual sealed packages for each type of item. It shall also include rubber gloves to prevent the transfer of infectious diseases. Provisions shall also be made to provide for quick drenching or flushing of the eyes should any person be working around corrosive materials. Eye flushing will be done with water containing no additives. The contents of the kit shall be checked before being sent out on each job and at least weekly to ensure the expended items are replaced.

1.190 Fire Prevention and Protection

A "fire plan" shall be set up prior to beginning a demolition job. This plan shall outline the assignments of key personnel in the event of a fire and provide an evacuation plan for employees on the site. Common sense shall be the general rule in all fire prevention planning.

1.191 Preparatory Operations

When employees are required to work within a structure to be demolished which has been damaged by fire, flood, explosion, or other cause, the walls or floor shall be shored or braced.

All electric, gas, water, steam, sewer, and other service lines shall be shut off, capped, or otherwise controlled, outside the building line before demolition work is started. In each case, any utility company that is involved shall be notified in advance.

If it is necessary to maintain any power, water or other utilities during demolition, such lines shall be temporarily relocated, as necessary, and protected.

It shall also be determined if any type of hazardous chemicals, gases, explosives, flammable materials, or similarly dangerous substances have been used in any pipes, tanks, or other equipment on the property. When the presence of any such substances is apparent or suspected, testing and purging shall be performed, and the hazard eliminated before demolition is started.

Where a hazard exists from fragmentation of glass, such hazards shall be removed.

Where a hazard exists to employees falling through wall openings, the opening shall be protected to a height of approximately 42 inches.

When debris is dropped through holes in the floor without the use of chutes, the area onto which the material is dropped shall be completely enclosed with barricades not less than 42 inches high and not less than 6 feet back from the projected edge of the opening above. Signs, warning of the hazard of falling materials, shall be posted at each level. Removal shall not be permitted in this lower area until debris handling ceases above.

All floor openings, not used as material drops, shall be covered over with material substantial enough to support the weight of any load that may be imposed. Such material shall be properly secured to prevent its accidental movement.

Except for the cutting of holes in floors for chutes, holes through which to drop materials, preparation of storage space, and similar necessary preparatory work, the demolition of exterior walls and floor construction shall begin at the top of the structure and proceed downward. Each story of exterior wall and floor construction shall be removed and dropped into the storage space before commencing the removal of exterior walls and floors in the story next below.

Employee entrances to multi-story structures being demolished shall be completely protected by sidewalk sheds or canopies, or both, providing protection from the face of the building for a minimum of 8 feet. All such canopies shall be at least 2 feet wider than the building entrances or openings (1 foot wider on each side thereof) and shall be capable of sustaining a load of 150 pounds per square foot.

During the planning stage of the job, all safety equipment needs shall be determined. The required number and type of respirators, lifelines, warning signs, safety nets, special face and eye protection, hearing protection, and other employee protection.

In the absence of an infirmary, clinic, hospital, or physician that is reasonably accessible in terms of time and distance to the work site, a person who has a valid certificate in first aid training shall be available at the work site to render first aid.

A comprehensive first aid plan is necessary for any confined space entry.

1.192 Pre-Demolition Considerations

- All potential sources of ignition shall be evaluated, and the necessary corrective measures taken.
- When storing debris or combustible material inside a structure, such storage shall not obstruct or adversely affect the means of exit.
- Electrical wiring and equipment for providing light, heat, or power shall be installed by a competent person and inspected regularly.
- A suitable location at the job site shall be designated and provided with plans, emergency information, and equipment, as needed.
- Access for heavy fire-fighting equipment shall be provided on the immediate job site at the start of the job and well maintained until the job is completed.
- Equipment powered by an internal combustion engine shall be located so that the exhausts discharge away from combustible materials and away from employees.
- When the exhausts are piped outside the building, clearance of at least six inches shall be maintained between such piping and combustible material.

- Free access from the street to fire hydrants and to outside connections for standpipes, sprinklers, or other fire extinguishing equipment, whether permanent or temporary, should be provided and maintained at all times.
- All internal combustion equipment shall be shut down prior to refueling. Fuel for this equipment shall be stored in a safe location.
- Pedestrian walkways should not be so constructed as to impede access to hydrants.
- Sufficient firefighting equipment shall be located near any flammable or combustible liquid storage area.
- Only approved containers and portable tanks shall be used for the storage and handling of flammable combustible liquids.
- No material or construction should interfere with access to hydrants, Siamese connections, or fire-extinguishing equipment.
- A temporary or permanent water supply of volume, duration, and pressure sufficient to operate the fire-fighting equipment properly shall be made available.
- Standpipes with outlets should be provided on large multistory buildings to provide for fire protection on upper levels. If the water pressure is insufficient, a pump shall also be provided.
- Heating devices shall be situated so they are not likely to overturn and shall be installed in accordance with their listing, including clearance to combustible material or equipment.
- Temporary heating equipment, when utilized, shall be maintained by competent personnel.
- An ample number of fully charged portable fire extinguishers should be provided throughout the operation. All motor driven mobile equipment shall be equipped with an approved fire extinguisher.
- Roadways between and around combustible storage piles shall be at least 15 feet wide and maintained free from accumulation of rubbish, equipment, or other materials.
- Smoking shall be prohibited at or in the vicinity of hazardous operations or materials. Where smoking is permitted, safe receptacles shall be provided for smoking materials.
- An alarm system, (e.g., telephone system, siren, two-way radio, etc.) shall be established in such a way that employees on the site and the local fire department can be alerted in case of an emergency. The alarm code and reporting instructions shall be conspicuously posted, and the alarm system shall be serviceable at the job site during the demolition. Fire

cutoffs shall be retained in the buildings undergoing alterations or demolition until operations necessitate their removal.

1.193 Stairs, Passageways, and Ladders

Only those stairways, passageways, and ladders, designated as means of access to the structure of a building, shall be used. Other access ways shall be entirely closed at all times.

All stairs, passageways, ladders and incidental equipment thereto, which are covered by this section, shall be periodically inspected and maintained in a clean safe condition.

In a multistory building, when a stairwell is being used, it shall be properly illuminated by either natural or artificial means, and completely and substantially covered over at a point not less than two floors below the floor on which work is being performed, and access to the floor where the work is in progress shall be through a properly lighted, protected, and separate passageway.

1.194 Chutes

No material shall be dropped to any point lying outside the exterior walls of the structure unless the area is effectively protected.

All materials chutes or sections thereof, at an angle of more than 45 degrees from the horizontal, shall be entirely enclosed, except for openings equipped with closures at or about floor level for the insertion of materials. The openings shall not exceed 48 inches in height measured along the wall of the chute. At all stories below the top floor, such openings shall be kept closed when not in use.

A substantial gate shall be installed in each chute at or near the discharge end. A competent employee shall be assigned to control the operation of the gate, and the backing and loading of trucks.

When operations are not in progress, the area surrounding the discharge end of a chute shall be securely closed off.

Any chute opening, into which employees dump debris, shall be protected by a substantial guardrail approximately 42 inches above the floor or other surface on which the employees stand to dump the material. Any space between the chute and the edge of openings in the floors through which it passes shall be solidly covered over.

Where the material is dumped from mechanical equipment or wheelbarrows, a securely attached toe board or bumper, not less than 4 inches thick and 6 inches high, shall be provided at each chute opening.

Chutes shall be designed and constructed of such strength as to eliminate failure due to impact of materials or debris loaded therein.

1.195 Removal of Materials through Floor Openings

Any openings cut in a floor for the disposal of materials shall be no larger in size than 25% of the aggregate of the total floor area, unless the lateral supports of the removed flooring remain in place. Floors weakened or otherwise made unsafe by demolition operations shall be shored to carry safely the intended imposed load from demolition operations.

1.196 Removal of Walls, Masonry Sections, and Chimneys

Masonry walls, or other sections of masonry, shall not be permitted to fall upon the floors of the building in such masses as to exceed the safe carrying capacities of the floors.

No wall section, which is more than one story in height, shall be permitted to stand alone without lateral bracing, unless such wall was originally designed and constructed to stand without such lateral support, and is in a condition safe enough to be self-supporting. All walls shall be left in a stable condition at the end of each shift.

Employees shall not be permitted to work on the top of a wall when weather conditions constitute a hazard.

Structural or load-supporting members on any floor shall not be cut or removed until all stories above such a floor have been demolished and removed. This provision shall not prohibit the cutting of floor beams for the disposal of materials or for the installation of equipment, provided that the jurisdictional requirements (Local, State, Federal, or Provincial) are met.

Floor openings within 10 feet of any wall being demolished shall be planked solid, except when employees are kept out of the area below.

In buildings of "skeleton-steel" construction, the steel framing may be left in place during the demolition of masonry. Where this is done, all steel beams, girders, and similar structural supports shall be cleared of all loose material as the masonry demolition progresses downward.

Walkways or ladders shall be provided to enable employees to safely reach or leave any scaffold or wall.

Walls, which serve as retaining walls to support earth or adjoining structures, shall not be demolished until such earth has been properly braced or adjoining structures have been properly underpinned.

Walls, which are to serve as retaining walls against which debris will be piled, shall not be so used unless capable of safely supporting the imposed load.

1.197 Manual Removal of Floors

Openings cut in a floor shall extend the full span of the arch between supports.

Before demolishing any floor arch, debris and other material shall be removed from such arch and other adjacent floor area. Planks not less than 2 inches by 10 inches in cross section, full size undressed, shall be provided for, and shall be used by employees to stand on while breaking down floor arches between beams. Such planks shall be so located as to provide a safe support for the employee should the arch between the beams collapse. The open space between planks shall not exceed 16 inches.

Safe walkways, not less than 18 inches wide, formed of planks not less than 2 inches thick of wood, or of equivalent strength if metal, shall be provided and used by employees when necessary to enable them to reach any point without walking upon exposed beams.

Stringers of ample strength shall be installed to support the flooring planks, and the ends of such stringers shall be supported by floor beams or girders, and not by floor arches alone.

Planks shall be laid together over solid bearings with the ends overlapping at least 1 foot.

When floor arches are being removed, employees shall not be allowed in the area directly underneath, and such an area shall be barricaded to prevent access to it.

Demolition of floor arches shall not be started until they, and the surrounding floor area for 20 feet, have been cleared of debris and any other unnecessary materials.

1.198 Removal of Walls, Floors, and Material with Equipment

Mechanical equipment shall not be used on floors or working surfaces unless such floors or surfaces are of sufficient strength to support the imposed load.

Floor openings shall have curbs or stop-logs to prevent equipment from running over the edge.

For use of cranes, derricks, and other mechanical equipment, the requirements specified in subparts N, O, and CC must be met.

1.199 Storage

The storage of waste material and debris on any floor shall not exceed the allowable floor loads.

In buildings having wooden floor construction, the flooring boards may be removed from not more than one floor above grade to provide storage space for debris, provided falling material is not permitted to endanger the stability of the structure.

When wood floor beams serve to brace interior walls or freestanding exterior walls, such beams shall be left in place until other equivalent support can be installed to replace them.

Floor arches, to an elevation of not more than 25 feet above grade, may be removed to provide storage area for debris, provided that such removal does not endanger the stability of the structure.

Storage space into which material is dumped shall be blocked off, except for openings necessary for the removal of material. Such openings shall be kept closed at all times when material is not being removed.

1.200 Removal of Steel Construction

When floor arches have been removed, planking in accordance with jurisdictional requirements (Local, State, Federal, or Provincial) shall be provided for the employees engaged in razing the steel framing.

Cranes, derricks, and other hoisting equipment used shall meet the requirements specified in other chapters of this Program.

Steel construction shall be dismantled column length by column length, and tier by tier (columns may be in two-story lengths).

Any structural member being dismembered shall not be overstressed.

1.201 Mechanical Demolition

No employees shall be permitted in any area, which can be adversely affected by demolition operations, when balling or clamming is being performed. Only those employees necessary for the performance of the operations shall be permitted in this area at any other time.

The crane boom and load line shall be as short as possible.

The weight of the demolition ball shall not exceed 50% of the crane's rated load, based on the length of the boom and the maximum angle of operation at which the demolition ball will be used, or it shall not exceed 25% of the nominal breaking strength of the line by which it is suspended, whichever results in a lesser value.

The ball shall be attached to the load line with a swivel-type connection to prevent twisting of the load line and shall be attached by positive means in such manner that the weight cannot become accidentally disconnected.

When pulling over walls or portions thereof, all steel members affected shall have been previously cut free.

All roof cornices or other such ornamental stonework shall be removed prior to pulling walls over.

During demolition, continuing inspections by a competent person shall be made as the work progresses to detect hazards resulting from weakened or deteriorated floors, or walls, or loosened material. No employee shall be permitted to work where such hazards exist until they are corrected by shoring, bracing, or other effective means.

1.202 Safe Work Practices when Demolishing a Chimney, Stack, Silo, or Cooling Tower

1.202.1 Inspection and Planning

When preparing to demolish any chimney, stack, silo, or cooling tower, the first step will be a careful, detailed inspection of the structure by an experienced person. If possible, architectural / engineering drawings shall be consulted. Particular attention shall be paid to the condition of the chimney or stack. Employees shall be on the lookout for any structural defects such as weak or acid-laden mortar joints, and any cracks or openings. The interior brickwork in some sections of industrial chimney shafts can be extremely weak. If the stack has been banded with steel straps, these bands shall be removed only as the work progresses from the top down. Sectioning of the chimney by water, etc. shall be considered.

1.202.2 Safe Work Practices

When hand demolition is required, it shall be carried out from a working platform.

Experienced personnel shall install a self-supporting tubular scaffold, suspended platform, or knee-braced scaffolding around the chimney.

Particular attention shall be paid to the design, support, and tie-in (braces) of the scaffold.

A competent person shall be present at all times during the erection of the scaffold.

It is essential that there be adequate working clearance between the chimney and the work platform.

Access to the top of the scaffold shall be provided by means of portable walkways.

The platforms shall be decked solid and the area from the work platform to wall bridged with a minimum of 2-inch-thick lumber.

A top rail 42 inches above the platform, with a mid-rail covered with canvas or mesh, shall be installed around the perimeter of the platform to prevent injury to employees below. Debris netting may be installed below the platform.

Excess canvas or plywood attachments can form a windsail that could collapse the scaffold.

When working on the work platform, all personnel shall wear hard hats, long-sleeve shirts, eye and face protection, such as goggles and face shields, respirators, and safety belts, as required.

Care shall be taken to assign the proper number of employees to the task.

Too many people on a small work platform can lead to accidents.

An alternative to the erection of a self-supporting tubular steel scaffold to "climb" the structure with a creeping bracket scaffold. Careful inspection of the masonry and a decision as to the safety of this alternative will be made by a competent person. It is essential that the masonry of the chimney be in good enough condition to support the bracket scaffold.

The area around the chimney shall be roped off or barricaded and secured with appropriate warning signs posted. No unauthorized entry shall be permitted to this area. It is also good practice to keep an employee (i.e., a supervisor, operating engineer, another employee, or a "safety person,") on the ground with a form of communication to the employees above.

Special attention shall be paid to weather conditions when working on a chimney. No work shall be done during inclement weather such as during lightning or high wind situations. The work site shall be wet down, as needed, to control dust.

1.202.3 Debris Clearance

If debris is dropped inside the shaft, it can be removed through an opening in the chimney at grade level.

The opening at grade will be kept relatively small in order not to weaken the structure.

If a larger opening is desired, a professional engineer shall be consulted.

When removing debris by hand, an overhead canopy of adequate strength shall be provided.

If machines are used for removal of debris, proper overhead protection for the operator shall be used.

Excessive debris shall not be allowed to accumulate inside or outside the shaft of the chimney as the excess weight of the debris can impose pressure on the wall of the structure and might cause the shaft to collapse.

The foreman shall determine when debris is to be removed, halt all demolition during debris removal, and make sure the area is clear of cleanup employees before continuing demolition.

1.202.4 Demolition by Deliberate Collapse

Another method of demolishing a chimney or stack is by deliberate collapse. Deliberate collapse requires extensive planning and experienced personnel and shall be used only when conditions are favorable.

There shall be a clear space for the fall of the structure of at least 450 feet on each side of the intended fall line and 1½ times the total height of the chimney. Considerable vibration may be set up when the chimney falls, so there should be no sewers or underground services on the line of the fall. Lookouts must be posted on the site and warning signals must be arranged. The public and other employees at the job site must be kept well back from the fall area.

1.202.5 Demolition of Prestressed Concrete Structures

The different forms of construction used in a number of more or less conventional structures built during the last few decades will give rise to a variety of problems when the time comes for them to be demolished. Prestressed concrete structures fall in this general category. The most important aspect of demolishing a prestressed concrete structure takes place during the pre-demolition survey or engineering survey. During the survey, a qualified person shall determine if the structure to be demolished contains any prestressed members.

It is the responsibility of the Company to inform all employees on the demolition job site of the presence of prestressed concrete members within the structure. The Company shall also instruct them in the safe work practice that will be followed to safely perform the demolition. Employees shall be informed of the hazards of deviating from the prescribed procedures and the importance of following their supervisor's instruction.

There are four main categories of prestressed members. The category or categories shall be determined before attempting demolition, bearing in mind that any prestressed structure may contain elements of more than one category.

- Category 1- Members are prestressed before the application of the superimposed loads, and all cables or tendons are fully bonded in the concrete or grouted within ducts.
- Category 2- Like Category 1, but the tendons are left ungrouted. This type of construction can sometimes be recognized from the access points that may have been provided for

inspection of the cables and anchors. More recently, unbonded tendons have been used in the construction of beams, slabs, and other members; these tendons are protected by grease and surrounded by plastic sheathing, instead of the usual metal duct.

- Category 3- Members are prestressed progressively as building construction proceeds and the dead load increases, using bonded tendons as in Category 1.
- Category 4 Like Category 3 but using unbonded tendons as in Category 2.

Examples of construction using members of Categories 3 or 4 are relatively rare. However, they may be found, for example, in the podium of a tall building or some types of bridges. They require particular care in demolition.

1.202.6 Pretensioned Members

These usually do not have any end anchors, the wires being embedded or bonded within the length of the member. Simple pretensioned beams and slabs of spans up to about 7 meters (23 feet) can be demolished in a manner similar to ordinary reinforced concrete. Pretensioned beams and slabs may be lifted and lowered to the ground as complete units after the removal of any composite concrete covering to tops and ends of the units. To facilitate breaking up, the members shall be turned on their sides. Lifting from the structure shall generally be done from points near the ends of the units or from lifting point positions. Reuse of lifting eyes, if in good condition, is recommended whenever possible. When units are too large to be removed, consideration shall be given to temporary supporting arrangements.

1.202.7 Ducts

Before breaking up, units of this type shall be lowered to the ground, if possible. It is advisable to seek the counsel of a professional engineer before carrying out this work, especially where there are ungrouted tendons. In general, this is true because grouting is not always 100% efficient. After lowering, the units can be turned on their side with the ends up on blocks after any composite concrete is removed. This may suffice to break the unit and release the prestress; if not, a sandbag screen, timbers, or a blast mat as a screen shall be erected around the ends and demolition commenced, taking care to clear the area of any personnel. It shall be borne in mind that the end blocks may be heavily reinforced and difficult to break up.

1.202.8 Monolithic Structures

The advice of the professional engineer experienced in prestressed work shall be sought before any attempt is made to expose the tendons or anchorages of structures in which two or more members have been stressed together. It will usually be necessary for temporary supports to be provided so the tendons and the anchorage can be cautiously exposed. In these circumstances it is essential that indiscriminate attempts to expose and de-stress the tendons and anchorages are not made.

1.202.9 Progressively Prestressed Structures

In the case of progressively prestressed structures, it is essential to obtain the advice of a professional engineer, and to demolish the structure in strict accordance with the engineer's method of demolition. The stored energy in this type of structure is large. In some cases, the inherent properties of the stressed section may delay failure for some time, but the presence of these large prestressing forces may cause sudden and complete collapse with little warning.

1.202.10 Safe Work Practices when Working in Confined Spaces

Confined spaces can be generally categorized in two major groups: those with open tops and a depth that restricts the natural movement of air, and enclosed spaces with very limited openings for entry. Examples of these spaces include storage tanks, vessels, degreasers, pits vaults, casing, and silos. The hazards encountered when entering and working in confined spaces can cause bodily injury, illness, and death. Accidents occur among employees because of failure to recognize that a confined space is a potential hazard. It shall therefore be considered that the most unfavorable situation exists in every case and that the danger of explosion, poisoning, and asphyxiation will be present at the onset of entry.

1.203 Blasting General Safe Work Practices

1.203.1 Blasting Survey and Site Preparation

Prior to the blasting of any structure or portion thereof, a complete written survey will be made by a qualified person of all adjacent improvements and underground utilities. When there is a possibility of excessive vibration due to blasting operations, seismic or vibration tests shall be taken to determine proper safety limits to prevent damage to adjacent or nearby buildings, utilities, or other property.

The preparation of a structure for demolition by explosives may require the removal of structural columns, beams, or other building components. This work shall be directed by a structural engineer or a competent person qualified to direct the removal of these structural elements. Extreme caution will be taken during this preparatory work to prevent the weakening and premature collapse of the structure.

The use of explosives to demolish smokestacks, silos, cooling towers, or similar structures shall be permitted only if there is a minimum of 900 feet of open space extended for at least 150% of the height of the structure or if the explosives specialist can demonstrate consistent previous performance with tighter constraints at the site.

1.203.2 Fire Precautions

The presence of fire near explosives presents a severe danger. Every effort shall be made to ensure that fires or sparks do not occur near explosive materials. Smoking, matches, firearms, open flame lamps, and other fires, flame, or heat-producing devices will be prohibited in or near explosive magazines or in areas where explosives are being handled, transported, or used. Persons working near explosives shall not carry matches, lighters, or other sources of sparks or flame. Open fires or flames shall be prohibited within 100 feet of any explosive materials. In the event of a fire which is in imminent danger of contact with explosives, all employees will be removed to a safe area.

Electrical detonators can be inadvertently triggered by stray RF (radio frequency) signals from two-way radios. RF signal sources shall be restricted from or near to the demolition site if electrical detonators are used.

1.203.3 Personnel Selection

A blaster is a competent person who uses explosives. A blaster will be qualified by reason of training, knowledge, or experience in the field of transporting, storing, handling, and using explosives. In addition, the blaster shall have a working knowledge of state and local regulations that pertain to explosives. Training courses are often available from manufacturers of explosives and blasting safety manuals are offered by the Institute of Makers of Explosives (IME) as well as other organizations.

Blasters shall be required to furnish satisfactory evidence of competency in handling explosives and in safely performing the type of blasting required. A competent person shall always be in charge of explosives and shall be held responsible for enforcing all recommended safety precautions in connection with them.

1.204 Transportation of Explosives

1.204.1 Vehicle Safety

Vehicles used for transporting explosives shall be strong enough to carry the load without difficulty and shall be in good mechanical condition. All vehicles used for the transportation of explosives shall have tight floors, and any exposed spark-producing metal on the inside of the body shall be covered with wood or some other non-sparking material.

Vehicles or conveyances transporting explosives shall only be driven by, and shall be under the supervision of, a licensed driver familiar with the local, state, and Federal regulations governing the transportation of explosives. No passengers shall be allowed in any vehicle transporting explosives.

Explosives, blasting agents, and blasting supplies shall not be transported with other materials or cargoes. Blasting caps shall not be transported with other materials or cargoes.

Blasting caps shall not be transported in the same vehicle with other explosives. If an open-bodied truck is used, the entire load shall be completely covered with a fire and water-resistant tarpaulin to protect it from the elements. Vehicles carrying explosives shall not be loaded beyond the manufacturer's safe capacity rating, and in no case shall the explosives be piled higher than the closed sides and ends of the body.

Every motor vehicle or conveyance used for transporting explosives shall be marked or placarded with warning signs required by OSHA and the DOT.

Each vehicle used for transportation of explosives shall be equipped minimally with at least 10 pound rated serviceable ABC fire extinguisher. All drivers shall be trained in the use of the extinguishers on their vehicle.

In transporting explosives, congested traffic and high density population areas shall be avoided, where possible, and no unnecessary stops shall be made. Vehicles carrying explosives, blasting agents, or blasting supplies shall not be taken inside a garage or shop for repairs or servicing. No motor vehicle transporting explosives shall be left unattended.

1.205 Storage of Explosives

1.205.1 Inventory Handling and Safe Handling

All explosives shall be accounted for at all times and all not being used will be kept in a locked magazine. A complete detailed inventory of all explosives received and placed in, removed from, and returned to the magazine shall be maintained at all times. Appropriate authorities will be notified of any loss, theft, or unauthorized entry into a magazine.

Manufacturers' instructions for the safe handling and storage of explosives are ordinarily enclosed in each case of explosives. The specifics of storage and handling are best referred to these instructions and the IME manuals. They shall be carefully followed. Packages of explosives shall not be handled roughly. Sparking metal tools shall not be used to open wooden cases. Metallic slitters may be used for opening fiberboard cases, provided the metallic slitter does not come in contact with the metallic fasteners of the case.

The oldest stock shall always be used first to minimize the chance of deterioration from long storage. Loose explosives or broken, defective, or leaking packages can be hazardous and shall be segregated and properly disposed of in accordance with the specific instructions of the manufacturer. If the explosives are in good condition it may be advisable to repack them. In this case, the explosives supplier shall be contacted. Explosives cases shall not be opened, or explosives packed or repacked while in a magazine.

1.205.2 Storage Conditions

Providing a dry, well ventilated place for the storage of explosives to make sure mats and other protection do not disturb the explosives is one of the most important and effective safety measures. Every precaution shall be taken to keep them dry and relatively cool. Dampness or excess humidity may be the cause of misfires resulting in injury or loss of life. Explosives shall be stored in properly constructed fire and bullet-resistant structures, located according to the IME American Table of Distances and kept locked at all times except when opened for use by an authorized person. Explosives shall not be left, kept, or stored where children, unauthorized persons, or animals have access to them, nor shall they be stored in or near a residence.

Detonators shall be stored in a separate magazine located according to the IME American Table of Distances. Detonators shall never be stored in the same magazine with any other kind of explosive.

Ideally, arrangements shall be made whereby the supplier delivers the explosives to the job site in quantities which will be used up during the workday. An alternative would be for the supplier to return to pick up unused quantities of explosives. If it is necessary for the Company to store explosives, they shall be familiar with all local requirements for such storage.

1.205.3 Proper Use of Explosives

Blasting operations shall be conducted between sunup and sundown, whenever possible. Adequate signs shall be sounded to alert to the hazard presented by blasting. Blasting mats or other containment shall be used where there is danger of rocks or other debris being thrown into the air or where there are buildings or transportation systems nearby. Care shall be taken to make sure mats and other protections do not disturb the connections to electrical blasting caps.

Radio, television, and radar transmitters create fields of electrical energy that can, under exceptional circumstances, detonate electric blasting caps. Certain precautions will be taken to prevent accidental discharge of electric blasting caps from current induced by radar, radio transmitters, lightning, adjacent power lines, dust storms, or other sources of extraneous or static electricity.

These precautions shall include:

- Ensuring that mobile radio transmitters on the job site which are less than 100 feet away from electric blasting caps, in other than original containers, shall be de-energized and effectively locked.
- The prominent display of adequate signs, warning against the use of mobile radio transmitters, on all roads within 1,000 feet of the blasting operations.
- Maintaining the minimum distances recommended by the IME between the nearest transmitter and electric blasting caps.
- The suspension of all blasting operations and removal of persons from the blasting area during the approach and progress of an electric storm.
- After loading is completed, there shall be as little delay as possible before firing. Each blast shall be fired under the direct supervision of the blaster, who shall inspect all connections before firing and who shall personally see that all persons are in the clear before giving the order to fire. Standard signals, which indicate that a blast is about to be fired and a later all clear signal have been adopted. It is important that everyone working in the area be familiar with these signals and that they be strictly obeyed.

1.206 Procedures After Blasting

1.206.1 Inspection After the Blast

Immediately after the blast has been fired, the firing line shall be disconnected from the blasting machine and short-circuited. Where power switches are used, they shall be locked open or in the off position. Sufficient time shall be allowed for dust, smoke, and fumes to leave the blasted area before returning to the location.

An inspection of the area and the surrounding rubble shall be made by the blaster to determine if all charges have been exploded before employees are allowed to return to the operation. All wires shall be traced and the search for unexploded cartridges made by the blaster.

1.206.2 Disposal of Explosives

Explosives, blasting agents, and blasting supplies that are obviously deteriorated or damaged shall not be used; they shall be properly disposed of. Explosives distributors will usually take back old stock. Local fire marshals or representatives of the United States Bureau of Mines may also arrange for disposal. Under no circumstances shall any explosives be abandoned.

Wood, paper, fiber, or other materials that have contained high explosives shall not be used again for any purpose but shall be destroyed by burning. These materials shall not be burned in a stove, fireplace, or other confined space. Rather, they shall be burned at an isolated outdoor location, at a safe distance from thoroughfares, magazines, and other structures. It is important to check that the containers are entirely empty before burning. During burning, the area shall be adequately protected from intruders and all persons kept at least 100 feet from the fire.

18. DISCIPLINARY PROGRAM

1.207 Purpose and Scope

The purpose of this program is to establish the minimum requirements to ensure proper action is taken when an employee violates written procedures and other known safety policies or goals.

This program applies to all T&T Industrial, Inc. employees.

1.208 Resources

Number	Title
29 CFR 1926 Subpart C	General Safety and Health Provisions
CMS-FM-0021	Safety Violation Form

1.209 Disciplinary Program

It is Company policy to provide a safe and healthy place of employment. A vital part of any program is employee participation and commitment to the safety program. In order to ensure compliance with established, communicated safety procedures, employee violations of those safety procedures shall be dealt with according to this program.

1.210 Roles and Responsibilities

The Safety Coordinator is responsible for enforcement of this disciplinary program.

All employees are responsible for following Company safety policies, procedures, and safe work practices.

1.211 Safety Violations

Safety violations include but are not limited to:

- Not following verbal or written safety policies, procedures, or safe work practices
- Not following guidelines or rules
- Horseplay
- Failure to wear or abuse of selected personal protective equipment (PPE)
- Substance abuse

1.212 Disciplinary Action

T&T Industrial, Inc.
Disciplinary Program

If a safety violation is issued, employees are subject to:

1) Verbal reprimand
2) Written warning if the issue continues
3) Suspension without pay and/or termination

T&T Industrial, Inc.
Disciplinary Program

Appendix 12 **Safety Violation Form**

General Information			
Employee Name:		Employee No.:	
Date of Violation:		Time of Violation:	
Location:			

Violation Description

Disciplinary Action

Employee Statement

Signatures			
I, ______________________________________, have read/been read and understand the safety rules of this company. I agree to act in accordance with the safety rules and understand that the violation of any rule is cause for disciplinary action, up to and including termination of employment.			
Employee Signature:		Date:	
Supervisor Signature:		Date:	

19. DRIVING SAFETY PROGRAM

1.213 Purpose and Scope

The purpose of this program is to protect employees, reduce the frequency and severity of accidents involving motor vehicles, and prevent environmental damage.

This program applied to all T&T Industrial, Inc. employees.

1.214 Resources

Number	Title
CMS-FM-0022	Vehicle Inspection Form

1.215 Driving Safety Program

Motor vehicle accidents are recognized as a leading cause of work and non-work-related serious injuries and fatalities. Therefore, the operation of motor vehicles must not be perceived as a routine activity. Successful implementation of the following elements will result in fewer driving related incidents, injuries, and fatalities.

Driver abstracts shall be obtained and reviewed for all drivers of company owned vehicles. A driver abstract contains information on the operator's license, conviction information, demerit points, and suspensions.

Drivers must perform pull-through parking (pulling through a space, so the vehicle is facing outwards in the next space) when available, or backing into a parking space if necessary. This provides the operator an easier exit from the parking area as well as a quick exit in case of an emergency. When backing, it is recommended that a spotter be stationed outside the vehicle to ensure the driver backs safely, whenever practicable.

1.216 Roles and Responsibilities

Drivers are responsible for possessing a valid driver’s license for the type of motor vehicle they operate. Training in the safe operation of motor vehicles will be provided to all employees who drive company vehicles.

Employees driving, along with their passengers, are responsible for ensuring seat belts are worn while the vehicle is in operation. Seat belt use is mandatory.

Drivers shall obey all traffic laws including possessing a valid driver's license, speed limits, signaling when changing lanes, obeying traffic lights, etc.

Drivers shall avoid distractions, such as adjusting the radio or other controls, eating, or drinking, and using the phone.

Drivers shall not drive with illegal drugs, including carrying or smoking marijuana, in the vehicle.

1.217 Substance Abuse

Employees are strictly prohibited from operating a motor vehicle while under the influence of drugs or alcohol. This includes:

- Blood alcohol level at or above the local legal limit,
- Illegal drugs, including marijuana, and
- Prescription medications that cause drowsiness or other conditions that may cause impairment. Employees taking prescription medication that may impact their safety shall report this to their supervisor.

1.218 Incidents

Motor vehicle incidents occurring while on company business shall be reported immediately, regardless of the severity. Emergency services should be called first (if necessary). All incidents shall be reported to the insurance company, the employee's supervisor, and law enforcement as soon as feasible. All incidents shall be reviewed / investigated to determine the cause and corrective action.

1.219 Cargo

Any cargo on or in motor vehicles shall be adequately stored and secured to prevent unintentional movement of tools and equipment which could cause spillage, damage to the vehicle, damage to the environment, or injury to the operator.

1.220 Vehicle Maintenance

Vehicles shall be in a safe and working condition.

Pre-use inspections shall be performed before operating a vehicle. This consists of a walk-around the vehicle to check for any defects to the vehicle and ensure there are no barriers blocking the path. Company-owned vehicles shall have a maintenance program in place meeting the minimum manufacturer's recommendation.

In the event employees are driving personal vehicles for company business, pre-use inspections and regular vehicle maintenance shall still be completed.

1.221 Fatigue Management

Employees shall be well rested, alert, and sober on the road. Drivers shall continually search the roadway to be alert to situations requiring quick action. Drivers are required to stop about every 2 hours for a break and get out to stretch, take a walk, and get refreshed.

T&T Industrial, Inc.

Driving Safety Program

Appendix 13 **Vehicle Inspection Form**

General Information			
Driver's Name:		Date:	
Vehicle ID:		Odometer:	

Inspection Checklist

Walk Around Inspection	✓	⚐	N/A	Start Up (Cont.)	✓	⚐	N/A
Tags and locks				Taillights			
Fluid Levels (oil/water/leaks)				Turn signals			
High visibility (flag/ID/tape)				Hazard lights			
Wheels (tire/rims/spare/tools)				Headlights			
Windshield and windows				Reverse alarm			
Seatbelts and seats				Horn			
Current registration/inspection				Two-way radio			
Current insurance				Unusual noises			
Vehicle condition				Unusual smells			
First aid kit				Unusual vibrations			
Fire extinguisher							

T&T Industrial, Inc.

Driving Safety Program

Start Up	✓	🏳	**N/A**
Mirrors			
Windshield wipers			
Washer fluid			
Fuel level			
Controls/gauges			
Warning lights			
Foot brake			
Parking brake			
Trailer brake			
Brake lights			

Other	✓	🏳	**N/A**

Driver Signature:		Date:	

20. Electrical Safety Awareness Program

1.222 Purpose and Scope

The purpose of this program is to define the minimum requirements that all employees shall adhere to when using electrical equipment.

This program applies to all T&T Industrial, Inc. employees working with electrical equipment.

1.223 Resources

Number	Title
29 CFR 1910 Subpart S	Electrical
Cal/OSHA T8 CCR Subchapter 5	Electrical Safety
Cal/OSHA T8 CCR Subchapter 4	Construction Safety

1.224 Electrical Safety Awareness Program

Electricity is dangerous when used without proper training, knowledge, and planning. The following basic safety principles provide guidance to use electricity safely prior to the start of work.

Safe work practices shall be employed to prevent electric shock or other injuries resulting from either direct or indirect electrical contacts when work is performed near or on equipment or circuits which are or may be energized.

1.225 Training / Qualification

Employees performing electrical work shall be trained in electrical safety-related work practices that pertain to their respective job assignments. All employees shall be trained in any electrically related safety practices which are necessary for their safety.

Only qualified persons may work on electric circuit parts or equipment that have not been deenergized. Such persons shall be capable of working safely on energized circuits and shall be familiar with the proper use of special precautionary techniques, personal protective equipment (PPE), insulating and shielding materials, and insulated tools.

Employees who face a risk of electric shock but who are not qualified persons shall be trained and familiar with electrically related safety practices. Employees shall be trained in safety related work practices including clearance distances that pertain to their respective job assignments.

1.226 Safe Work Practices

Safe work practices shall be employed to prevent electric shock or other injuries resulting from either direct or indirect electrical contacts when work is performed near or on equipment or circuits which are or may be energized.

Live parts to which an employee may be exposed must be deenergized before the employee works on or near them unless deenergizing the parts introduces additional or increased hazards or is unfeasible due to equipment design or operational limitations. Examples of increased or additional hazards include interruption of life support equipment, deactivation of emergency alarm systems, shutdown of hazardous location ventilation equipment, or removal of illumination for an area. Live parts that operate at less than 50 volts to ground need not be deenergized if there are no increased exposures to electrical burns or to explosions due to electric arcs.

Conductors and parts of electrical equipment that have been de-energized but not locked or tagged out shall be treated as live parts.

While any employee is exposed to contact with parts of fixed electric equipment or circuits which have been deenergized, the circuits energizing the parts shall be locked out or tagged or both.

Employees shall not enter spaces containing exposed energized parts unless illumination is provided which enables the employees to work safely.

Protective shields, protective barriers, or insulating materials as necessary shall be provided.

Portable ladders shall have nonconductive siderails if they are used where the employee or the ladder could contact exposed energized parts.

Conductive articles of jewelry and clothing (such as watch bands, bracelets, rings, key chains, necklaces, metalized aprons, cloth with conductive thread, or metal headgear) shall not be worn if they might contact exposed energized parts. However, such articles may be worn if they are rendered nonconductive by covering, wrapping, or other insulating means.

1.227 Power Lines

Assume that all overhead wires are energized at lethal voltages. Never assume that a wire is safe to touch even if it is down or appears to be insulated.

The lines shall be de-energized and grounded or other protective measures shall be provided before work is started.

Never touch a fallen overhead power line. Call the electric utility company to report fallen electrical lines.

Stay at least 10 feet away from overhead wires during cleanup and other activities. If working at heights or handling long objects, survey the area before starting work for the presence of overhead wires.

If an overhead wire falls across your vehicle while you are driving, stay inside the vehicle and continue to drive away from the line. If the engine stalls, do not leave your vehicle. Warn people not to touch the vehicle or the wire. Call or ask someone to call the local electric utility company and emergency services.

21. Emergency Action Plan Program

1.228 Purpose and Scope

The purpose of this program is to provide a framework for planning for and responding appropriately to emergency situations related to health, safety, the environment, or security. The key to preparedness is to have an effective plan, well trained responders, and informed and responsive employees.

This program applies to all T&T Industrial, Inc. employees.

1.229 Resources

Number	Title
29 CFR 1910 Subpart E	Exit Routes and Emergency Planning - Emergency Action Plans
	Emergency Drill Log

1.230 Emergency Action Plan

The emergency action plan shall establish guidelines for all reasonably foreseeable workplace emergencies. Thoughtful actions based on situation assessment are required when responding to an emergency.

The emergency action plan shall be kept in the workplace and made available to employees for review. Employees shall be informed of the plan orally.

1.231 Bridging Documents

When required, emergency response procedures of the Company and its clients or subcontractors shall be bridged to clarify the responsibilities for control of the emergency to ensure there is:

- An Ultimate Work Authority (UWA) over the work site established.
- A clear and direct line of communication set up between the work site and the Company / project / external parties.
- A clear definition of personnel roles and responsibilities.
- Only one authoritative source of information to external organizations, media, and relatives.

1.232 Training

Employees shall undergo emergency action plan training as applicable to the authority having jurisdiction.

1.233 Responsibilities

Roles and responsibilities for employees such as fire wardens and supervisors during emergency situations shall be designated in accordance with jurisdictional requirements.

1.234 Plan Elements

1.234.1 Reporting Fire or Other Emergency

Employees will report fires by first calling 911 and pulling the alarm. Emergencies must be reported to management.

1.234.2 Emergency Evacuation

In the event of an emergency evacuation, employees will follow the evacuation routes.

1.234.3 Critical Operations

Employees who remain to operate critical operations before they evacuate must be trained in evacuation procedures specific to their responsibilities.

1.234.4 Accounting for Employees

Employees will be accounted for after evacuation by a roll call or checking in with their manager.

1.234.5 Medical or Rescue Duties

Employees who perform medical or rescue duties must be trained in the specific rescue duties.

1.234.6 More Information

The Safety Coordinator may be contacted by other employees for more information about the plan or an explanation of their duties under the plan.

1.235 Emergency Drills

Drills will be conducted at regular intervals and will include the following scenarios:

- Fire Drills – Evacuation in case of fire or smoke hazards
- Severe Weather Drills – Procedures for tornadoes, hurricanes, or other natural disasters
- Hazardous Material Spill Drills – Response to fuel or chemical spills
- Medical Emergency Drills – First aid and response to medical crises
- Active Shooter/Violence Drills – Emergency actions for workplace violence

1.235.1 Responsibilities

- Safety Coordinator: Oversees emergency drill planning and execution.
- Department Managers: Ensure employees participate and understand emergency procedures.
- Employees: Follow drill instructions and report any safety concerns.
- Emergency Response Team: Trained personnel responsible for guiding others during an emergency.

1.235.2 Drill Frequency and Documentation

- Quarterly fire drills
- Annual severe weather drills
- Semi-annual hazardous material spill drills
- Annual active shooter/violence training
- Each drill will be documented, including the date, time, response time, and areas for improvement.

1.235.3 Procedures During Drills

- Alert and Communication: Notification will be given via alarms, intercoms, or verbal instructions.

- Evacuation or Shelter-in-Place: Employees follow pre-designated evacuation routes or shelter areas.
- Accountability: Designated personnel ensure all employees are accounted for at assembly points.
- Post-Drill Review: Feedback is collected to identify improvements.

1.235.4 Training and Continuous Improvement

- All new employees must complete emergency response training within their first 30 days.
- Refresher training will be conducted annually.
- Drills will be evaluated, and adjustments will be made based on feedback and changing conditions.

1.236 Evacuation Procedures

Upon hearing the alarm or when directed by a warden:

- Prepare to evacuate.
- Get your workplace ready to be left unattended. Shut down computers; turn off gas and electrical equipment, if safe to do so.
- For fire, close the doors as you go – do not lock them. In the case of a bomb threat, leave doors open.
- Assist any person in immediate danger.
- Leave the building via the nearest safe route.
- Obey all directions from wardens.
- Move calmly to the assembly point or other advised area and stay there until the All Clear has been given.
- Follow closely the instructions of emergency services personnel.
- Wait for the OK to re-enter the building.

1.237 Fire

- Call 911
- Assist any person in immediate danger (only if safe to do so).
- If safe to do so, close doors to minimize spread of the fire.
- Attack the fire only if safe to do so.
- Contact the nearest supervisor and follow their directions.
- Assist with the evacuation of mobility impaired occupants.
- Move to the evacuation assembly point or other safe location, and stay there until the All Clear has been given.
- Follow closely the instructions of emergency services personnel.

1.238 Medical Emergency

- Assess the situation.
- Do not move a victim unless they are exposed to a life-threatening situation.
- Contact the nearest first aid officer.
- In extreme emergency situations contact the ambulance service by dialing 911
- Arrange for the ambulance to be met at the front or other nominated area.
- Remain with the victim and administer first aid as appropriate until assistance arrives.
- Follow closely the instructions of emergency services personnel.

1.239 Bomb Threat

On receipt of a telephone bomb threat:

- Keep the caller talking (do not hang up).
- Remain calm and do not say or do anything that may encourage irrational behavior.
- Ask someone else to call 911.
- Do not use mobile phones. Turn them all off.
- Evacuate the building via alternate exits, leaving doors and windows open.

- Take personal belongings with you, noting any suspicious parcels in your area as you leave.
- Move to assembly point maintaining a clear distance from parked vehicles.
- Follow closely the instructions of emergency services personnel.

1.240 Civil Disturbance

- Keep well clear of the disturbance and do not say or do anything that may encourage irrational behavior.
- Consider “locking down” the building to prevent unauthorized entry.
- Follow closely the instructions of emergency services personnel.
- Evacuate the building only if instructed to do so by emergency services personnel.

1.241 Attack Or Armed Threat

- Keep well clear of the intruder and do not say or do anything that may encourage irrational behavior.
- Notify 911.
- Note as many details as possible.
- Follow closely the instructions of emergency services personnel.
- Evacuate the building only if instructed to do so by emergency services personnel.
- Stay clear of windows.

1.242 Personal Preparation

- Know the location of emergency exits in your building.
- Plan an escape route from your office to each exit.
- Familiarize yourself with the location of any fire alarms in your building.
- Note the location of fire extinguishers.
- Familiarize yourself with the identity and location of the first aid officers and first aid kits.

1.243 Tornadoes

Preparing for a tornado requires identifying a place to take shelter, being familiar with and monitoring your community's warning system, and creating procedures to account for personnel.

Underground areas, such as a basement or storm cellar, are the recommended places to shelter from a tornado. If an underground shelter is unavailable, you should:

- Seek a small interior room or hallway on the lowest floor possible.
- Stay away from doors, windows, and outside walls.
- Stay in the center of the room and avoid corners because they attract debris.
- Avoid auditoriums and other buildings that have flat, wide-span roofs.

1.244 Floods

If you are in an area that could flood, you should monitor National Oceanic and Atmospheric Administration (NOAA) Weather Radio or commercial radio and television stations for information about flood watches and warnings.

Be prepared to move to higher ground immediately if you receive information about the potential for flash flooding. You should be prepared to evacuate before water levels rise and potentially cut off evacuation routes.

Do not drive through flooded areas. As little as 6 inches of water can cause a vehicle to lose control or stall. A foot of water is enough to float many cars.

1.245 Earthquakes

If you are in an area where earthquakes are a potential threat, you should identify safe places to shelter in your workplace and home, such as under a sturdy table or desk or against an interior wall away from windows or tall objects that could fall on you.

The shorter the distance you must move to get to safety, the less likely you are to be injured.

Practice "drop, cover, and hold on" in each safe place so that they become an automatic response:

- Drop under a sturdy desk or table.
- Hold on to one leg of the table or desk.

- Protect your eyes by keeping your head down.

1.246 Alert System

The alarm system shall be distinctive and recognizable as a signal to evacuate the work area or perform actions designated under the emergency action plan. For those employers with 10 or fewer employees in a particular workplace, direct voice communication is an acceptable procedure for sounding the alarm provided all employees can hear the alarm.

1.247 Plan Review

The emergency action plan shall be reviewed when the plan is developed, when the employee is initially assigned to a job, when the employee's responsibilities under the plan change, and when the plan is changed.

T&T Industrial, Inc.

Emergency Drill Log

Location/Site:	
Comments:	

Date	Time	Type of Drill	Evacuation Time	Shelter-In-Place Time	Issued Identified? (Y/N)	Corrective Actions Needed? (Y/N)	Drill Coordinator Name

Additional Notes or Observations

22. Hazard Analysis Program (SEMS JSA)

1.248 Purpose and Scope

The purpose of this program is to provide requirements for hazard analysis to identify, evaluate and, where applicable, reduce the likelihood and/or minimize the consequences of hazards.

This program applies to all T&T Industrial, Inc. employees.

1.249 Resources

Number	Title
30 CFR 250 Subpart S	Oil and Gas and Sulphur Operations in the Outer Continental Shelf – Safety and Environmental Management Systems (SEMS)
CMS-FM-0040	Job Safety Analysis (JSA) Form
CMS-FM-0048	Hazard Identification Worksheet

1.250 Hazard Analysis Program

A hazard analysis (facility level) and a job safety analysis (JSA) (operations / task level) shall be conducted for all facilities and activities identified in the safety and environmental management systems (SEMS) program.

The hazard analysis shall be appropriate for the complexity of the operation and must identify, evaluate, and manage the hazards involved.

A Job Safety Analysis (JSA) be conducted daily before each job task is started.

1.251 JSA General Requirements

The following must be addressed and be on the JSA:

- The location of the worksite
- The name of the supervisor
- Supervisory approval
- Weather hazards
- Work activity
- Personal protective equipment (PPE)

- Planned job steps
- Potential hazards
- Actions to mitigate hazards
- Hazards associated with SIMOPS (simultaneous operations)
- The mitigations implemented during execution of work
- Clearly defined individual responsibilities

The JSA must be developed, reviewed, and signed by the entire work crew and all visitors to the site.

The JSA must be updated (red-lined) based upon additional hazards being discovered and corresponding changes when applicable.

Emergency procedures must be clearly defined on the JSA for the task, such as nearest hospital with directors, first aid personnel, drivers of vehicles to transport injured, doctor, and phone numbers for ambulance, police, fire, etc.

Emergency procedures for work site preparations for egress and muster points must be clearly defined on the JSA for the task.

The JSA must identify task specific requirements including personnel, equipment/tools, process controls, permits, etc.

A specific means of communication must be identified and be addressed on the JSA.

A quality review of the JSAs must be conducted by supervision / leadership on at least a quarterly basis.

Employees must receive adequate JSA training.

1.252 Hazard Analysis (facility level)

The hazards analysis shall address the following:

- Hazards of the operation.
- Previous incidents related to the operation being evaluated, including any incident in which an Incident of Noncompliance or a civil or criminal penalty was issued.
- Control technology applicable to the operation the hazards analysis is evaluating.
- A qualitative evaluation of the possible safety and health effects on employees, and potential impacts to the human and marine environments, which may result if the control technology fails.

The hazards analysis shall be performed by a person(s) with experience in the operations being evaluated. These individuals also need to be experienced in the hazards analysis methodologies being employed.

Recommendations in the hazard analysis shall be resolved and the resolution documented.

A single hazard analysis can be performed to fulfill the requirements for simple and nearly identical facilities, such as well jackets and single well caissons. This single hazard analysis can also be applied to simple and nearly identical facilities after verification that any site-specific deviations are addressed in each SEMS program.

1.253 Job Safety Analysis

A JSA shall be developed and implemented for each identified operation and task in the SEMS program.

Operational JSA's shall be maintained at the job site for the life of the operation at the facility and shall be readily accessible to employees.

The JSA shall include all personnel involved with the job activity.

The JSA shall identify, analyze, and record:

- The steps involved in performing a specific job.
- The existing or potential safety, health, and environmental hazards associated with each step.
- The recommended action(s) and/or procedure(s) that will eliminate or reduce these hazards, the risk of a workplace injury or illness, or environmental impacts.

The immediate supervisor of the crew performing the job onsite shall conduct the JSA, sign the JSA, and ensure that all personnel participating in the job understand and sign the JSA.

The individual designated as being in charge of the facility shall approve and sign all JSAs before personnel start the job.

If a particular job is conducted on a recurring basis, and if the parameters of these recurring jobs do not change, then the person in charge of the job may decide that a JSA for each individual job is not required. The parameters to consider in making this determination include, but are not limited to, changes in personnel, procedures, equipment, and environmental conditions associated with the job.

All personnel, which includes contractors, shall be trained in accordance with the requirements of SEMS training criteria. Prior to performing a job, contractor training according to SEMS training criteria shall be verified.

Stop Work Authority (SWA) procedures and expectations must be included as a statement on all JSAs.

T&T Industrial, Inc.

Hazard Analysis Program (SEMS JSA)

Appendix 14 **Job Safety Analysis (JSA) Form**

<table>
<tr><th colspan="5">General Information</th></tr>
<tr><td>Prepared By:</td><td></td><td colspan="3" rowspan="2">Work instruction to be discussed with work group prior to the start of the activity / task.</td></tr>
<tr><td>Location:</td><td></td></tr>
<tr><td>Work Activity / Task:</td><td colspan="2"></td><td>Date:</td><td></td></tr>
<tr><th colspan="5">Required PPE</th></tr>
</table>

T&T Industrial, Inc.

Hazard Analysis Program (SEMS JSA)

Item	Job Step *Break the job down into steps*	Potential Hazards *What can cause injury?*	Controls to be Implemented *What will be done to make the step as safe as possible?*	Who Will Ensure This Happens

Signatures	
Printed Name	Signature

T&T Industrial, Inc.

Hazard Analysis Program (SEMS JSA)

Appendix 15 **Hazard Identification Worksheet**

General Information			
Prepared By:		Date:	
Location:			
Operation:			

Hazard Description	Potential Consequence	Initial Risk H/M/L	Existing Control	Recommended Control	Residual Risk H/M/L	Consequence Mitigation	Person/Discipline Responsible
	☐ Injury / Death ☐ Harm to Environment ☐ Asset Damage / Loss ☐ Production Loss / Downtime ☐ Reputation						

	☐ Injury / Death ☐ Harm to Environment ☐ Asset Damage / Loss ☐ Production Loss / Downtime ☐ Reputation						
	☐ Injury / Death ☐ Harm to Environment ☐ Asset Damage / Loss ☐ Production Loss / Downtime ☐ Reputation						
	☐ Injury / Death ☐ Harm to Environment ☐ Asset Damage / Loss ☐ Production Loss / Downtime						

	☐ Reputation						
	☐ Injury / Death ☐ Harm to Environment ☐ Asset Damage / Loss ☐ Production Loss / Downtime ☐ Reputation						

23. Excavations and Trenching Program

1.254 Purpose and Scope

The purpose of this program is to provide specific requirements and safety principles to ensure that excavation and trenching is conducted safely and effectively.

This program applies to all T&T Industrial, Inc. employees in excavation and trenching operations.

1.255 Resources

Number	Title
29 CFR 1926 Subpart P	OSHA Excavation Standards
Cal/OSHA T8 CCR Subchapter 4	Construction Safety Orders - Excavations General Requirements
CMS-FM-0023	Excavation Permit / Competent Person Checklist
CMS-FM-0024	Excavation / Trenching Daily Inspection Form
CMS-FM-0025	Soil Classification Worksheet

1.256 Definitions

Acronym/Term	Definition
Trench	A narrow excavation (in relation to its length) made below the surface of the ground. In general, the depth of a trench is greater than its width, but the width of a trench (measured at the bottom) is not greater than 15 feet.
Excavation	Any man-made cut, cavity, trench, or depression in the Earth's surface formed by earth removal.
Competent Employee / Person	A person who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.

1.257 Excavation and Trenching

Specific requirements shall be met and enforced when trenching or excavating takes place. It shall be the responsibility of management overseeing the operations to assure that the individuals who are involved in and exposed to trenching or excavating have been properly trained and instructed whether they be in-house or contract employees. Training shall include excavator notification and excavation practices.

The minimum requirements of this procedure, as set forth in the 29 CFR 1926 Construction Excavation Standards by the Federal Occupational Safety and Health Administration (OSHA), are quite extensive and therefore shall be reviewed thoroughly by those responsible for enforcement, administration, monitoring, and implementation prior to commencement of any work task involving excavation.

1.258 Roles and Responsibilities

A competent person shall document the layout and diagram of the trench and/or excavation prior to start-up and conduct daily documented site inspections. Inspections shall also be documented after natural events such as rainstorms or other hazard-increasing occurrences such as nearby traffic or simultaneous work.

The competent person overseeing the trenching or excavation project is responsible for the following:

- Obtaining and completing an excavation permit form.
- Contacting utility companies or owners within established or customary local response time, advising of proposed work, and asking to establish the exact location of the underground utility installations prior to the start of actual excavation.
- Locating and marking the location of underground lines before project begins.
- Assessing the area and erecting proper barricading.
- Obtaining all personal protective equipment (PPE) necessary.
- Ensuring proper benching, shoring, or shielding are in place. In excavations greater than 4 feet, test for oxygen deficiency or hazardous atmospheres. A trench excavation may present many of the hazards of a permit-required confined space. In general practice, all trench excavations over 4 feet in depth shall be considered confined spaces until a competent person has ruled out all the potential hazards associated with them.

- If hazardous condition could exist or develop, ensure the ready availability of emergency rescue equipment such as breathing apparatus, safety harness, or basket stretcher. This equipment shall be attended when in use.
- In excavations 4 feet or greater, provide means of access and egress not to exceed 25 feet from workers.

1.259 Underground Installations

The estimated location of utility installations, such as sewer, telephone, fuel, electric, water lines, or any other underground installations that reasonably may be expected to be encountered during excavation work, shall be determined prior to opening an excavation.

Utility companies or owners shall be contacted within established or customary local response times, advised of the proposed work, and asked to establish the location of the utility underground installations prior to the start of actual excavation. When utility companies or owners cannot respond to a request to locate underground utility installations within 24 hours (unless a longer period is required by state or local law) or cannot establish the exact location of these installations, the Company may proceed, provided this is done so with caution, and provided detection equipment or other acceptable means to locate utility installations are used.

When excavation operations approach the estimated location of underground installations, the exact location of the installations shall be determined by safe and acceptable means. While the excavation is open, underground installations shall be protected, supported, or removed as necessary to safeguard employees.

Energized lines shall be located by manual digging when there is a possibility of damaging them with mechanical excavation equipment. Some of the lines encountered may have to be locked and tagged out for employee safety. Tools and equipment may have to be grounded and insulation provided for employees when electrical exposure is possible. Should the work involve the excavation of an area to uncover a gas leak, good ventilation, the use of a multi-gas monitor, and the use of non-sparking tools shall be required.

Backhoes or other digging machines shall not be allowed to excavate close to underground facilities that must be left in place. A proximity limit for machine operations shall be established and the excavation completed by hand digging.

When hand excavation is conducted, employees shall be warned about driving picks or using other powered tools that may break through the envelope of buried facilities.

1.260 Access and Egress

A stairway, ladder, ramp, or other safe means of egress shall be located in trench excavations that are 4 feet or more in depth so as to require no more than 25 feet of lateral travel for employees.

Structural ramps that are used solely by employees as a means of access or egress from an excavation shall be designed by a designated competent person.

Structural ramps used for access or egress of equipment shall be designed by a competent person qualified in structural design and shall be constructed in accordance with the design.

Ramps and runways constructed of two or more structural members shall have the structural members connected together to prevent displacement.

Structural ramps used in lieu of steps shall be provided with cleats or other surface treatments on the top surface to prevent slipping.

Structural members used for ramps and runways shall be of uniform thickness. Cleats or other appropriate means used to connect runway structural members shall be attached in a manner to prevent tripping.

1.261 Vehicular Traffic

Employees exposed to public vehicular traffic shall be provided with, and shall wear, warning vests or other suitable garments marked with or made of reflectorized or high-visibility material.

1.262 Exposure to Falling Loads

No employee shall be permitted underneath loads handled by lifting or digging equipment. Employees shall be required to stand away from any vehicle being loaded or unloaded to avoid being struck by any spillage or falling materials. Operators may remain in the cabs of vehicles being loaded or unloaded when the vehicles are equipped to provide adequate protection for the operator during loading and unloading operations.

1.263 Warning Systems for Mobile Equipment

When mobile equipment is operated adjacent to an excavation, or when such equipment is required to approach the edge of an excavation, and the operator does not have a clear and direct view of the edge of the excavation, a warning system, such as barricades, hand or mechanical signals, or stop logs shall be utilized. Where possible, the grade should be away from the excavation. Material excavated by machine or hand shall be kept at least 2 feet from the edge of the excavation unless adequate barricading is provided.

1.264 Hazardous Atmospheres

To prevent employee exposure to harmful levels of atmospheric contaminants and to ensure acceptable atmospheric conditions within excavations, the following requirements shall be met:

- Prior to allowing employees to enter an excavation where oxygen deficiency or a hazardous atmosphere exists or could reasonably be expected to exist, such as in landfill areas or excavations in areas where hazardous substances are stored nearby, the atmospheres in excavations greater than 4 feet in depth shall be tested.
- It shall be ascertained by air sampling performed by a competent person that the atmospheres in the excavations contain an adequate quantity of oxygen and that harmful contaminants have been diluted to safe concentrations prior to, and periodically during, occupancy.
- Tests for gas and toxic substances shall be made whenever there is a possibility that the excavation contains or may have contained a toxic substance. Test results shall be within the limits established by OSHA's permissible exposure level (PEL) or American Conference of Governmental Industrial Hygienists (ACGIH) threshold limit values (TLVs).
- If any test conducted indicates that the atmosphere is unsafe, before any employee is permitted to enter the excavation, the space shall be ventilated until the concentration of hazardous substance is reduced to a safe level or removed. Ventilation shall be continued as long as recurrence of the hazard is probable.
- Emergency rescue equipment, such as breathing apparatus, a safety harness and line, or a basket stretcher, shall be readily available where hazardous atmospheric conditions exist or may reasonably be expected to develop during work in an excavation. This equipment shall be attended when in use.
- Employees entering bell bottom pier holes, or other similar deep and confined footing excavations, shall wear a harness with a lifeline securely attached to it. The lifeline shall be separate from any line used to handle materials and shall be individually attended at all times while the employee wearing the lifeline is in the excavation.

1.265 Protection from Water Accumulation

Employees shall not work in excavations in which there is accumulated water or in excavations in which water is accumulating, unless adequate precautions have been taken to protect employees against the hazards posed by water accumulation. The precautions necessary to protect employees adequately vary with each situation but could include:

- Special support or shield systems to protect from cave-ins.
- Water removal to control the level of accumulating water.
- Use of a safety harness and lifeline.

If water is controlled or prevented from accumulating by the use of water removal equipment, the water removal equipment and operations shall be monitored by a competent person to ensure proper operation. If excavation work interrupts the natural drainage of surface water (such as streams), diversion ditches, dikes, or other suitable means shall be used to prevent surface water from entering the excavation and to provide adequate drainage of the area adjacent to the excavation. Excavations subject to runoff from heavy rains will require an inspection by a competent person.

1.266 Stability of Adjacent Structures

All surface encumbrances that are located so as to create a hazard to employees shall be removed or supported, as necessary, to safeguard employees.

Where the stability of adjoining buildings, walls, or other structures is endangered by excavation operations, support systems shall be provided, such as shoring, bracing, or underpinning to ensure the stability of such structures for the protection of employees.

Excavation below the level of the base or footing of any foundation or retaining wall that could be reasonably expected to pose a hazard to employees shall not be done without meeting one of the following criteria:

a) A support system, such as underpinning, is provided to ensure the safety of employees and the stability of the structure.
b) The excavation is in stable rock.
c) A registered professional engineer has determined that the structure is sufficiently removed from the excavation so that it will be unaffected by the excavation activity.
d) A registered professional engineer has determined that such excavation work will not pose a hazard to employees.

Sidewalks, pavements, and appurtenant structures shall not be undermined unless a support system or another method of protection is provided to protect employees from the possible collapse of such structures.

A level area extending at least three feet (one meter) from the upper edge of each wall of an excavation shall be kept clear of equipment, excavated soil, rock and construction material.

1.267 Subsurface Installations

The approximate location of subsurface installations, such as sewer, telephone, fuel, electric, water lines, or any other subsurface installations that reasonably may be expected to be encountered during excavation work, shall be determined by the excavator prior to opening an excavation.

Excavation shall not commence until:

1. The excavation area has been marked by the excavator; and
2. The excavator has received a positive response from all known owner/operators of subsurface installations within the boundaries of the proposed project; those responses confirm that the owner/operators have located their installations, and those responses either advise the excavator of those locations or advise the excavator that the owner/operator does not operate a subsurface installation that would be affected by the proposed excavation.

When the excavation is proposed within 10 feet of a high priority subsurface installation, the excavator shall be notified by the facility owner/operator of the existence of the high priority subsurface installation before the legal excavation start date and time, and an onsite meeting involving the excavator and the subsurface installation owner/operator's representative shall be scheduled by the excavator and the owner/operator at a mutually agreed on time to determine the action or activities required to verify the location of such installations. High priority subsurface installations are high pressure natural gas pipelines with normal operating pressures greater than 415 kPA gauge (60 p.s.i.g.), petroleum pipelines, pressurized sewage pipelines, conductors or cables that have a potential to ground of 60,000 volts or more, or hazardous materials pipelines that are potentially hazardous to employees, or the public, if damaged.

Only qualified persons shall perform subsurface installation locating activities, and all such activities shall be performed in accordance with regulations. Persons who complete a training program in accordance with the requirements of Section 1509, Injury and Illness Prevention Program (IIPP), that meets the minimum training guidelines and practices of the Common Ground Alliance (CGA) Best Practices, Version 3.0, published March 2006, or the standards of the National Utility Locating Contractors Association (NULCA), Standard 101: Professional

Competence Standards for Locating Technicians, 2001, First Edition, which are incorporated by reference, shall be deemed qualified for the purpose of this section.

Employees who are involved in the excavation operation and exposed to excavation operation hazards shall be trained in the excavator notification and excavation practices required by regulation.

All Regional Notification Centers as defined by Government Code Section 4216(j) in the area involved and all known owners of subsurface facilities in the area who are not members of a Notification Center shall be advised of the proposed work at least 2 working days prior to the start of any digging or excavation work. Except for repair work to subsurface facilities done in response to an emergency.

When excavation or boring operations approach the approximate location of subsurface installations, the exact location of the installations shall be determined by safe and acceptable means that will prevent damage to the subsurface installation.

While the excavation is open, subsurface installations shall be protected, supported, or removed as necessary to safeguard employees.

An excavator discovering or causing damages to a subsurface installation shall immediately notify the facility owner/operator or contact the Regional Notification Center to obtain subsurface installation operator contact information immediately after which the excavator shall notify the facility operator. All breaks, leaks, nicks, dents, gouges, grooves, or other damages to an installation's lines, conduits, coatings, or cathodic protection shall be reported to the subsurface installation operator. If damage to a high priority subsurface installation results in the escape of any flammable, toxic, or corrosive gas or liquid or endangers life, health or property, the excavator responsible shall immediately notify 911, or if 911 is unavailable, the appropriate emergency response personnel having jurisdiction. The facility owner/operator shall also be contacted.

1.268 Loose Rock or Soil

Employees shall be adequately protected from loose rock or soil that could pose a hazard by falling or rolling from an excavation face. Such protection shall consist of scaling to remove loose material, installing protective barricades at intervals as necessary on the face to stop and contain falling material, or providing other means of equivalent protection.

Employees shall be protected from excavated or other materials or equipment that could pose a hazard by falling or rolling into excavations. Protection shall be provided by placing and keeping such material or equipment a minimum of 2 feet from the edge of excavations, by the use of retaining devices that are sufficient to prevent materials or equipment from falling or rolling into excavations, or by a combination of both if necessary.

1.269 Inspections

Daily inspections of excavations, the adjacent areas, and protective systems shall be made by a competent person for evidence of a situation that could result in possible cave-ins, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions.

An inspection shall be conducted by the competent person prior to the start of work and as needed throughout the shift. Inspections shall also be made after every rainstorm or other hazard increasing occurrence. These inspections are only required when employee exposure can be reasonably anticipated.

Where the competent person finds evidence of a situation that could result in a possible cave-in, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions, exposed employees shall be removed from the hazardous area until the necessary precautions have been taken to ensure their safety.

1.270 Fall Protection

Where employees or equipment are required or permitted to cross over excavations over 6-feet in depth and wider than 30 inches, walkways or bridges with standard guardrails shall be provided.

Adequate barrier physical protection shall be provided at all remotely located excavations. All wells, pits, shafts, etc., shall be barricaded or covered. Upon completion of exploration and other similar operations, temporary wells, pits, shafts, etc., shall be backfilled.

1.271 Crossings and Walkways

Where employees or equipment are required or permitted to cross over excavations over 6 feet in depth and wider than 30 inches, walkways or bridges with standard guardrails shall be provided.

Where the potential exists for exposed ends of a walkway or bridge to create a tripping hazard, it shall be appropriately provided with beveled cleating.

Adequate barrier physical protection shall be provided at all remotely located excavations. All wells, pits, shafts, etc., shall be barricaded or covered. Upon completion of exploration and similar operations, temporary wells, pits, shafts, etc., shall be backfilled.

1.272 Soil Classifications and Employee Protection

Soil classifications shall be determined by testing and protective systems designed according to soil classifications.

The most stable type of soil is Type A. It is dense and heavy and consists primarily of clay. Type B has a medium level of stability and is made of soils such as silt, sandy loam, and medium clay. The least stable soil is Type C, which consists of gravel, loamy sand, and soft clay.

Adequate protection shall be provided to protect employees from loose rock or soil or excavated other materials or equipment that could pose a hazard by falling or rolling from into excavations.

1.273 Protective Systems

All protective systems shall be inspected by a competent person before work begins.

Types of protective systems:

- Benching - a method of protecting workers from cave-ins by excavating the sides of an excavation to form one or a series of horizontal levels or steps, usually with vertical or near-vertical surfaces between levels. Benching cannot be done in Type C soil.
- Sloping - cutting back the trench wall at an angle inclined away from the excavation. Shoring requires installing aluminum hydraulic or other types of supports to prevent soil movement and cave-ins.
- Shielding - protects workers by using trench boxes or other types of supports to prevent soil cave-ins. Designing a protective system can be complex because many factors must be considered such as soil classification, depth of cut, water content of soil, changes caused by weather or climate, surcharge loads (e.g., spoil, other materials to be used in the trench) and other operations in the vicinity. Shields shall be used to protect against water accumulation.

1.274 Design of Sloping and Benching Systems

The slopes and configurations of sloping and benching systems shall be selected and constructed in accordance with one of the following options:

- **Option 1:**

If no attempt is made to determine soil type, excavations shall be sloped at an angle not steeper than one and one-half horizontal to one vertical (34 degrees measured from the horizontal). This angle represents the worst soil condition (Type C) and therefore requires the use of configurations that are in accordance with the slopes shown for Type C soil.

- **Option 2**

Maximum allowable slopes and allowable configurations for sloping and benching systems shall be determined in accordance with the conditions and requirements set forth.

- **Option 3**

Design of sloping or benching systems shall be selected from and be in accordance with tabulated data. The tabulated data shall be in written form and shall include all the following:

  - Identification of the parameters that affect the selection of a sloping or benching system drawn from such data.
  - Identification of the limits of use of the data, to include the magnitude and configuration of slopes determined to be safe.
  - Explanatory information as may be necessary to aid the user in making a correct selection of a protective system from the data.

A copy of the tabulated data, which identifies the registered professional engineer who approved the data, shall be maintained at the jobsite during construction and use of the protective system.

- **Option 4**

Sloping and benching systems not utilizing Options 1, 2, or 3 above shall be approved by a registered professional engineer. All such designs shall be in written form and include all the following:

  - The magnitude of the slopes that were determined to be safe for the particular project.
  - The configurations that were determined to be safe for the particular project.
  - The identity of the registered professional engineer approving the design.

A copy of the design shall be maintained at the jobsite during construction and use of the system.

1.275 Materials and Equipment

Ensure that all materials and equipment used for protective systems are free from damage or defects that might impair their proper function.

Manufactured materials and equipment used for protective systems shall be used and maintained in a manner that is consistent with the recommendations of the manufacturer and in a manner that will prevent employee exposure to hazards.

Any materials or equipment that are damaged shall be assessed by the supervisor or engineer designated as the competent person as to its suitability for continued use.

1.276 Installation and Removal of Supports

Members of support systems shall be securely connected together to prevent sliding, falling, kickouts, or other predictable failure.

Support systems shall be installed and removed in a manner that protects employees from cave-ins, structural collapses, or from being struck by members of the support system.

Individual members of support systems shall not be subjected to loads exceeding their designed capacity.

Before removal of individual members begins, the need for additional precautionary measures to ensure the safety of employees shall be evaluated by a competent person. Such measures may include the installation of other structural members to carry the loads imposed on the support system.

Removal operations shall begin at and progress from the bottom of the excavation. Members shall be released slowly to note any indication of possible failure of the remaining members or possible cave-in of the sides of the excavation.

Backfilling shall progress together with the removal of support systems from excavations.

1.277 Additional Requirements for Support Systems for Trench Excavations

The depth of trench excavations may extend to a maximum of 2 feet below the bottom of the members of a support system, provided the system is designed to resist the forces calculated for the full depth of the trench and there are no indications, while the trench is open, of a possible loss of soil from behind or below the bottom of the support system.

Installation of support systems shall be closely coordinated with the excavation of trenches.

Employees shall not be permitted to work on the faces of sloped or benched excavations at levels above other employees unless employees at the lower levels are protected from the hazards of falling, rolling, or sliding material or equipment.

1.278 Identifying underground installations

1.278.1 Identification

An excavation permit / checklist shall be utilized to prepare for excavation work in the event the identity of an underground installation(s) is unknown.

Contact the relevant local One Call Center telephone number for requesting mark-out of buried public utilities, such as gas lines, electrical lines, telephone / cable lines, sewer lines, and water lines. This number is typically called a minimum of 72 hours prior to subsurface activities depending on the area in which the work will be conducted. The One Call Center will notify the local public utilities of a request for a line location mark-out for the location. The individual public utilities must locate and mark-out the utilities upon request. In most cases, the mark-outs will not be performed on private property. A confirmation number is typically established, and confirmation report generated and submitted to the requester.

Once the underground installation has been identified, proper surface marking shall be made in accordance with the guidelines contained within this procedure.

1.278.2 Surface Markings

Color-coded surface marks (paints or similar coatings) shall be used to indicate the type, location, and route of buried installations. Additionally, to increase visibility, color-coded vertical markers (temporary stakes or flags) shall supplement surface marks.

All marks and markers shall indicate the name, initials, or logo of the company that owns or operates the installation and the width of the installation if it is greater than 2 inches.

If the surface over the buried installation is to be removed, supplemental offset marking shall be used. Offset markings shall be on a uniform alignment and shall clearly indicate that the actual installation is a specific distance away.

1.278.3 Uniform Color-Coding

The following color codes are based on American Public Works Association's (APWA) Utility Location and Coordination Council (ULCC) Uniform Color Code based on ANSI Standard Z53.1 - Safety Colors. The colors and corresponding installation type are as follows:

Color	Type of Installation
RED	Electric Power Lines, Cables, Conduit, and Lighting Cables
YELLOW	Gas, Oil, Steam, Petroleum, or Gaseous Materials
ORANGE	Communication, Alarm or Signal Lines, Cables, or Conduit
BLUE	Water, Irrigation, and Slurry Lines
GREEN	Sewers and Drain Lines
WHITE	Proposed Excavation

1.279 Recordkeeping

All information regarding the identification of underground installations shall be transferred to the appropriate drawings and/or prints and shall be available onsite. Drawings and/or prints shall be maintained for the life of the project.

Appendix 16 **Sloping and Benching**

Table B-1 Maximum Allowable Slopes

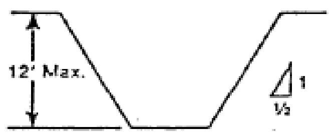
Soil or Rock Type	Maximum Allowable Slopes (H:V)[1] for Excavations Less Than 20 Feet Deep[3]	
Stable Rock	Vertical	(90°)
Type A[2]	3/4:1	(53°)
Type B	1:1	(45°)
Type C	1½:1	(34°)

Notes:

[1] Numbers shown in parentheses next to maximum allowable slopes are angles expressed in degrees from the horizontal. Angles have been rounded off.

[2] A short-term maximum allowble slope of 1/2H:1V (53°) is allowed in excavations in Type A soil that are 12 feet (3.67 m) or less in depth. Short-term maximum allowable slopes for excavations greater than 12 feet (3.67 m) in depth shall be 3/4H:1V (53°).

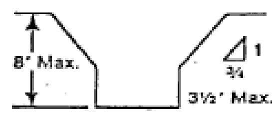
[3] Sloping or benching for excavations greater than 20 feet deep shall be designed by a registered professional engineer.



Simple Slope—Short Term

Exception: Simple slope excavations which are open 24 hours or less (short term) and which are 12 feet or less in depth shall have a maximum allowable slope of ½:1.

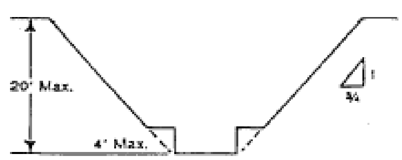
Exception: Simple slope excavations which are open 24 hours or less (short term) and which are 12 feet or less in depth shall have a maximum allowable slope of ½:1.



Unsupported Vertically Sided Lower Portion—Maximum 8 Feet in Depth

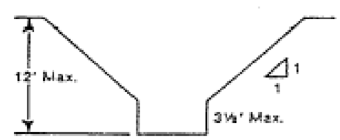
All excavations more than 8 feet but not more than 12 feet in depth which unsupported vertically sided lower portions shall have a maximum allowable slope of 1:1 and a maximum vertical side of 3½ feet.

All excavations more than 8 feet but not more than 12 feet in depth which unsupported vertically sided lower portions shall have a maximum allowable slope of 1:1 and a maximum vertical side of 3½ feet.



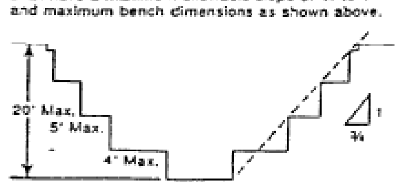
Simple Bench

2. All benched excavations 20 feet or less in depth shall have a maximum allowable slope of ¾ to 1 and maximum bench dimensions as shown above.



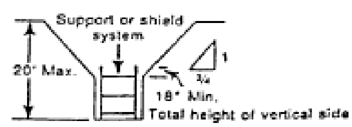
Unsupported Vertically Sided Lower Portion—Maximum 12 Feet in Depth

All excavations 20 feet or les in depth which have vertically sided lower portions that are supported or shielded shall have a maximum allowable slope of ¾:1. The support or shield system must extend at least 18 inches above the top of the vertical side.



Multiple Bench

3. All excavations 8 feet or less in depth which have unsupported vertically sided lower portions shall have a maximum vertical side of 3½ feet.



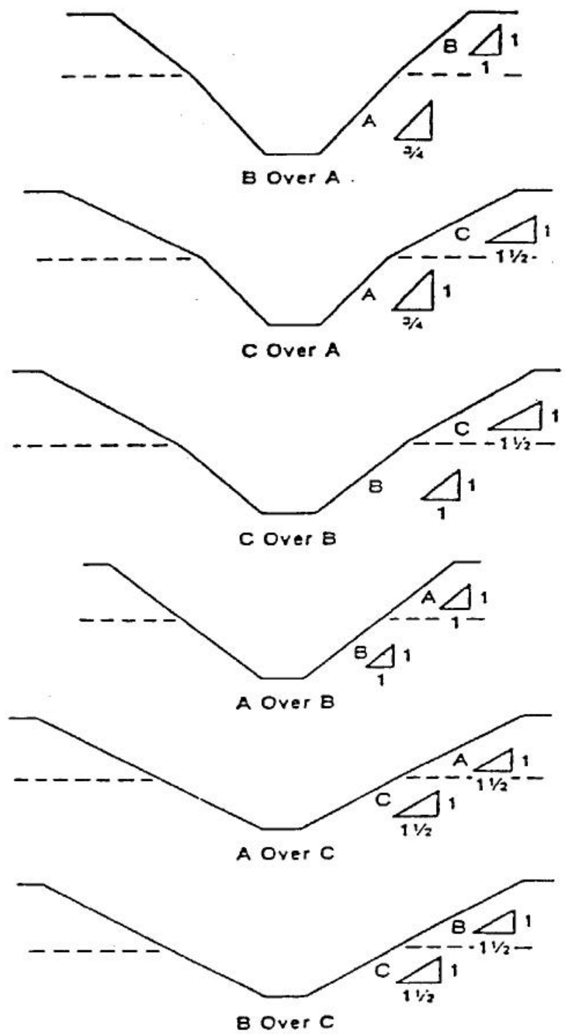
Supported or Shielded Vertically Sided Lower Portion

4. All other simple slope, compound slope, and vertically sided lower portion excavations shall be in accordance with the other options permitted under § 1926.652(b).

Appendix 1 continued

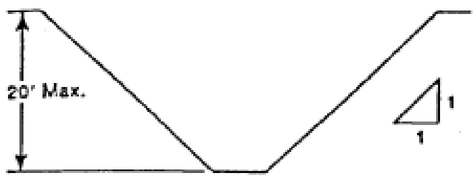
B—1.4 Excavations Made In Layered Soils

1. All excavations 20 feet or less in depth made in layered soils shall have a maximum allowable slope for each layer as set forth below.



Appendix 1 continued

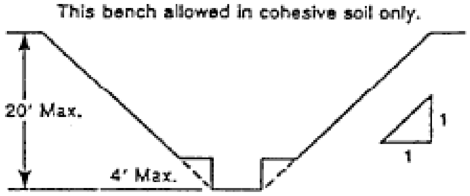
B—1.2 Excavations Made In Type B Soil



Simple Slope

1. All simple slope excavations 20 feet or less in depth shall have a maximum allowable slope of 1:1.

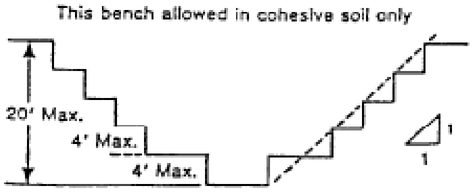
This bench allowed in cohesive soil only.



Single Bench

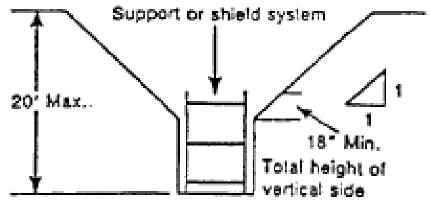
2. All benched excavations 20 feet or less in depth shall have a maximum allowable slope of 1:1 and maximum bench dimensions as shown above.

This bench allowed in cohesive soil only



Multiple Bench

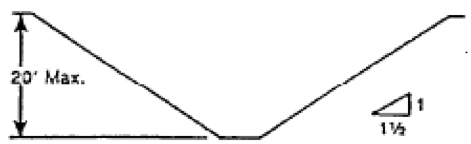
3. All excavations 20 feet or less in depth which have vertically sided lower portions shall be shielded or supported to a height at least 18 inches above the top of the vertical side. All such excavations shall have a maximum allowable slope of 1:1.



Vertically Sided Lower Portion

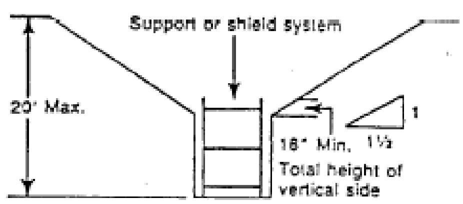
4. All other sloped excavations shall be in accordance with the other options permitted in § 1926.652(b).

B—1.3 Excavations Made In Type C Soil



Simple Slope

1. All simple slope excavations 20 feet or less in depth shall have a maximum allowable slope of 1½:1.



Vertical Sided Lower Portion

2. All excavations 20 feet or less in depth which have vertically sided lower portions shall be shielded or supported to a height at least 18 inches above the top of the vertical side. All such excavations shall have a maximum allowable slope of 1½:1.

3. All other sloped excavations shall be in accordance with the other options permitted in § 1926.652(b).

Appendix 17 **Soil Classification**

SCOPE

This attachment describes a method of classifying soil and rock deposits based on site and environmental conditions and on the structure and composition of earth deposits. Contained herein are definitions, set forth requirements, and acceptable visual and manual tests for use in classifying soils.

Classification of soil by a competent person is a prerequisite to designing protective systems for excavations.

DEFINITIONS

Cemented Soil - A soil in which the particles are held together by a chemical agent, such as calcium carbonate, such that a hand-size sample cannot be crushed into powder or individual soil particles by finger pressure.

Cohesive Soil - Clay (fine grained soil) or soil with a high clay content, which has cohesive strength. Cohesive soil does not crumble, can be excavated with vertical side slopes, and is plastic when moist. Cohesive soil is hard to break up when dry and exhibits significant cohesion when submerged. Examples include clayey silt, sandy clay, silty clay, clay, and organic clay.

Dry Soil - Soil that does not exhibit visible signs of moisture content.

Fissured - A soil material that tends to break along definite planes of fracture with little resistance or a material that exhibits open cracks, such as tension cracks, in an exposed surface.

Granular Soil - Gravel, sand, or silt (coarse grained soil) with little or no clay content. Granular soil has no cohesive strength. Some moist granular soils exhibit apparent cohesion. Granular soil cannot be molded when moist and crumbles easily when dry.

Layered System - Two or more distinctly different soil or rock types arranged in layers. Micaceous seams or weakened planes in rock or shale are considered layered.

Moist Soil - A condition in which a soil looks and feels damp. Moist, cohesive soil can easily be shaped into a ball and rolled into small diameter threads before crumbling. Moist granular soil that contains some cohesive material will exhibit signs of cohesion between particles.

Plastic - A property of a soil which allows the soil to be deformed or molded without cracking or appreciable volume change.

Saturated Soil - A soil in which the voids are filled with water. Saturation does not require flow. Saturation, or near saturation, is necessary for the proper use of instruments such as a pocket penetrometer or shear vane.

Soil Classification System - A method of categorizing soil and rock deposits in a hierarchy of Stable Rock, Type A, Type B, and Type C, in decreasing order of stability. The categories are determined based on an analysis of the properties and performance characteristics of the deposits and the environmental conditions of exposure.

Stable Rock - Natural solid mineral matter that can be excavated with vertical sides and remain intact while exposed.

Submerged Soil - Soil which is underwater or is free-seeping.

Type A - Cohesive soil with an unconfined compressive strength of 1.5 tons per square foot (tsf) (144 kPa) or greater. Examples of cohesive soils are clay, silty clay, sandy clay, clay loam, and, in some cases, silty clay loam and sandy clay loam. Cemented soils such as caliche and hardpan are also considered Type A. However, no soil is Type A if:

1) The soil is fissured; or
2) The soil is subject to vibration from heavy traffic, pile driving, or similar effects; or
3) The soil has been previously disturbed; or
4) The soil is part of a sloped, layered system where the layers dip into the excavation on a slope of four horizontal to one vertical (4H:1V) or greater; or
5) The material is subject to other factors that would require it to be classified as a less stable material.

Type B – Means:

1) Cohesive soil with an unconfined compressive strength greater than 0.5 tsf (48 kPa) but less than 1 .5 tsf, or:
2) Granular, cohesionless soils, including angular gravel, silt, silt loam, sandy loam, and, in some cases, silty clay loam and sandy clay loam.
3) Previously disturbed soil, except those which would otherwise be classed as Type C soil.
4) Soil that meets the unconfined compressive strength or requirements of Type A but is fissured or subject to vibration; or
5) Dry rock that is not stable; or
6) Material that is part of a sloped, layered system where the layers dip into the excavation on a slope less steep than 4H:1V, but only if the material would otherwise be classified as Type B.

Type C – Means:

1) Cohesive soil with an unconfined compression strength of 0.5 tsf or less; or
2) Granular soils including gravel, sand, and loamy sand; or
3) Submerged soil or soil from which water is free-seeping; or
4) Submerged rock that is not stable; or
5) Material in a sloped, layered system where the layers dip into the excavation on a slope of four horizontal to one vertical (4H:1V) or steeper.

Unconfined Compressive Strength - The load per unit area at which a soil will fail in compression. It can be determined by laboratory testing or estimated in the field using a pocket penetrometer, by thumb penetration tests, and other methods.

Wet Soil - Soil that contains significantly more moisture than moist soil, but in such a range of values that cohesive material will slump or begin to flow when vibrated. Granular material that would exhibit cohesive properties when moist will lose those cohesive properties when wet.

REQUIREMENTS

Each soil and rock deposit shall be classified by a competent person as Stable Rock, Type A, Type B, or Type C. The classification of deposits shall be made based on the results of at least one visual and at least one manual analysis using the tests described in this attachment or in other recognized methods of soil classification and testing, such as those adopted by the American Society of Testing Materials.

In a layered system, the system shall be classified in accordance with its weakest layer. However, each layer may be classified individually where a more stable layer lies under a less stable layer.

If, after classifying a deposit, the properties, factors, or conditions affecting its classification change in any way, the deposit shall be reclassified to reflect the changed conditions.

Acceptable Visual and Manual Tests

Visual Tests - Visual analysis is conducted to determine qualitative information regarding the excavation site in general, the soil adjacent to the excavation, the soil forming the sides of the open excavation, and the soil taken as samples from excavated material.

1) Observe samples of soil that are excavated and soil in the sides of the excavation. Estimate the range of particle sizes and the relative amounts of the particle sizes. Soil that is primarily composed of fine-grained material is cohesive material. Soil composed primarily of coarse-grained sand or gravel is granular material.
2) Observe soil as it is excavated. Soil that remains in clumps when excavated is cohesive. Soil that breaks up easily and does not stay in clumps is granular.
3) Observe the side of the opened excavation and the surface area adjacent to the excavation. Crack-like openings such as tension cracks could indicate fissured material.

If chunks of soil spall off a vertical side, the soil could be fissured. Small spalls are evidence of moving ground and are indications of potentially hazardous situations.

4) Observe the area adjacent to the excavation and the excavation itself for evidence of existing utility and other underground structures and to identify previously disturbed soil.
5) Observe the opened side of the excavation to identify layered systems. Examine layered systems to identify whether the layers slope toward the excavation. Estimate the degree of slope of the layers.
6) Observe the area adjacent to the excavation and the sides of the opened excavation for evidence of surface water, water seeping from the sides of the excavation, or the location of the level of the water table.
7) Observe the area adjacent to the excavation and the area within the excavation for sources of vibration that may affect the stability of the excavation face.

Manual Tests - Manual analysis of soil samples is conducted to determine quantitative as well as qualitative properties of soil and to provide more information in order to classify soil properly.

1) Plasticity - Mold a moist or wet sample of soil into a ball and attempt to roll it into threads as thin as 1/8 inch in diameter. Cohesive material can be successfully rolled into threads without crumbling. For example, if at least a 2 inch/50 millimeter (2"/50mm) length of 1/8 inch thread can be held on one end without tearing, the soil is cohesive.

2) Dry Strength - If the soil is dry and crumbles on its own or with moderate pressure into individual grains or fine powder, it is granular (any combination of gravel, sand, or silt). If the soil is dry and falls into clumps which break up into smaller clumps, but the smaller clumps can only be broken up with difficulty, it may be clay in any combination with gravel, sand, or silt. If the dry soil breaks into clumps which do not break up into small clumps and which can only be broken with difficulty, and there is no visual indication the soil is fissured, the soil may be considered unfissured.

3) Thumb Penetration - The thumb penetration test can be used to estimate the unconfined compressive strength of cohesive soils. (This test is based on the thumb penetration test described in American Society for Testing and Materials (ASTM) Standard designation D2488 - "Standard Recommended Practice for Description of Soils (Visual-Manual Procedure)"). Type A soils with an unconfined compressive strength of 1.5 tsf can be readily indented by the thumb; however, they can be penetrated by the thumb only with very great effort. Type C soils with an unconfined compressive strength of 0.5 tsf can be easily penetrated several inches by the thumb and can be molded by light finger pressure. This test should be conducted on an undisturbed soil sample, such as a large clump of soil, as soon as practical after excavation to keep to a minimum the effects of exposure to drying influences. If the excavation is later exposed to wetting influences (rain, flooding), the classification of the soil shall be changed accordingly.

4) Other Strength Tests - Estimates of unconfined compressive strength of soils can also be obtained by use of a pocket penetrometer or by using a hand-operated Torvane shear device.

5) Drying Test - The basic purpose of the drying test is to differentiate between cohesive material with fissures, unfissured cohesive material, and granular material. The procedure for the drying test involves drying a sample of soil that is approximately 1 inch/2.54 centimeters thick (1”/2.54cm) and 6 inches/15.24 centimeters (6”/15.24cm) in diameter until it is thoroughly dry:

   - If the sample develops cracks as it dries, significant fissures are indicated.
   - Samples that dry without cracking are to be broken by hand. If considerable force is necessary to break a sample, the soil has significant cohesive material content. The soil can be classified as an unfissured cohesive material and the unconfined compressive strength should be determined.
   - If a sample breaks easily by hand, it is either a fissured cohesive material or a granular material. To distinguish between the two, pulverize the dried clumps of the sample by hand or by stepping on them. If the clumps do not pulverize easily, the material is cohesive with fissures. If they pulverize easily into very small fragments, the material is granular.

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Appendix 18 **Excavation Permit / Competent Person Checklist**

Instructions	
1.	Complete permit before excavation begins.
2.	Post permit and JSA at the job site until work is complete.
3.	Retain permit on file for two years.

Site Details	
Supervisor’s Name:	Job Site:
Permit Begins: Date__________ Time:___________a.m./p.m. Permit Expires: Date:_________ Time:_________a.m./p.m.	
Location & Dimensions of Excavation: Soil Type:	
Describe Nature of Work:	

Inspection Checklist							
Check yes, no, or N/A:	**Yes**	**No**	**N/A**		**Yes**	**No**	**N/A**
Inspection of all open trenches?				Was traffic in area adequately away from trenching operations with barricades?			
Preplanning for emergencies and trench rescue.				Was there any evidence of caving or sloughing of soil since the last field inspection?			
Tension cracks observed along top of any slopes?				Were hydraulic shores pumped to design pressure?			
Were slopes cut at design angle and repose?				Access and egress provided?			

Was any water seepage noted in trench walls or bottom?				Atmospheric testing required?			
Was there evidence of significant fracture planes in soil or rock?				System lockout / tagout?			
Confined space entry permit required? (attach)				Stability of adjacent structures checked?			
Underground utilities/piping located and flagged?				Exposure from falling loads controlled (material at least 2’ from edge)?			
Were trench box(es) certified?				Protection from water accumulation?			
Was bracing system installed in accordance with design? Shoring system installed and maintained in accordance with manufacturer’s instruction?				Emergency Services: Phone Number: Method of Communication:			
Hot work permit required? (attach)							

Protective Systems					
Sloping and Benching: What is the angle or slope ratio?	Option:	1	2	3	4
Support System (shoring):	Option:	1	2	3	4
Shielding System:	Option:	1	2	3	4

Atmospheric Testing

Oxygen Safe Range of 20.5% to 22% required
Combustibles < 10% LEL

% Oxygen	_____%	Test Time:______	Flammables/	_____LEL	Test Time:______
	_____%	Test Time:______	Combustibles	_____LEL	Test Time:______
	_____%	Test Time:______		_____LEL	Test Time:______
	_____%	Test Time:______		_____LEL	Test Time:______
Toxicity Test:	_____PPM	Test Time:______			
	_____PPM	Test Time:______	Other:		
	_____PPM	Test Time:______			
	_____PPM	Test Time:______			

Signature of Competent Person Authorizing Entry			
I hereby attest that the following conditions existed and that the following items were checked or reviewed during this inspection.			
Competent Person:		Date:	
Safety Representative:		Date:	

Appendix 19 **Excavation / Trenching Daily Inspection Form**

General Information			
Project:		Date:	
Weather:		Trench Depth:	
Soil Type:		Width:	
		Length:	
Type of Protective System:			
Type of Protective System:			

Checklist			
Excavation	**Yes**	**No**	**N/A**
Excavations and Protective Systems inspected by Competent Person daily before start of work.			
Competent Person has authority to remove workers from excavation immediately.			
Surface encumbrances supported or removed.			
Employees protected from loose rock or soil.			
Hard hats worn by all employees.			
Spoils, materials, and equipment set back a minimum of 2' from edge of excavation.			
Barriers provided at all remote excavations, wells, pits, shafts, etc.			
Walkways and bridges over excavations 6' or more in depth equipped with guardrails.			
Warning vests, or other highly visible PPE provided and worn by all employees			

exposed to vehicular traffic.			
Employees prohibited from working or walking under suspended loads.			
Employees prohibited from working on faces of sloped or benched excavations above other employees.			
Warning system established and used when mobile equipment is operating near edge of excavation.			
Utilities	**Yes**	**No**	**N/A**
Utility companies contacted and/or utilities located.			
Exact location of utilities marked when near excavation.			
Underground installations protected, supported, or removed when excavation is open.			
Wet Conditions	**Yes**	**No**	**N/A**
Precautions taken to protect employees from accumulation of water.			
Water removal equipment monitored by Competent Person.			
Surface water controlled or diverted.			
Inspection made after each rainstorm.			

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Hazardous Atmosphere	Yes	No	N/A
Atmosphere tested when there is a possibility of oxygen deficiency or build-up of hazardous gases.			
Oxygen content is between 19.5% and 21%.			
Ventilation provided to prevent flammable gas build-up to 20% of lower explosive limit of the gas.			
Continuous testing conducted to ensure that atmosphere remains safe.			
Emergency Response Equipment readily available where a hazardous atmosphere could or does exist.			
Employees trained in the use of Personal Protective and Emergency Response Equipment.			
Safety harness and lifeline individually attended when employees enter deep confined excavation.			

Signature			
Competent Person Signature:		Date:	

Appendix 20 Soil Classification Worksheet

Instructions
This worksheet outlines the visual and manual tests that the competent person shall perform at least once, and each time soil conditions change. At least one visual and one manual test shall be performed; however, performing several tests is recommended so that the condition of the excavation is thoroughly examined.

General Information			
Project:		Project Location:	
Date:		Time:	
From where was the sample taken?:			

Visual Tests	
One or more visual tests are required for each classification and each time conditions change.	
1. Estimate range of particle sizes:	a. Primarily fine-grained = cohesive material b. Primarily coarse-grained = granular material
2. Observe excavated soil:	a. Clumps = cohesive material b. Breaks up easily – granular material
3. Observe sides and adjacent surface area of opened excavation:	a. Cracks like openings = fissured material b. Soil spills of vertical sides = possible fissured material
4. Previous excavation activities:	a. Previously disturbed soil b. Not previously disturbed soil
5. Observe opened side of excavation:	a. Layered systems estimated degree of slope of layers b. Layers sloped towards excavation
6. Water condition:	a. Evidence of surface water b. Water seeping from sides. c. Depth of water table.
7. Vibration present:	a. Area adjacent to excavation b. Area within excavation

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Manual Tests	
One or more manual tests are required for classification and each time soil conditions change.	
1. Plastically – soil is cohesive if following is true:	a. Mold soil samples into a small ball b. Roll ball into thread ___ " diameter c. Pick up 2" length of ___ " thread by one end without breaking
2. Dry Soil Strength:	a. Crumbles on its own or with moderate pressure = granular b. Falls into clumps which break into smaller clumps that are only broken with difficulty = clay with gravel, sand, or silt. c. Breaks into clumps which do not break into smaller clumps and can only be broken with difficulty with no visual indication of fissures = unfissured.

Signature			
Competent Person Signature:		Date:	

24. Fall Protection Program

1.280 Purpose and Scope

The purpose of this program is to provide specific requirements and safety principles to ensure that work at heights is conducted safely and effectively.

This program applies to all T&T Industrial, Inc. employees exposed to working at heights.

1.281 Resources

Number	Title
29 CFR 1926 Subpart M	Fall Protection
29 CFR 1910 Subpart D	Walking-Working Surfaces
29 CFR 1917 Subpart F	Terminal Facilities - Guarding of Edges
29 CFR 1915 Subpart E	Scaffolds, Ladders, and Other Working Surfaces - Guarding of Deck Openings and Edges
29 CFR 1926 Subpart R	Steel Erection-Fall Protection
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders - Injury and Illness Prevention Program
Cal/OSHA T8 CCR Subchapter 4	Construction Safety Orders - Fall Protection

1.282 Definitions

Acronym/Term	Definition
Competent Employee / Person	A person who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.

1.283 Fall Protection Program

The standards for regulating fall protection systems and procedures are intended to prevent employees from falling off, onto, or through working levels and to protect employees from falling objects. Fall protection requirements under the regulations require considerable planning and preparation.

1.284 Roles and Responsibilities

A competent person shall be assigned to act as the safety monitoring system. Responsibilities are to:

- Recognize fall hazards.
- Warn employees if they are unaware of a fall hazard or are acting in an unsafe manner.
- Be on same working surface and in visual sight.
- Stay close enough for verbal communication.
- Not have other assignments that would take monitors attention from the monitoring function.

Mechanical equipment must not be used or stored in areas where safety monitoring systems are being used to monitor employees engaged in roofing operations on low-slope roofs. No employee other than an employee engaged in roofing work [on low-sloped roofs] or an employee covered by a fall protection plan shall be allowed in an area where an employee is being protected by a safety monitoring system. Each employee working in a controlled access zone shall be directed to comply promptly with fall hazard warnings from safety monitors.

1.285 Training

Training shall be provided for each employee who might be exposed to fall hazards. Upon first employment, they shall be given instructions regarding the hazards and safety precautions applicable to the type of work in question and directed to read the OSHA Code of Safe Practices. Training must enable each employee to recognize the hazards of falling and the procedures to follow to minimize these hazards. Records showing participants, training dates, and signatures of instructors shall be maintained.

Only qualified persons are permitted to operate equipment and machinery. Where employees are subject to known job site hazards, such as, flammable liquids and gases, poisons, caustics, harmful plants and animals, toxic materials, confined spaces, falls, etc., they shall be instructed in the recognition of the hazard, in the procedures for protecting themselves from injury, and in the first aid procedure in the event of injury.

Re-training shall be provided when there are deficiencies in training, inadequacies in an affected employee's knowledge or use of fall protection systems or equipment, when work practices are changed, or when fall protection equipment is modified.

1.286 Fall Protection Requirement

Fall protection is required whenever employees are potentially exposed to falls from heights that exceed applicable regulatory thresholds. Guard rails, safety nets, or personal or fall arrest systems should be used. Applicable regulatory thresholds include:

- General Industry 1910.28(b)(1)(i) - Protection for wall openings and holes. Every wall opening from which there is a drop of more than 4 feet shall be guarded.
- Construction Industry 1926.501(b)(1) - Unprotected sides and edges. Each employee on a walking/working surface (horizontal and vertical surface) with an unprotected side or edge which is 6 feet or more above a lower level shall be protected from falling by the use of guardrail systems, safety net systems, or personal fall arrest systems.
- Marine Terminals 1917.112(b)(1) - Guardrails shall be provided at locations where employees are exposed to floor or wall openings or waterside edges, including bridges or gangway-like structures leading to pilings or vessel mooring or berthing installations, which present a hazard of falling more than 4 feet or into the water.
- Shipyard Industry 1915.73(d) - When employees are exposed to unguarded edges of decks, platforms, flats, and similar flat surfaces, more than 5 feet above a solid surface, the edges shall be guarded by adequate guardrails.
- Steel Erection 1926.760(a)(1) - Each employee engaged in a steel erection activity who is on a walking/working surface with an unprotected side or edge more than 15 feet above a lower level shall be protected from fall hazards by guardrail systems, safety net systems, personal fall arrest systems, positioning device systems, or fall restraint systems.
- Cal/OSHA Construction CCR 1670(a) - ANSI Approved personal fall arrest, personal fall restraint or positioning systems shall be worn by those employees whose work exposes them to falling in excess of 7 1/2 feet from the perimeter of a structure, unprotected sides and edges, leading edges, through shaftways and openings, sloped roof surfaces steeper than 7:12, or other sloped surfaces steeper than 40 degrees not otherwise adequately protected.

1.287 Incidents

Incidents shall be investigated, and corrective actions must be developed and implemented. The investigation shall look at the fall protection plan to see if any updates are needed to prevent recurrence.

1.288 Fall Protection Plan

100% tie off is required.

A fall protection plan shall be prepared by a qualified person and developed specifically for the site (leading edge work, precast concrete work, or residential construction) where the work is being performed. The plan shall be maintained up to date and include the minimum qualifications of the person(s) preparing the plan by identifying that person by name or title.

1.289 Rescue Plan

Equipment and services for prompt rescue of fallen workers, including self-rescue, shall be available before elevated work begins. Local fire departments may not have the means to perform safe and efficient rescue so do not assume they are able to do so.

1.290 Fall Arrest Systems

Personal fall arrest systems and components subjected to impact loading shall be immediately removed from service and shall not be used again for employee protection until inspected and determined by a competent person to be undamaged and suitable for reuse.

1.290.1 Personal Fall Arrest Systems

Personal fall arrest systems, when stopping a fall, shall:

- Limit maximum arresting force on an employee to 1,800 pounds when used with a body harness.
- Be rigged such that an employee can neither free fall more than 6 feet, nor contact any lower level, and, where practicable, the anchor end of the lanyard shall be secured at a level not lower than the employee's waist.
- Bring an employee to a complete stop and limit maximum deceleration distance an employee travels to 3.5 feet.
- Have sufficient strength to withstand twice the potential impact energy of an employee free falling a distance of 6 feet, or the free fall distance permitted by the system, whichever is less.

1.290.2 Anchorages

Anchorages used for attachment of personal fall arrest equipment shall be independent of any anchorage being used to support or suspend platforms and capable of supporting at least 5,000 pounds per employee attached, or shall be designed, installed, and used as follows:

- As part of a complete personal fall arrest system which maintains a safety factor of at least two.
- Under the supervision of a qualified person.

1.290.3 Positioning Systems

Positioning device systems and their use shall conform to the following provisions:

- Positioning devices shall be rigged such that an employee cannot free fall more than 2 feet.
- Position devices shall be inspected prior to each use for wear, damage, and other deterioration, and defective components shall be removed from service.
- The use of non-locking snaphooks shall be prohibited after January 1, 1998.
- Anchorage points for positioning device systems shall be capable of supporting two times the intended load or 3,000 pounds, whichever is greater.

1.290.4 Fall Protection Equipment

All safety belts, harnesses, and lanyards placed in service or purchased on or before February 1, 1997, shall be labeled as meeting the requirements contained in ANSI A10.14-1975, Requirements for Safety Belts, Harnesses, Lanyards, Lifelines and Drop Lines for Construction and Industrial Use.

All personal fall arrest, personal fall restraint, and positioning device systems purchased or placed in service after February 1, 1997, shall be labeled as meeting the requirements contained in ANSI A10.14-1991 American National Standard for Construction and Demolition Use, or ANSI Z359.1-1992 American National Standard Safety Requirements for Personal Fall Arrest Systems, Subsystems and Components.

When purchasing equipment and raw materials for use in fall protection systems, applicable standards and requirements shall be met.

1.291 Access Control

Access to dangerous areas where safety monitoring systems are in place shall be controlled. When used to control access to areas where leading edge and other operations are taking place, the controlled access zone shall be defined by a control line or by any other means that restricts access. Signs shall be posted to warn unauthorized employees to stay out of the controlled access zone.

1.292 Guardrails

A standard guardrail shall consist of top rail, mid-rail or equivalent protection, and posts, and shall have a vertical height within the range of 42 inches to 45 inches from the upper surface of the top rail to the floor, platform, runway, or ramp level.

25. Fatigue Management Program

1.293 Purpose and Scope

The purpose of this program is to define the minimum requirements established regarding managing employee fatigue and ensure that mental or physical fatigue does not lead to injury to personnel or damage to property.

This program applies to all T&T Industrial, Inc. employees.

1.294 Fatigue Management Program

Fatigue can significantly affect the ability to communicate clearly, work safely and productively, and react optimally in an emergency. Fatigue and related consequences such as unintentionally falling asleep can be significant factors in incidents. Even without incidents occurring, fatigue impairment can significantly impact efficiency and productivity. One of the critical consequences of fatigue is that the ability to assess fitness for duty becomes impaired. This means employees may not be fully aware of their fatigue related impairment, which can make communicating about potential issues more difficult.

Workers must never operate motor vehicles and/or heavy equipment while excessively fatigued.

1.295 Training

Employees shall be trained on the components of this fatigue management program.

Initial and annual training shall be provided on how to recognize fatigue, how to control fatigue through appropriate work and personal habits and reporting of fatigue to supervision.

1.296 Work Hour Limitations

The following work hour limitations have been set to control job rotation schedules to control fatigue, allow for sufficient sleep, and increase mental fitness to control employee turnover and absenteeism:

- A planned work shift not to exceed 12 hours.
- Normal work schedules shall not exceed 28 consecutive days.

1.297 Ergonomic Equipment

Equipment and workstations must be evaluated based on ergonomic principles to ensure that work tasks are designed to control employee fatigue and increase mental alertness.

Ergonomic equipment shall be used to improve workstation conditions such as anti-fatigue mats for standing, lift assist devices for repetitive lifting, proper lighting, control of temperature, and other ergonomic devices as deemed appropriate.

1.298 Work Tasks

Work tasks and schedules to control fatigue that affect amount of sleep, timing, and quality of sleep each day, amount of time since last sleep period, time of day, and workload and time on task shall be analyzed and evaluated periodically.

1.299 Rest Breaks

Rest breaks shall be provided to control fatigue and increase mental fitness.

Rest breaks are required, sleeping in areas such as service trucks, base camps, hotels, trailers, rig/platform rest and sleeping areas.

Chairs shall be provided for employees to sit periodically and will provide periodic rest breaks for personnel.

1.300 Reporting

Employees are expected to be fit for duty. Employees in safety critical positions shall report fatigue / tiredness and lack of mental acuity to supervision. Supervisors shall make safety critical decisions and take appropriate actions to prevent loss.

1.301 Drug Use

Employees shall not chronically use over the counter or prescription drugs to increase mental alertness. Employees shall not take any substance known to increase fatigue, including fatigue that sets in after the effects of the drug wear off.

1.302 Review

The Fatigue Management Program shall undergo periodic assessments of its effectiveness with a continuous improvement plan created to close any gaps.

26. Fire Protection Program

1.303 Purpose and Scope

The purpose of this program is to describe a framework for fire hazards commonly occurring at work sites and provide protocols and procedures to control these hazards.

This program applies to all T&T Industrial, Inc. employees.

1.304 Resources

Number	Title
29 CFR 1926 Subpart F	Fire Protection
29 CFR 1910 Subpart L	Fire Protection and Prevention
Cal/OSHA T8 Subchapter 7	General Industry Safety Orders
Cal/OSHA T8 Subchapter 4	Construction Safety Orders
NFPA 10	Portable Fire Extinguishers

1.305 Fire Protection Program

Ignition sources can include any material, equipment, or operation that emits a spark or flame including obvious items, such as torches, as well as less obvious items, such as static electricity and grinding operations. Equipment or components that radiate heat, such as kettles, catalytic converters, and mufflers, also can be ignition sources.

Fuel sources include combustible materials, such as wood, paper, trash, and clothing; flammable liquids, such as gasoline or solvents; and flammable gases, such as propane or natural gas.

1.306 Training

Where portable fire extinguishers are provided for employee use, employees shall be trained to familiarize themselves with the general principles of fire extinguisher use and the hazards associated with basic firefighting.

Employees who are expected to use fire extinguishers in case of emergency shall be trained during their orientation upon initial assignment. Refresher training shall occur at least annually.

1.307 Inspection

Portable fire extinguishers shall be subjected to monthly visual inspections and annual servicing / maintenance. Monthly inspections involve a visual check to ensure the pin is in place, it is adequately charged, and not corroded. Visual inspections shall be noted on the tag. The annual servicing and inspection are more thorough. The annual inspection is typically performed by a third-party professional, as it may involve re-charging the extinguisher and maintenance / servicing.

The annual maintenance date shall be recorded and retained for 1 year after the last entry or life of the shell, whichever is less.

1.308 Selection and Distribution

Portable fire extinguishers shall be provided for employee use and selected and distributed based on the classes of anticipated workplace fires and on the size and degree of hazard which would affect their use.

Management shall examine its premises and processes thoroughly and repeatedly, correcting threatening situations as soon as they are identified. In addition, adequate fire protection equipment shall be provided.

Particular guidance on fire codes and standards may be sought from manufacturers of fire protection equipment and systems, local municipal fire departments, the company insurance agency, the National Fire Protection Association (NFPA), and other fire protection agencies whose services may be secured on a consulting or staff basis.

1.309 Number of Extinguishers Required

See NFPA 10 for classification of occupancy hazards.

Class A Hazards (ordinary combustibles such as wood, cloth, paper, rubber, and many plastics)

Criteria	Light Hazard Occupancy	Ordinary Hazard Occupancy	Extra Hazard Occupancy
Minimum rated single extinguisher	2-A	2-A	4-A
Maximum floor area per unit of A	3000 sq ft	1500 sq ft	1000 sq ft
Maximum floor area per extinguisher	11,250 sq ft	11,250 sq ft	11,250 sq ft
Maximum travel distance to extinguisher	75 ft	75 ft	75 ft

Class B Hazards (flammable or combustible liquids, petroleum greases, tars, oils, oil-based paints, alcohols, solvents, lacquers, flammable gases, and similar materials)

Basic Minimum Extinguisher Rating for Area Specified	Basic Minimum Extinguisher Rating	Maximum Travel Distance to Extinguishers
Light (low)	5B 10B	30 ft 50 ft
Ordinary (moderate)	10B 20B	30 ft 50 ft
Extra (high)	40B 80B	30 ft 50 ft

Class C Hazards (energized electrical equipment)

Live electrical fires require certain types of extinguishing agents (e.g., CO^2, powder, halon). If the power can be turned off the fire hazard is reclassified as a class A or B. However, certain types of electrical equipment, such as capacitors, retain electrical charge even when electricity is turned off.

Class D Hazards (combustible metals such as magnesium, titanium, zirconium, sodium, lithium, and potassium)

Distribution of portable fire extinguishers for Class D hazards, such as combustible metal powders, flake, or shavings are required in the work areas so that the maximum travel distance is 75 feet or less in any direction.

Class K Hazards (cooking appliances that involve combustible cooking media such as vegetable or animal oils and fats)

Fire extinguishers provided for the protection of cooking grease fires shall be of an approved type compatible with the automatic fire-extinguishing system agent. Class K portable fire extinguishers must be located 30 feet of travel distance from the hazard to the extinguishers.

1.310 Requirements of Portable Extinguishers

- Extinguishers shall be fully charged and kept in their designated areas.
- Extinguishers shall be conspicuously located, clearly marked as to their intended use, and not be unobscured from view.
- The top shall not be more than 5 feet above the floor if the extinguisher weighs less than 40 lbs. The top must not be more than 3.5 feet above the floor if the extinguisher weighs more than 40 lbs. Clearance between the floor and the bottom of the extinguisher shall not be less than 4 inches.
- Extinguishers shall be thoroughly examined and/or recharged or repaired as needed at regular intervals not more than 1 year apart.
- Extinguishers shall be hydrostatically tested at the specified interval.

1.311 Reporting of Hostile Fires

All hostile fires resulting in property damage, no matter how small, shall be investigated to prevent recurrence and to develop loss experience data upon which preventative measures can be based.

- Investigation – fires shall be investigated by the appropriate personnel (HSE, incident commander, etc.).
- Reporting – copies of the completed investigation reports shall be distributed to management.

27. First Aid Program

1.312 Purpose and Scope

The purpose of this program is to ensure the safety of employees and describe a framework for administering first aid.

This program applies to all T&T Industrial, Inc. employees.

1.313 Resources

Number	Title
29 CFR 1910 Subpart K	Medical and First Aid
29 CFR 1926 Subpart D	Occupational Health and Environmental Controls
Cal/OSHA T8 Subchapter 4	Construction Safety Orders – Emergency Medical Services
Cal/OSHA T8 Subchapter 4	General Industry Safety Orders-Personal Safety Services and Safeguards
CMS-FM-0026	First Aid Kit Inspection Form

1.314 First Aid

Medical facilities will be made available where possible. In the absence of an infirmary, clinic, or hospital in near proximity to the workplace, a person or persons shall be available and adequately trained to render first aid.

1.315 Training

A person who has a valid certificate in first-aid training from the American Red Cross or equivalent that can be verified by documentary evidence shall be available at the worksite to render first aid.

1.316 First Aid Supplies

The first aid equipment and supplies shall be determined by the potential occupational injuries and illnesses of personnel. First aid supplies shall be easily accessible when required.

The items and amounts of each item needed on site will depend on the following variables:

- Size of work force
- Type of work
- Availability of medical services
- Types of injuries and illnesses
- Scope and environment of the work location

Adequate first aid supplies shall be available and periodically reassessed for the demand for supplies with inventories adjusted.

First aid kits shall be placed in a weatherproof container with individual sealed packages of each type of item. See the Appendix for suggested contents of a first aid kit.

For construction operations, first aid kits shall be checked before being sent out to each job and at least weekly to ensure that the expended items are replaced.

1.317 Location

Where there is a first aid facility provided, it shall be located as close as possible to the main work area to provide prompt first aid care to injured and ill employees. Distance should not hamper the prompt reporting of minor injuries. See the Appendix for suggested first aid supplies.

Location of the first aid facility shall also be near water and sanitary sewer lines. The first aid facility shall be easily accessible to ambulance service.

The first aid facility should be designed to eliminate noise, vibration, and other disturbances insofar as is practical.

1.318 Transport

Proper equipment for prompt transportation of the injured person to a physician or hospital or a communication system for contacting necessary ambulance service shall be provided.

If an ambulance service is not readily available to the work site, or if travel conditions are not normal, other transportation must be available that:

- Is suitable, considering the distance to be travelled and the types of acute illnesses or injuries that may occur at the work site
- Protects occupants from the weather
- Has systems that allow the occupants to communicate with the health care facility to which the injured or ill worker is being taken
- Can accommodate a stretcher and an accompanying person if required to

1.319 Emergency Eye Washing

Where the eyes or body of any person may be exposed to injurious corrosive materials, suitable facilities shall be provided within the work area.

1.320 Emergency Phone Numbers

In areas where 911 is not available, the telephone numbers of the physicians, hospitals, or ambulances shall be conspicuously posted. The site Safety Coordinator is responsible for posting these telephone numbers.

1.321 Documentation

The company must keep a record of all circumstances respecting an accident as described by the injured worker, the date and time of its occurrence, the names of witnesses, the nature and exact location of the injuries to the worker and the date, time, and nature of each first aid treatment given.

Appendix 21 Suggested First Aid Kit Contents

First Aid Supply	Minimum Quantity		Minimum Size or Volume	
	Class A Kits	Class B Kit	(US)	(metric)
Adhesive Bandage	16	50	1 x 3 in.	2.5 x 7.5 cm
Adhesive Tape	1	2	2.5 yd (total)	2.3 m
Antibiotic Application	10	25	1/57 oz	0.5 g
Antiseptic	10	50	1/57 oz	0.5 g
Breathing Barrier	1	1	N/A	N/A
Burn Dressing (gel soaked)	1	2	4 x 4 in.	10 x 10 cm
Burn Treatment	10	25	1/32 oz	0.9 g
Cold Pack	1	2	4 x 5 in.	10 x 12.5 cm
Eye Covering, with attachment	2	2	2.9 sq. in.	19 sq. cm
Eye/Skin Wash	1	0	1 fl. oz total	29.6 ml
	0	1	4 fl. oz total	118.3 ml
Foil Blanket	1	1	52 x 84 in.	132 x 213 cm
First Aid Guide	1	1	N/A	N/A
Hand Sanitizer	10	20	1/32 oz	0.9 g
Medical Exam Gloves	2 pair	4 pair	N/A	N/A
Roller Bandage	1	2	2 in. x 4 yd	5 cm x 3.66 m
	0	1	4 in. x 4 yd	10 cm x 3.66 m
Scissors	1	1	N/A	N/A
Splint	0	1	4.0 x 4 yd	10.2 x 61 cm

Sterile Pad	2	4	3 x 3 in.	7.5 x 7.5 cm
Tourniquet	0	1	1.5 in. (width)	3.8 cm (width)
Trauma Pad	2	4	5 x 9 in.	12.7 x 22.9 cm
Triangular Bandage	1	2	40 x 40 x 56 in.	101 x 101 x 142 cm

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Appendix 22 **First Aid Kit Inspection Form**

General Information			
Inspector Name:		Inspection Date:	
Kit Location:			

Required First Aid Items		Stocked	Needs Restocking
1.	Gauze pads		
2.	Large gauze pads		
3.	Adhesive bandages (Band-Aids)		
4.	Gauze roller bandages		
5.	Triangular bandages		
6.	Wound cleaning agent		
7.	Scissors		
8.	Blanket		
9.	Tweezers		
10.	Adhesive tape		

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11.	Latex gloves		
12.	Resuscitation bag / pocket mask		
13.	Elastic wraps		
14.	Splint		
15.	Ibuprofen		
16.	Saline solution		
17.	Directions for requesting emergency assistance		

Signatures			
Employee Signature:		Date:	

28. Fit for Duty Program

1.322 Purpose and Scope

The purpose of this program is to ensure employees are fit to perform their assigned tasks safely and reliably.

This program applies to all T&T Industrial, Inc. employees.

1.323 Fit for Duty Program

Employees shall be physically capable of performing their assigned job functions.

Depending on the job function, pre-employment physicals should be included in the hiring process, also when changing into certain job functions and different environments.

Employees shall be assessed and monitored to confirm fitness to wear required personal protective equipment (PPE) for their job function.

Employees must be trained to safely perform their assigned tasks.

Employees must follow safe work procedures identified in the Company health and safety program.

Ensure that workers are educated on the company's Fit For Duty policies and procedures.

If an employee is determined to be unfit for duty, the company has a process in place to provide reasonable assistance to the employee. This may include, but is not limited to, transferring the worker to another role, providing a leave of absence, Employee Assistance Programs, etc.

1.324 Drug and Alcohol Screening

Drug and alcohol screening shall be conducted for pre-employment, post-accident, and randomly as required by Department of Transportation (DOT) regulations or site requirements.

Employees shall report all medications they are taking. Over-the-counter medications such as allergy or cold and flu medications could also impair one's ability to perform safely and shall also be reported to their supervisor.

Employee's activities and behaviors shall be monitored to determine if removal from the work site is necessary.

1.325 Fatigue

Employees are responsible for ensuring they are physically and mentally fit to perform their job functions safely. If an employee is not able to perform their duties safely due to their physical or mental state, they are responsible for notifying their supervisor. Employees shall take responsibility for their own safety as well as not report to work in a condition as to endanger the safety of their fellow workers.

29. Forklifts & Powered Industrial Trucks Safety Program

1.326 Purpose and Scope

The purpose of this program is to provide safety procedures for forklifts and powered industrial trucks to maintain a safe workplace for employees and prevent or mitigate incidents.

This program applies to all T&T Industrial, Inc. employees that work with or around forklifts or powered industrial trucks.

1.327 Resources

Number	Title
29 CFR 1910 Subpart N	Materials Handling and Storage
29 CFR 1926 Subpart C	General Safety and Health Provisions
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders
CMS-FM-0027	Forklift/Powered Industrial Truck Inspection Checklist

1.328 Definitions

Acronym/Term	Definition
Competent Employee / Person	A person who can identify existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.
Free Rigging	The direct attachment to or placement of rigging equipment (slings, shackles, rings, etc.) onto the tines of a forklift/powered industrial truck for a below-the-forks lift. This type of lift does not use an approved lifting attachment.
Rated Capacity	The maximum working load that a forklift/powered industrial truck is designed, by the manufacturer, to carry at a specified load height.
Truck	Forklift or any powered industrial truck.
Unattended Forklift	When the forklift operator is 25 feet or more away from the forklift even if it remains in view or whenever the forklift operator leaves the forklift, and it is not in view.

1.329 Forklifts and Powered Industrial Trucks Program

A forklift or powered industrial truck is a powerful tool that allows one person to precisely lift and place large heavy loads with little effort. Using a tool such as a forklift, cart, or hand truck instead of lifting and carrying items by hand can reduce the risk of back injury. However, there is a greater risk of injury or death when an operator has not been trained properly, is not familiar with the way a particular forklift operates, operates carelessly, or operates a malfunctioning forklift.

1.330 Operation Qualification

Powered industrial truck operators shall be competent to operate the equipment safely. A competent operator has the necessary education / knowledge, training, and experience to safely perform the job.

1.331 Training

Training shall consist of a combination of formal instruction (e.g., lecture, discussion, interactive computer learning, video, written material), practical training (demonstrations performed by the trainer and practical exercises performed by the trainee), and evaluation of the operator's performance in the workplace. Practical training involves instructor demonstrations and trainee exercises.

All operator training and evaluation shall be conducted by persons who have the knowledge, training, and experience to train powered industrial truck operators and evaluate their competence. The training content shall include forklift operating instructions, use of controls, capacity, and load stability as a minimum.

Refresher training in relevant topics shall be provided to the operator when:

- The operator has been observed to operate the vehicle in an unsafe manner,
- The operator has been involved in an accident or near-miss incident,
- The operator has received an evaluation that reveals that the operator is not operating the truck safely,
- The operator is assigned to drive a different type of truck, or
- A condition in the workplace changes in a manner that could affect safe operation of the truck.

Recertification: An evaluation of each powered industrial truck operator's performance shall be conducted at least once every 3 years.

1.332 Trainees

Trainees may operate powered industrial trucks only:

- Under the direct supervision of persons who have the knowledge, training, and experience to train operators and evaluate their competence; and
- Where such operation does not endanger the trainee or other employees.

1.333 Inspection

Industrial trucks shall be examined before being placed in service and shall not be placed in service if the examination shows any condition adversely affecting the safety of the vehicle. Such examination shall be made at least daily. Operators must ensure the equipment is safe prior to operating.

Where industrial trucks are used on around-the-clock basis, they shall be examined after each shift. Defects when found shall be immediately reported and corrected.

1.334 Vehicle Certifications

Name plates indicating the capacity are to be current and visible.

Forklifts approved for use in flammable vapor or dust-hazardous areas shall bear a label or some other identifying mark indicating approval by the testing laboratory.

High-lift rider trucks shall be fitted with an overhead guard unless operating conditions make this impossible. If the type of load presents a hazard, the truck shall be equipped with a vertical load backrest extension.

1.335 Truck Operations

Trucks shall not be driven up to anyone standing in front of a bench or other fixed object.

No person is allowed to stand or pass under the elevated portion of any truck, whether loaded or empty.

Unauthorized personnel shall not be permitted to ride on powered industrial trucks. A safe place to ride shall be provided where riding of trucks is authorized.

Arms or legs shall not be placed between the uprights of the mast or outside the running lines of the truck.

Forklifts shall not be used for non-lifting tasks such as pushing or pulling unless recommended by the manufacturer or with the appropriate use of an attachment.

When a powered industrial truck is left unattended, load engaging means shall be fully lowered, controls neutralized, power shut off, and brakes set. Wheels shall be blocked if the truck is parked on an incline.

A powered industrial truck is unattended when the operator is 25 feet or more away from the vehicle which remains in view, or whenever the operator leaves the vehicle, and it is not in view.

A safe distance shall be maintained from the edge of ramps or platforms while on any elevated dock, platform, or freight car. Trucks shall not be used for opening or closing freight doors.

Brakes shall be set with wheel blocks in place to prevent movement of trucks, trailers, or railroad cars while loading or unloading. Fixed jacks may be necessary to support a semitrailer during loading or unloading when the trailer is not coupled to a tractor. The flooring of trucks, trailers, and railroad cars shall be checked for breaks and weakness before they are driven onto.

There shall be sufficient headroom under overhead installations, lights, pipes, sprinkler systems, etc.

An overhead guard shall be used as protection against falling objects. It should be noted that an overhead guard is intended to offer protection from the impact of small packages, boxes, bagged material, etc., representative of the job application, but not to withstand the impact of a falling capacity load.

A load backrest extension shall be used whenever necessary to minimize the possibility of the load or part of it from falling rearward.

Only approved industrial trucks may be used in hazardous locations.

Fire aisles, access to stairways, and fire equipment shall be kept clear.

1.336 Spotters / Banksman

A risk assessment shall be performed to determine when operators will use spotters / banksman. As guidance, the following shall be considered when determining the need for spotters:

- Pedestrian proximity
- Adjacent traffic / simultaneous operations (SIMOPS)
- Lateral, overhead, or other obstructions exist in the work area
- Blind spots in the vicinity (equipment layout, buildings, trucks, vehicles, etc.)

- Blind spots due to the type of equipment or the load being carried
- Overall visibility (nighttime, rain, etc.)
- Ease of picking up and setting down the load
- Any other reason deemed necessary by the operator(s) or supervisor.

Spotters shall wear a high visibility vest and be in direct visual contact with the operator at all times. In the event the operator loses visual reference of the banksman / spotter they shall stop all equipment movement until they regain line of sight.

Banksman / spotters shall also have a radio that allows them to speak directly with the operator. The channel to be used shall be determined in the toolbox talk prior to the commencement of the task. The channel shall be one that is free from excessive use.

1.337 Traveling

The wearing of a safety belt is mandatory for the driver of a self-propelled vehicle equipped with a roll-over protective structure as well as for any worker in the vehicle while it is in motion. Any persons other than the driver are prohibited from being on a self-propelled vehicle if it is not equipped with a seat and a belt to accommodate each person.

All traffic regulations shall be observed, including authorized speed limits. A safe distance shall be maintained approximately three truck lengths from the truck ahead, and the truck shall be kept under control at all times.

The right of way shall be yielded to ambulances, fire trucks, or other vehicles in emergency situations.

Other trucks traveling in the same direction at intersections, blind spots, or other dangerous locations shall not be passed.

Drivers are required to slow down and sound the horn at cross aisles and other locations where vision is obstructed. If the load being carried obstructs forward view, the driver shall travel with the load trailing.

Railroad tracks shall be crossed diagonally wherever possible. Parking closer than 8 feet from the center of railroad tracks is prohibited.

Drivers are required to look in the direction of and keep a clear view of the path of travel.

Grades shall be ascended or descended slowly.

When ascending or descending grades more than 10%, loaded trucks shall be driven with the load upgrade.

On all grades, the load and load engaging means shall be tilted back, if applicable, and raised only as far as necessary to clear the road surface.

Under all travel conditions the truck shall be operated at a speed that will permit it to be brought to a stop in a safe manner.

Stunt driving and horseplay is not permitted.

Drivers are required to slow down for wet and slippery floors.

Dockboards or bridge plates, shall be properly secured before they are driven over. Dockboards or bridge plates shall be driven over carefully and slowly, and their rated capacity never exceeded.

Elevators shall be approached slowly and then entered squarely after the elevator car is properly leveled. Once on the elevator, the controls shall be neutralized, power shut off, and the brakes set.

Motorized hand trucks shall enter elevator or other confined areas with load end forward.

Running over loose objects on the roadway surface shall be avoided.

While negotiating turns, speed shall be reduced to a safe level by means of turning the hand steering wheel in a smooth, sweeping motion. Except when maneuvering at a very low speed, the hand steering wheel shall be turned at a moderate, even rate.

A load engaging means shall be placed under the load as far as possible; the mast shall be carefully tilted backward to stabilize the load.

Extreme care shall be used when tilting the load forward or backward, particularly when high tiering. Tilting forward with load engaging means elevated is prohibited except to pick up a load. An elevated load shall not be tilted forward except when the load is in a deposit position over a rack or stack. When stacking or tiering, only enough backward tilt to stabilize the load shall be used.

The operator shall verify trailer chocks, supports, and dock plates prior to loading or unloading.

1.338 Maintenance

If at any time a powered industrial truck is found to be in need of repair, defective, or in any way unsafe, the truck shall be taken out of service until it has been restored to safe operating condition by authorized personnel.

Repairs to the fuel and ignition systems of industrial trucks which involve fire hazards shall be conducted only in locations designated for such repairs.

Trucks in need of repairs to the electrical system shall have the battery disconnected prior to such repairs.

All parts of any such industrial truck requiring replacement shall be replaced only by parts equivalent as to safety with those used in the original design.

Industrial trucks shall not be altered so that the relative positions of the various parts are different from what they were when originally received from the manufacturer, nor may they be altered either by the addition of extra parts not provided by the manufacturer or by the elimination of any parts. Additional counterweighting of fork trucks shall not be done unless approved by the truck manufacturer.

Water mufflers shall be filled daily or as frequently as is necessary to prevent depletion of the supply of water below 75% of the filled capacity. Vehicles with mufflers having screens or other parts that may become clogged shall not be operated while such screens or parts are clogged. Any vehicle that emits hazardous sparks or flames from the exhaust system shall immediately be removed from service and not returned to service until the cause for the emission of such sparks and flames has been eliminated.

When the temperature of any part of any truck is found to be in excess of its normal operating temperature, thus creating a hazardous condition, the vehicle shall be removed from service and not returned to service until the cause for such overheating has been eliminated.

Industrial trucks shall be kept in a clean condition, free of lint, excess oil, and grease. Noncombustible agents shall be used for cleaning trucks. Low flash point (below 100 °F.) solvents shall not be used. High flash point (at or above 100 °F.) solvents may be used. Precautions regarding toxicity, ventilation, and fire hazard shall be consonant with the agent or solvent used.

1.339 Refueling Stations

Refueling stations shall be designated, properly equipped, maintained, and provided with instructions for each type of forklift.

1.340 Battery Charging Station Equipment and Precautions

A carboy tilter or siphon for handling electrolyte and a hose with running water to flush and neutralize a spill are required. When charging batteries, acid shall be poured into the water.

Fire protection in accordance with the size of the forklift shall be provided.

Safety shower and eyewash station shall be provided.

Adequate ventilation for dispersal or removal of hydrogen gas shall be provided.

Non-sparking or coasted battery rack supports, and an overhead hoist or equivalent battery-handling equipment shall be provided.

Trucks shall be properly positioned with brakes applied.

Vent caps shall be kept in place and battery compartment or cover shall be open.

Smoking and open flames shall not be permitted, and efforts shall be made to prevent sparks or electric arcs.

Tools and other metallic objects shall be kept clear of the top of uncovered batteries.

1.341 Gasoline and Diesel Precautions

Fuel tanks shall not be filled while the engine in running.

The tank shall not be filled to the top. Spillage of oil or fuel shall be carefully cleaned up or completely evaporated and the fuel tank cap replaced before restarting engine.

Open flames shall not be used for checking gasoline level in fuel tanks.

Trucks shall not be operated with a leak in the fuel system until the leak has been corrected.

An appropriate portable fire extinguisher shall be provided at the refueling station.

1.342 LP Gas (propane) Precautions

LPG-powered trucks shall not be refueled in confined areas where LPG vapors could collect if a leak occurs.

LPG-powered trucks shall not be left near heat sources, stairways, exits, or other egress areas.

When parking LPG-powered trucks for a long period of time, the service valve shall be turned off.

Only trained and authorized personnel are authorized to replace LPG containers.

Spare tanks shall be stored in an outside shelter with adequate ventilation and an appropriate portable fire extinguisher shall be provided.

Appendix 23 **Forklift/Powered Industrial Truck Inspection Checklist**

General Information (Internal Combustion Engine Truck – Gas / LPG / Diesel Truck)			
Operator Name:		Date.:	
Truck No.:		Hours Worked:	
Hour Meter Start of Shift:		End of Shift:	
Record of Fluid Added			
Battery Water:		Hydraulic Oil:	
Fuel:		Engine Oil:	
Radiator Coolant:		Other:	
Meter Reading			
Drive Hour:		Hoist Hour:	
Safety & Operational Checks			
Mark any defective item with an X and give details below. Have a qualified mechanic correct all problems.			

Engine Off Checks	**OK**	**Maintenance Required**
Leaks – fuel, hydraulic oil, engine oil, or radiator coolant		
Tires – condition and pressure		
Forks, top clip retaining pin and heel – check condition		
Load backrest – securely attached		
Hydraulic hoses, mast chains, cables and stops – check visually		
Overhead guard – attached		
Finger guards – attached		
Propane tank (LP gas truck) – rust corrosion, damage		
Safety warnings – attached (refer to parts manual for location)		
Battery – check water / electrolyte level and charge		
All engine belts – check visually		
Hydraulic fluid level – check level		

Engine oil level – dipstick		
Transmission fluid level – dipstick		
Engine air cleaner – squeeze rubber dirt trap or check the restriction alarm (if equipped)		
Fuel sedimentor (diesel)		
Radiator coolant – check level		
Operator’s manual – in container		
Nameplate – attached and information matches model, serial number, and attachments		
Seatbelt – functioning smoothly		
Hood latch – adjusted and securely fastened		
Brake fluid – check level		
Engine On Checks	**OK**	**Maintenance Required**
Accelerator or direction control pedal – functioning smoothly		
Service brake – functioning smoothly		
Parking brake – functioning smoothly		
Steering operation – functioning smoothly		
Drive control – forward / reverse – functioning smoothly		
Tilt control – forward and back – functioning smoothly		
Hoist and lowering control – functioning smoothly		
Attachment control – operation		
Horn and lights – functioning		
Cab (if equipped) – heater, defroster, wipers - functioning		
Gauges: ammeter, engine oil pressure, hour meter, fuel level, instrument monitors - functioning		

Signatures			
Inspector Signature:		Date:	

T&T Industrial, Inc.

Forklifts & Powered Industrial Trucks Safety Program

General Information (Electric Truck)			
Operator Name:		Date.:	
Truck No.:		Hours Worked:	
Hour Meter Start of Shift:		End of Shift:	
Record of Fluid Added			
Battery Water:		Hydraulic Oil:	
Fuel:		Engine Oil:	
Radiator Coolant:		Other:	
Meter Reading			
Drive Hour:		Hoist Hour:	
Safety & Operational Checks			
Mark any defective item with an X and give details below. Have a qualified mechanic correct all problems.			

Motor Off Checks	**OK**	**Maintenance Required**
Leaks – hydraulic oil, battery		
Tires – condition and pressure		
Forks, top clip retaining pin and heel – condition		
Load backrest extension – attached		
Hydraulic hoses, mast chains, cables and stops – check visually		
Finger guards – attached		
Overhead guard – attached		
Safety warnings – attached (refer to parts manual for location)		
Battery – water / electrolyte level and charge		
Hydraulic fluid level – dipstick		
Transmission fluid level – dipstick		
Operator's manual in container		
Capacity plate attached – information matches model, serial number,		

and attachments		
Battery restraint system – adjust and fasten		
Operator protection • Sit-down truck – seatbelt – functioning smoothly • Man-up truck – fall protection / restraining means – functioning smoothly		
Brake fluid – check level		
Motor On Checks	**OK**	**Maintenance Required**
Accelerator linkage – functioning smoothly		
Parking brake – functioning smoothly		
Service brake – functioning smoothly		
Steering operation – functioning smoothly		
Drive control – forward / reverse – functioning smoothly		
Tilt control – forswear and back – functioning smoothly		
Hoist and lowering control – functioning smoothly		
Attachment control – operation		
Horn – functioning		
Lights and alarms (where present) – functioning		
Hour meter – functioning		
Battery discharge indicator – functioning		
Instrument monitors - functioning		

Signatures			
Inspector Signature:		Date:	

30. General Waste Management Program

1.343 Purpose and Scope

The purpose of this program is to provide general guidelines on an awareness level basis for the proper management, handling, and storage of waste, trash, or garbage to prevent the discharge of pollutants.

This program applies to all T&T Industrial, Inc. employees.

1.344 Resources

Number	Title
29 CFR 1910 Subpart H	Materials Handling, Storage, Use, and Disposal

1.345 Waste Management

The volume of waste is reduced through the use of sound waste minimization practices utilizing a reduce, reuse, and recycle approach.

Affected employees must be informed of site-specific waste management procedures prior to initial assignment and upon any changes in the site-specific waste management plan.

Coordinate with the client to ensure proper disposal of wastes or scrap materials. For example, the company must ensure the client is aware of whether wastes and scrap materials will be taken off site by the company or will be disposed of on the client's site.

The company must assign person(s) accountable for disposition of wastes generated at the work site.

1.346 Assessment

The amount of waste that will be generated shall be estimated prior to work being performed so that the need for containers and waste removal, if necessary, can be determined. If the same wastes or scrap materials are generated regularly, the estimation may be done initially.

1.347 Handling and Storage

Waste materials shall be properly stored and handled to minimize the potential for spills or impact to the environment. During outdoor activities, receptacles shall be covered to prevent dispersion of waste materials and to control the potential for run-off.

Proper segregation of waste materials is encouraged to ensure opportunities for reuse or recycling. Waste should be recycled whenever possible.

Waste storage areas must be inspected regularly.

Safe practices related to the immediate storage and handling of waste, scrap, or leftover materials must be followed. Gloves and/or other PPE must be worn when handling waste.

1.348 Training

Employees shall be instructed on the proper handling, storage, and disposal of wastes. This may include general instruction on disposal of non-hazardous wastes, trash, or scrap materials. If wastes generated are classified as hazardous, employees shall be trained to ensure proper disposal.

1.349 Construction Activities

When materials are dropped more than 20 feet to any point lying outside the exterior walls of a building, an enclosed chute of wood, or equivalent material, shall be used. An enclosed chute is a slide, closed in on all sides, through which material is moved from a high place to a lower one.

When debris is dropped through holes in the floor without the use of chutes, the area onto which the material is dropped shall be completely enclosed with barricades not less than 42 inches high and not less than 6 feet back from the projected edge of the opening above. Signs warning of the hazard of falling materials shall be posted at each level. Removal shall not be permitted in this lower area until debris handling ceases above.

All scrap lumber, waste material, and rubbish shall be removed from the immediate work area as the work progresses.

Disposal of waste material or debris by burning shall comply with local fire regulations.

All solvent waste, oily rags, and flammable liquids shall be kept in fire resistant covered containers until removed from the worksite.

31. Ground Fault Circuit Interrupters (GFCI) Program

1.350 Purpose and Scope

The purpose of this program is to provide regulations required when ground fault circuit interrupters (GFCI) are not in place.

This program applies to all T&T Industrial, Inc. employees involved in electrical work in construction applications.

1.351 Resources

Number	Title
29 CFR 1926 Subpart K	Electrical
Cal/OSHA T8 CCR Subchapter 5	Electrical Safety Orders

1.352 Ground Fault Circuit Interrupters (GFCI) Program

The GFCI is designed to protect people from severe or fatal shocks but because a GFCI detects ground faults, it can also prevent some electrical fires and reduce the severity of other fires by interrupting the flow of electric current. Any equipment that has not met the requirements of this program shall not be made available or permitted to be used by employees. If GFCIs are not used on all 120-volt, single phase 15 and 20-ampere temporary wiring on construction sites with information covered in the Company Electrical Program then the following shall be in place:

1.353 Assured Grounding Conductor

An assured grounding conductor program shall be implemented on sites covering all cord sets, receptacles which are not part of the building or structure, and equipment connected by cord and plug which are available for use or used by employees.

1.354 Competent Person(s)

There shall be a designated competent person who can identify existing and predictable electrical hazards and who has authorization to take prompt corrective measures to eliminate them.

1.355 Inspection

Each cord set, attachment cap, plug and receptacle of cord sets, and any equipment connected by cord and plug, except cord sets and receptacles which are fixed and not exposed to damage, shall be visually inspected before each day's use for external defects, such as deformed or missing pins or insulation damage, and for indications of possible internal damage. Equipment found damaged or defective shall not be used until repaired. Damaged items shall be tagged 'DO NOT USE' and be removed from service until repaired and tested.

1.356 Testing

All equipment grounding conductors shall be tested for continuity and shall be electrically continuous.

Each receptacle and attachment cap or plug shall be tested for correct attachment of the equipment grounding conductors.

The equipment grounding conductor shall be connected to its proper terminal:

- Before each use.
- Before equipment is returned to service following any repairs.
- Before equipment is used such as when a cord has been run over.
- At intervals not to exceed 3 months.
- Cord sets and receptacles which are fixed and not exposed to damage shall be tested at intervals not exceeding 6 months.

Tests performed as required by this program shall be recorded as to the identity of each receptacle, cord set, and cord and plug connected equipment that passed the test and shall indicate the last date tested or interval for which it was tested. This record shall be kept by means of logs, color coding, or other effective means and shall be maintained until replaced by a more current record. These records shall be made available at the job site for inspection by the Assistant Secretary and any affected employees.

1.357 Defective Electrical Equipment

Any equipment which has not met the requirements of this program shall not be available or permitted to be used. Damaged items shall not be used until repaired.

32. Hand and Power Tools Safety Program

1.358 Purpose and Scope

The purpose of this program is to ensure all employees are aware of safe work practices for the use of hand and power tools to prevent or mitigate incidents that may arise from the improper handling of the tools.

This program applies to all T&T Industrial, Inc. employees that use hand and power tools on the job.

1.359 Resources

Number	Title
29 CFR 1926 Subpart I	Tools-Hand and Power
29 CFR 1910 Subpart P	Hand and Portable Powered Tools and Equipment, General
Cal/OSHA T8 CCR Subpart 7	General Industry Safety Standards
29 CFR 1910 Subpart I	Personal Protective Equipment
29 CFR 1910 Subpart J	General Environmental Controls
29 CFR 1926 Subpart C	General Safety and Health Provisions

1.360 Definitions

Acronym/Term	Definition
Point of Operation	The area on a machine where work is actually performed upon the material being processed.
Competent Employee / Person	A person who can identify existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.

1.361 Hand and Power Tools Program

Hand tools are tools that are powered manually. Some examples of hand tools include anvils, axes, chisels, files, hammers, hand boring tools, planes, pliers, punches, saws, industrial scissors, screw drivers, tin snips, and wrenches.

Power tools must be equipped with safety switches and guards (if provided by the manufacturer). Types of power tools are determined by their power source: electric, pneumatic, liquid fuel, hydraulic, and powder actuated.

There is a variety of hazards associated with hand and power tools.

A power tool may only be operated by a competent person. A person must not be authorized to operate a power tool until the person has been adequately instructed and trained, and has demonstrated an ability to safely operate it.

1.362 General

The proper tools for the job shall be selected and employees in the proximity of work shall be alerted. Tools shall be used in accordance with manufacturer's recommendations.

Tools must be inspected prior to use.

An effective written or other permanent recording system or log must be immediately available to the operator, and to any other person involved with inspection and maintenance of the tool.

Employees shall be alert to any potential hazards in the area such as flammable or explosive gases, vapors, dusts, etc. that may ignite if a spark is generated by the tool or from work being done with the tool. Use of tools in an area where flammable gases are normally present which may be explosive shall be electrically rated for service in that area.

All power-driven tools shall be stopped when not in use.

Accidental start-ups shall be avoided. Ensure that the switch or other activating mechanism on the tools is in the "off" position before connecting to the power source.

Operators shall be capable of inspecting the tools to ensure safe operating condition prior to use and be aware of the tool's limitations and potential hazards.

Before undertaking any maintenance, repair, or unjamming work in a machine's danger zone, the following safety precautions shall be taken:

- Turn the machine's power supply switch to the off position
- Bring the machine to a complete stop
- Each person exposed to danger locks off all the machine's sources of energy in order to avoid any accidental startup of the machine for the duration of the work.

In areas where there is a danger of contact with moving parts, workers shall comply with the following standards:

- Their clothing shall fit well and have no loose flaps
- Necklaces, bracelets, or rings shall not be worn, with the exception of medical alert bracelets
- Anyone with long hair shall tuck it under a bonnet, a hat, or a hairnet

1.363 Training

Training on the safe and proper use of all hand and power tools shall be provided to employees that operate hand and power tools.

Only authorized, trained employees shall operate hand and power tools.

In-house power tool repairs shall be performed by trained technicians.

1.364 Hand and Power Tool Condition and Location

All hand and power tools and similar equipment, whether furnished by the Company or the employee, shall be maintained in a safe condition.

Hand tools and portable power tools shall be examined regularly and if found defective, be repaired or replaced.

The use of any machinery, tool, material, or equipment which is not in compliance with any applicable requirement of this document is prohibited. Such machine, tool, material, or equipment shall either be identified as unsafe by tagging or locking the controls to render them inoperable or shall be physically removed from its place of operation.

Compressed air shall not be used for cleaning purposes except where reduced to less than 30 psi and then only with effective chip guarding and personal protective equipment (PPE).

Machines designed for a fixed location shall be securely anchored to prevent walking or moving.

1.365 Ergonomics

Awkward postures are postures that strain the neck, shoulders, elbows, wrists, hands or back. Bending, stooping, twisting, and reaching are examples of awkward postures. Tool use and body positioning the work piece will affect your shoulder, elbow, wrist, hand or back posture.

Choose an ergonomic tool requiring the least continuous force and repetitive motion and which can be used without awkward postures. The right tool will help you to minimize pain and fatigue by keeping your neck, shoulders, and back relaxed and your arms at your sides. Avoid raising your shoulders and elbows; relaxed shoulders and elbows are more comfortable and will make it easier to drive downward.

Ergonomic Tools

- A tool becomes “ergonomic” only when it fits the task you are performing, and it fits your hand without causing awkward postures, harmful contact pressures or other safety and health risks.

- If you select and use a tool that does not fit your hand or use the tool in a way it was not intended, you might develop an injury such as carpal tunnel syndrome, tendonitis, or muscle strain.
- These injuries do not happen because of a single event but result from repetitive movements performed over time.

These repetitive movements may result in damage to muscles, tendons, nerves, ligaments, joints, cartilage, spinal discs, or blood vessels.

1.365.1 Selecting and Using the Right Tool

- Make and take the time, before you pick up a tool and begin working to think about the requirements of the job. Do you have the right tools for the job? For optimum safety, find the correct tool or should you rent a specialized tool or hire a professional?
- Select tools designed for the intended and specific use purpose.
- Using a tool for something other than its intended purpose often damages the tool and could cause you pain, discomfort, or injury.
- Assess your workspace to determine which tool will work efficiently and safely in that space.

1.366 Guarding

When power operated tools are designed to accommodate guards, they shall be in place and operable at all times while the tool is in use. The guard shall not be manipulated in such a way that will compromise its integrity or compromise the protection intended. Guarding shall meet the requirements set forth in ANSI B15.1-1953 (R1958), Safety Code for Mechanical Power-Transmission Apparatus.

Belts, gears, shafts, pulleys, sprockets, spindles, drums, fly wheels, chains, or other reciprocating, rotating, or moving parts of equipment shall be guarded if such parts are exposed to contact by employees or otherwise create a hazard.

One or more methods of machine guarding shall be provided to protect the operator and other employees in the machine area from hazards such as those created by point of operation, ingoing nip points, rotating parts, flying chips and sparks. Examples of guarding methods are - barrier guards, two-hand tripping devices, electronic safety devices, etc.

The point of operation of machines whose operation exposes an employee to injury, shall be guarded. The guarding device shall be in conformity with any appropriate standards therefore, or, in the absence of applicable specific standards, shall be so designed and constructed as to prevent the operator from having any body part in the danger zone during the operating cycle.

Special hand tools for placing and removing material shall be such as to permit easy handling of material without the operator placing a hand in the danger zone. Such tools shall not be in lieu of other guarding required but can only be used to supplement protection provided.

When the periphery of the blades of a fan is less than 7 feet above the floor or working level, the blades shall be guarded. The guard shall have openings no larger than 1/2 inch.

Safety guards for bench and floor stands and cylindrical grinders, where the operator stands in front of the opening, shall be constructed so that the peripheral protecting member can be adjusted to the constantly decreasing diameter of the wheel. The maximum angular exposure above the horizontal plane of the wheel spindle shall never be exceeded, and the distance between the wheel periphery and the adjustable tongue or the end of the peripheral member at the top shall never exceed 1/4 inch.

1.367 Bench and Floor Stands

The angular exposure of the grinding wheel periphery and sides for safety guards used on machines known as bench and floor stands shall not exceed 90 degrees or one-fourth of the periphery. This exposure shall begin at a point not more than 65 degrees above the horizontal plane of the wheel spindle.

1.368 Cylindrical Grinders

The maximum angular exposure of the grinding wheel periphery and sides for safety guards used on cylindrical grinding machines shall not exceed 180 degrees. This exposure shall begin at a point not more than 65 degrees above the horizontal plane of the wheel spindle.

1.369 Positive Accessory Holding Means

All hand-held powered platen sanders, grinders with wheels 2-inch diameter or less, routers, planers, laminate trimmers, nibblers, shears, scroll saws, and jigsaws with blade shanks 1/4 inch wide or less shall be equipped with only a positive "on-off" control.

All hand-held powered drills, tappers, fastener drivers, horizontal, vertical, and angle grinders with wheels greater than 2 inches in diameter, disc sanders, belt sanders, reciprocating saws, saber saws, and other similar operating powered tools shall be equipped with a momentary contact "on-off" control and shall have a lock-on control provided that turnoff can be accomplished by a single motion of the same finger or fingers that turn it on.

All other hand-held powered tools, such as circular saws, chain saws, and percussion tools without positive accessory holding means, shall be equipped with a constant pressure switch that will shut off the power when the pressure is released.

1.370 Personal Protective Equipment (PPE)

Employees using hand and power tools and exposed to the hazard of falling, flying, abrasive, and splashing objects, or exposed to harmful dust, fumes, mists vapors, or gases shall be provided with and use particular PPE necessary to protect them from the hazard.

33. Hazard Communication Program

1.371 Purpose and Scope

The purpose of this program is to provide information concerning the hazards associated with the work activities, including but not limited to the health, safety, environmental, and security hazards where employees may be exposed to hazardous substances under normal working conditions or during emergency situations.

This program applies to all T&T Industrial, Inc. employees exposed to hazardous situations.

1.372 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances – Hazard Communication
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders – Control of Hazardous Substances
CMS-FM-0028	Inventory of Hazardous Chemicals Form

1.373 Hazard Communication Program

Potential hazards include materials that cause fire or explosion or result in injury by inhalation, skin or eye contact, or ingestion. One of the benefits of this program is that employees will know the hazards of the chemicals with which they are working.

1.374 General

A written hazard communication program shall be developed, implemented, and maintained at each workplace. The safety department has full authority for program implementation and execution. The program shall describe how the requirements for labels and other forms of warning, safety data sheets, and employee information and training will be met.

Examples of qualities which make a chemical "hazardous" include but are not limited to:

- Flammable, combustible, and/or explosive
- Corrosive (acids/caustics)
- Irritating/damaging to the eyes and/or skin on contact
- Poses health hazard through inhalation, ingestion, or body contact
- Any known or suspected carcinogen

1.375 Training

Employees shall be trained on the dangers of the hazardous chemicals with which they work. This training shall be given when the employee starts work and when a new chemical is used in the workplace. This training shall cover types of hazards (e.g., flammability or carcinogenicity) or specific chemicals. Chemical-specific information shall always be available through labels and safety data sheets (SDS).

On job sites with multiple employers / companies performing work, information concerning hazardous chemicals in use, methods of providing SDSs, methods of precautionary measures to be taken and methods of providing information on labeling systems shall be provided.

Documentation of safety and health training shall include:

- Employee name or another identifier
- Training dates
- Type(s) of training
- Training providers

This documentation shall be maintained for at least one year.

1.376 Inventory of Hazardous Chemicals

An up-to-date working inventory of all chemicals in stock along with all chemicals sent to other destinations shall be compiled. The list may be compiled for the workplace as a whole or for individual work areas.

This inventory shall include:

- The full chemical name or identity that is referenced on the appropriate SDS
- CAS number
- Approximate amount of the chemical with suitable units of measurement
- Physical state
- Responsible party
- Location
- Expiration date if applicable

1.377 Non-routine Tasks

Before employees perform non-routine or special tasks that may expose them to hazardous chemicals, they shall be trained on the hazards associated with those chemicals. This training shall be documented and maintained, including how and by whom the employees were trained.

Methods will be documented as how and by whom employees will be informed of the hazards of non-routine tasks (i.e., the cleaning of reactor vessels, etc.) and the hazards associated with chemicals contained in unlabeled pipes in their work areas. Employees will be informed of the hazards by conducting a hazard assessment with their supervisor before work begins.

1.378 Multiple Worksites

If hazardous chemicals are produced, used, or stored in such a way that the employees of other employers may be exposed, the hazard communication program shall include:

- The methods used to share SDS either by posting at the site or through electronic means before work begins.
- The methods used to inform other employers of any precautionary measures that need to be taken to protect employees during the workplace's normal operating conditions and in foreseeable emergencies by conducting hazard assessments before work begins.
- The methods used to inform other employers of the labeling system used in the workplace by toolbox talks before work begins.

The program shall be made available, upon request, to employees, their designated representatives, the OSHA Assistant Secretary, and the OSHA Director. Where employees must travel between workplaces during a work shift (multi job sites), the written program may be kept at a primary job site.

1.379 Labeling

Each container of hazardous chemicals shall be labeled with information regarding the hazards of the chemicals, and which, in conjunction with the other information immediately available to employees under the hazard communication program, will provide employees with the specific information regarding the physical and health hazards of the hazardous chemical. Labels shall not be removed or defaced on incoming containers of hazardous chemicals.

Container labels shall contain at least the following information:

- Product identifier
- Signal word
- Hazard statement(s)
- Pictogram(s)
- Precautionary statement(s)
- And name, address, and telephone number of the chemical manufacturer, importer, or other responsible party

Labels or other forms of warning shall be legible, in English, and prominently displayed on the container, or readily available in the work area throughout each work shift. Employees shall not remove or deface labels on incoming containers of hazardous chemicals. If there are employees who speak other languages information in that language may be added to the material presented, as long as the information is presented in English as well.

The label for bulk shipments of hazardous chemicals must be on the immediate container, transmitted with the shipping papers or the bills of lading or, with the agreement of the receiving entity, transmitted by technological or electronic means so that it is immediately available to workers in printed form on the receiving end of the shipment.

For a container less than or equal to 100 ml capacity, the chemical manufacturer, importer, or distributor must include, at a minimum, the following information on the label of the container:

- Product identifier
- Pictogram(s)
- Signal word
- Chemical manufacturer's name and phone number

- A statement that the full label information for the hazardous chemical is provided on the immediate outer package

For a container less than or equal to 3 ml capacity, where the chemical manufacturer, importer, or distributor can demonstrate that any label interferes with the normal use of the container, no label is required, but the container must bear, at a minimum, the product identifier.

For all small containers, the immediate outer package must include:

- The full label information required for each hazardous chemical in the immediate outer package. The label must not be removed or defaced.
- A statement that the small container(s) inside must be stored in the immediate outer package bearing the complete label when not in use.

1.380 Safety Data Sheets

Safety data sheets (SDS) following the Globally Harmonized System of Classification and Labelling of Chemicals (GHS) standard are required for each hazardous chemical used in the workplace.

Chemical manufacturers are responsible for developing SDSs. An SDS for each chemical used shall be on hand. If an SDS is not provided, appears inadequate, or the composition of the chemical is unknown or questionable, then the manufacturer, supplier, and/or client shall be contacted for more details.

The manufacturer, importer, or employer preparing the safety data sheet shall ensure that the information recorded accurately reflects the scientific evidence used in making the hazard determination.

If the manufacturer, importer, or employer becomes aware of any significant information regarding the hazards of a substance, or ways to protect against the hazards, this new information shall be added to the safety data sheet within 3 months. If the substance is not currently being produced or imported, the manufacturer or importer shall add the information to the safety data sheet before the substance is introduced into the workplace again.

SDSs shall be readily accessible during each work shift to employees when they are in their work area(s). Where employees must travel between workplaces during a work shift (i.e., their work is carried out at more than one geographical location) the safety data sheets may be kept at the primary workplace facility.

1.381 Accidents, Incidents, and Emergencies Involving Hazardous Substances

All spills shall be cleaned up as soon as possible.

Disposal of any clean up materials shall be carried out in a responsible manner and any discharge shall be immediately reported according to Company policy and legislation.

1.382 Trade Secrets

The identity of a trade secret chemical, including the name and other specific identification, may be withheld from the written list of hazardous chemicals, the label, and the SDS, provided that the Company:

- Can support the claim that the chemical's identity is a trade secret
- Identifies the chemical in a way that it can be referred to without disclosing the secret
- Indicates in the SDS that the chemical's identity is withheld as a trade secret
- Discloses in the SDS information on the properties and effects of the hazardous chemical

Make the chemical's identity available to employees, designated representatives, and health professionals in accordance with requirements.

The standard does not require the Company to disclose process or percentage of mixture information, which is a trade secret, under any circumstances.

Appendix 24 **Inventory of Hazardous Chemicals Form**

Facility Identification			
Facility Name:		Phone No.:	
Responsible Party:		Date of Last Update:	
Address:			

Chemical Name	CAS Number	Quantity	Physical State	Location	Expiration Date
			Solid ☐ Liquid ☐ Gas ☐		
			Solid ☐ Liquid ☐ Gas ☐		

			Solid ☐ Liquid ☐ Gas ☐		
			Solid ☐ Liquid ☐ Gas ☐		
			Solid ☐ Liquid ☐ Gas ☐		
			Solid ☐ Liquid ☐ Gas ☐		

			Solid ☐ Liquid ☐ Gas ☐		
			Solid ☐ Liquid ☐ Gas ☐		

34. HEAT ILLNESS PREVENTION PROGRAM

1.383 Purpose and Scope

The purpose of this program is to provide information on the recognition, evaluation, and control of potential heat stress conditions to prevent heat related illnesses.

This program applies to all T&T Industrial, Inc. employees.

1.384 Heat Illness Prevention

Heat stress occurs when the heat load on the body exceeds the body's capacity to cool itself.

Being uncomfortable is not the major problem with working in high temperature and humidity. Employees who are suddenly exposed to working in a hot environment face additional and generally avoidable hazards to their safety and health.

Specific measures must be in place to controls the effects of environmental factors related to heat illnesses and heat related thermal stressors.

Methods of electrolyte replacement must be provided during physical activities in how climates where such activities could bring on heat related illnesses.

1.385 Workplace and Task Evaluation

A thorough evaluation of the workplace may be necessary to identify tasks and conditions that present a potential heat stress hazard. This analysis is limited to job classifications where a majority of employees have occupational exposure to heat illness for more than 30 minutes of any 60-minute period, excluding breaks. This evaluation should include observations, discussions with employees and supervisors, and the review of any reported heat-related disorders.

Physical and other factors that can contribute to heat related illness shall be taken into consideration prior to performing tasks in a heat stress environment, which include but are not limited to:

- Job Location
- Work Duration
- Schedule
- Staffing
- Clothing type, weight, and breathability
- Metabolism
- Environmental conditions (ambient temperature, relative humidity)
- Fitness for duty
- Existing controls (e.g., proper tools and equipment, work/rest ratio, ventilation, cool zones, cool vests, fluid and electrolyte replacement, buddy system, etc.).

1.386 Risk Factors

Physical factors that contribute to heat related illness shall be taken into consideration before performing a task. The most common physical factors that can contribute to heat related illness are type of work, level of physical activity and duration, and clothing color, weight, and breathability.

Personal risk factors include medical conditions, lack of physical fitness, previous episodes of heat-related illness, alcohol consumption, drugs, and use of certain medication.

Supervisors shall ensure personal factors that contribute to heat related illness are taken into consideration before assigning a task where there is the possibility of a heat-related illness occurring.

1.387 Recognition

When the human body cannot maintain the internal body temperature or electrolyte balance, this leads to heat related illness such as heat edema, heat cramps, heat exhaustion, and heat stroke.

The body temperature must be maintained near the normal body temperature of 98.6 °F to function properly. The body is capable of removing excess heat, leading to the following heat related illnesses:

1.387.1 Heat Stroke

Heat stroke is the most serious heat-related illness. It occurs when the body becomes unable to control its temperature: the body's temperature rises rapidly, the sweating mechanism fails, and the body is unable to cool down. When heat stroke occurs, the body temperature can rise to 106 °F or higher within 10 to 15 minutes. Heat stroke can cause death or permanent disability if emergency treatment is not given. Symptoms include confusion, altered mental status, slurred speech, loss of consciousness, seizures, very high body temperature.

Take the following steps to treat an employee with heat stroke:

- Call 911 for emergency medical care.
- Stay with the employee until emergency medical services arrive.
- Move the employee to a shaded, cool area and remove outer clothing.
- Cool the employee quickly with a cold water or ice bath if possible; wet the skin, place cold wet cloth on skin, or soak clothing with cool water.
- Circulate the air around the employee to speed cooling.
- Place cold wet cloth or ice on head, neck, armpits, and groin; or soak the clothing with cool water.

1.387.2 Heat Exhaustion

Heat exhaustion is the body's response to an excessive loss of the water and salt, usually through excessive sweating. Employees most prone to heat exhaustion are those that are elderly, have high blood pressure, and those working in a hot environment. Symptoms include headache, nausea, dizziness, weakness, irritability, thirst, heavy sweating, elevated body temperature, and decreased urine output.

Treat an employee suffering from heat exhaustion with the following:

- Take them to a clinic or emergency room for medical evaluation and treatment.
- If medical care is unavailable, call 911.
- Someone should stay with employee until help arrives.
- Remove the employee from hot area and give liquids to drink.
- Remove unnecessary clothing, including shoes and socks.
- Cool the employee with cold compresses or have the employee wash head, face, and neck with cold water.

- Encourage frequent sips of cool water.

1.387.3 Rhabdomyolysis

Rhabdomyolysis is a medical condition associated with heat stress and prolonged physical exertion, resulting in the rapid breakdown, rupture, and death of muscle. When muscle tissue dies, electrolytes and large proteins are released into the bloodstream that can cause irregular heart rhythms and seizures and damage the kidneys. Symptoms include muscle cramps / pain, abnormally dark urine, weakness, exercise intolerance, or be asymptomatic.

Employee with symptoms of rhabdomyolysis should:

- Stop activity.
- Increase oral hydration (water preferred).
- Seek immediate care at the nearest medical facility.
- Ask to be checked for rhabdomyolysis (i.e., blood sample analyzed for creatine kinase).

1.387.4 Heat Syncope

A fainting (syncope) episode or dizziness that usually occurs with prolonged standing or sudden rising from a sitting or lying position. Factors that may contribute to heat syncope include dehydration and lack of acclimatization.

Employees with heat syncope should sit or lie down in a cool place and slowly drink water, clear juice, or a sports drink.

1.387.5 Heat Cramps

Heat cramps usually affect employees who sweat a lot during strenuous activity. This sweating depletes the body's salt and moisture levels. Low salt levels in muscles causes painful cramps. Heat cramps may also be a symptom of heat exhaustion.

Employees should drink water and have a snack and/or carbohydrate-electrolyte replacement liquid every 15 to 20 minutes. Salt tablets should be avoided. Get medical help if the employee has heart problems, is on a low sodium diet, or if cramps do not subside within one hour.

1.387.6 Heat Edema

Heat causes the blood vessels to expand, so body fluid moves into the hand or legs by gravity.

Mild edema usually goes away on its own, particularly if the affected limb is raised higher than the heart.

1.387.7 Heat Rash

Heat rash is a skin irritation caused by excessive sweating during hot, humid weather. Symptoms include what looks like red cluster of pimples or small blisters that usually appear on the neck, upper chest, groin, under the breasts, and in elbow creases.

Employees experiencing heat rash should:

- When possible, a cooler, less humid work environment is best treatment.
- Keep rash area dry.
- Powder may be applied to increase comfort.
- Ointments and creams should not be used.

1.388 Evaluation

The Heat Index should be used as a reference or indicator to define the general overall heat stress conditions. The Heat Index is also known as the “effective / feels like temperature”.

The Heat Index is based on relative humidity and air temperature. It is predictive of heat stress in circumstances in which the relationship has been established for a particular environment. Heat Index indicates thermal comfort.

1.389 Controls

Employees shall have access to fresh, pure, and suitably cool potable drinking water at no charge. Where it is not plumbed or otherwise continuously supplied, it shall be provided in sufficient quantity at the beginning of the work shift, but not less than one quart per hour per employee.

At or below 80 degrees Fahrenheit employees shall have timely access to shade that is either open to the air or provided with ventilation or cooling upon request. For temperatures at or above 80 degrees Fahrenheit, one or more areas with shade shall be provided at all times while employees are present. Shade shall accommodate the number of employees on recovery or rest periods at all times.

In high temperatures, the following shall be considered:

- Effective communication by voice or electronic means.
- Observation of employees for alertness and signs / symptoms of heat illness.

- Designation of employees on each worksite to call for emergency medical services.
- Reminders to drink water throughout the shift.
- Pre-shift meetings before beginning work.
- Reminders to employees of their right to take a cool-down rest when necessary.

Vehicles with enclosed cabs must have air conditioning no later than May 1, 2025, per jurisdiction.

An individual at the worksite shall be responsible for monitoring conditions and implementing the heat plan throughout the workday. This individual can be a foreman, jobsite supervisor, safety director, or anyone else with the proper training.

Engineering controls such as air conditioning, with cooled air, and increased air flow shall be employed. During their first few days in warm or hot environments, employees should consume adequate fluids, work shorter shifts, take frequent breaks, and quickly identify any heat illness symptoms.

Work processes that may generate additional heat or humidity must be identified and mitigated.

1.389.1 Acclimation

Acclimatization is the beneficial physiological adaptations that occur during repeated exposure to a hot environment. These practices must be followed for newly assigned or reassigned work.

To acclimatize employees, gradually increase their exposure time in hot environmental conditions over a 7-to-14-day period. New employees will need more time to acclimatize than employees who have already had some exposure.

For new employees, the schedule should be no more than a 20% exposure on day one and an increase of no more than 20% on each additional day.

For employees who have had previous experience with the job, the acclimatization regimen should be no more than a 50% exposure on day one, 60% on day two, 80% on day three, and 100% on day four.

In addition, the level of acclimatization each employee reaches is relative to the initial level of physical fitness and the total heat stress experienced by the individual.

1.390 Training

Training must be available and understandable. The program shall include:

- The environmental and personal risk factors for heat illness.

- Company procedures for complying with the requirements of the standard.
- The importance of frequent consumption of small quantities of water, up to four cups per hour, when the work environment is hot, and employees are likely to be sweating more than usual in the performance of their duties.
- The importance of acclimatization.
- The different types of heat illness and the common signs and symptoms of heat illness.
- How to recognize the hazards of heat illness.
- Procedures to be followed to minimize the hazards of heat illness.
- The importance to employees of immediately reporting to the employer, directly or through the employee's supervisor, symptoms, or signs of heat illness in themselves, or in co-workers.
- Company procedures for responding to symptoms of possible heat illness, including how emergency medical services will be provided should they become necessary
- Company procedures for contacting emergency medical services, and if necessary, for transporting employees to a point where they can be reached by an emergency medical service provider.
- Company procedures for ensuring that, in the event of an emergency, clear and precise directions to the work site can and will be provided as needed to emergency responders.
- Environmental and personal risk factors, prevention, how to recognize and report signs and symptoms of heat illness and injury, how to administer appropriate first aid measures, and how to report heat illness and injury to emergency medical personnel.

Supervisors shall be trained in heat related illness prior to supervision of employees working in the heat. Training shall include:

- The procedures the supervisor is to follow to implement the applicable procedures to prevent heat illness.
- The procedures the supervisor is to follow when an employee exhibits symptoms consistent with possible heat illness, including emergency response procedures.

1.391 Emergency Response

When any heat-related illness symptom is present, promptly provide first aid to the affected employee. First aid for heat related illness includes:

- Taking the affected employee to a cooler area,
- Cooling the employee immediately,
- Never leave an employee with heat-related illness alone, and
- When in doubt, call 911.

An individual who will contact emergency services if an employee is experiencing signs of heat illness must be designated.

1.392 Climate Controlled Environments

The requirements of the regulation do not apply to employees who work indoors in climate-controlled environments, including motor vehicles with a properly functioning climate control system.

If the climate control system becomes non-functional or ineffective, make a good faith effort to reestablish an effective climate control system as soon as practicable.

Until the climate control system is rendered effective, mitigate the potential hazards that could cause heat illness.

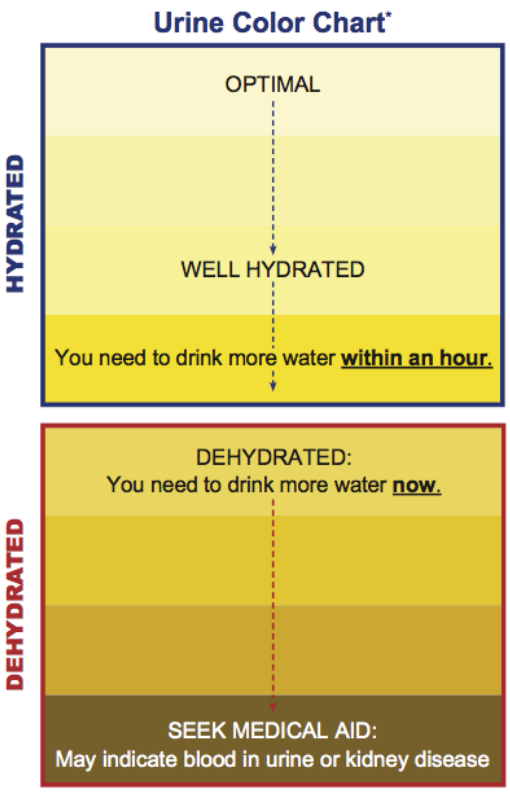
1.393 Collective Bargaining Agreements

The Company can exceed the requirements of the regulation on its own or through collective bargaining agreements.

Collective bargaining agreements cannot waive or reduce the requirements of the regulation, and the regulation does not relieve the Company of contractual obligations under a collective bargaining agreement.

T&T Industrial, Inc.
Heat Illness Prevention Program

Appendix 25 Hydration Chart and Heat Index



*This color chart is not for clinical use.

NWS Heat Index

Temperature (°F)

Relative Humidity (%)	80	82	84	86	88	90	92	94	96	98	100	102	104	106	108	110
40	80	81	83	85	88	91	94	97	101	105	109	114	119	124	130	136
45	80	82	84	87	89	93	96	100	104	109	114	119	124	130	137	
50	81	83	85	88	91	95	99	103	108	113	118	124	131	137		
55	81	84	86	89	93	97	101	106	112	117	124	130	137			
60	82	84	88	91	95	100	105	110	116	123	129	137				
65	82	85	89	93	98	103	108	114	121	128	136					
70	83	86	90	95	100	105	112	119	126	134						
75	84	88	92	97	103	109	116	124	132							
80	84	89	94	100	106	113	121	129								
85	85	90	96	102	110	117	126	135								
90	86	91	98	105	113	122	131									
95	86	93	100	108	117	127										
100	87	95	103	112	121	132										

Likelihood of Heat Disorders with Prolonged Exposure or Strenuous Activity

Caution | Extreme Caution | Danger | Extreme Danger

Heat Category	WBGT Index, °F	Easy Work *Water Intake (Quart/Hour)*	Moderate Work *Water Intake (Quart/Hour)*	Hard Work *Water Intake (Quart/Hour)*
1	78° - 81.9°	½	¾	¾
2	82° - 84.9°	½	¾	1
3	85° - 87.9°	¾	¾	1
4	88° - 89.9°	¾	¾	1
5	> 90°	1	1	1

35. Heavy Equipment Safety Program

1.394 Purpose and Scope

The purpose of this program is to mitigate risks, prevent accidents, and protect the well-being of individuals operating and working around heavy equipment.

More than 100 people each year are killed by mobile heavy equipment.

These are the main causes of death:

- Workers on foot are struck by equipment, usually when it is backing up or changing direction.
- Equipment rolls over and kills the operator while on a slope or when equipment is loaded or unloaded from a flatbed/lowboy truck.
- Operators or mechanics are run over or caught in equipment when the brakes are not set, equipment is left in gear, wheel chocks are not used, or the equipment and controls are not locked out.
- Workers on foot or in a trench are crushed by falling equipment loads, backhoe buckets, or other moving parts.

This program applies to all T&T Industrial, Inc. employees that work with heavy equipment.

1.395 Training

Training will be provided for employees whose job activities involve the use of heavy equipment.

Training and competency requirements for heavy mobile equipment operators are crucial to ensure safety, efficiency, and compliance with regulations in industries such as construction, mining, agriculture, and logistics. The specific requirements may vary depending on the type of equipment and local regulations. Following are general training and competency requirements in accordance with local jurisdiction:

- Basic Training:
  Attend a formal training program specific to the type of heavy mobile equipment to be operated. This training should cover the fundamentals of equipment operation, safety procedures, and maintenance.
- Classroom Instruction:
  Classroom instruction should cover topics such as:

- Equipment components and controls
- Safety regulations and standards
- Hazard identification and risk assessment
- Pre-operation inspection and maintenance
- Emergency procedures and response

- Practical Training:
  Hands-on training under the supervision of an experienced operator or instructor is essential. Trainees should become proficient in operating the equipment safely and effectively.
- Certification and Licensing:
  Depending on the jurisdiction, heavy equipment operators may need to obtain a license or certification to operate specific types of equipment. This may require passing written and practical exams.
- Equipment-Specific Training:
  Operators should receive equipment-specific training for the type of heavy mobile equipment they will operate, such as bulldozers, excavators, cranes, forklifts, or dump trucks.
- Site-Specific Training:
  Training should include knowledge of the specific work site, including potential hazards, safety protocols, and site-specific rules and regulations.
- Safety Training:
  Comprehensive safety training is essential and should cover:
  - Safe equipment operation techniques
  - Load handling and securement
  - Traffic control and work zone safety
  - Personal protective equipment (PPE) usage
  - Emergency shutdown procedures
  - Communication protocols
- Maintenance and Inspection Training:
  Operators should be trained in routine equipment maintenance, including daily inspections and basic troubleshooting. This ensures that operators can identify and address minor issues before they become major problems.

- Continuous Education:
  Operators should stay updated with the latest safety regulations, equipment advancements, and industry best practices through ongoing training and education programs.
- Competency Assessment:
  Regularly assess operators' competency through practical evaluations, written tests, and performance reviews. This ensures that operators continue to meet safety and proficiency standards.

1.396 General Operation

All employees must follow specific procedures when working on, with, or around heavy equipment, such as:

- Dump trucks,
- Backhoes,
- Trenching machines,
- Side booms,
- Bulldozers,
- Gas shovels,
- Air compressors, and
- Front-end loaders.

1.397 General Operation Procedures

All employees must obtain and keep current the proper operator license for each type of vehicle that is operated.

The Company will provide adequate training on vehicle operation, maintenance and safety to the employees performing such work.

Before using heavy equipment, test and check it to be sure it is in proper operating condition as per manufacturer specifications.

When leaving heavy equipment, lower suspended equipment, shut off the power, and set the brakes.

Be sure that people can hear the truck's warning signals and only use them only when necessary.

Make sure to stop a safe distance from other trucks or pedestrians.

Always give pedestrians the benefit of the doubt, as they may be deaf or wearing hearing protection.

Only authorized maintenance and repairmen are allowed to work on the equipment mechanism.

When transporting a load, always keep it as low as possible and tilted toward the body of the equipment.

All equipment left unattended at night, adjacent to a highway in normal use, or adjacent to construction areas where work is in progress, must have appropriate lights or reflectors, or barricades equipped with appropriate reflectors, to identify the location of the equipment.

Where traffic is diverted on to dusty surfaces, good visibility must be maintained by the suppression of dust, through the periodic application of brine or water to the grade surface, as required.

Do not operate heavy equipment, a vehicle or power tools, within 10 feet of any power line or electrical distribution.

Personnel must not get on or off the machine while the machine is in motion.

Riders, except mechanics and persons in training to operate equipment, must not be allowed on equipment unless a seat with a seatbelt is provided and used.

1.398 General Safety and Maintenance

All employees are responsible for the following:

Placing substantial blocking under any chain-hoist-suspended or jack supported equipment under which people must work. (The operator of the equipment should never leave the controls while shovels are suspended without blocking).

Work must not occur in areas where passing automobiles or moving machinery result in a hazardous condition.

All work areas must be provided with proper ventilation.

Employees must not work in areas where they are exposed to excessive carbon monoxide gas from exhausts of running engines.

Do not keep gasoline in open containers or pits.

Use a reasonably nontoxic solvent with a high flash point for cleaning parts. Never use gasoline or tetrachloride.

Ask for help or use a hoist to lift unusually heavy loads.

Keep wrenches or tools clean and in safe working condition

Secure unbolted heavy parts or engines if necessary, to leave the work.

Wear goggles when eye protection is needed.

Aisles and open spaces must be kept free of tools and parts.

Change clothes that become soaked with oil or gasoline to prevent risk of a fire.

Make sure that all lock washers and cotter pins are in place before deeming the maintenance work complete.

Always keep a suitable fire extinguisher ready. Inspect fire extinguishers regularly and keep them in good operating order.

Enforce no-smoking rules.

Ground electric appliances and keep them in good working condition. Ensure that sparking will not ignite gases or vapors. Do not permit live cords to touch workers.

Put oily rags in closed metal containers for disposal after use.

Review personal-protection provisions for arc welders before attempting such work.

No repairs on a blade or dozer equipment will be permitted unless the motor has been stopped and the dozer blade is resting on the ground or securely blocked.

Tire rack, cage, or equivalent protection shall be provided and used when inflating, mounting, or dismounting tires installed on split rims, or rims equipped with locking rings or similar device.

Heavy machinery, equipment, or parts thereof, which are suspended or held aloft by use of slings, hoists, or jacks shall be substantially blocked or cribbed to prevent falling or shifting before employees are permitted to work under or between them.

1.399 Heavy Equipment Engineering Controls and Work Practices

All vehicles must have:

- A service brake system, an emergency brake system, and a parking brake system.
- Working headlights, taillights, and brake lights
- An audible warning device (horn).
- Intact windshield with working windshield wipers.

Ensure that all operators have been trained on the equipment they will use.

Check vehicles at the beginning of each shift to ensure that the parts, equipment, and accessories are in safe operating condition. Repair or replace any defective parts or equipment prior to use.

Do not operate vehicle in reverse with an obstructed rear view unless it has a reverse signal alarm capable of being heard above ambient noise levels or a signal observer indicates that it is safe to move.

Vehicles loaded from the top (e.g., dump trucks) must have cab shields or canopies to protect the operator while loading.

Ensure that vehicles used to transport workers have seats, with operable seat belts, firmly secured and adequate for the number of workers to be carried.

Equipment should have roll-over protection and protection from falling debris hazards as needed.

Prior to permitting construction equipment or vehicles onto an access roadway or grade, verify that the roadway or grade is constructed and maintained to safely accommodate the equipment and vehicles involved.

Do not modify the equipment's capacity or safety features without the manufacturer's written approval.

Where possible, do not allow debris collection work or other operations involving heavy equipment under overhead lines.

1.400 Material Falling from Vehicles Engineering Controls and Work Practices

Do not overload vehicles.

Ensure that loads are balanced and are fully contained within the vehicle. Trim loads, where necessary, to ensure loads do not extend beyond the sides or top of the vehicle.

Cover and secure loads before moving the vehicle.

1.401 Fueling Engineering Controls and Work Practices

Ensure that ignition sources are at least 25 feet away from fueling areas.

Prohibit smoking in fueling areas.

Ensure that vehicles are attended while being fueled.

36. HEXAVALENT CHROMIUM SAFETY PROGRAM

1.402 Purpose and Scope

The purpose of this program is to provide proper health and safety guidelines pertaining to work with hexavalent chromium to prevent adverse health effects.

This program applies to all T&T Industrial, Inc. employees with possible exposure to hexavalent chromium.

1.403 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances-Chromium
29 CFR 1926 Subpart Z	Toxic and Hazardous Substances-Chromium in Construction

1.404 Hexavalent Chromium Program

Hexavalent chromium is a potential health hazard associated with the welding and cutting of stainless and alloy steels and many non-ferrous alloys, specialty paints and pigments, chrome plating, and in chemical synthesis.

Hexavalent chromium exposure is associated with increased risk of lung cancer, asthma, nasal septum ulcerations and perforations, skin ulcerations, and allergic and contact dermatitis.

1.405 Identification

Scopes of work with the potential for hexavalent chromium exposures shall be reviewed to identify any potential sources. This includes safety data sheet (SDS) reviews of metal and paint compositions for welding, cutting, grinding, and abrasive blasting operations as well as process operations, chrome plating operations, and painting where hexavalent chromium may be present (e.g., chromate pigments in paint formulations).

Note that chromium in most metal alloys is not in a hexavalent chromium state; however, during heating operations, such as welding, the chromium fume or dust may be converted to hexavalent chromium. Thus, SDSs may not identify hexavalent chromium as a constituent although chromium in another valence state is present.

1.406 Roles and Responsibilities

The competent person plays a critical role in protecting workers from exposure to hexavalent chromium and ensuring compliance with relevant regulations. The competent person must have the necessary knowledge, expertise, and training according to local jurisdiction to effectively carry out their responsibilities. Typical roles and responsibilities of a competent person include:

- Risk Assessment: Conducting comprehensive risk assessments to identify potential sources of hexavalent chromium exposure in the workplace and evaluating the associated risks to workers.
- Exposure Monitoring: Implementing a monitoring program to measure and assess workers' exposure levels to hexavalent chromium. This may involve air sampling, surface sampling, or other appropriate methods to determine exposure levels.
- Engineering Controls: Recommending and overseeing the implementation of engineering controls to minimize or eliminate hexavalent chromium exposure. This may include ventilation systems, enclosure or isolation of processes, and substitution of less hazardous materials.
- Work Practices: Developing and enforcing safe work practices and procedures for handling hexavalent chromium, including proper storage, handling, use, and disposal methods. This may involve establishing restricted areas and implementing decontamination procedures.
- Training and Education: Providing training and education programs to workers on the hazards of hexavalent chromium exposure, safe work practices, and the proper use of personal protective equipment (PPE). Ensuring that workers are informed about the risks associated with hexavalent chromium and understand the necessary precautions.
- Medical Surveillance: Implementing a medical surveillance program to monitor the health of workers exposed to hexavalent chromium. This may include pre-placement and periodic medical examinations, biological monitoring, and maintaining medical records.
- Personal Protective Equipment (PPE): Assessing the need for and recommending appropriate PPE, such as respirators, gloves, protective clothing, and eyewear. Ensuring that workers are trained in the proper use, maintenance, and limitations of PPE.
- Hazard Communication: Ensuring proper labeling of containers containing hexavalent chromium and providing adequate safety data sheets (SDS) to workers. Communicating information about the hazards, safe handling procedures, and emergency response protocols associated with hexavalent chromium.

- Recordkeeping and Documentation: Maintaining accurate records of exposure monitoring results, medical surveillance records, training documentation, and any incidents or accidents related to hexavalent chromium exposure. This includes documenting control measures implemented and reviewing the effectiveness of the program.
- Regulatory Compliance: Staying informed about relevant regulations and standards related to hexavalent chromium exposure and ensuring compliance with applicable laws, regulations, and guidelines.
- Program Evaluation and Improvement: Regularly reviewing and evaluating the effectiveness of the hexavalent chromium program, identifying areas for improvement, and implementing corrective actions as necessary.

1.407 Training

Training shall be provided to affected employees on the hazards of hexavalent chromium exposure, methods of control, and the medical surveillance program.

1.408 Exposure

No employee shall be exposed to an airborne concentration of chromium (VI) in excess of 5 micrograms per cubic meter of air (5 $\mu g/m^3$), calculated as an 8-hour time-weighted average (TWA).

1.408.1 Exposure Monitoring

Initial monitoring shall be performed to determine the 8-hour TWA exposure for each employee to characterize full shift exposure on each shift, for each job classification, in each work area.

If monitoring reveals employee exposures to be at or above the action level, periodic monitoring shall be performed at least every 6 months. If monitoring reveals employee exposures to be above the permissible exposure limit (PEL), periodic monitoring shall be performed at least every 3 months and a written notification shall be included with the corrective action being taken to reduce exposure to or below the PEL.

Additional monitoring shall be performed when there has been any change in the production process, raw materials, equipment, personnel, work practices, or control methods that may result in new or additional exposures to chromium, or when there is reason to believe that new or additional exposures have occurred.

1.408.2 Medical Surveillance

Medical surveillance shall be provided when an employee experiences signs or symptoms of the adverse health effects of hexavalent chromium (dermatitis, asthma, bronchitis, etc.), who are or may be exposed at or above the action level for 30 or more days in a year, or exposed in an emergency. Medical evaluations shall be provided at no cost to employees at a reasonable time and place. Examinations shall be performed by or under the supervision of a physician or other licensed health care professional.

1.409 Regulated Areas

Regulated areas shall be established when an employee's exposure is or is expected to be in excess of the PEL. Regulated areas shall be marked with warning signs to alert employees. Access is restricted to authorized persons required by work duties to be present in the regulated areas.

1.410 Safe Work Practices

Engineering and work practice controls shall be used to reduce and maintain employee exposure to chromium to or below the PEL unless it can be demonstrated that such controls are not feasible. Wherever feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PEL, these controls shall be used to reduce employee exposure to the lowest levels achievable and shall supplement them by the use of respiratory protection.

1.410.1 Respiratory Protection

Respirators shall be used when engineering controls and work practices cannot reduce employee exposure, during work operations where engineering controls and work practices are not feasible, and emergencies.

1.410.2 Personal Protective Equipment

Personal protective equipment (PPE) shall be provided when there is a hazard from skin or eye contact (e.g., gloves, aprons, coveralls, goggles, foot covers etc.). Contaminated PPE shall be removed at the end of the work shift. Protective clothing shall be cleaned, laundered, repaired, and replaced as needed.

Appropriate personal protective equipment (PPE) must be issued to employees at no cost when engineering and work practice controls cannot reduce affected employee exposures to levels below the permissible exposure limit (PEL). Appropriate respiratory protection must be issued where engineering and work practice controls are not practicable and in the event of emergencies.

1.410.3 Housekeeping

Surfaces shall be maintained as free as practicable of accumulation of chromium. All spills and releases of chromium shall be cleaned promptly. Methods of cleaning include HEPA filtered vacuums, dry or wet sweeping, shoveling, or other methods to minimize exposure.

1.410.4 Decontamination

Change rooms for decontamination shall be provided and facilities shall prevent cross-contamination. Washing facilities shall be readily accessible for removing chromium from the skin. Employees shall wash their hands and face or any other potentially exposed skin before eating, drinking, or smoking.

1.411 Recordkeeping

An accurate record of all employee exposure monitoring, medical surveillance, and training records shall be maintained and made available.

37. Housekeeping Program

1.412 Purpose and Scope

The purpose of this program is to instill a culture of proactive housekeeping practices, where cleanliness, organization, and proper storage are prioritized to prevent accidents, injuries, and property damage.

This program applies to all T&T Industrial, Inc. employees.

1.413 Safe Housekeeping

Housekeeping is an important part of maintaining a safe and healthful workplace.

Good housekeeping can reduce workplace injuries and illnesses and promote positive behavior, habits, and attitudes.

Tidy working conditions should be maintained at all times in both work and office areas.

Properly managing housekeeping allows for effective operation and use of tools, equipment, storage facilities, supplies, and waste material.

Using safe housekeeping practices can lead to lower incident rates and increased efficiency.

All places of employment shall be kept clean to the extent that the nature of the work allows.

Portable drinking water dispensers shall be designed, constructed, and serviced so that sanitary conditions are maintained, shall be capable of being closed, and shall be equipped with a tap.

No employee shall be allowed to consume food or beverages in a toilet room nor in any area exposed to a toxic material.

Washing facilities (restrooms, break areas, etc.) shall be maintained in a sanitary condition.

1.414 Working Surfaces

Occupational Safety and Health Administration (OSHA) regulations require that each working surface be cleared of debris, including solid and liquid waste, at the end of each work shift or job.

It is recommended that good housekeeping be maintained throughout the course of the job and workday.

Each workplace must be assessed before work begins to identify potential hazards and determine ways to remove those hazards.

Walkways and working surfaces should be inspected often for hazards before they become a danger to employees.

Slips or falls can happen when employees have to walk or work on oily or slippery facility floors or other potentially hazardous working surfaces. To help prevent slips and falls:

- Keep floors free from grease and oil.
- Keep walkways free from ice, snow, and standing water.

Any spilled materials such as oil, grease, and water must be immediately cleaned from walkways and working surfaces to remove slip hazards.

The floor of every workroom shall be maintained, so far as practicable, in a dry condition. Where wet processes are used, drainage shall be maintained and false floors, platforms, mats, or other dry standing places shall be provided, where practicable, or appropriate waterproof footgear shall be provided.

To facilitate cleaning, every floor, working place, and passageway shall be kept free from protruding nails, splinters, loose boards, and unnecessary holes and openings.

All sweepings, solid or liquid wastes, refuse, and garbage shall be removed in such a manner as to avoid creating a menace to health and as often as necessary or appropriate to maintain the place of employment in a sanitary condition.

Every enclosed workplace shall be so constructed, equipped, and maintained, so far as reasonably practicable, as to prevent the entrance or harborage of rodents, insects, and other vermin. A continuing and effective extermination program shall be instituted where their presence is detected.

Potable water shall be provided in all places of employment, for drinking, washing of the person, cooking, washing of foods, washing of cooking or eating utensils, washing of food preparation or processing premises, and personal service rooms.

1.415 Flammable and Hazardous Materials

Fires can be caused by oily rags left in an area where hot work is performed or the accumulation of combustible dust. Reduce fire risks by:

- Properly disposing of flammable waste
- Keeping surfaces free from accumulated dust, including elevated locations

Poor storage and improper labeling of hazardous chemicals can lead to employees being exposed to hazardous substances.

Make sure all materials and chemicals are stored and labeled properly.

1.416 Lighting

Inadequate lighting can increase the chances of employee injuries, such as lacerations and amputations, because employees cannot see those hazards effectively.

Reduce the chances of these kinds of injuries by making sure areas have enough lighting for the work being done.

1.417 Walkways

Cords in walkways and work areas can cause employees to trip or fall over them. To prevent employees from tripping or falling:

- Properly arrange hoses and cords
- Keep machinery and equipment neat and orderly
- Use cord and wire covers

Another part of safe housekeeping is properly identifying passageways and making sure there is unobstructed access to all entrances, exits, and fire extinguishers.

This allows employees to safely respond and evacuate during a fire or other emergency.

1.418 Equipment Maintenance

Poorly maintained equipment or energy sources, such as broken, cracked, or damaged insulation and wiring connections, can cause electric shock. Protect employees from electric shock by:

- Inspecting electrical wiring for damage
- Removing and replacing any damaged parts or components

1.419 Written Policy

Jobsites should have written polices for housekeeping procedures. These policies should detail how frequently housekeeping tasks should be performed.

Written housekeeping policies should also contain specific instructions for how each housekeeping task should be performed, such as:

- Listing which chemicals to use for cleaning up spills
- Explaining how to properly dispose of flammable waste

All employees should be trained on safe housekeeping practices and all employees should participate in good housekeeping because safety is part of everyone's job.

38. HOUSEKEEPING PROGRAM – SLIPS, TRIPS, AND FALLS

1.420 Purpose and Scope

The purpose of this program is to mitigate the risks associated with slip, trip, and fall accidents by implementing effective housekeeping practices and promoting awareness.

This program applies to all T&T Industrial, Inc. employees.

1.421 Housekeeping – Slips, Trips, and Falls

Many workers are injured annually due to falls on walking and working surfaces. These injuries account for a significant percentage of lost-time injuries. Not only are slips, trips, and falls an economical loss, but they also account for a lot of pain and suffering and sometimes even death. It is important to understand how slips, trips, and falls happen, how to identify hazards, and how to eliminate or minimize these hazards.

1.422 Slips

Slips happen because of a lack of friction or traction between the footwear we are wearing and the walking surface. Some common causes of slips are:

- Spills
- Hazards created from weather (e.g., puddles, ice)
- Surfaces that are wet or oily
- Loose rugs or mats

1.423 Trips

Trips occur when your foot strikes or hits an object which causes you to lose your balance. Common causes of tripping are:

- Clutter on the floor (e.g., power cords, boxes)
- Poor lighting
- Uneven walking surfaces (e.g., carpeting, steps, thresholds)
- Sudden change in slip resistance properties of walking surfaces (e.g., wet floor or stepping from tiled to thick pile carpeted floors)

1.424 Falls

Falls can occur from a height or on surfaces that are on the same level. A fall can be the result of a slip or a trip where your center of gravity is shifted causing you to lose your balance. Preventive measures should be taken to avoid slips and trips.

1.425 Preventing Slips, Trips, and Falls

When there is an unexpected change in the contact between your feet and the ground or walking surface, the result is usually a slip or trip. This demonstrates the importance of training and educating employees about the hazards, selecting suitable walking surfaces, having proper housekeeping standards, and wearing proper footwear to prevent falls. Here are a few different methods of controlling hazards leading to slips, trips, and falls.

1.425.1 Training and Education

It is very important that all employees be trained on recognizing hazards related to slips, trips, and falls in their workplace. Many employees are not aware that they can contribute to the risks of a slip, trip, and fall hazard through typical work tasks. A couple of examples of an employee creating hazards through work habits are:

- Leaving a mess behind after completing a task by not following workplace housekeeping standards after receiving training to do so.
- Putting boxes in walkways, on the stairs, or in high traffic areas where there are designated storage spaces and racks.

Workplace policies and employee behavior have significant impact on the incidence of slips, trips, and falls. Time pressures for completing tasks can cause behaviors such as rushing, not paying attention, and being unaware of hazards due to a lack of training can mistakenly be seen as carelessness. Workplaces should identify potentially dangerous hazards and behaviors and control or eliminate them through education and communication.

Employees need to know how to properly identify, report, or eliminate any hazards that can be encountered in their work; this may require specific standards and training. For example:

- Good housekeeping standards should be set with provided supervision, information, and training.
- Employees should be trained on spill cleanup and proper disposal of spilled materials such as chemicals, oils, inks, coolants, grease, etc.

- Employees should be trained on how to prevent falls on icy, wet, and unstable conditions (loose gravel or sand).

1.425.2 Walking Surfaces

The quality of walking surfaces is critical in preventing slip and trips. Flooring should be regularly maintained to eliminate tripping hazards, such as bunched carpet, chipped tile or hardwood, missing tiles, etc. Replacing floors, installing mats, or resurfacing floors can help to improve safety and reduce the risk of falling. However, it is important to remember that improving the quality of the flooring also requires good housekeeping practices to be effective.

Weather is also a significant factor in relation to slips, trips, and fall hazards. Rain, snow, ice, leaves, mud, etc., can all become hazards. Parking lots, walkways, stairs, and other high traffic areas should be monitored frequently for any of the identified hazards and control measures should be put in place to remove / eliminate these hazards.

1.425.3 Lighting

Any lighting that is not working should be repaired immediately. Any identified dark areas should be well lit to avoid tripping over hazards or slipping due to a change in floor condition.

1.425.4 Housekeeping

Good housekeeping is very important when working to prevent falls due to slips and trips.

Without having good housekeeping practices, preventive measures (e.g., specialty footwear or floor surfaces) will not be fully effective. Good housekeeping includes:

- Provide housekeeping standards training for employees and supervisors before starting work.
- Clean up any spills immediately and investigate the cause to prevent reoccurrence.
- Immediately correct any hazard that might cause a fall or report it to a supervisor.
- Keep walkways and floors clear of boxes, extension cords, and litter.
- Sweep debris from floors.
- Move anything that is stored on or near stairways or report the hazard to a supervisor.
- Mark any temporarily made wet areas with signs or limit pedestrian access.
- Secure mats, rugs, and carpets to prevent slippage and overlaps.
- Make sure to always close file cabinet or storage drawers.

- Cover cables that cross over walkways.
- Keep walkways and work areas well-lit for good visibility

1.425.5 Footwear

When selecting proper footwear, it is important that it be appropriate safe footwear for the work environment (e.g., slip-resistant safety shoes or boots in an agricultural work environment, factory or warehouse). Footwear that fits properly, increases comfort and helps to prevent fatigue, which also improves safety for employees.

1.425.6 Working on the Same Level

- Take your time and pay attention to where you are going.
- Adjust your pace to suit the walking surface (e.g., wet, rough, icy, sloped or cluttered).
- Make wide turns at corners.
- Use a flashlight if you enter a dark room where there is no light.
- When carrying a load, be sure that there is clear visibility over or around the load.
- Close cabinet doors and drawers.
- Hold handrail when going up or down stairs.
- Floor openings should be guarded by a standard fixed railing surrounding the hole.
- Walk when using stairways – do not run.
- Open, exposed stairways should have a railing – be sure to use it.
- Closed stairways should have at least one handrail.
- Keep stairways uncluttered.
- Keep platforms or steps on machinery clean and dry
- Use handholds, handrails, and steps provided on riding machinery (e.g., lift trucks, tractors) when mounting or dismounting, using the 3-point system (both hands and one foot or one hand and two feet on the machine at all times).

1.425.7 Pictograms

Below are pictograms that can be used when setting workplace standards and providing training to help warn against specific hazards related to slips, trips, and falls or when using stairs. For each hazard warning pictogram there is a corresponding one that demonstrates control information for that hazard.

Trip Hazard

Control

Slip Hazard

Control

Fall on Stair Hazard

Control

39. HYDROGEN SULFIDE SAFETY PROGRAM

1.426 Purpose and Scope

The purpose of this program is to provide specific requirements and safety principles for exposure to hydrogen sulfide (H_2S).

This program applies to all T&T Industrial, Inc. employees with possible exposure to hydrogen sulfide.

1.427 Resources

Number	Title
29 CFR 1926 Subpart D	Occupational Health and Environmental Controls-Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances-Air Contaminants

1.428 Hydrogen Sulfide Program

Hydrogen sulfide (H_2S) may be encountered during drilling operations. The gas may be associated with recycled drilling mud, water from sour crude wells, blowouts, tank gauging, field maintenance, and tank batteries and wells. Hydrogen sulfide may also be present in refineries and is associated with decaying material in natural settings.

Hydrogen sulfide is colorless and has the odor of rotten eggs. H_2S is toxic, flammable, corrosive, soluble in water, and creates toxic by-products when burned. At higher concentrations, H_2S is odorless and smell alone cannot be used to detect its presence.

It is heavier than air and can collect in low-lying and enclosed, poorly ventilated areas such as basements, manholes, sewer lines, and underground telephone / electrical vaults. At the same time, since it is only 18% denser than air, it should not be assumed that personnel are safe from the hazard in an elevated location. H_2S can and will spread in all directions away from the source.

In addition, H_2S is a highly flammable gas and gas / air mixtures can be explosive. It may travel to sources of ignition and flash back. If ignited, the gas burns to produce toxic vapors and gases, such as sulfur dioxide.

Further information can be found in the Appendix.

1.429 Exposure Health Effects

The health effects of hydrogen sulfide include irritation of the eyes, nose, throat, respiratory system, headache, cough, nausea, loss of sense of smell, blistering, central nervous system weakness, difficulty breathing, unconsciousness, and possible death. Hydrogen sulfide is both an irritant and a chemical asphyxiant with effects on both oxygen utilization and the central nervous system. Its health effects can vary depending on the level and duration of exposure.

Chronic exposures to low doses of H_2S may lead to high blood pressure, chronic headaches, nausea, and sleep disorders.

Contact with liquid H_2S causes frostbite. If clothing becomes wet with the liquid, avoid ignition sources, remove the clothing, and isolate it in a safe area to allow the liquid to evaporate.

1.430 Air Monitoring

In areas where H_2S may be present monitors shall be used. Monitors shall be bump tested at a minimum as required by manufacturer, if monitor fails a bump test a full calibration is required. Monitors shall be calibrated according to manufacturer's recommendations.

Personal or area monitors that alarm when the permissible exposure limit (PEL) exceeds the preset level of 20 ppm for General Industry or 10 ppm for the Construction Industry shall be used.

Employees shall immediately put on supplied air respiratory equipment (SCBA) and/or evacuate the area upon sounding of H_2S alarms. Notify or contact necessary personnel and do not return to work area until clearance is given for re-entry.

1.431 Protection Against Exposure

Before entering areas where H_2S may be present:

- Air shall be tested for the presence and concentration of H_2S by a qualified person using air monitoring equipment, such as H_2S detector tubes or a multi-gas meter that detects the gas.
- Testing should also determine if fire / explosion precautions are necessary.
- If the gas is present, the space / area must be ventilated continually to remove the gas.
- If the gas cannot be removed, the person entering the space / area shall use appropriate respiratory protection and any other necessary personal protective equipment, rescue, and communication equipment.

1.432 Training

Employees shall be trained prior to working in H_2S environments with a minimum of three to four hours of instructor led classroom training. Training programs shall adhere to the ANSI/ASSE Z390.1-2017 Accepted Practices for H_2S Training Programs. Employees are required to refresh training annually. Training must also include site specific emergency action plans and evacuation procedures.

Employees who have the potential to be exposed to H_2S above the occupational exposure limit (OEL) or the permissible exposure limit (PEL) must be trained in:

- The operation and maintenance of the portable and personal gas detection equipment they are expected to use
- How to bump test the portable and personal gas detection equipment they are expected to use
- How to accurately calibrate the portable and personal gas detection equipment they are expected to use
- The required elements of OSHA's Respiratory Protection standard, 29 CFR 1910.134, to include medical evaluations, fit testing, and selected respirator training

1.433 Contingency Plan

Contingency plan provisions shall be developed and communicated.

1.434 Respirators

For concentrations exceeding 10ppm, supplied air respirators of a self-contained breathing apparatus shall be used.

1.435 Confined Spaces

A written confined space program shall be developed per 29 CFR 1910.146 and employees shall be trained under 1910.146(g).

1.436 First Aid Measures

Take the following steps in the event of a release according to the type of release:

Inhalation: Take precautions to prevent a fire (e.g., remove sources of ignition). Take precautions to ensure your own safety before attempting rescue (e.g., wear appropriate protective equipment). Move victim to fresh air. Keep at rest in a position comfortable for breathing. If breathing is difficult, trained personnel should administer emergency oxygen. Do not allow victim to move about unnecessarily. Symptoms of pulmonary edema may be delayed. If breathing has stopped, trained personnel should begin artificial respiration (AR). If the heart has stopped, trained personnel should start cardiopulmonary resuscitation (CPR) or automated external defibrillation (AED). Avoid mouth-to-mouth contact by using mouth guards or shields. Immediately call a Poison Center or doctor. Treatment is urgently required. Transport to a hospital. Victims may pose a threat to responders due to the release of hydrogen sulfide from their clothing, skin, and exhaled air.

Skin Contact: Liquefied gas: quickly remove victim from source of contamination. Do not attempt to rewarm the affected area on site. Do not rub area or apply direct heat. Gently remove clothing or jewelry that may restrict circulation. Carefully cut around clothing that sticks to the skin and remove the rest of the garment. Loosely cover the affected area with a sterile dressing. Do not allow victim to drink alcohol or smoke. Immediately call a Poison Center or doctor. Treatment is urgently required. Transport to a hospital. Double bag, seal, label and leave contaminated clothing, shoes, and leather goods at the scene for safe disposal.

Eye Contact: Gas: immediately flush the contaminated eye(s) with lukewarm, gently flowing water for 15-20 minutes, while holding the eyelid(s) open. Liquefied gas: immediately and briefly flush with lukewarm, gently flowing water. Do not attempt to rewarm. Cover both eyes with a sterile dressing. Do not allow victim to drink alcohol or smoke. Immediately call a Poison Center or doctor. Treatment is urgently required. Transport to a hospital.

Appendix 26 **H_2S Information**

Hydrogen sulfide	CAS 7783-06-4
H_2S	RTECS **MX1225000** **https://www.cdc.gov/niosh/topics/hydrogensulfide/**
Synonyms & Trade Names • Hydrosulfuric acid • Sewer gas • Sulfuretted hydrogen	DOT ID & Guide 1050 117 **https://www.fmcsa.dot.gov/regulations/hazardous-materials**

Physical Description Colorless gas with a strong odor of rotten eggs. [Note: Sense of smell becomes rapidly fatigued and cannot be relied upon to warn of the continuous presence of H_2S. Shipped as a liquefied compressed gas.]			
MW: 34.1	BP: -77°F	FRZ: -122°F	Sol: 0.4%
VP: 17.6 atm	IP: 10.46 eV	RGasD: 1.19	
Fl.P: NA (Gas)	UEL: 44.0%	LEL: 4.0%	
Flammable Gas			
Incompatibilities & Re-activities: Strong oxidizers, strong nitric acid, metals			

Measurement Methods **NIOSH** 6013 (https://www.cdc.gov/niosh/docs/2003-154/pdfs/6013.pdf
OSHA ID141 (http://www.osha.gov/dts/sltc/methods/inorganic/id141/id141.html)

See: NMAM (http://www.cdc.gov/niosh/nmam/)or
OSHA Methods

Personal Protection & Sanitation: See protection (https://www.cdc.gov/niosh/npg/	First Aid: See procedures (https://www.cdc.gov/niosh/npg/
Skin: Frostbite Eyes: Frostbite Wash skin: No recommendation Remove: When wet (flammable) Change: No recommendation Provide: Frostbite wash	Eye: Frostbite Skin: Frostbite Breathing: Respiratory support

Respirator Recommendations
NIOSH
Up to 100 ppm:
(APF = 25) Any powered, air-purifying respirator with cartridge(s) providing protection against the compound of concern
(APF = 50) Any air-purifying, full-facepiece respirator (gas mask) with a chin-style, front- or back-mounted canister providing protection against the compound of concern
(APF = 10) Any supplied-air respirator*
(APF = 50) Any self-contained breathing apparatus with a full facepiece
Emergency or planned entry into unknown concentrations or IDLH conditions:
(APF = 10,000) Any self-contained breathing apparatus that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode
(APF = 10,000) Any supplied-air respirator that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode in combination with an auxiliary self-contained positive-pressure breathing apparatus
Escape:
(APF = 50) Any air-purifying, full-facepiece respirator (gas mask) with a chin-style, front- or back-mounted canister providing protection against the compound of concern/Any appropriate escape-type, self-contained breathing apparatus
Important additional information about respirator selection (http://www.cdc.gov/niosh/npg/pgintrod.html#mustread)

Exposure Routes: inhalation, skin and/or eye contact

Symptoms:

H_2S is both an irritant and a chemical asphyxiant with effects on both oxygen utilization and the central nervous system.

Irritation eyes, respiratory system; apnea, coma, convulsions; conjunctivitis, eye pain, lacrimation (discharge of tears), photophobia (abnormal visual intolerance to light), corneal vesiculation; dizziness, headache, lassitude (weakness, exhaustion), irritability, insomnia; gastrointestinal disturbance; liquid: frostbite

Target Organs: Eyes, respiratory system, central nervous system

40. Illumination Program

1.437 Purpose and Scope

The purpose of this program is to establish and maintain an effective illumination program that focuses on providing adequate lighting levels and optimal visibility in the workplace, with the goal of promoting a safe and productive environment for employees.

This program applies to all T&T Industrial, Inc. employees.

1.438 Illumination Program

Construction areas, ramps, runways, corridors, offices, shops, and storage areas must be lighted to not less than the minimum illumination intensities listed in the table below while any work is in progress.

Foot-Candles	Area of Operation
5	General construction area lighting.
3	General construction areas, concrete placement, excavation and waste areas, access ways, active storage areas, loading platforms, refueling, and field maintenance areas.
5	Indoors: warehouses, corridors, hallways, and exit ways.
5	Tunnels, shafts, and general underground work areas: (Exception: minimum of 10 foot- candles is required at tunnel and shaft heading during drilling, mucking, and scaling. Bureau of Mines approved cap lights shall be acceptable for use in the tunnel heading)
10	General construction plant and shops (e.g., batch plants, screening plants, mechanical and electrical equipment rooms, carpenter shops, rigging lofts and active storerooms, mess halls, and indoor toilets and workrooms.)
30	First aid stations, infirmaries, and offices.

41. Incident Investigation and Reporting Program

1.439 Purpose and Scope

The purpose of this program is to define the company incident investigation and reporting procedures.

This program applies to all T&T Industrial, Inc. employees.

1.440 Resources

Number	Title
30 CFR 250	SEMS-Incident Investigations
29 CFR 1926 Subpart C	General Safety and Health Provisions-Recording and Reporting of Injuries
N/A	OSHA Incident Investigations Guide for Employers
CMS-FM-0029	Incident Reporting Form
CMS-FM-0030	Incident Investigation Report
CMS-FM-0042	Injury / Illness Recording Flowchart
CMS-FM-0037	Incident Workplace Related Flowchart
CMS-FM-0038	Injury and Illness Classification Chart

1.441 Definitions

Acronym/Term	Definition
Hazard	A situation or an inherent property with the potential to cause harm to personnel, assets, the environment, or the company's reputation.
Hazard Observation	An unsafe condition that could lead to an incident involving people, environment, or property.
Incident	An unplanned, undesired event that can result in physical harm and/or property A work-related event in which an injury or ill-health (regardless of severity) or fatality occurred or could have occurred.
Investigation	To derive a level of understanding from a systematic gathering and subsequent analysis of information.
Near Miss/Near Hit	Incidents with no injury or property or environmental damage but having the potential to cause injury or property or environmental damage under slightly different circumstances. This could include an unsafe act.
Work Related Injury or Illness	An event or exposure in the work environment that caused or contributed to the condition or significantly aggravated a pre-existing condition.

1.442 Incident Investigations

When notification of a work-related incident is received, qualified personnel shall be appointed to complete an investigation of the incident. Qualified personnel shall be knowledgeable in investigation techniques, processes involved, and other relevant specialties. The investigation should take place as soon as possible after the incident occurs. While all incidents, regardless of size and impact, including fatalities, injuries, illness, and near misses should be investigated, the extent of such investigation shall reflect the seriousness of the incident. First aid incidents shall be investigated but minimal resources may be required.

Prior to an incident occurring, assignments shall be made establishing responsibility for how and when management is to be notified; who will conduct investigations and what training they should have received; who will receive investigation recommendations; and, who is responsible for implementing corrective actions.

Before investigating, all emergency response needs shall be completed, and the incident site shall be safe and secure for entry and investigation.

At a minimum, the incident investigation program shall address:

- The nature of the incident.
- Human or other contributing factors leading to the incident.
- Recommended changes identified as a result of the investigation.

1.443 Reporting

When an employee is involved in a work-related incident or is aware of a condition that may cause one, the employee must report the incident to Management as soon as possible. Incidents include near misses, injuries, illnesses, property damage, etc.

Fatalities shall be reported to OSHA within 8 hours of their discovery. Inpatient hospitalizations, amputations, and loses of an eye shall be reported to OSHA within 24 hours. Incidents shall also be reported to the host client / site operator as soon as possible, or in a timely manner (within 24 hours of incident).

The company must report severe injuries and/or fatalities using one of the following methods: (a) by telephone or in person to the OSHA Area Office that is nearest to the site of the incident, (b) by telephone to the OSHA toll-free central telephone number, 1-800-321-OSHA (1-800-321-6742), or by electronic submission using the reporting application located on OSHA's public web site at www.osha.gov.

1.444 Training

Members of the incident investigation team shall be qualified / competent individuals. Training shall be provided on investigation techniques used during an incident investigation. Personnel shall be trained in their roles and responsibilities for incident response and incident investigation techniques.

1.445 Roles and Responsibilities

Incident Investigator: The primary role of the incident investigator is to lead the investigation process. Their responsibilities include:

- Planning and organizing the investigation.
- Gathering evidence and conducting interviews.
- Analyzing data and identifying root causes.
- Developing recommendations for prevention and mitigation.
- Preparing and presenting investigation reports.

Team Members: Depending on the complexity of the incident, the investigator may work with a team. Team members' responsibilities include:

- Assisting with evidence collection and analysis.
- Conducting interviews and documenting statements.
- Reviewing relevant documents and records.
- Providing technical expertise or subject matter knowledge.
- Collaborating with the investigator to identify root causes.

Subject Matter Experts: In certain investigations, subject matter experts may be involved to provide specialized knowledge or expertise related to the incident. Their responsibilities include:

- Providing technical guidance and support.
- Analyzing specific aspects of the incident.
- Assisting with data interpretation and analysis.
- Recommending corrective actions based on their expertise.

Interviewers: Interviewers play a crucial role in gathering information from individuals involved in the incident. Their responsibilities include:

- Conducting interviews with witnesses, victims, and involved parties.
- Documenting interviewee statements accurately and objectively.
- Probing for additional details and clarifications.
- Maintaining confidentiality and professionalism during interviews.

Evidence Collectors: Individuals responsible for collecting evidence have the following responsibilities:

- Identifying and documenting physical evidence at the incident site.
- Taking photographs, measurements, and samples, if applicable.
- Ensuring proper handling and preservation of evidence.
- Maintaining a chain of custody for the collected evidence.

Documentation Specialists: Accurate and comprehensive documentation is crucial for incident investigations. Documentation specialists are responsible for:

- Recording all relevant information, actions, and findings.
- Organizing and cataloging collected evidence and documentation.
- Maintaining a centralized repository for investigation records.
- Preparing clear and concise reports.

Legal and Compliance Advisors: Depending on the nature of the incident, legal and compliance advisors may be involved. Their responsibilities include:

- Providing guidance on legal requirements and compliance obligations.
- Assessing the potential legal implications of the incident.
- Ensuring the investigation adheres to legal and regulatory standards.
- Reviewing investigation findings and recommendations for legal compliance.

1.446 Evidence Collection

Initial identification of evidence may include:

- A listing of people, equipment, and materials involved.

- A recording of environmental factors such as weather, illumination, temperature, noise, ventilation.
- Physical factors such as fatigue, age, and medical conditions.

Incident information can be collected through interviews, document reviews and other means including: equipment manuals, industry guidance documents, Company policies and records, maintenance schedules, records and logs, training records, audit and follow-up reports, enforcement policies and records, and previous corrective action recommendations.

Witness interviews and statements shall be collected. Evidence shall be preserved using cones, tapes, and/or guards, secured, and collected through notes, photographs, witness statements, flagging, and impoundment of documents and equipment.

1.447 Corrective Action

Corrective action programs shall be established based on the findings of the incident investigation. The investigation shall be expedited, and findings and recommendations resolved in a timely manner. Corrective action programs shall analyze incidents for root causes and shall, at a minimum:

- Retain incident investigation findings for future hazard analysis or 2 years; whichever is greater.
- Determine and document responses to findings to ensure corrective action plans are completed.
- Implement a system to distribute incident investigation findings to appropriate personnel and/or similar facilities throughout the organization.

1.448 Incident Investigation Report

Incident investigations shall be documented. Participants shall prepare a written report including the description of the incident, any evidence collected during the investigation, an explanation of the causes of the incident, and corrective actions.

The written incident investigation report shall include any immediate corrective actions that were taken as well as any long-term actions that are required to prevent the recurrence of the incident.

Incident facts may include:

- The injured employee's name
- Injury description

- Whether they are temporary or permanent
- The date and location of the incident

Document the investigation such as date of the investigation and who is investigating. Investigators may also document the scene by video recording, photographing, and sketching.

1.449 Lessons Learned

Lessons learned shall be reviewed and communicated. Changes to processes shall be placed into effect to prevent reoccurrence or similar events. Corrective actions shall be supported by senior management.

Appendix 27 **Incident Reporting Form**

General Information		
Incident No. / Title:	**Facility / Project Name:**	☐ **Near Miss**
		☐ **Equipment Damage / Production Loss**
		☐ **Employee Injury / Illness**
		☐ **Environmental**

Logistics of Incident							
Incident Date:		**Location:**		**Weather:**		**Meet/assist notified?**	
Incident Time:		**Project /Contract:**		**Wind:**		**How injured transporter?**	
Shift		**Specific Activity:**		**Other:**		**Location/time of meet?**	
						Destination?	

Relevant Personnel					
Manager:		**Other:**		**Witness:**	
First Responder:					
Medic:					
Supervisor:					

Persons Involved in Incident									
Name	**Company**	**Title**	**Employee ID**	**Days on Shift**	**Length of Service**	**Type of Incident**	**Body Part**	**Side of Body**	**Primary Diagnosis (Doctor)**

T&T Industrial, Inc.

Incident Investigation Program

Description of Incident

Initial Actions Taken to Mitigate Risks and/or Prevent Escalation or Recurrence

Notifications Made

Entity Name	Name of Individual	Individual's Title	Date	Time

Signatures			
Printed Name:		**Date:**	
Signature:		**Comments:**	

T&T Industrial, Inc.

Incident Investigation Program

Appendix 28 **Incident Investigation Report**

General Information			
Company Name:		**Date:**	
Team Member Name:		**Title:**	

Incident Description / Injury Information					
Name of Injured Employee:					
Employee's Age:					
Employee's Job Title:					
Job at Time of Injury:					
Type of Employment:	☐ **Full-time**	☐ **Part-time**	☐ **Temporary**	☐ **Seasonal**	☐ **Other:**
Date of Incident:		**Time of Incident:**			
Length of Time with Company:					
Time in Current Position:					
Location of Incident:					
Description of Injury:					

Detailed Description of Incident *Include relevant events leading up to, during, and after the incident, preferably provided by the injured employee.*
Eyewitness Description of Incident *Include relevant events leading up to, during, and after the incident. Include names of persons interviewed, job titles, and date and time of interviews.*

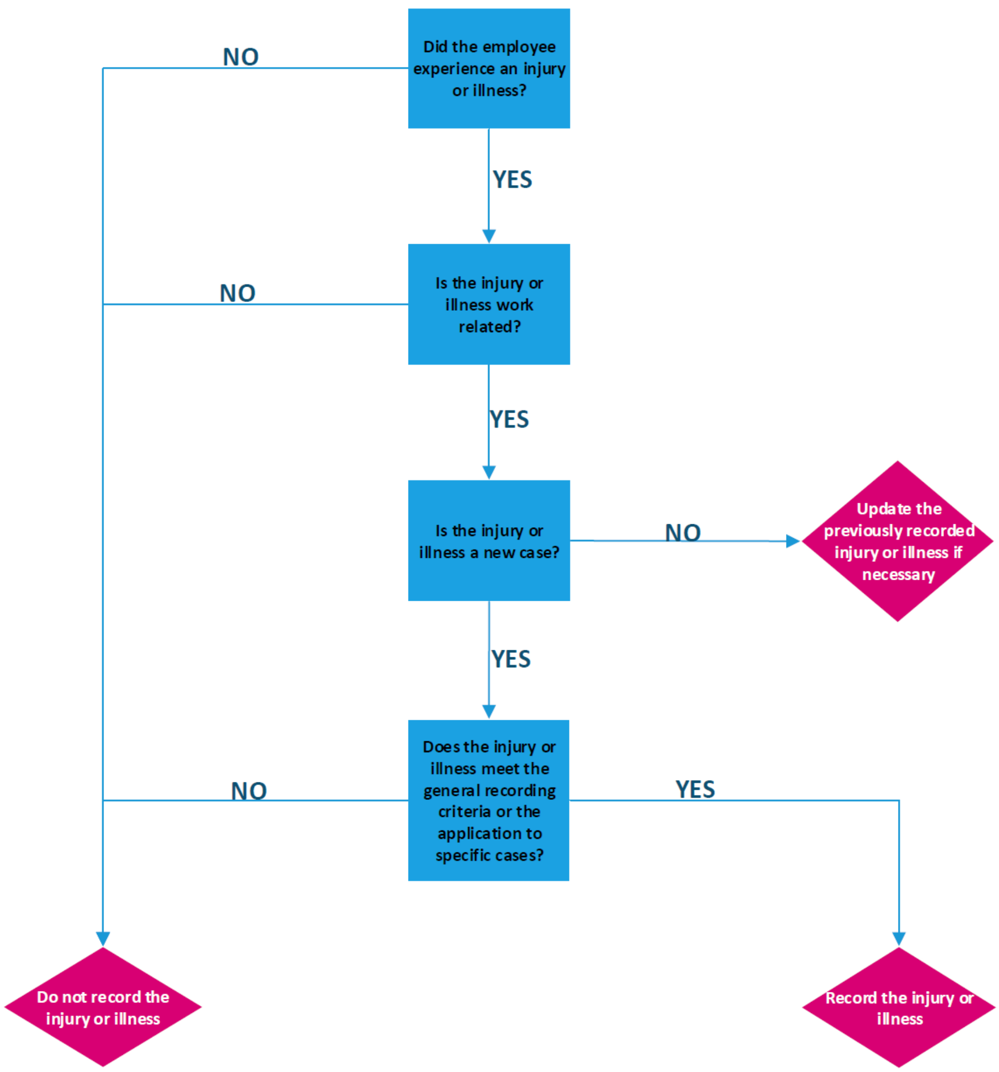
Description of Incident from Additional Employees with Knowledge *Include relevant events leading up to, during, and after the incident. Include names of persons interviewed, job titles, and date and time of interviews.*
Identify the Root Cause *What caused or allowed the incident to happen?*
The root causes are the underlying reasons the incident occurred and are the factors that need to be addressed to prevent future incidents. If safety procedures were not being followed, why were they not being followed? If a machine was faulty or a safety device failed, why did it fail? It is common to find factors that contributed to the incident in several of these areas: equipment/machinery, tools, procedures, training or lack of training, and work environment. If these factors are identified, it must be determined why these factors were not address before the incident.

T&T Industrial, Inc. **Incident Investigation Program**

Recommended Corrective Actions to Prevent Future Incidents
Corrective Actions Taken / Root Causes Addressed

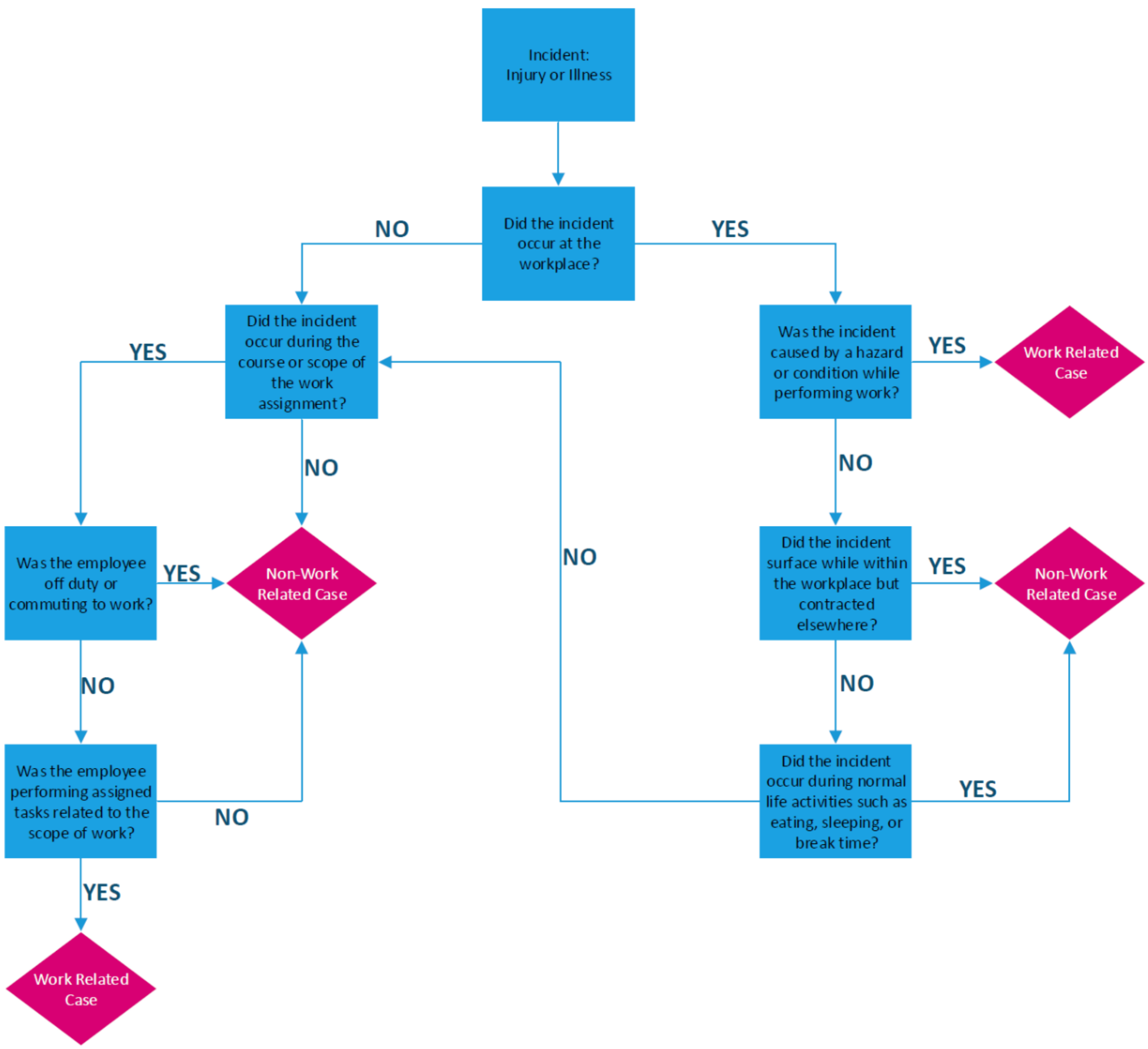
Appendix 29 **Injury / Illness Recording Flowchart**

Injury / Illness Recording Flowchart

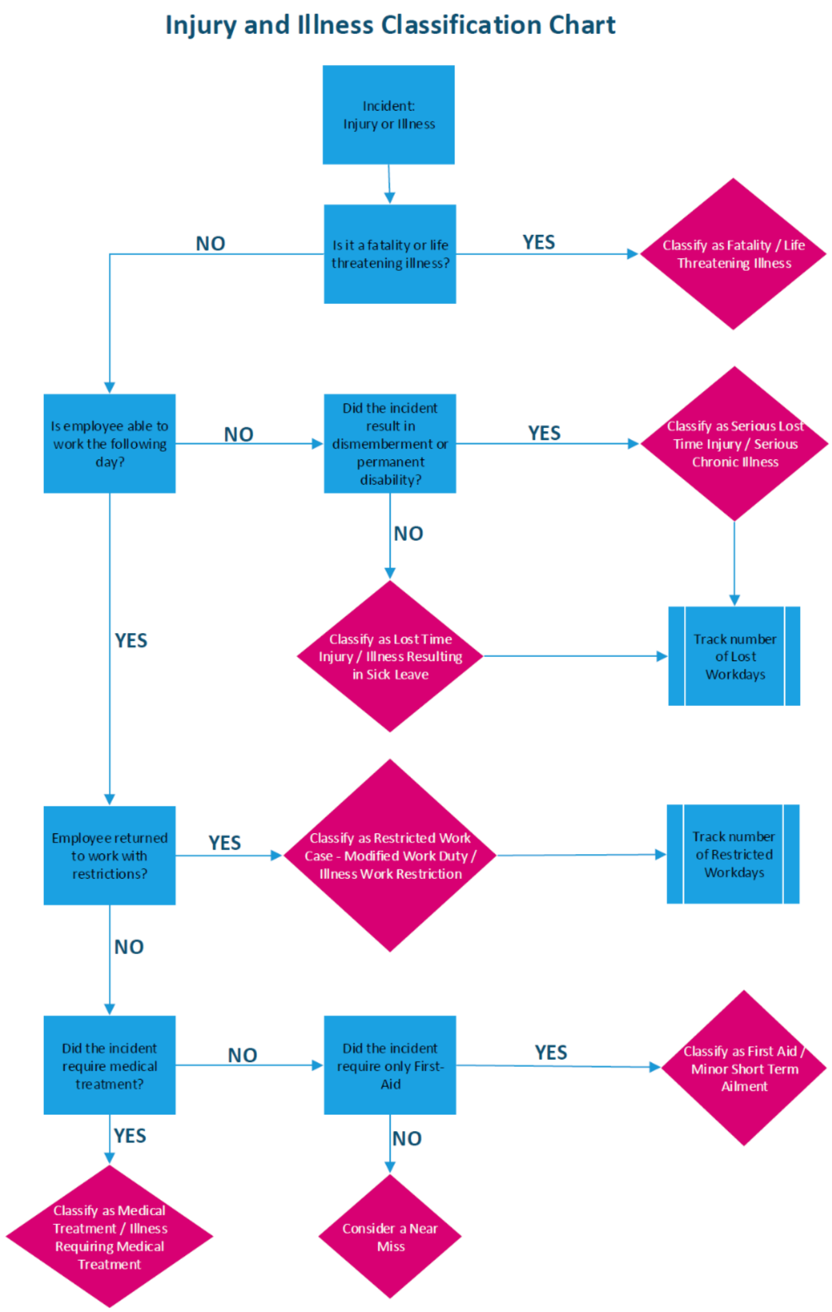


Appendix 30 **Incident Workplace Related Flowchart**

Work Related or Non-Work Related Decision Chart



Appendix 31 Injury and Illness Classification Chart



42. INJURY / ILLNESS RECORDKEEPING PROGRAM

1.450 Purpose and Scope

The purpose of this program is to define injury and illness recordkeeping requirements.

This program applies to all T&T Industrial, Inc. employees.

1.451 Resources

Number	Title
29 CFR 1904	Recording and Reporting Occupational Injuries and Illnesses

1.452 Injury / Illness Recordkeeping Requirements

Recordkeeping and reporting are required and appropriate for developing information regarding the causes and prevention of occupational incidents and illnesses and for maintaining a program of collection, compilation, and analysis of occupational safety and health statistics.

1.453 Records

Records shall be kept of fatalities, injuries, and illnesses that:

- Are work-related
- Are new cases
- Meet one or more of the general recording criteria (death, days away from work, restricted work or transfer to another job, medical treatment beyond first aid, loss of consciousness, a significant injury or illness diagnosed by a physician or other licensed health care professional, needlesticks and sharps injury cases, tuberculosis cases, hearing loss cases, medical removal cases, musculoskeletal cases).

1.454 OSHA 300 Log

Each recordable injury or illness shall be entered on an OSHA 300 Log and 301 Incident Report, or other equivalent form, within 7 calendar days of receiving information that a recordable injury or illness has occurred.

A company executive shall certify that the OSHA 300 Log has been examined and when found to be correct, sign the OSHA 300A Summary.

A copy of the annual summary shall be posted in each establishment in a conspicuous place or places where notices to employees are customarily posted. The posted annual summary shall not be altered, defaced, or covered by other material.

The annual summary shall be posted no later than February 1st of the year following the year covered by the records and the posting kept in place through April 30th.

The OSHA 300 Log, the privacy case list (if one exists), the annual summary, and the OSHA 301 Incident Report forms shall be retained for 5 years following the end of the calendar year that these records cover.

1.455 Restricted Work or Job Transfer

Restricted work occurs when, as the result of a work-related injury or illness:

- The employee is kept from performing one or more of the routine functions of their job, or from working the full workday that they would otherwise have been scheduled to work; or
- A physician or other licensed health care professional recommends that the employee not perform one or more of the routine functions of their job, or not work the full workday that they would otherwise have been scheduled to work.

1.456 Medical Treatment Beyond First Aid

Medical treatment means the management and care of a patient to combat disease or disorder. Medical treatment does not include:

- Visits to a physician or other licensed health care professional solely for observation or counseling,
- The conduct of diagnostic procedures, such as x-rays and blood tests, including the administration of prescription medications used solely for diagnostic purposes (e.g., eye drops to dilate pupils), or
- First aid.

1.457 First Aid

First aid means:

- Using a non-prescription medication at nonprescription strength (for medications available in both prescription and non-prescription form, a recommendation by a physician or other licensed health care professional to use a non-prescription medication at prescription strength is considered medical treatment for recordkeeping purposes).

- Administering tetanus immunizations (other immunizations, such as Hepatitis B vaccine or rabies vaccine, are considered medical treatment).
- Cleaning, flushing, or soaking wounds on the surface of the skin.
- Using wound coverings such as bandages, plastic strip bandages, gauze pads, etc. or using butterfly bandages (other wound closing devices such as sutures, staples, etc., are considered medical treatment).
- Using hot or cold therapy.
- Using any non-rigid means of support, such as elastic bandages, wraps, non-rigid back belts, etc. (devices with rigid stays or other systems designed to immobilize parts of the body are considered medical treatment for recordkeeping purposes).
- Using temporary immobilization devices while transporting an accident victim (e.g., splints, slings, neck collars, back boards, etc.).
- Drilling of a fingernail or toenail to relieve pressure or draining fluid from a blister.
- Using eye patches.
- Removing foreign bodies from the eye using only irrigation or a cotton swab.
- Removing splinters or foreign material from areas other than the eye by irrigation, tweezers, cotton swabs, or other simple means.
- Using finger guards.
- Using massages (physical therapy or chiropractic treatment are considered medical treatment for recordkeeping purposes).
- Drinking fluids for relief of heat stress.

43. Ionizing Radiation Exposure Safety Program

1.458 Purpose and Scope

The purpose of this program is to protect employees from the potential health hazards associated with exposure to ionizing radiation, while ensuring compliance with applicable regulations and standards.

This program applies to all T&T Industrial, Inc. employees.

1.459 Ionizing Radiation Exposure Safety

An unstable atom is said to be "radioactive," and the energy it releases is referred to as "radiation." When the radiation has enough energy to ionize other atoms (i.e., remove negatively charged particles called "electrons") in its path, it is referred to as "ionizing radiation." Types of ionizing radiation include alpha, beta, and neutron particles; gamma rays; and X-rays.

1.460 Sources of Ionizing Radiation

Sources of ionizing radiation include radioactive materials and radiation-generating machines.

Radioactive materials can be naturally occurring (such as uranium and radium found in the earth) or manmade in an accelerator or reactor.

Radiation-generating machines, such as medical X-ray machines, produce ionizing radiation electronically and stop producing radiation when turned off. Equipment that contains radioactive material, such as some industrial radiography equipment, cannot be turned off because the radioactive source emits ionizing radiation. These sources must be shielded (i.e., surrounded by a material that can block radiation) to prevent or reduce radiation exposure.

All individuals working in or frequenting any portion of a radiation area shall be informed of the occurrence of radioactive materials and shall be instructed in the safety problems associated with exposure, precautions, and devices to minimize exposure, including but not limited to time, distance, shielding, and keeping exposure limits as low as possible (ALARA). They must also be instructed in the applicable provisions of this section for the protection of employees from exposure to radiation or radioactive materials and shall be advised of reports of radiation exposure, which employees may request a copy of.

1.461 Exposure

When ionizing radiation interacts with humans, it is capable of damaging living cells in the human body. Humans can be exposed: 1) to external radiation from a radiation source outside of the body, such as an X-ray from an X-ray machine; or 2) through internal exposure following inhalation (breathing in), ingestion (swallowing), or wound uptake (i.e., through non-intact skin) of radioactive material. In addition, the skin can become contaminated with radioactive materials when proper controls are not in place to prevent contamination or following an emergency.

No sources of ionizing radiation shall be used or transferred in such a manner as to cause any individual in a restricted area to receive in any period of one calendar quarter from sources in the employer's possession or control a dose in excess of the limits specified in Table G-18.

- Rems per calendar quarter
- Whole body: Head and trunk; active blood-forming organs; lens of eyes; or gonads: 1 1/4
- Hands and forearms; feet and ankles: 18 3/4
- Skin of whole body: 7 1/2

Type of Ionizing Radiation	How it Travels and Penetrates	How it Delivers Dose to the Body
Alpha Particles	Alpha particles cannot penetrate most other materials. A piece of paper, the dead outer layers of skin, or even a few inches of air are sufficient to stop alpha particles.	Radioactive material that emits alpha particles can be very harmful to living cells when alpha particles are inhaled, ingested, or absorbed into the blood stream (e.g., through a cut in or area of non-intact skin).
Beta Particles	Beta particles can travel up to several feet in the air. Beta particles can be stopped by some plastics, aluminum, or a block of wood. Beta particles should never be shielded with lead or other high atomic number shields, which could result in X-rays being released.	Some beta particles are capable of penetrating the skin and causing radiation damage, such as skin burns. Beta particles are most harmful to living cells when they are inhaled or ingested.
Gamma rays and X-rays (electromagnetic radiation)	Gamma rays and X-rays are very penetrating and can travel great distances. Lead or concrete is able to reduce the intensity of gamma rays and X-rays.	Gamma rays and X-rays can easily pass completely through the human body; however, a fraction of the energy can be absorbed by tissue and can damage living cells.
Neutron Particles	Neutrons have an exceptional ability to penetrate materials. Hydrogen-containing materials (concrete or water) are best for shielding neutrons.	Neutrons can contribute significantly to radiation dose.

1.462 Health Effects

Workers may be exposed to ionizing radiation in several ways, depending on their job tasks. The health effects of radiation dose depend on the type of radiation emitted, the radiation dose received by a worker, and the parts of the body that are exposed, among other factors. Radiation dose depends on the duration of exposure, the amount of radiation generated from the radiation source, the distance from the radiation source, and the amount and type of shielding in place. In general, radiation dose is received when a worker is:

- In close proximity to an unshielded or partially shielded radiation source.
- Unprotected when near unshielded radiation-generating machines (e.g., X-ray machine, accelerator, etc.) in operation.
- Unprotected when handling radioactive materials (e.g., radionuclides).
- In close proximity to surfaces or areas contaminated with radioactive materials (e.g., from small spills or leaks).
- Contaminated with radioactive materials.

1.463 Safe Work Practices

Engineering controls and safe work practices to reduce/maintain ionizing radiation exposure below applicable limits include, at a minimum:

Qualified staff / Competent Person (e.g., RSO, health physicist) to provide oversight and responsibility for radiation protection policies and procedures. The competent person is responsible for the implementation, coordination, and day-to-day oversight. In addition, the competent person has the authority to enforce radiation policies and procedures regarding radiation safety and regulatory compliance of the use of ionizing radiations.

ALARA stands for As Low As Reasonably Achievable (ALARA). It is a guiding principle in radiation protection used to eliminate radiation doses that have no direct benefit.

A **dosimetry program** in which personal exposure monitoring is conducted, as required by federal or state regulations, for external dose and, as needed, for internal dose.

Medical surveillance and area monitoring to document radiation levels, contamination with radioactive materials, and potential worker exposures.

Radiological controls, including entry and exit controls, receiving, inventory control, storage, and disposal.

Surveys must be taken. Survey means an evaluation of the radiation hazards incident to the production, use, release, disposal, or presence of radioactive materials or other sources of radiation under a specific set of conditions. When appropriate, such evaluation includes a physical survey of the location of materials and equipment, and measurements of levels of radiation or concentrations of radioactive material present.

Symbols shall use the conventional radiation caution colors (magenta or purple on yellow background). The symbol is the conventional three-bladed design.

Each radiation area shall be conspicuously posted with a sign or signs bearing the radiation caution symbol with the words: "Caution - Radiation Area".

Worker training on radiation protection, including health effects associated with ionizing radiation dose, and radiation protection procedures and controls to minimize dose and prevent contamination. Employees that are exposed to airborne concentrations of ionizing radiation shall be trained in accordance with local jurisdictional requirements. Employees must receive training and instruction on the hazards associated with exposure to radiation, precautions to be taken and devices to be used to minimize exposure to include, but not limited to: time, distance, shielding, and keeping exposure limits to as low as reasonably achievable (ALARA). Employee training must include the provisions of 29 CFR 1910.1096(i)(2-3). Affected employees must be advised of radiation exposure and request copies of exposure reports.

Appropriate **personnel monitoring equipment** shall be provided and used, such as film badges, pocket chambers, pocket dosimeters, or film rings.

Emergency procedures to identify and respond to radiological emergency situations.

Recordkeeping and reporting programs to maintain all records and provide dosimetry reports and notifications, as required by federal or state regulations. Exposure to radiation records must be maintained for all affected employees for whom personal monitoring is required. Each affected employee must be informed of their individual radiation exposure on at least an annual basis.

Internal audit procedures to annually audit all aspects of the radiation protection program.

44. Ladder Safety Program

1.464 Purpose and Scope

The purpose of this program is to establish guidelines for the safe use of ladders to minimize hazards and ensure employee safety.

This program applies to all T&T Industrial, Inc. employees that use ladders.

1.465 Resources

Number	Title
29 CFR 1926 Subpart X	Stairways and Ladders-Ladders
29 CFR 1910 Subpart D	Walking Working Surfaces-Ladders

1.466 Ladder Safety Program

Ladders are a very handy tool, both at work and around the home. Ladders are such simple tools that many people forget the dangers involved when using a ladder.

1.467 Ladder Specifications

Ladders shall meet the following OSHA / ANSI specifications:

- Ladder rungs, cleats, and steps shall be parallel, level, and uniformly spaced, when the ladder is in position for use.
- Rungs, cleats, and steps of portable ladders (except as provided below) and fixed ladders (including individual rung/step ladders) shall be spaced not less than 10 inches apart, nor more than 14 inches apart, as measured between center lines of the rungs, cleats, and steps.
- Rungs, cleats, and steps of step stools shall be not less than 8 inches apart, nor more than 12 inches apart, as measured between center lines of the rungs, cleats, and steps.
- Rungs, cleats, and steps of the base section of extension trestle ladders shall not be less than 8 inches nor more than 18 inches apart, as measured between center lines of the rungs, cleats, and steps. The rung spacing on the extension section of the extension trestle ladder shall be not less than 6 inches nor more than 12 inches, as measured between center lines of the rungs, cleats, and steps.
- The minimum clear distance between the sides of individual rung / step ladders and the minimum clear distance between the side rails of other fixed ladders shall be 16 inches.

The minimum clear distance between side rails for all portable ladders shall be 11 1/2 inches.

- The rungs of individual rung / step ladders shall be shaped such that employees' feet cannot slide off the end of the rungs. The rungs and steps of portable metal ladders shall be corrugated, knurled, dimpled, coated with skid-resistant material, or otherwise treated to minimize slipping.

1.468 Load Limits

Ladders are built to hold a limited amount of weight. Ladders shall not be loaded beyond the maximum intended load for which they were built, nor beyond their manufacturer's rated capacity.

Ladder shall be capable of supporting the following loads without failure:

- Each self-supporting portable ladder: At least four times the maximum intended load, except that each extra-heavy-duty type 1A metal or plastic ladder shall sustain at least 3.3 times the maximum intended load. The ability of a ladder to sustain the loads indicated shall be determined by applying or transmitting the requisite load to the ladder in a downward vertical direction. Ladders built and tested in conformance with the applicable provisions of the Appendix of this subpart will be deemed to meet this requirement.
- Each portable ladder that is not self-supporting: At least four times the maximum intended load, except that each extra-heavy-duty type 1A metal or plastic ladder shall sustain at least 3.3 times the maximum intended load. The ability of a ladder to sustain the loads indicated shall be determined by applying or transmitting the requisite load to the ladder in a downward vertical direction when the ladder is placed at an angle of 75 1/2 degrees from the horizontal. Ladders built and tested in conformance with the applicable provisions of the Appendix will be deemed to meet this requirement.
- Each fixed ladder: At least two loads of 250 pounds each, concentrated between any two consecutive attachments (the number and position of additional concentrated loads of 250 pounds each, determined from anticipated usage of the ladder, shall also be included), plus anticipated loads caused by ice buildup, winds, rigging, and impact loads resulting from the use of ladder safety devices. Each step or rung shall be capable of supporting a single concentrated load of at least 250 pounds applied in the middle of the step or rung. Ladders built in conformance with the applicable provisions of the Appendix will be deemed to meet this requirement.

1.469 Ladder Usage

Ladders shall be used only for the purpose for which they were designed. Never use a ladder in a horizontal position or as scaffolding. Do not place ladders on top of boxes, barrels, crates, etc.

1.470 Inspection

Ladders shall be inspected by a competent person for defects that can be seen and after any event that could make them unsafe to use. Ladders shall be inspected before initial use in each work shift and more frequently, as necessary, to identify any visible defects that could cause employee injury. Documentation shall be maintained.

Portable and fixed ladders that are unsafe to use shall be immediately marked to show they are unsafe to use or be tagged with "Do Not Use" or similar language. They shall be taken out of service until they are repaired or replaced.

Examples of problems that make a ladder unsafe to use include:

- Broken or missing rungs, cleats, or steps.
- Broken or split rails, corroded components.
- Other faulty or defective components.

1.471 Safe Work Practices

When portable ladders are used to reach a surface above the employee, the ladder side rails shall extend at least 3 feet above the surface the employees will be stepping onto. A ladder that cannot be extended shall be secured to a rigid support that will not cause it to slip or move and a grasping device, such as a grabrail, shall be provided to assist employees in mounting and dismounting the ladder.

A ladder that does not support itself shall be placed at an angle that is safe. This angle is defined by OSHA as being about one-quarter of the working length of the ladder, which is usually known as a 4:1 ratio. This means the ladder shall be 1 foot from the wall for every 4 feet it reaches up.

Employees shall face the ladder when climbing up or down, using at least one hand to grasp the ladder. Employees shall not carry any object or load that could cause the employee to lose balance and fall while climbing up or down the ladder.

Ladders shall only be used on stable and level surfaces unless they are secured or stabilized to prevent accidental displacement.

Appendix 32 **Guideline for Ladder Loading and Strength Requirements**

This appendix serves as a non-mandatory guideline to assist in complying with the ladder loading and strength requirements. A ladder designed and built in accordance with the applicable national consensus standards, as set forth below, will be considered to meet the OSHA requirements:

- Manufactured portable wood ladders:

  American National Standards Institute (ANSI) A14.1-1982 - American National Standard for Ladders-Portable Wood-Safety Requirements.

- Manufactured portable metal ladders:

  ANSI A14.2-1982 - American National Standard for Ladders - Portable Metal - Safety Requirements.

- Manufactured fixed ladders:

  ANSI A14.3-1984 - American National Standard for Ladders - Fixed - Safety Requirements.

- Job-made ladders:

  ANSI A14.4-1979 - Safety Requirements for Job-Made Ladders.

- Plastic ladders:

  ANSI A14.5-1982 - American National Standard for Ladders - Portable Reinforced Plastic - Safety Requirements

45. Lead Awareness Program

1.472 Purpose and Scope

The purpose of this program is to protect employees against potential exposure to Lead in accordance with regulation on Lead standards. The procedure defines the requirements to manage potential exposure of employees to Lead where it may be encountered.

This program applies to all T&T Industrial, Inc. employees with the possibility of exposure to lead.

1.473 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances-Lead
29 CFR 1926 Subpart D	Occupational Health and Environmental Controls-Lead
Cal/OSHA T8 CCR Subchapter 4	Construction Safety Orders-Dusts, Fumes, Mists, Vapors, and Gases

1.474 Lead Awareness Program

Lead enters the body primarily through inhalation and ingestion. Today, adults are mainly exposed to lead by breathing in lead-containing dust and fumes at work or from hobbies that involve lead.

Lead passes through the lungs into the blood where it can harm many of the body's organ systems. While inorganic lead does not readily enter the body through the skin, it can enter the body through accidental ingestion (eating, drinking, and smoking) via contaminated hands, clothing, and surfaces.

1.475 Training

Lead awareness training shall be provided for employees whose work activities may contact lead containing materials but do not disturb the material during their work activities or are exposed to airborne concentrations. Lead awareness training is required at time of hire, during orientation, or before assignment to areas containing lead. Refresher training shall be given annually. Lead awareness training shall be documented including dates of training, employee name, and trainer name.

The lead awareness training includes, where employees may find lead containing materials such as in lead paints, leaded solders, pipes, batteries, leaded glass, circuit boards, cathode tubes, and salvage materials.

Lead awareness training must include the health effects of lead and include at a minimum, the acute symptoms of lead poisoning which are loss of appetite, nausea, vomiting, stomach cramps, constipation, difficulty sleeping, fatigue, moodiness, headaches, joint or muscle aches, and anemia. The must also include the potential chronic health effects from long term exposure to lead such as severe damage to the blood forming, nervous, urinary, and reproductive systems.

1.476 Lead Containing Material

Lead containing materials may be found in lead paints, leaded solders, pipes, batteries, circuit boards, cathode ray tubes, leaded glass, and demolition / salvage materials.

Employees shall abide by any signs / labels / assessment reports indicating the presence of lead containing materials. Appropriate work practices shall be followed to ensure the lead containing materials are not disturbed.

Employees' hands and faces shall be washed if lead containing materials are contacted.

Employees must not disturb lead containing materials and must abide by any signs, labels, and assessment reports that indicate the presence of lead.

1.477 Exposure Health Effects

Common symptoms of acute lead poisoning are:

- Loss of appetite
- Nausea
- Vomiting
- Stomach cramps
- Constipation
- Difficulty in sleeping
- Fatigue
- Moodiness
- Headache
- Joint or muscle aches
- Anemia

Long term (chronic) overexposure to lead may result in severe damage to the blood-forming, nervous, urinary, and reproductive systems.

1.478 Roles and Responsibilities

The competent person plays a crucial role in ensuring the health and safety of workers who may be exposed to lead in the workplace. The competent person must have the necessary knowledge, training, and experience to effectively fulfill their responsibilities according to local jurisdiction. Typical roles and responsibilities of a competent person include:

- Risk Assessment: Conducting thorough risk assessments to identify potential sources of lead exposure in the workplace and evaluating the associated risks to workers.
- Exposure Monitoring: Implementing a monitoring program to measure and assess workers' exposure levels to lead. This may involve air sampling, biological monitoring, or other appropriate methods to determine lead exposure levels.
- Hazard Communication: Developing and implementing a comprehensive hazard communication program to ensure that workers are aware of the risks associated with lead exposure. This may include providing training, safety data sheets (SDS), warning signs, and labels for lead-containing materials.
- Training and Education: Providing training and education programs to workers on the hazards of lead exposure, including information about sources of exposure, health effects, safe work practices, and the proper use of personal protective equipment (PPE). Ensuring that workers understand the risks associated with lead and the necessary precautions to protect themselves.
- Medical Surveillance: Implementing a medical surveillance program to monitor the health of workers exposed to lead. This may include pre-placement and periodic medical examinations, blood lead level monitoring, and maintaining medical records.
- Engineering Controls: Recommending and overseeing the implementation of engineering controls to minimize or eliminate lead exposure. This may include ventilation systems, enclosure or isolation of lead processes, and substitution of lead-containing materials with less hazardous alternatives.
- Work Practices: Developing and enforcing safe work practices and procedures for handling lead-containing materials, including proper storage, handling, use, and disposal methods. This may involve establishing restricted areas, implementing decontamination procedures, and promoting good hygiene practices.
- Personal Protective Equipment (PPE): Assessing the need for and recommending appropriate PPE, such as respirators, gloves, protective clothing, and eyewear. Ensuring that workers are trained in the proper use, maintenance, and limitations of PPE.

- Recordkeeping and Documentation: Maintaining accurate records of exposure monitoring results, medical surveillance records, training documentation, and any incidents or accidents related to lead exposure. This includes documenting control measures implemented and reviewing the effectiveness of the program.
- Regulatory Compliance: Staying up to date with relevant regulations and standards regarding lead exposure and ensuring compliance with applicable laws, regulations, and guidelines.
- Program Evaluation and Improvement: Regularly reviewing and evaluating the effectiveness of the lead awareness program, identifying areas for improvement, and implementing corrective actions as necessary.

1.479 Engineering Controls

Implementing effective engineering controls is crucial to reduce and maintain exposure levels below applicable limits. Measures include:

- Local Exhaust Ventilation: Install and maintain local exhaust ventilation systems to capture and remove lead dust or fumes at the source. This can include hoods, booths, or other collection devices that effectively capture airborne contaminants.
- General Ventilation: Ensure adequate general ventilation in the workplace to dilute and remove any airborne lead particles. This involves providing a sufficient supply of clean air and exhausting contaminated air.
- Enclosure and Isolation: Enclose or isolate processes that generate lead dust or fumes to prevent their dispersion into the work area. Physical barriers, such as curtains or walls, can be used to contain the lead-containing materials or operations.
- Substitution of Materials: Whenever possible, substitute lead-containing materials with less hazardous alternatives. This can help eliminate or minimize the generation of lead dust or fumes in the first place.
- Process Automation: Automate processes or use mechanical devices to reduce or eliminate manual handling of lead-containing materials. This reduces the potential for lead dust generation during activities like handling, mixing, or pouring.

1.480 Work Practices

Implementing effective work practices is crucial to reduce and maintain exposure levels below applicable limits. Measures include:

- Good Hygiene Practices: Implement strict hygiene practices to minimize the spread of lead dust. This includes regular cleaning of work surfaces, equipment, and personal protective equipment (PPE). Use wet methods or HEPA vacuuming for cleaning to prevent lead dust from becoming airborne.
- Personal Hygiene: Promote good personal hygiene among workers, including regular handwashing, proper use of protective clothing, and avoiding eating, drinking, or smoking in areas where lead exposure is possible.
- Contaminated Clothing and PPE: Establish procedures for the proper removal, storage, and cleaning of contaminated clothing and PPE to prevent cross-contamination. Provide dedicated storage areas and laundry facilities for contaminated items.
- Restricted Areas and Signage: Identify and clearly mark areas where lead exposure is likely to occur. Restrict access to authorized personnel only and post appropriate warning signs to inform workers and visitors about potential hazards.
- Training and Education: Provide comprehensive training to workers on lead hazards, safe work practices, proper use of PPE, and emergency response procedures. Regularly reinforce training to ensure ongoing awareness and compliance.
- Waste Management: Properly manage and dispose of lead-contaminated waste and materials. Use sealed containers or bags to prevent the release of lead dust and follow applicable regulations for disposal.
- Regular Maintenance and Inspection: Conduct routine maintenance and inspection of engineering controls, ventilation systems, and PPE to ensure their effectiveness. Promptly address any issues or malfunctions that could lead to increased lead exposure.

46. Lockout Tagout Program

1.481 Purpose and Scope

This program outlines safeguards to prevent the unexpected energization or startup of machinery and equipment or release of hazardous energy or material that could cause injury to personnel.

This program applies to all T&T Industrial, Inc. employees working with or around hazardous energy.

1.482 Resources

Number	Title
29 CFR 1910 Subpart J	General Environmental Controls-The Control of Hazardous energy (lockout/tagout)
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders-Safe Practices and Personal Protection
CMS-FM-0033	Lockout Tagout Form
CMS-FM-0035	LOTO Certification of Inspection

1.483 Lockout / Tagout (LOTO) Program

Energy sources including electrical, mechanical, hydraulic, pneumatic, chemical, thermal, steam, tension, gravity, or other sources in machines and equipment can be hazardous to employees. During the servicing and maintenance of machines and equipment, the unexpected startup or release of stored energy can result in serious injury, including electrocution, burns, being crushed, lacerations, amputations, fracture limbs, or death.

Management is responsible for the implementation of the LOTO program.

1.484 Requirements

LOTO devices shall indicate the identity of the employee applying the device(s) and shall be durable, standardized, substantial, and identifiable.

LOTO shall be performed only by the authorized employees who are performing the servicing or maintenance.

A hazard risk assessment must be conducted to identify potential hazards associated with machinery and equipment, and appropriate control measures must be implemented to minimize risks. Alternative procedures must be considered and addressed and controls implemented to provide adequate protection of hazardous energy sources when LOTO or tagout is not feasible.

1.485 Training

Training shall be provided to ensure that the purpose and function of the energy control program are understood by employees and that the knowledge and skills required for the safe application, usage, and removal of the energy controls are acquired by employees.

Accomplishment of employee training shall be certified and kept up to date. The certification shall contain each employee's name and dates of training.

Retraining shall be provided for all authorized and affected employees whenever there is a change in their job assignments, a change in machines, equipment or processes that present a new hazard, or when there is a change in the energy control procedures. Additional retraining shall also be conducted whenever a periodic inspection reveals, or whenever the Company has reason to believe that there are deviations from or inadequacies in the employee's knowledge or use of the energy control procedures.

Employee training must be accomplished and kept up to date. The certification shall contain each employee's name and dates of training.

1.486 Inspection

A periodic inspection of the energy control procedure shall be conducted at least annually to ensure the procedure is being followed. The periodic inspection shall be performed by an authorized employee other than the ones(s) utilizing the energy control procedure being inspected. The periodic inspection shall be conducted to correct any deviations or inadequacies identified.

Where lockout is used for energy control, the periodic inspection shall include a review, between the inspector and each authorized employee, of that employee's responsibilities under the energy control procedure being inspected.

Where tagout is used for energy control, the periodic inspection shall include a review, between the inspector and each authorized and affected employee, of that employee's responsibilities under the energy control procedure.

Performance of periodic inspections shall have certification. The certification shall identify the machine or equipment on which the energy control procedure was being utilized, the date of the inspection, the employees included in the inspection, and the person performing the inspection.

1.487 Shutdown Procedures

Before an authorized or affected employee turns off a machine or equipment, the authorized employee shall have knowledge of the type and magnitude of the energy, the hazards of the energy to be controlled, and the methods or means to control the energy.

The machine or equipment shall be turned off or shutdown using the procedures established for the machine or equipment. An orderly shutdown shall be utilized to avoid any additional or increased hazard(s) to employees as a result of the equipment stoppage.

1.488 Isolation

Zero Energy State means the hazard has been eliminated; thus, no hazardous energy exists. A state of zero energy must be verified after a lockout device is installed.

Verification of a zero energy state must be accomplished prior to the operation.

Attempt to re-start the equipment to verify that the energy sources have been de-energized. Turn on switches, open valves, push start buttons, etc. If an energy release occurs during this verification, work cannot proceed until this source is located, isolated, and verified as de-energized. Turn switches off and close valves once de-energized state is verified.

All energy isolating devices that are needed to control the energy to the machine or equipment shall be physically located and operated in such a manner as to isolate the machine or equipment from the energy source(s).

If maintenance, cleaning, or adjustments will be performed on a piece of equipment while it is in operation, safe work procedures must be in place that include how to complete the job safely. Employees must be trained on these safe work procedures and the procedures must be easily accessible.

1.489 Application

LOTO devices shall be singularly identified and affixed to each energy isolating device by authorized employees.

Lockout devices, where used, shall be affixed in a manner that will hold the energy isolating devices in a safe or off position.

Tagout devices, where used, shall be affixed in such a manner as will clearly indicate that the operation or movement of energy isolating devices from the safe or off position.

Tagout only procedures must have additional means of hazardous energy control.

Where tagout devices are used with energy isolating devices designed with the capability of being locked, the tag attachment shall be fastened at the same point at which the lock would have been attached.

Where a tag cannot be affixed directly to the energy isolating device, the tag shall be located as close as safely as possible to the device in a position that will be immediately obvious to anyone attempting to operate the device.

1.490 Stored Energy

Following the application of LOTO devices to energy isolating devices, all potentially hazardous stored or residual energy shall be relieved, disconnected, restrained, or otherwise rendered safe. If there is a possibility of re-accumulation of stored energy, verification of isolation shall be continued until the servicing or maintenance is completed or until the possibility of such accumulation no longer exists.

1.491 Verification

Prior to starting work on machines or equipment that have been locked or tagged out, the authorized employee shall verify that isolation and de-energization of the machine or equipment have been accomplished.

1.492 Removal without Authorized Employee

The device may be removed under the direction of the Company, provided that specific procedures and training for such removal have been developed, documented, and incorporated into the energy control program. First, verify the authorized employee is not at the facility, make reasonable efforts to contact the authorized employee to inform them of removal, and ensure that the authorized employee has this knowledge before they resume work at the facility.

1.493 Notification

Affected employees shall be notified by the Company or authorized employee of the application and removal of LOTO devices. Notification shall be given before the controls are applied and after they are removed from the machine or equipment.

1.494 Safety Testing

The following sequence of actions shall be followed to safety test machines when the LOTO devices must be temporarily removed:

1) Clear away tools,

2) Remove employees,
3) Remove the LOTO device,
4) Energize and proceed with testing,
5) De-energize and reapply control measures.

1.495 Group LOTO

Primary responsibility is vested in an authorized employee. The authorized employee ascertains the exposure status of group members.

Each employee shall affix a personal LOTO device to the group LOTO device before engaging in the servicing and maintenance operation. The supervisor in charge of the group LOTO shall not remove the group LOTO device until each employee in the group has removed the personal device. When more than one crew, craft, department, etc. is involved, overall job associated LOTO control shall be assigned.

1.496 Shift or Personnel Changes

Specific procedures shall be utilized during shift or personnel changes to ensure the continuity of LOTO protection including provision for the orderly transfer of LOTO device protection between off going and oncoming employees to minimize exposure to hazards from the unexpected energization or startup of the machine or equipment or the release of stored energy.

1.497 Outside Personnel / Contractors

Outside personnel / contractors shall be advised that the Company has and enforces the use of LOTO procedures. They shall be informed of the use of locks and tags and notified about the prohibition of attempts to restart or re-energize machines or equipment that are locked out or tagged out.

The company shall obtain information from the outside personnel / contractor about their LOTO procedures and advise affected employees of this information.

The outside personnel / contractor shall be required to sign a certification form. If outside personnel / contractor has previously signed a certification that is on file, additional signed certification is not necessary.

Appendix 33 **Lockout Tagout Form**

Compliance
All employees are required to comply with the restrictions and limitations imposed upon them during the use of LOTO. Authorized employees are required to perform equipment isolation in accordance with procedure. Disciplinary action will be taken against all violators.

General Information	
Department:	
Equipment / System Name:	
Location of Equipment:	
Reason for Isolation:	

Authorized Personnel	**Affected Personnel**

Hazard Assessment (check all that apply)

☐	Electrical	☐	Thermal (cryogenic)
☐	Chemical	☐	Radiation, ionizing
☐	Pressure (hydraulic, pneumatic)	☐	Radiation, non-ionizing
☐	Mechanical	☐	Stored Energy
☐	Thermal (heat)	☐	Other:

T&T Industrial, Inc.

Lockout Tagout Program

LOTO Sequence			
Step 1 – Notify affected employees that servicing will take place under LOTO.			
Step 2 – Shut down (i.e., turn off) the equipment or system following the normal stopping or shut down procedures.			
Step 3 – Follow the steps below to properly isolate each energy source, apply LOTO device, and verify that the equipment is in a zero-energy state.			
Hazard	**Magnitude**	**Method of Isolation**	**Verification Check**
☐ Addendum – Procedure steps continued on separate page.			
Step 4 – Release / control stored energy (list):			
Step 5 – Attempt to restart equipment; verify that equipment will not start. Return operating control(s) to neutral of "off" position after verifying the isolation of the equipment.			

Return equipment to Service
Step 1 – Check the equipment and the immediate area around the machine to ensure that nonessential items have been removed and that the machine or equipment components are operationally intact.
Step 2 – Check the work area to ensure that all employees have been safely positioned or removed from the area.
Step 3 – Verify that the controls are in neutral or "off" position.
Step 4 – Remove the lockout devices and reenergize the machine or equipment.
Step 5 – Notify the affected employees that the servicing or maintenance is complete, and the machine or equipment is ready to use.

Document Review and Approval				
Authorized Employee Signature:		Date:		Next Review Due:
Supervisor Signature:		Date:		

Appendix 34 **LOTO Certification of Inspection**

General Information			
Authorized Employee:		Supervisor:	
Item or Equipment:		Date:	
Location:			

Inspection			
Equipment Inspected	**Inspector Signature**	**Date**	**Authorized Employee**

Comments / Corrective Actions Taken:

Signatures			
Employee Signature:		Date:	
Supervisor Signature:		Date:	

47. MACHINE GUARDING PROGRAM

1.498 Purpose and Scope

The purpose of this program is to prevent accidents, injuries, and occupational hazards associated with the use of machines by implementing effective machine guarding measures.

This program applies to all T&T Industrial, Inc. employees.

1.499 Machine Guarding

Safeguards are essential for protecting employees from preventable injuries caused by moving machine parts. Any machine part, function, or process that may cause injury must be safeguarded.

Machine guards are barriers that prevent access to the danger area of a machine.

Moving machine parts can cause injuries, such as crushed fingers or hands, amputations, burns, or blindness.

Employees who operate and maintain machinery experience approximately 18,000 amputations, lacerations, crushing injuries, and abrasions, and more than 800 deaths per year.

One or more methods of machine guarding shall be provided to protect the operator and other company employees in the machine area from hazards such as those created by point of operation, ingoing nip points, rotating parts, flying chips, and sparks.

1.500 Injuries

Amputations are one of the most severe and crippling types of injuries in the occupational workplace, and often result in permanent disability.

Amputations are widespread and involve a variety of activities and equipment.

Amputations often occur when the following equipment is unguarded or inadequately safeguarded:

- Mechanical power presses
- Power press brakes
- Roll-forming and roll-bending machines
- Band saws
- Drill presses
- Milling machines

- Shears, grinders, and slitters
- Table and portable saws

1.501 Safeguards

Employees must be protected from amputation hazards through adequate guarding and employee training on how to do the job safely.

The best way to prevent amputations caused by stationary or portable machinery is with machine safeguarding.

Safeguards should be secure and strong, and employees should not be able to bypass, remove, or tamper with them.

Safeguards should not obstruct the operator's view or prevent others from working.

Safeguards must prevent hands, arms, and any other part of an employee's body from contacting dangerous moving parts.

Machine guards protect employees from:

- The point of operation
- In-running nip points
- Rotating parts
- Flying chips and sparks

1.502 Types of Machine Guards

Acceptable guarding methods can include barrier guards, two-hand tripping devices, electronic safety devices, and similar methods.

Types of machine guards include:

- Fixed
- Interlocked
- Adjustable
- Self-adjusting

Fixed guarding is a permanent part of the machine. They may be constructed of sheet metal, screen, wire cloth, bars, or plastic. Fixed guards provide a barrier and allow for stock feeding but do not allow the operator to reach the danger area.

Interlocked guards stop the machine before an employee can reach their hand into the danger area. When an interlocked machine guard is opened or removed, the:

- Tripping mechanism and/or power automatically shuts off or disengages.

- Moving parts of the machine are stopped.
- Machine cannot cycle or be started until the guard is back in place.

Adjustable guards are useful because they allow flexibility in accommodating different sizes of stock. They provide a barrier that may be adjusted to work with a variety of production operations.

Self-adjusting guards open and close to admit stock. They provide a barrier that moves according to the size of the stock entering the danger area.

1.503 Training

Machine operators should receive specific and detailed training in safeguarding against mechanical hazards.

Training should explain:

- All hazards in the work area, including machine-specific hazards.
- Safe work practices and machine operating procedures.
- The purpose and proper use of machine safeguards.
- All procedures for responding to safeguarding problems.

1.504 Personal Protective Equipment (PPE)

When engineering controls cannot fully protect employees, machine operators must wear personal protective equipment (PPE).

Appropriate PPE may include hard hats, eye protection, hearing protection, protective coveralls, special sleeves and gloves, and protective footwear.

Sometimes PPE can create a hazard, like a protective glove becoming caught between rotating parts or loose clothing becoming entangled in rotating spindles or other kinds of moving equipment.

1.505 Safe Practices

Employees shall not wear loose-fitting clothing or jewelry or other items that could become entangled in machinery when working around or with moving parts.

Long hair should be worn under a cap or otherwise contained to prevent entanglement in moving machinery.

The point of operation of machines whose operation exposes an employee to injury, shall be guarded. The guarding device shall be in conformity with any appropriate standards therefore, or, in the absence of applicable specific standards, shall be so designed and constructed as to prevent the operator from having any part of his body in the danger zone during the operating cycle.

When the periphery of the blades of a fan is less than 7 feet above the floor or working level, the blades must be guarded. The guard shall have openings no larger than 1⁄2 inch.

Other safe work practices when working around machinery include:

- Remove slip, trip, and fall hazards from the areas around machines.
- Use drip pans when oiling equipment.
- Remove waste stock as it is generated.
- Make the work area large enough for machine operation and maintenance.
- Place machines away from high traffic areas to reduce employee distraction.
- Guards shall be affixed to the machine where possible and secured elsewhere if for any reason attachment to the machine is not possible.
- Guards shall be such that it does not offer an accident hazard in itself.

48. Manual Lifting Program

1.506 Purpose and Scope

The purpose of this program is to provide guidelines and training to ensure that manual lifting tasks are performed in a manner that minimizes strain on the body and maximizes efficiency.

This program applies to all T&T Industrial, Inc. employees.

1.507 Manual Lifting Program

Anyone may be involved in manually lifting objects while at work or at home. Whether lifting is an everyday job or an occasional task, lifting improperly can cause serious injury regardless of the weight of the object or the physical condition of the person lifting the object.

Always follow proper lifting procedures to reduce the risk of injury. Being physically ready to do the job can further reduce the risk of injury

1.508 Hazard Assessment

Before manual lifting is performed, a hazard assessment shall be completed. The assessment must consider size, bulk, and weight of the object(s), if mechanical lifting equipment is required, if two-man lift is required, whether vision is obscured while carrying, the walking surface and path where the object is to be carried, awkward posture, high-frequency and long duration lifting, inadequate handholds, and environmental factors.

Management must evaluate employee work conditions and work techniques and procedures to assess the risk of injuries and to prevent such injuries in the design phase of the work.

1.509 Training

Training shall include general principles of ergonomics, recognition of hazards and injuries, procedures for reporting hazardous conditions, and methods and procedures for early reporting of injuries. Additionally, job specific training shall be given on safe lifting and work practices, hazards, and controls.

1.510 Injury Investigation

Musculoskeletal injuries caused by improper lifting shall be investigated and documented. Incorporation of investigation findings into work procedures shall be accomplished to prevent future injuries.

Incident investigations and root cause analyses must be performed when injuries related to lifting occur, and corrective actions must be taken

1.511 Controls

Manual lifting equipment shall be used instead of manual lifting where possible. Supervisors shall enforce the use of lifting equipment.

Supervisors must periodically monitor employees for improper manual lifting techniques and provide positive corrections to prevent injuries. When new operations are planned, proper engineering controls such as lift assists, mechanical lifting devices, and other suitable engineering controls must be evaluated to engineer out the hazards caused by manual lifting.

Where use of lifting equipment is impractical or not possible, two-man lifts shall be used.

Engineering controls such as worktable height, ergonomic layout of the workplace, and use of lifts, jacks, and other machinery shall be used to lessen the physical burden of lifting.

Manual lifting equipment such as dollies, hand trucks, lift-assist devices, jacks, carts, and hoists shall be provided for employees.

When moving materials manually, employees shall attach handles or holders to loads. In addition, employees shall wear appropriate personal protective equipment (PPE) and use proper lifting techniques.

1.512 Evaluation

Supervision shall periodically evaluate work areas and employees' work techniques to assess the potential for and prevention of injuries. New operations shall be evaluated to engineer out hazards before work processes are implemented.

49. Manual Material Handling and Storage Program

1.513 Purpose and Scope

The purpose of this program is to establish minimum requirements for safe handling and storage of materials.

This program applies to all T&T Industrial, Inc. employees conducting abrasive blasting.

1.514 Roles and Responsibility

Supervisors shall ensure that only trained personnel perform material handling and handle hazardous materials.

1.515 Procedure

Before a worker manually lifts, lowers, pushes, pulls, carries, handles, or transports a load that could injure the worker, a hazard assessment must be performed that considers: (a) the weight of the load, (b) the size of the load, (c) the shape of the load, (d) the number of times the load will be moved, and (e) the manner in which the load will be moved.

- Supervisors shall give advance consideration to the size, shape, and weight of materials to be handled and plan the most efficient and safest method to accomplish the task. Proper tools shall be provided for the job, and alternate methods should be considered.
- Supervisors shall ensure that the work fits the employee in terms of knowledge and physical abilities.
- When the manual moving of loads or persons compromises the worker's safety, mechanical devices shall be put at their disposal.
- Mobile Lift wheels shall be locked and chocked when in operation.
- Safe lifting techniques shall be employed according to the Back Injury Prevention Program.
- Both temporary and permanent storage shall be neat and orderly. When planning material storage, a minimum of 36 inches of clearance shall be allowed under sprinkler heads. Automatic sprinkler controls and electrical panel boxes shall be kept free and unobstructed.
- There shall be a 3-foot unobstructed access way to fire hoses and extinguishers. Clear access to emergency exits and aisles shall be maintained. Areas immediately outside emergency exits shall be left clear for egress.
- Material shall not be placed within 6 feet of any hoist way or inside floor opening or within 10 feet of an exterior wall which does not extend above the top of the material stored.

- Storage bins and racks that are in good condition shall be used to make storing materials easier and reduce hazards. As necessary, storage racks shall be secured to the wall and/or floor as well as to each other. Damaged racks shall not be used for storage. Employees shall not be allowed to climb racks. Where warranted, racks and bins shall have a weight limit and shall be labeled accordingly. Elevated floors shall also be labeled when necessary, indicating their load capacities.
- The safety of a manual lift depends not only on the weight of the load, but also the horizontal distance of the load from the employee, the vertical distance traveled, the frequency of the lift, and the dimension and weight distribution of the load. All loads that will be handled often and/or manually shall be stored as close to "waist" height as possible

1.516 Training and Documentation

This program will be communicated to all new employees during New Employee Orientation.

Training documentation shall contain employee's name and signature, the name and signature of the trainers, and the dates of training.

50. Mobile Equipment Safety Program

1.517 Purpose and Scope

The purpose of this program is to prevent accidents, injuries, and occupational hazards associated with the use of mobile equipment by implementing effective safety measures and promoting a culture of safety and compliance.

This program applies to all T&T Industrial, Inc. employees who work with or near mobile equipment.

1.518 Mobile Equipment Program

The safe operation of mobile equipment on and around the jobsite is essential.

A vast majority of incidents can be avoided when employees follow some simple safety rules.

1.519 Operators

Operators of mobile equipment shall have sufficient education, training, and experience. No employee is allowed to operate mobile equipment unless the employee:

- Is trained to safely operate the equipment,
- Has demonstrated competency in operating the equipment to a designated competent employee,
- Is familiar with the equipment's operating instructions, and
- Is authorized by the company to operate the equipment.

1.520 Inspection

An inspection shall be completed before mobile equipment is used on each shift to ensure that it is in safe operating condition and that no one will be endangered by the start-up of the equipment. A checklist shall be used for pre-use inspections.

1.521 Safe Work Practices

Mobile equipment shall have a working signal alarm while backing up. The operator shall make sure the warning signal is operating when the equipment is backing up.

If so equipped, before starting the engine and while the equipment is in use, the operator shall fasten seat belts and adjust them for a proper fit.

The operator shall not load mobile equipment beyond its established load limit and shall not move loads which because of the length, width, or height that have not been centered and secured for safe transportation.

51. Noise Awareness, Exposure, and Hearing Conservation Program

1.522 Purpose and Scope

The purpose of this program is to provide requirements to minimize employee hearing loss caused by excessive occupational exposure to noise.

This program applies to all T&T Industrial, Inc. employees working in or near high noise areas.

1.523 Resources

Number	Title
29 CFR 1910 Subpart G	Occupational Health and Environmental Control-Occupational Noise Exposure
29 CFR 1926 Subpart D	Occupational Health and Environmental Control-Occupational Noise Exposure
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders-Occupational Noise and Ergonomics

1.524 Definitions

Acronym/Term	Definition
Audiometric Testing	Detection by the person being tested of a series of pure tones. For each tone, the person indicates the lowest level of intensity that they are able to perceive.
Decibels	The sound energy measured by a sound level meter using the "A" scale. The "A" scale is electronically weighted to simulate the response of the human ear to high and low frequency noise.
Standard Threshold Shift	A change in hearing threshold relative to the baseline audiogram of an average of 10 dB (corrected for age) at 2000, 3000 and 4000 Hz in either ear.

1.525 Noise Awareness, Exposure, and Hearing Conservation Program

Occupational hearing loss is a cumulative result of repeated or continued absorption of sound energy by the ear; employee protection is based on reduction of the noise level at the ear or limiting the employee's exposure time.

A continuing effective hearing conservation program shall be administered when employees are exposed to sound levels greater than 85 decibels on an 8-hour time-weighted average (TWA) basis.

Employees exposed to noise at or above 85 dBA as an eight-hour time weighted average must be included in the company's hearing conservation program.

Whenever practical, noise levels identified as exceeding 85 decibels shall be reduced by means of engineering or administrative controls, including isolation, enclosure, and application of noise-reduction materials.

1.526 Training

A training program shall be provided for all employees who are exposed to noise at or above an 8-hour TWA of 85 decibels. The training shall be provided initially and repeated annually for each employee. Employees shall be trained and have access to information and training materials regarding hearing protection.

Training shall include:

- The effects of noise on hearing,
- The purpose of hearing protection,
- The advantages and disadvantages and attenuation of various types of protection,
- Instruments on selection, fitting, use, and care of protection, and
- Techniques for selection, use, and the purpose of audiometric testing along with an explanation of the test procedures.
- The company is required to provide affected company workers exposed to the action level of the Occupational Safety and Health Administration's (OSHAs) of 85dBA of noise as calculated as an eight-hour time-weighted average (TWA), with appropriate hearing protectors at no cost to the company employees

Training shall be updated to stay current with any changes in processes.

1.527 Hearing Protection

Hearing protection shall be available to and worn by all employees exposed to an 8-hour TWA of 85 decibels or greater at no cost to the employee. Hearing protection shall be replaced, as necessary.

Each company employee who is exposed to noise at or above 85dBA as calculated as an eight-hour time weighted average (TWA) must be included in the hearing conservation program.

Hearing protection shall be evaluated for the specific noise environments in which the protection will be used.

Employees shall be given the opportunity to select their hearing protection from a variety of suitable hearing protectors provided by the Company.

Hearing protection shall be worn by any employee that has been provided hearing protection. Employees shall wear hearing protection in signed areas while at a host facility.

Unless a physician determines that the standard threshold shift is not work related or aggravated by occupational noise exposure, the following steps shall be taken:

- Employees not using hearing protectors shall be fitted with hearing protectors, trained in their use and care, and required to use them.
- Employees already using hearing protectors shall be refitted and retrained in the use of hearing protectors and provided with hearing protectors offering greater attenuation if necessary.
- Employees shall be referred for a clinical audiological evaluation or an ontological examination, as appropriate, if additional testing is necessary or if it is suspected that a medical pathology of the ear is caused or aggravated by the wearing of hearing protectors.

The company is responsible for ensuring proper initial fitting of hearing protection and for supervising the correct use of all hearing protection.

1.528 Monitoring

Noise measuring shall be done to see if employees are being exposed to noise that is 85 decibels or louder on an 8-hour TWA. This measuring can be either sampling performed when needed or monitoring performed all the time.

Monitoring shall be repeated whenever a change in production, process, equipment, or controls increases noise exposures to the extent that additional employees may be exposed at or above the action level or the attenuation provided by hearing protectors being used by employees may be rendered inadequate.

1.529 Audiometric Testing

Employees who are exposed to noise that is 85 decibels or louder on an 8-hour TWA shall have hearing tests (also called audiograms or audiometric testing) available to them at no cost to the employees. These tests help by showing any hearing loss that might be happening and shall be done every year after the baseline test.

A baseline, or initial, hearing test shall be done within 6 months if an employee has been exposed to noise that is 85 decibels or louder on an 8-hour TWA. Testing to establish a baseline audiogram shall be preceded by at least 14 hours without exposure to workplace noise. Hearing protection may be used to meet the requirement. Employees shall be notified of the need to avoid high levels of noise.

If a mobile testing van for hearing tests is used, the baseline can be established within one year.

At least annually after obtaining the baseline audiogram, a new audiogram shall be obtained for each employee exposed at or above an 8-hour TWA of 85 decibels. Each employee's annual audiogram shall be compared to that employee's baseline audiogram to determine if the audiogram is valid and if a standard threshold shift has occurred. If a comparison of the annual audiogram to the baseline audiogram indicates a standard threshold shift, the employee shall be informed of this fact in writing within 21 days of the determination.

1.530 Signage

Equipment or areas with noise levels equal to or exceeding 85 decibels shall be identified with labels or signs, which are posted on the individual pieces of equipment (whether owned or leased) or at the entrance to noisy areas. The sign or label shall state either "Hearing Protection Is Required While the Equipment Is Operating" or "Hearing Protection Is Required While Working in the Area" or similar wording, as appropriate.

Equipment typically requiring labels includes but is not limited to compressors, forklifts, generators, and pneumatic tools.

Labels shall be placed where the operator can readily see the warning, such as next to power switches.

1.531 Recordkeeping

Accurate records of all employee exposure and audiometric measurements shall be maintained as required by the regulation. The record shall include:

- Name and job classification of the employee.
- Date of the audiogram.
- The examiner's name.
- Date of the last acoustic or exhaustive calibration of the audiometer.
- Employee's most recent noise exposure assessment.

Noise exposure measurement records shall be retained for 2 years. Audiometric test records shall be retained for the duration of the affected employee's employment.

All records shall be provided upon request to employees, former employees, representatives designated by the individual employee, and the Assistant Secretary.

If the company ceases to do business, all records shall be transferred to the successor company for retention.

52. NON-DOT DRUG AND ALCOHOL POLICY

1.532 Purpose and Scope

The purpose of this policy is to eliminate drug and alcohol abuse among employees and to prevent the threat to the health and safety of employees and to the security of the company's equipment and facilities caused by drug and alcohol abuse.

This policy applies to all T&T Industrial, Inc. employees.

1.533 Non-DOT Drug and Alcohol Policy

The use, sale, purchase, transfer, possession, or presence in one's system of any controlled substance (except medically prescribed drugs) by any employees while on the Company's premises, engaged in Company business, operating Company equipment, or while under the authority of the Company is strictly prohibited.

The unauthorized use, sale, purchase, transfer, possession, or presence in one's system of alcohol or any other intoxicating agent by any employee while on the Company's premises, engaged in Company business, operating Company equipment, or while under the authority of the Company is strictly prohibited.

The possession, use, distribution, or sale of illegal drugs or alcohol on Company property, in Company vehicles, or while conducting Company business is strictly prohibited. Any employee or individual found in violation of this policy will be subject to immediate removal from Company property and may face disciplinary action up to and including termination of employment.

1.534 Pre-Employment Testing

Applicants being considered for hire shall pass a drug test before beginning work or receiving an offer of employment. Refusal to submit to testing will result in disqualification of further employment consideration.

1.535 Random Testing

Drug and alcohol testing shall be administered at random times for all employees. Employees shall be chosen through an unbiased selection process. No qualifiers or exclusions can be made to limit the applicability of the random testing requirement.

1.536 Reasonable Suspicion

If a company official or competent person has determined that there is reasonable cause or suspicion that an individual is performing work under the influence, then that individual shall be required to submit to a drug and alcohol test.

1.537 Incidents

Employees involved in a work-related incident where drug or alcohol use can be reasonably suspected as contributing to the incident shall be tested. This does not require testing for all work-related incidents.

1.538 Site Removal

Any employee that receives unacceptable drug and alcohol test results shall not be allowed to work on a client / host site or facility.

1.539 Return to Duty and Follow Up

Employees retained by the Company after a positive test result or a test refusal shall be subject to return-to-duty drug and alcohol testing. No employee shall be permitted to perform any safety-sensitive function until they have received a verified negative drug and alcohol test result. Thereafter, such employees shall be subject to certain follow-up drug and alcohol testing as established by a Substance Abuse Professional (SAP).

1.540 Drug Test Refusal

Drug testing refusal includes:

- Failure to appear at a collection site for any test (except a pre-employment test) within a reasonable time, as determined by the Company. This includes the failure of the employee to appear for a test when called by the Company's third-party administrator.
- Failure to remain at the collection site until the testing process is complete, provided that a person who leaves the testing site before the testing process commences for a pre-employment test is not deemed to have refused to test.
- Failure to provide a specimen.
- Failure to permit a monitored or observed collection if the Company ordered or if the collector required the collection to be monitored or observed or if instructions during monitoring are not followed.

- Failure to provide a sufficient amount of urine specimen, provided the Medical Review Officer (MRO) finds there was no medical reason for the employee to provide insufficient amount of urine.
- Failure or declination to take an additional drug test that the Company or collector has directed.
- Failure to undergo a medical examination or evaluation the MRO or the Company has directed.
- Failure to cooperate with any part of the specimen collection process.
- Possessing or wearing a prosthetic or other device that could be used to interfere with the collection process if the employee is found to have or wear a prosthetic or other device designed to carry clean urine or a urine substitute.
- Admitting to the collector to having adulterated or substituted the specimen.
- Adulterated or substituted a urine specimen.
- Admitting to the MRO to having adulterated or substituted the specimen.

1.541 Alcohol Testing Refusal

Alcohol testing refusal includes:

- Failure to appear at an alcohol test site for any test within a reasonable time, as determined by the Company. This includes the failure of the employee to appear for a test when called by the Company's third-party administrator.
- Failure to remain at the alcohol test site until the testing process is complete.
- Failure to provide an adequate amount of saliva or breath.
- Failure to provide a sufficient breath specimen, provided the physician finds that there was no medical reason for the employee to provide an insufficient amount of breath.
- Failure to undergo a medical examination or evaluation as the Company has directed as part of the insufficient breath procedures.
- Failure to cooperate with any part of the testing process.

1.542 Searches

To maintain a safe, productive, and drug-free workplace, the Company reserves the right to conduct searches of all employees, visitors, contractors, and their personal property while on company premises. This includes, but is not limited to, bags, lockers, desks, vehicles, and other containers brought onto company property.

By entering Company property, all individuals consent to such searches when requested. Refusal to comply with a search request may result in denial of access to company property, disciplinary action up to and including termination of employment, or removal from the premises.

The purpose of this policy is to ensure compliance with our drug and alcohol policy, protect the safety of all personnel, and maintain a secure work environment.

53. Pandemic Preparedness Program

1.543 Purpose and Scope

The purpose of this program is to ensure employees are informed, prepared, and protected from potential pandemic exposure and to ensure early precaution and preventive actions are in place to stop the possible threat in the workplace.

This program applies to all T&T Industrial, Inc. employees.

1.544 Resources

Number	Title
N/A	Centers for Disease Control and Prevention
CMS-FM-0046	Pandemic Health Screening Tool for Employees

1.545 Definitions

Acronym/Term	Definition
Pandemic	A disease prevalent throughout an entire country, continent, or the whole world; epidemic over a large area.

1.546 Pandemic Preparedness Program

Disease can spread quickly from sick employees to others who are nearby in the workplace. Employees are often in close contact, sharing the same space, supplies, and equipment for long periods of time. As a result, there is an increased risk that employees will spread illnesses to each other.

1.547 Roles and Responsibilities

In the event of a disease outbreak, ownership of this program shall be assigned to the Safety Coordinator who shall be responsible for dealing with disease issues and their impact at the workplace. Should it be necessary, health care providers shall be contracted in advance to develop and implement protocols for response to ill employees.

1.548 Prevention of Spread

A workplace hazard assessment specific to pandemic purposes shall be conducted.

Any employee, in the event of possible exposure shall self-quarantine and seek medical screening and testing as appropriate and notify their manager or supervisor.

Hand washing and use of hand sanitizers shall be encouraged by company supervision. Hand washing facilities, hand sanitizers, tissues, no touch trash cans, hand soap, and disposable towels shall be provided at no cost to employees.

Consideration shall be given to social distancing including increasing space between employee work areas and decreasing the possibility of contact by limiting large or close contact gatherings.

Flexible work schedules shall be established when possible. Employees are encouraged to stay home when ill, when having to care for others that are ill, or when under such orders from governmental authorities.

Areas that are likely to have frequent hand contact such as doorknobs, faucets, handrails, etc. shall be cleaned periodically and when visibly soiled. Employee work surfaces shall also be cleaned frequently with normal cleaning products.

Upon return to work, the incident management team shall devise a plan to prevent further spread and authorization for employees to return.

Employees are encouraged to obtain appropriate immunizations, when available, to help avoid disease.

An effective internal and external communications plan with contact information for employees and customers shall be developed, including a method for managing local regulatory reporting requirements.

1.549 Training

Employees shall be trained on:

- Health issues of the pertinent disease to include prevention of illness.
- Initial disease symptoms.
- Preventing the spread of the disease.
- When it is appropriate to return to work after illness.

Disease containment plans and expectations shall be shared with employees. Communicating information with non-English speaking employees or those with disabilities shall be considered.

1.550 Business Continuity

Business continuity plans shall be prepared so that if significant absenteeism or changes in business practices are required business operations can be effectively maintained. Tele-commuting or other work-at-home strategies should be developed. The plan shall include a chain of communications, contact information for employees, and a process for tracking business and employee status.

If the company's ability to perform regular services is impacted, key contacts such as customers and suppliers shall be notified of this impact and when operations resume.

1.551 Testing and Lessons Learned

The plan and emergency communication strategies shall be periodically tested to ensure it is effective and workable.

Following a pandemic event, the program owner shall identify lessons learned and take action to implement any corrective actions.

Appendix 35 **Signs and Symptoms of COVID-19**

People with COVID-19 have had a wide range of symptoms reporting – ranging from mild symptoms to severe illness.

These symptoms may appear 2-14 days after exposure of the virus. People with these symptoms may have COVID-19:

- Fever
- Cough
- Shortness of breath or difficulty breathing
- Fatigue
- Muscle of body aches
- Headache
- New loss of taste or smell
- Sore throat
- Congestion or runny nose
- Nausea or vomiting
- Diarrhea

Look for emergency warning signs for COVID-19. If someone is showing any of these signs, seek emergency medical care immediately:

- Trouble breathing
- Persistent pain or pressure in the chest
- New confusion
- Inability to wake or stay awake
- Bluish lips or face

Appendix 36 **Pandemic Health Screening Tool for Employees**

General Information			
Employee Name:		Date:	
Department:		Time In:	
Escort if Visitor:			
In the past 48 hours, have you experienced?			
Subjective fever (felt feverish) or chills:	☐ Yes	☐ No	
New or worsening cough:	☐ Yes	☐ No	
Shortness of breath:	☐ Yes	☐ No	
Fatigue:	☐ Yes	☐ No	
Muscle or body aches:	☐ Yes	☐ No	
Headache:	☐ Yes	☐ No	
New loss of taste or smell:	☐ Yes	☐ No	
Sore throat:	☐ Yes	☐ No	
Congestion or runny nose:	☐ Yes	☐ No	
Nausea or vomiting:	☐ Yes	☐ No	
Diarrhea:	☐ Yes	☐ No	
Current temperature:			

If you answer "yes" to any of the symptoms listed above, or your temperature is 100.4 °F or higher, notify your manager and do not go into work. Self-isolate at home and contact your healthcare provider for advice.

- You should isolate at home for 14 days since symptoms first appeared.
- You must also have 3 days without fever and improvement in respiratory symptoms.

Have you?		
Tested positive for COVID-19 in the past 10 days?	☐ Yes	☐ No
Are you currently awaiting results from a COVID-19 test?	☐ Yes	☐ No
Been diagnosed with COVID-19 by a licensed healthcare provider (for example, a doctor, nurse, pharmacist, or other) in the past 10 days?	☐ Yes	☐ No
Been told that you are suspected to have COVID-19 by a licensed healthcare provider in the past 10 days?	☐ Yes	☐ No
In the past 14 days, had close contact with an individual diagnosed with COVID-19?	☐ Yes	☐ No
In the past 14 days, traveled via airplane internationally or domestically (unless exempt)?	☐ Yes	☐ No

If you answered "yes" to either of these questions, notify your manager and do not go into work (unless a travel exempt employee). Self-quarantine at home for 14 days.

Daily monitoring for potential COVID-19 symptoms is important to track your current health status. If you experience new symptoms, consider seeing your healthcare provider or getting a test for COVID19, especially where you may have had potential exposures to COVID-19.

You should also monitor your health and consider consulting your primary care physician after testing positive for COVID-19.

Contact your supervisor or Human Resources to report health symptoms.

You MUST inform your supervisor if you:

- Receive a confirmed positive COVID-19 test result
- Have been diagnosed with COVID-19 by a licensed healthcare provider
- Have been told you are suspected to have COVID-19 by a licensed healthcare provider
- Experience new loss of taste and/or smell with no other explanation
- Experience both fever (≥100.4° F) and new unexplained cough associated with shortness of breath

54. Personal Protective Equipment (PPE) Program

1.552 Purpose and Scope

The purpose of this program is to establish minimum requirements to protect employees through the use of personal protective equipment (PPE).

This program applies to all T&T Industrial, Inc. employees.

1.553 Resources

Number	Title
29 CFR 1910 Subpart I	Personal Protective Equipment
29 CFR 1926 Subpart E	Personal Protective and Life Saving Equipment
Cal/OSHA T8CCR Subchapter 7	General Industry Safety Orders-Safe Practices and Personal Protection
CMS-FM-0047	Personal Protective Equipment (PPE) Hazard Assessment Form

1.554 Personal Protective Equipment (PPE) Program

Personal Protective Equipment (PPE) is equipment worn to minimize exposure to serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards.

PPE shall be provided at no cost to the employee with the exception of non-specialty safety-toe footwear and non-specialty prescription safety eyewear if permitted to be worn off the jobsite.

1.555 Training

Training shall be provided to each employee who is required to use PPE. Proper training includes at least:

- When PPE is necessary.
- What PPE is necessary.
- How to properly don, doff, adjust, and wear PPE.
- The limitations of PPE.
- The proper care, maintenance, useful life, and disposal of PPE.

Training shall be documented including the employee name, the dates of training, and the training content.

Retraining is required when the workplace changes, making the earlier training obsolete, the type of PPE changes, or when the employee demonstrates lack of use, improper use, or insufficient skill or understanding. The certification must include the employee's name, the dates of training, and the certification subject.

1.556 PPE Maintenance

PPE shall be provided, used, and maintained in a sanitary and reliable condition wherever it is necessary by reason of hazards, of processes, or environment to protect body parts from inhalation, absorption, or physical contact. This applies to Company issued PPE or employee-owned PPE, if employee-owned PPE is allowed based on jurisdiction.

1.557 Hazard Assessment

A workplace hazard assessment shall be conducted to determine if hazards are present or are likely to be present, which necessitate the use of PPE. Verification shall be conducted to ensure the required workplace hazard assessment has been performed through a written certification that identifies the workplace evaluated, the person certifying that the evaluation has been performed, the date(s) of the hazard assessment, and identification of assessment documents.

If hazards are present, or likely to be present, the Company shall select and have each affected employee use the type of PPE necessary, communicate selection decisions, and select PPE that properly fits each affected employee.

Consideration shall be given to comfort and fit. PPE that fits poorly will not afford the necessary protection. Continued wearing of the PPE is more likely if it fits the wearer comfortably. PPE is generally available in a variety of sizes. Care should be taken to ensure that the right size is selected.

1.558 PPE Inspection

All PPE shall meet the appropriate safety standards and regulations.

PPE shall be inspected regularly to identify any defects, damage, or signs of wear that may compromise its effectiveness.

Any damaged, defective, or expired PPE shall be immediately removed from service and replaced with new or properly functioning equipment.

1.558.1 Initial Inspection

Upon receipt of new PPE, it shall be inspected to ensure it is in good condition and free from defects.

Inspections should include checking for any visible damage, missing components, or signs of wear.

1.558.2 Pre-Use Inspection

Before using any PPE, employees shall inspect it for any visible damages or defects.

Inspections should include checking straps, buckles, lenses, shields, or any other components for proper functionality.

If any issues are identified during the pre-use inspection, the PPE should not be used, and the employee shall report it to their supervisor.

1.558.3 Routine Inspections

Regular inspections of PPE shall be conducted at predetermined intervals.

The frequency of routine inspections may vary depending on the type of PPE and the nature of the work environment.

Inspections should be documented, and records of inspections shall be maintained for future reference.

1.558.4 Post-Incident Inspections

Following any incident or accident where PPE was involved, the PPE shall be inspected to determine its integrity and effectiveness.

Damaged or compromised PPE shall be immediately removed from service and replaced.

1.559 Defective PPE

Defective or damaged PPE shall not be used. PPE that is in disrepair shall be discarded or removed from service until repaired.

1.560 Employee-Owned PPE

Employee-owned PPE is allowed, and the Company is responsible for the assurances of its adequacy, maintenance, and sanitation.

1.561 Types of Protection

The correct level of PPE determined by the assessment, shall be worn at all times. PPE may include:

- Coveralls
- Flame Resistant Clothing (FRC)
- Hand Protection (gloves-chemical resistant, anti-impact, leather, rubber)
- Foot Protection (steel toe boots, shoes, toe covers)
- Head Protection
- Eye and Face Protection (safety glasses, shields)
- Hearing Protection
- Respiratory Protection
- Fall Prevention and Protection

Appendix 37 Personal Protective Equipment (PPE) Hazard Assessment Form

General Information			
Employee Name:		Date:	
Location:			

Instructions

1) *Complete this form for each task to document evaluation of the workplace hazards that necessitate the use of PPE.*

2) *Provide training and document on the training roster.*

NOTE: These PPE controls should be used in conjunction with other controls (engineering, administrative, and work practices).

Hazard Assessment

Task	Hazard	PPE Required
		☐ Eye / Face:
		☐ Body:
		☐ Hand:
		☐ Foot:
		☐ Other:
		☐ Eye / Face:
		☐ Body:
		☐ Hand:

		☐ Foot:
		☐ Other:
		☐ Eye / Face:
		☐ Body:
		☐ Hand:
		☐ Foot:
		☐ Other:
		☐ Eye / Face:
		☐ Body:
		☐ Hand:
		☐ Foot:
		☐ Other:

Certification

By signing this form, the individual certifies that a workplace hazard assessment has been performed in accordance with OSHA requirement.

Employee Signature:		Date:	

55. Preventative Maintenance Program

1.562 Purpose and Scope

The purpose of this program is to deliver preventative maintenance requirements to reduce employee injuries and equipment and facility damage.

This program applies to all T&T Industrial, Inc.

1.563 Resources

Number	Title
CMS-FM-0045	Preventative Maintenance Schedule

1.564 Preventative Maintenance Program

Well-developed preventive maintenance plans and proper employee training will result in fewer and less severe employee injuries and less equipment and facility damage.

1.565 Inventory

An inventory of machinery / equipment shall be established and kept current. When new machinery or equipment is acquired, it shall be added to the inventory.

The equipment inventory is established and maintained.

1.566 Schedule

A preventative maintenance schedule shall be established based on manufacturer requirements and industry standards.

1.567 Records

Preventative maintenance performed on machinery or equipment shall be documented and retained for the life of the machinery or equipment.

1.568 Defective Equipment

Defects observed in machinery or equipment shall be reported to a supervisor and shall be removed from service and repaired or replaced before being used again.

Appendix 38 **Preventative Maintenance Schedule**

Schedule							
Place a tick mark in each box according to the manufacturer recommendations.							
Equipment Name	**Daily**	**Weekly**	**Quarterly**	**Bi-annually**	**Annually**	**Biennial**	**As Needed**

56. RESPIRATORY PROTECTION PROGRAM

1.569 Purpose and Scope

The purpose of this program is to ensure that all employees required to wear respiratory protection are protected from respiratory hazards through the proper use of respirators.

This program applies to all T&T Industrial, Inc. employees that use respiratory protection.

1.570 Resources

Number	Title
29 CFR 1910 Subpart I	Personal Protective Equipment-Respiratory Protection
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders-Control of Hazardous Substances

1.571 Definitions

Acronym/Term	Definition
Employee Exposure	Exposure to a concentration of an airborne contaminant that would occur if the employee were not using respiratory protection.
Escape-Only Respirator	A respirator intended to be used only for emergency exit.
Fit Test	The use of a protocol to qualitatively or quantitatively evaluate the fit of a respirator on an individual.
Immediately Dangerous to Lift or Health (IDLH)	An atmosphere that poses an immediate threat to life, would cause irreversible adverse health effects, or would impair an individual's ability to escape from a dangerous atmosphere.
Qualitative Fit Test (QLFT)	A pass / fail fit test to assess the adequacy of respirator fit that relies on the individual's response to the test agent.
Quantitative Fit Test (QNFT)	An assessment of the adequacy of respirator fit by numerically measuring the amount of leakage into the respirator.
Tight-Fitting Facepiece	A respiratory inlet covering that forms a complete seal with the face.
User Seal Check	An action conducted by the respirator user to determine if the respirator is properly seated to the face.

1.572 Respiratory Protection Program

The Respiratory Protection Program is directed to ensure employees are protected from hazardous atmospheres through a comprehensive program of recognition, evaluation, engineering, administrative controls, work practice controls, and personal protective equipment. Hazard elimination and engineering, along with work practice controls, shall be employed to control employee exposure to within allowable exposure limits as much as possible. Respirators and other personal protective equipment (PPE) shall be provided to affected employees under this program.

1.573 Training

Effective training shall be provided to employees who are required to use respirators. Training shall be provided prior to requiring the employee to use a respirator in the workplace. Retraining shall be administered annually, and when the following situations occur:

- Changes in the workplace or the type of respirator render previous training obsolete.
- Inadequacies in the employee's knowledge or use of the respirator indicate that the employee has not retained the requisite understanding or skill.
- Any other situation arises in which retraining appears necessary to ensure safe respirator use.

Each employee shall demonstrate knowledge of at least the following:

- Why the respirator is necessary and how improper fit, usage, or maintenance can compromise the protective effect of the respirator.
- What the limitations and capabilities of the respirator are.
- How to use the respirator effectively in emergency situations, including situations in which the respirator malfunctions.
- How to inspect, put on and remove, use, and check the seals of the respirator.
- What the procedures are for maintenance and storage of the respirator.
- How to recognize medical signs and symptoms that may limit or prevent the effective use of respirators.

1.574 Respirator Use

Engineering controls shall be used (e.g., enclosure or confinement of the operation, ventilation, substitution of less toxic materials) to minimize exposure. For employees who are exposed to airborne contaminants above the permissible exposure limit (PEL) and when effective engineering controls are not feasible, or while they are being instituted, or during emergency situations, appropriate respirators shall be provided and used to protect employees against harmful vapors and oxygen-deficient atmospheres. A respirator shall be provided to each employee when such equipment is necessary to protect the health of such employee, at no cost to the employee. Respirators shall be provided which are applicable and suitable for the purpose intended.

Before employees voluntarily use a respirator, ensure that its use does not present a health hazard. Any employee using a respirator voluntarily must be medically able to use that respirator, and the respirator must be cleaned, stored, and maintained so that its use does not present a health hazard.

1.574.1 Precautions for Voluntary Use

Take the following precautions to ensure that the respirator itself does not present a hazard:

- Read and heed all instructions provided by the manufacturer on use, maintenance, cleaning and care, and warnings regarding the respirator limitations.
- Choose respirators certified for use to protect against the contaminant of concern. NIOSH, the National Institute for Occupational Safety and Health of the U.S. Department of Health and Human Services, certifies respirators. A label or statement of certification should appear on the respirator or respirator packaging. It will list what the respirator is designed for and the level of protection.
- Do not wear respirators into atmospheres containing contaminants for which it is not designed to protect against. For example, a respirator designed to filter dust particles will not protect against gases, vapors, or very small solid particles of fumes or smoke.
- Keep track of respirators so that they are not mistakenly used by someone else.

1.575 Operating Practices

Employees using atmosphere-supplying respirators (supplied-air and SCBA) must be supplied with breathing gases of high purity meeting all jurisdictional requirements. Compressed breathing air shall meet at least the requirements for Grade D breathing air described in ANSI/Compressed Gas Association Commodity Specification for Air, G-7.1

The Company must ensure that, on an annual basis, breathing air:

- Has 19.5 to 23.5% oxygen
- Has less than 5 milligrams per cubic meter of hydrocarbons (i.e., oils)
- Has less than 10 ppm of carbon monoxide
- Has less than 1000 ppm of carbon dioxide and
- Lack of "noticeable odor".

People shall not be cleaned off with compressed air. The regulations relate to the cleaning of objects or items, (e.g., blow drying parts, etc.).

It is prohibited to use compressed air to clean clothing while worn or any part of the body under any circumstances.

Compressed air shall not be used for cleaning or blow down activities unless air pressure is regulated to below 30 psi. and areas have been isolated from pedestrian traffic and effective chip guarding and personal protective equipment is implemented.

All compressed gas cylinders shall be handled, stored, received, and used in a safe manner consistent with OSHA requirements. Any material that is under pressure can be dangerous if it is not handled properly. If the material is a compressed gas it may be flammable, explosive, reactive, toxic, or any combination of these. Because of the hazards of compressed gasses, it is important to know what you are working with, what its hazardous properties are, and how to safely handle the compressed gas cylinder.

1.576 Program Administrator

The Safety Coordinator is qualified by appropriate training or experience that is commensurate with the complexity of the program to oversee the program and conduct the required evaluations of program effectiveness.

1.577 Program Evaluation

To verify written program effectiveness evaluations of the workplace shall be conducted as necessary to ensure implementation. Employees shall be regularly consulted about fit, selection, use, maintenance, etc., and overall program effectiveness.

Factors to be assessed include, but are not limited to:

- Respirator fit (including the ability to use the respirator without interfering with effective workplace performance).
- Appropriate respirator selection for the hazards to which the employee is exposed.
- Proper respirator use under the workplace conditions the employee encounters.
- Proper respirator maintenance.

1.578 Selection of Respirators

Appropriate respirators shall be selected and provided based on the respiratory hazards to which employees are exposed and workplace and user factors that affect respirator performance and reliability. National Institute for Occupational Safety and Health (NIOSH) certified respirators shall be selected. The respirator shall be used in compliance with the conditions of its certification.

Respirators shall be selected from a sufficient number of respirator models and sizes so that the respirator is acceptable to, and correctly fits, the user.

The respiratory hazard(s) in the workplace shall be identified and evaluated; this evaluation shall include a reasonable estimate of employee exposures to respiratory hazard(s) and an identification of the contaminant's chemical state and physical form. Where employee exposure cannot be identified or reasonably estimated, the atmosphere shall be considered as immediately dangerous to life or health (IDLH).

1.579 Medical Evaluation

A medical evaluation shall be provided to determine the employee's ability to use a respirator before fit tested or required use of the respirator. A physician or other licensed health care professional shall be identified to perform medical evaluations using a medical questionnaire or an initial medical examination that obtains the same information as the medical questionnaire. The medical questionnaire and examinations shall be administered confidentially during the employee's normal working hours.

When there is a need for a follow-up medical examination, it shall include any medical tests, consultations, or diagnostic procedures that the physician or licensed health care professional deems necessary to make a final determination.

1.580 Fit Test

Employees using a tight-fitting facepiece respirator shall pass an appropriate qualitative (QLFT) or quantitative (QNFT) fit test. Employees using tight-fitting facepiece respirators shall be fit tested prior to initial use of the respirator, whenever a different respirator facepiece (size, style, model, or make) is used, and at least annually thereafter.

Tight-fitting facepieces may not be used by employees who:

- Have facial hair that comes between the sealing surface of the facepiece and the face or that interferes with valve function.
- Any condition that interferes with the face-to-facepiece seal or valve function.
- Wear corrective glasses or goggles that cause seal leakage.

Employees shall perform a user seal check each time they put on the respirator.

1.581 Surveillance

Employees using respirators shall be under surveillance to ensure they leave the area to wash their faces and respirator facepieces, change cartridges, or if they detect vapor or gas breakthrough or breathing resistance.

1.582 IDLH Atmospheres

For all IDLH atmospheres, the following shall be in place:

- One or more employees are located outside the IDLH atmosphere.
- Visual, voice or signal line communication is maintained between the employees in the IDLH atmosphere and those located outside.
- The employees located outside the IDLH atmosphere are trained and equipped to provide effective emergency rescue.
- A company designate is notified before the employees located outside the IDLH atmosphere enter the IDLH atmosphere to provide emergency rescue.
- The company designate provides necessary assistance appropriate to the situation.
- Employees located outside the IDLH atmospheres are equipped with proper equipment for rescuing employees who enter these hazardous atmospheres.

1.583 Maintenance and Storage

Employees shall be provided respirators that are clean, sanitary, and in good working order. Respirators shall be cleaned and disinfected as often as necessary to be maintained in a sanitary condition. All respirators shall be stored to protect them from damage, contamination, dust, sunlight, extreme temperatures, excessive moisture, and damaging chemicals, and they shall be packed or stored to prevent deformation of the facepiece and exhalation valve.

Emergency respirators shall be kept accessible in the work area, stored in compartments or in covers that are clearly marked as containing emergency respirators, and stored in accordance with any applicable manufacturer instructions.

1.584 Inspection

All respirators used in routine situations shall be inspected before use and during cleaning. Respirators maintained for use in emergency situations shall be inspected at least monthly and shall be checked for proper function before and after each use. Emergency escape-only respirators shall be inspected before being carried into the workplace for use. Inspections shall include a check of respirator function, tightness of connections, and the condition of the various parts including, but not limited to, the facepiece, head straps, valves, connecting tube, and cartridges, canisters, filters, and elastomeric parts.

Respirators maintained for emergency use shall be certified by documenting the date the inspection was performed, the name (or signature) of the person who made the inspection, the findings, required remedial action, and a serial number or other means of identifying the inspected respirator. This information shall be provided on a tag or label that is attached to the storage compartment for the respirator, kept with the respirator, or included in inspection reports stored as paper or electronic files. This information shall be maintained until replaced following a subsequent certification.

Respirators that fail an inspection or are otherwise found to be defective shall be removed from service, and discarded or repaired or adjusted in accordance with the following procedures:

- Repairs or adjustments to respirators are to be made only by persons appropriately trained to perform such operations and shall use only the respirator manufacturer's NIOSH-approved parts designed for the respirator.
- Repairs shall be made according to the manufacturer's recommendations and specifications for the type and extent of repairs to be performed.
- Reducing and admission valves, regulators, and alarms shall be adjusted or repaired only by the manufacturer or a technician trained by the manufacturer.

1.585 Recordkeeping

Written information regarding medical evaluations, fit testing, and the respirator program shall be established and retained and made available.

Records of the qualitative and quantitative fit tests administered shall include:

- The name or identification of the employee tested.
- Type of fit test performed.
- Specific make, model, style, and size of respirator tested.
- Date of test.
- The pass / fail results for QLFTs or the fit factor and strip chart recording or other recording of the test results for QNFTs.

Fit test records shall be retained for respirator users until the next fit test is administered.

A written copy of the current respirator program shall be retained.

Required written materials shall be made available for examination and copying.

57. RIGGING PROGRAM

1.586 Purpose and Scope

The purpose of this program is to provide requirements and guidance to support safe rigging and handling of loads.

This program applies to all T&T Industrial, Inc. employees involved with rigging operations.

1.587 Resources

Number	Title
29 CFR 1926 Subpart H	Material Handling, Storage, Use, and Disposal-Rigging Equipment for Material Handling
29 CFR 1910 Subpart N	Materials Handling and Storage-Slings
29 CFR 1926 Subpart CC	Cranes and Derricks in Construction
Cal/OSHA T8 CCR Subchapter 7	General Industry Orders-Cranes and Other Hoisting Equipment

1.588 Rigging Program

The following requirements apply to slings used in conjunction with other material handling equipment for the movement of materials by hoisting.

1.589 Inspection

Each day and on each shift before being used, the slings and all fastenings and attachments shall be inspected for damage or defects by a qualified person. Additional inspections shall be performed during sling use, where service conditions warrant.

Slings and hooks that are damaged or defective shall not be used. Defective rigging equipment shall be removed from service.

1.590 Qualification

Only qualified riggers may be used during hoisting activities.

A rigger is anyone who attaches or detaches lifting equipment to loads or lifting devices. In order to be considered a qualified rigger, the person shall be qualified by the employer to perform specific rigging tasks and possess a recognized degree, certificate, or professional standing, or has extensive knowledge, training, and experience, and can successfully demonstrate the ability to solve problems related to rigging loads.

1.591 Rated Capacity and Identification

Rigging equipment shall have permanently affixed and legible identification markings as prescribed by the manufacturer that indicate the recommended safe working load. Rigging shall not be loaded in excess of its recommended safe working load as prescribed on the identification markings by the manufacturer; and shall not be used without affixed, legible identification markings.

Chains, wire ropes, synthetic or metal web slings, shackles or any other lifting attachments without permanently affixed and legible identification markings prescribed by the manufacturer shall not be used.

1.592 Safe Use of Rigging

Rigging equipment, when not in use, shall be removed from the immediate work area so as not to present a hazard to employees.

Tag lines shall be used unless their use creates an unsafe condition.

Hooks used in the connection between the hoist line and the personnel platform (including hooks on overhaul ball assemblies, lower load blocks, bridle legs, or other attachment assemblies or components) must be:

- Of a type that can be closed and locked, eliminating the throat opening.
- Closed and locked when attached.

Suspended loads shall be kept clear of all obstructions and all employees shall be kept clear of loads about to be lifted and of suspended loads.

Shock loading is prohibited.

Chain or wire rope slings shall not be shortened with knots or bolts or other makeshift devices. Slings shall not be kinked or knotted.

Slings used in a basket hitch shall have the loads balanced to prevent slippage.

Slings shall be padded or protected from the sharp edges of their loads.

Hands or fingers shall not be placed between the sling and its load while the sling is being tightened around the load.

A sling shall not be pulled from under a load when the load is resting on the sling and damage to the sling may result.

Slings shall be set to avoid slippage.

Rigging shall be used and maintained in accordance with manufacturer's recommendations.

Hooks and shackles shall only be used in a manner recommended by the manufacturer.

Proof coil steel chain, also known as common or hardware chain, or other chain not recommended for slinging or hoisting by the manufacturer, shall not be used for hoisting purposes.

Wrought iron chains in constant use shall be annealed or normalized at intervals not exceeding 6 months when recommended by the manufacturer. The chain manufacturer shall be consulted for recommended procedures for annealing or normalizing. Alloy chains shall not be annealed. Deformed hooks or rings shall be replaced or repaired and reshaped under proper metallurgical control and proof tested. Proper annealing or normalizing procedures done only in accordance with the chain manufacturer's specifications shall be followed.

1.592.1 Proof Testing

Special custom design grabs, hooks, clamps, or other lifting accessories for such units as modular panels, prefabricated structures and similar materials, shall be marked to indicate the safe working loads and shall be proof-tested to 125% of the rated load prior to use.

58. Risk Assessment Program

1.593 Purpose and Scope

The purpose of this program is to identify hazards and evaluate any associated risks to health, safety, and the environment arising from work activities, enabling informed decisions to be made to eliminate or minimize any risk of harm to those affected.

This program applies to all T&T Industrial, Inc. employees.

1.594 Risk Assessment

Risk assessments do not have to be complicated; the level of detail contained in them should be relevant to the level of the risks involved with the activity. Risk assessments may lead to clarification of procedures, identifying efficiencies in existing processes, and identification of training and supervision required for the activity.

1.595 Hazard Identification

Risk assessments shall be conducted prior to the beginning of work to formally identify and assess hazards and correct them in a timely manner. This can be accomplished through Job Safety Analysis (JSA), daily hazard assessments, pre-job hazard assessments, or hazard workplace inspection.

A site safety inspection must be conducted in a timely manner and the form must be signed, dated with site name, as required.

A JSA shall be developed for all routine tasks. Formal workplace inspections for safety hazards of all operations, equipment, work areas, and facilities shall be performed on a regular basis. Risk assessments and JSAs shall be updated whenever changes occur to processes, equipment, work areas, and facilities.

Information shall be collected, organized, and reviewed with employees to determine what types of hazards may be present and which employees may be exposed or potentially exposed. Information available in the workplace may include:

- Equipment and machinery operating manuals.
- Safety Data Sheets (SDS).
- Inspection reports.
- Records of previous injuries and illnesses.
- Incident investigation reports.
- Results of JSAs.

Employees shall be actively involved in the risk identification process. If subcontractors are performing work at the location, they should be included. Identified hazards and risks shall be reviewed with all affected employees.

Additional hazards from workers performing tasks with other trades or crafts must be evaluated.

1.595.1 Health Hazards

Identification of health hazards shall include chemical hazards, physical hazards, biological hazards, and ergonomic risk factors by conducting qualitative exposure assessments and reviewing employee medical records.

1.596 Training

All employees shall be trained on the hazard identification in the workplace, the risk assessment process, and how to report and control hazards using the hierarchy of controls.

1.597 Roles and Responsibilities

Those who lead and perform active roles in this process play a crucial role in identifying and mitigating potential hazards. Following are the key roles and responsibilities for individuals involved in the Hazard Identification process:

Hazard Identification Team Leader:

Overall Responsibility: The team leader is responsible for overseeing the entire Hazard Identification process and ensuring its effectiveness including:

- Planning
- Team Coordination
- Data Collection
- Risk Assessment
- Mitigation Planning
- Documentation
- Review and Continuous Improvement

Hazard Identification Team Members:

- Data Gathering
- Hazard Analysis
- Reporting
- Recommendations
- Follow-up

Subject Matter Experts (SMEs):

- Technical Expertise
- Risk Assessment
- Mitigation Strategies

Safety Officers:

- Compliance
- Training
- Auditing

Management and Leadership:

- Support
- Decision-Making
- Review

Workers and Employees:

- Reporting
- Participation
- Compliance

1.598 Incident Investigations

Workplace incidents including injuries, illnesses, near misses, and stop work interventions shall be investigated to identify the root cause in order to prevent future occurrences.

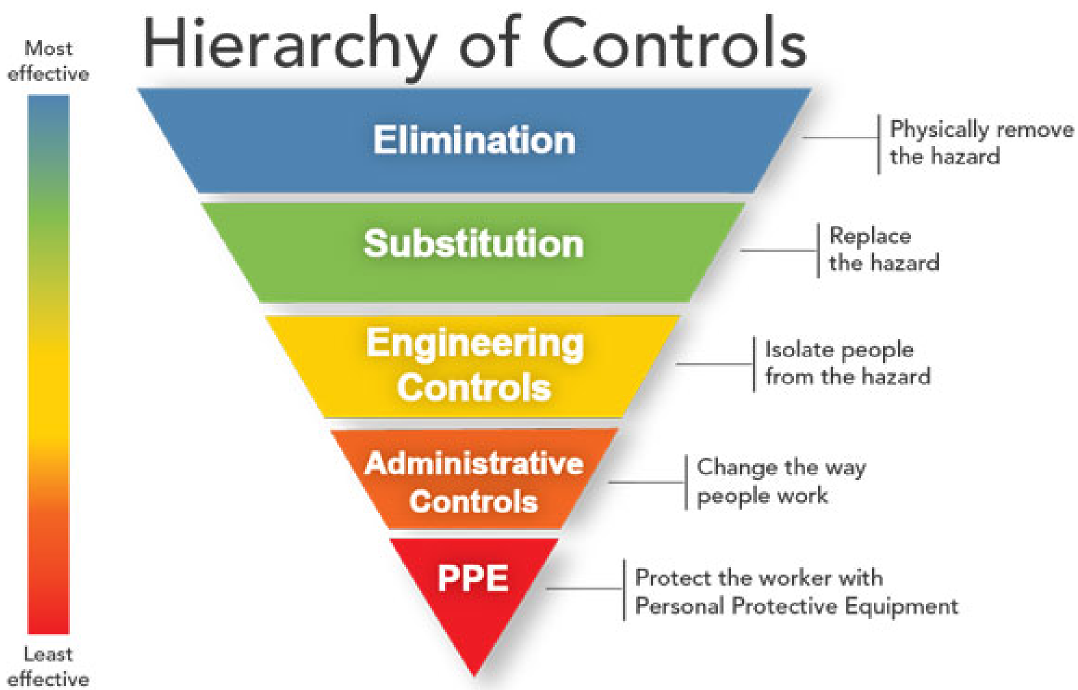
1.599 Hazard Classification and Rank

A formal system for classifying and ranking hazards according to risk shall be followed. Risk shall be determined by analyzing the probability of the hazard causing harm, the frequency the hazard is encountered, and the potential consequences of impact with the hazard. A risk matrix shall be followed to assist with the risk assessment.

Evaluation of each hazard by considering the severity of potential outcomes, the likelihood that an event or exposure will occur, and the number of employees who might be exposed shall be conducted to prioritize the hazards so that those presenting the greatest risk are addressed first.

1.600 Hierarchy of Controls

The hierarchy of controls shall be used to mitigate hazards before employees are exposed. When a hazard is identified, first attempt to eliminate the hazard. If elimination is not practicable, use engineering controls. If engineering controls are not practicable, implement administrative controls. If the hazard cannot be adequately controlled using engineering and/or administrative controls, employees shall use Personal Protective Equipment (PPE). A combination of engineering controls, administrative controls, and PPE is usually best.



1.601 Risk Evaluation and Estimation

1.602 Likelihood and Severity

Once hazards associated with activities have been identified, the likelihood of a hazardous event occurrence and the severity of that occurrence shall be estimated using the levels below:

Likelihood:
1. **Improbable** (unlikely to occur)
2. **Remote** (unlikely, though possible)
3. **Occasional** (likely to occur occasionally during standard operations)
4. **Probable** (not surprised, will occur in given time)
5. **Frequent** (likely to occur, to be expected)

Severity:
1. **Negligible** (the hazard will not result in serious injury or illness, or has a remote possibility of damage)
2. **Marginal** (the hazard could cause illness, injury, or equipment or environmental damage, but its effects would not be serious)
3. **Moderate** (the hazard can result in serious injury or illness, property, or equipment or environmental damage)
4. **Critical** (the hazard can result in serious injury, illness, property, or equipment or environmental damage)
5. **Catastrophic** (the hazard is capable of causing death or illness)

1.603 Risk Matrix

Multiply the hazard's probability and severity to calculate the level of risk. Use the following risk matrix to determine the level of risk. Reduce risk as described in the table below.

	Catastrophic: 5	Critical: 4	Moderate: 3	Marginal: 2	Negligible: 1
Frequent: 5	High – 25	High – 20	Serious – 15	Serious – 10	Medium – 5
Probable: 4	High – 20	Serious – 16	Serious – 12	Medium – 8	Medium – 4
Occasional: 3	Serious – 15	Serious – 12	Medium – 9	Medium – 6	Low – 3
Remote: 2	Serious – 10	Medium – 8	Medium – 6	Medium – 4	Low – 2
Improbable: 1	Medium – 5	Medium – 4	Low – 3	Low – 2	Low - 1

Low	The risk is tolerable, assuming all control measures are fully identified and effectively implemented.
Medium	The risk is tolerable assuming the risk levels have been reduced to as low as reasonably practicable (ALARP).
Serious	The risk is likely tolerable. Reduce to ALARP. May be acceptable where consequences are potentially high, but the likelihood has been reduced significantly.
High	The risk is intolerable. The risk must be reduced to medium or low before work begins.

1.604 Continuous Improvement

A continuous improvement process for lessons learned to be incorporated into hazard controls such as plan-do-check-act (PDCA) or other similar continuous improvement process must be used.

59. SCAFFOLDS PROGRAM

1.605 Purpose and Scope

The purpose of this program is to provide training, guidance, and resources to ensure the safe and efficient use of scaffolds, reduce the risk of falls, and prevent accidents and injuries.

This program applies to all T&T Industrial, Inc. employees that work with or around scaffolds.

1.606 Resources

Number	Title
29 CFR 1926 Subpart L	Scaffolds
29 CFR 1910 Subpart J	General Environmental Controls – Specifications for Accident Prevention Signs and Tags
29 CFR 1926 Subpart G	Signs, Signals, and Barricades – Accident Prevention Signs and Tags
Cal/OSHA T8 CCR Subchapter 4	Construction Safety Orders – General Requirements – Injury and Illness Prevention – Safety Instructions for Employees
Cal/OSHA T8 CCR Subchapter 7	General Industry Safety Orders – Accident Prevention Tags

1.607 Definitions

Acronym/Term	Definition
Competent Employee / Person	A person who can identify existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.
Guardrail System	A vertical barrier, consisting of, but not limited to, toprails, midrails, and posts, erected to prevent employees from falling off a scaffold platform or walkway to lower levels.
Lower Levels	Areas below the level where the employee is located and to which an employee can fall. Such areas include, but are not limited to, ground levels, floors, roofs, ramps, runways, excavations, pits, tanks, materials, water, and equipment.
Maximum Intended Load	The total load of all persons, equipment, tools, materials, transmitted loads, and other loads reasonably anticipated to be applied to a scaffold or scaffold component at any one time.
Personal Fall Arrest System	A system used to arrest an employee's fall. It consists of an anchorage, connectors, a body belt, or body harness and may include a lanyard, deceleration device, lifeline, or combinations of these.
Platform	A work surface elevated above lower levels. Platforms can be constructed using individual wood planks, fabricated planks, fabricated decks, and fabricated

Acronym/Term	Definition
	platforms.
Qualified	One who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated the ability to solve or resolve problems related to the subject matter, the work, or the project.
Rated Load	The manufacturer's specified maximum load to be lifted by a hoist or to be applied to a scaffold or scaffold component.
Supported Scaffold	One or more platforms supported by outrigger beams, brackets, poles, legs, uprights, posts, frames, or similar rigid support.

1.608 Scaffolds Program

Scaffolding has a variety of applications. It is used in new construction, routine maintenance, renovation, painting, repairing, removal, and performing arts activities. Scaffolding offers a safer and more comfortable work arrangement compared to leaning over edges, stretching overhead, and working from ladders. Scaffolding provides employees safe access to work locations, level and stable working platforms, and temporary storage for tools and materials for performing immediate tasks. Scaffolding accidents mainly involve personnel falls and falling materials caused by equipment failure, incorrect operating procedures, and environmental conditions. Additionally, scaffolding overloading is a frequent single cause of major scaffold failure.

1.609 Training and Qualification

Each employee who performs work while on a scaffold shall be trained by a person qualified in the subject matter to recognize the hazards associated with the type of scaffold being used and to understand the procedures to control or minimize those hazards.

Employees who work on scaffolds shall be trained on:

- The nature of electrical hazards, fall hazards, and falling object hazards.
- The correct procedures for dealing with electrical hazards.
- The correct procedures for erecting, maintaining, and disassembling the fall protection systems and falling object protection systems being used.
- Proper use and proper handling of materials on the scaffold.
- The maximum intended load and the load-carrying capacities of the scaffolds used.
- Any other pertinent information under the OSHA regulation.

In addition, employees involved in erection, disassembling, moving, operating, repairing, maintaining, or inspecting a scaffold shall be trained by a competent person to recognize any hazards associated with the work in question. They shall also be trained in the correct procedures for erecting, disassembling, moving, operating, repairing, inspecting, and maintaining the type of scaffold in question.

Only qualified and competent personnel shall modify scaffolding systems.

1.609.1 Retraining

When there is reason to believe an employee lacks the skill or understanding of the safe erection, use or dismantling of scaffolds, the employee shall be re-trained.

Retraining is required in at least the following situations:

- Where changes at the worksite present a hazard about which an employee has not been previously trained.
- Where changes in the types of scaffolds, fall protection, falling object protection, or other equipment present a hazard about which an employee has not been previously trained.
- Where inadequacies in an affected employee's work involving scaffolds indicate that the employee has not retained the requisite proficiency.

1.610 Competent Person Responsibilities

The competent person's responsibilities include (but are not limited to):

- Design and preplanning of the scaffold including weight limitations, scaffold type, fall protection, tie-offs, supports, etc.
- Overseeing the erection of the scaffold. Final inspection of the scaffold prior to initial occupation for use.
- Overseeing disassembly of the scaffold.

1.611 Inspection

Scaffolds and scaffold components shall be inspected for visible defects by a competent person before each work shift and after any occurrence which could affect a scaffold's structural integrity. Any part of a scaffold damaged or weakened shall be immediately repaired or replaced, braced, or removed from service until repaired.

Unsafe equipment or conditions shall be tagged out by a competent person. Employees shall comply with tagging of unsafe equipment or conditions.

1.612 Safe Working Capacity and Load Ratings

Each scaffold and scaffold component shall be capable of supporting, without failure, its own weight and at least four times the maximum intended load applied or transmitted to it.

Scaffolds and scaffold components shall not be loaded in excess of their maximum intended loads or rated capacities, whichever is less.

The design load of all scaffolds shall be calculated on the basis of:

- Light - designed and constructed to carry a working load of 25 pounds per square foot.
- Medium - designed and constructed to carry a working load of 50 pounds per square foot.
- Heavy - designed and constructed to carry a working load of 75 pounds per square foot.

The load limit must be on the scaffold itself.

1.613 Scaffold Platforms

Each platform on all working levels of scaffolds shall be fully planked or decked between the front uprights and the guardrail supports. All planking or platforms shall be overlapped (minimum 12 inches) or secured from movement.

The maximum work level height shall not exceed three times the least base dimension below the platform. Where the basic mobile unit does not meet this requirement, outrigger frames shall be employed to achieve this least base dimension, or provisions shall be made to guy or brace the unit against tipping.

1.614 Supported Scaffolds

Supported scaffolds with a height to base width (including outrigger supports, if used) ratio of more than four to one (4:1) shall be restrained from tipping by guying, tying, bracing, or equivalent means.

Supported scaffold poles, legs, posts, frames, and uprights shall bear on base plates and mud sills or other adequate firm foundation. Footings shall be level, sound, rigid, and capable of supporting the loaded scaffold without settling or displacement. Unstable objects shall not be used to support scaffolds or platform units.

Wheels or casters shall be properly designed for strength and dimensions to support four times the design working load. All scaffold wheels, casters, and swivels shall be provided with a positive locking device or other effective means to prevent movement of the scaffold.

Where leveling of the elevated work platform is required, screw jacks or other similar means for adjusting the height shall be provided in the base section of each mobile unit. The screw jack shall extend into its leg tube at least 1/3 its length but in no case shall the exposed portion of the screw jack exceed 12 inches.

1.615 Work Levels

All scaffold work levels 6 feet or higher above the ground or floor shall have a toe board at locations where persons are required to work or pass under the scaffold.

All scaffold work levels 30 inches or higher above the ground or floor shall have guardrail protection.

1.616 Access

When scaffold platforms are more than 2 feet above or below a point of access, portable ladders, hook-on ladders, attachable ladders, stair towers (scaffold stairways / towers), stairway-type ladders (such as ladder stands), ramps, walkways, integral prefabricated scaffold access, or direct access from another scaffold, structure, personnel hoist, or similar surface shall be used. Cross braces shall not be used as a means of access.

1.617 Power Lines

Scaffolds shall not be erected, used, dismantled, altered, or moved such that they or any conductive material handled on them might come closer to exposed and energized power lines than as follows:

- Less than 50 kv- 10 feet.
- More than 50 kv- 10 feet plus 0.4 inches for each 1 kv over 50 kv.

1.618 Environmental Conditions

Employees shall not work on scaffolds covered with snow, ice, or other slippery material except as necessary for removal of such materials.

Work on or from scaffolds is prohibited during storms or high winds unless a competent person has determined that it is safe for employees to be on the scaffold and those employees are protected by a personal fall arrest system or wind screens.

1.619 Fall Protection

Each employee on a scaffold more than 10 feet above a lower level shall be protected from falling to that lower level. Each employee shall be protected by the use of personal fall arrest systems or guardrail systems.

1.620 Falling Object Protection

In addition to wearing hardhats, each employee on a scaffold shall be provided with additional protection from falling hand tools, debris, and other small objects through the installation of toe boards, screens, or guardrail systems, or through the erection of debris nets, catch platforms, or canopy structures that contain or deflect the falling objects. When the falling objects are too large, heavy, or massive to be contained or deflected by any of the above-listed measures, such potential falling objects shall be placed away from the edge of the surface from which they could fall and shall be secured as necessary to prevent their falling.

1.621 Guardrail Systems

Guardrail systems shall be installed along all open sides and ends of platforms. Guardrail systems shall be installed before the scaffold is released for use by employees other than erection / dismantling crews.

60. Short Service Employee (SSE) Program

1.622 Purpose and Scope

The purpose of this program is to prevent work related injury and illness to new hires.

This program applies to all T&T Industrial, Inc. employees deemed as Short Service Employees (SSE).

1.623 Short Service Employee Program

All new employees face a period of transition into new surroundings and work processes. It is during this period that new employees are exposed to the greatest risk of personal injury.

An SSE is a permanent employee, temporary employee, contractor, subcontractor, or supplier that has less than 6 months experience in their craft or assigned job.

Mentors must be knowledgeable and experienced and must not allow an SSE to work alone. A work crew of less than five employees shall not have more than one SSE.

Prior to starting work, the contractor shall notify the host facility (project coordinator, contractor contact, and/or on-site supervisor) if SSEs are present on work crews.

Contractors, subcontractors, and suppliers shall manage their SSEs in accordance with the requirements of the SSE program.

1.624 Training

Management shall ensure that each SSE is assigned a mentor and properly trained per federal, state, industry, Company, and client requirements before starting work. The mentor selection for each SSE shall be based upon the area and craft in which the employee will perform work.

Training may include, but not be limited to:

- General safety rules.
- General and job specific requirements for personal protective equipment (PPE).
- Injury reporting and follow-up procedures.
- Regulatory and job skill training specific to job tasks.
- Safety meetings and pre-job job safety analysis (JSA) processes.
- Site specific procedures and hazard information.

1.625 Identification

SSEs shall be visibly identified using a different colored hardhat or other method of identification. The method used to identify SSEs should be communicated to the client.

1.626 Monitoring

SSEs shall be monitored for compliance with health, safety, and environmental (HSE) policies and procedures. Once the SSE has demonstrated competency and compliance with HSE policies and procedures, the hi-visibility identifier may be removed.

1.627 Mentor

A mentoring system shall be implemented to provide guidance to SSEs and assist with their development. A mentor may only be assigned to one crew that includes SSEs and shall remain on site with them.

The mentor shall provide on the job safety training and guidance and ensure that the SSE has demonstrated the necessary work ethic and safety knowledge necessary for their work.

The mentor shall communicate with management on the progress of the SSEs placed under their guidance.

61. Stop Work Authority Program

1.628 Purpose and Scope

The purpose of this program is to provide employees with the responsibility and obligation to stop work when a perceived unsafe condition or behavior may result in damage to the environment, equipment, or people.

This program applies to all T&T Industrial, Inc. employees.

1.629 Stop Work Authority (SWA) Program

No activity is so urgent or important that health, safety, or the environment (HSE) may be compromised. Stop work actions take precedence over all other priorities and procedures.

All employees have the authority and obligation to stop any task or operation where concerns or questions regarding the control of HSE risk exist.

Work shall not resume until all stop work concerns have been addressed and the designated individual with restart authority determines that the imminent risk does not exist or no longer exists.

Any form of retribution or intimidation directed at any individual or company for exercising their right to issue a stop work authority in good faith shall not be tolerated, even if deemed unnecessary.

1.630 Training

Employees shall receive Stop Work Authority training before initial assignment. The training shall be documented including the employee name, the dates of training, and subject.

1.631 Roles and Responsibilities

Senior management shall be responsible for creating a culture that promotes SWA and supports use of SWA without potential for retribution.

Supervisors and managers shall be responsible for honoring SWA requests and resolution before resuming operations.

The HSE department is responsible for providing training, support, and documentation and monitoring compliance of the SWA program.

Employees and contractors are responsible for initiating stop work and supporting stop work initiated by others.

1.632 SWA Steps

SWA is a several step process.

1) Stop - When an employee perceives conditions or behaviors that pose imminent danger, a stop work intervention shall be initiated immediately.
2) Notify - Affected personnel and supervision shall be notified of the stop work action.
3) Investigate - Affected personnel shall discuss the situation and come to an agreement on the stop work action.
4) Correct - Corrective actions shall be made according to the corrections agreed upon in the investigation.
5) Resume - All affected employees shall be notified of what corrective actions were implemented and work shall recommence by personnel with restart authority.
6) Follow Up - A root cause analysis to the stop work shall be completed to identify any potential opportunities for improvement.

1.633 Corrective Action

All stop work interventions shall be documented for lessons learned and corrective measures to be put into place.

1.634 Follow-Up

Stop work reports shall be reviewed by a supervisor or manager in order to measure participation, determine quality of interventions and follow-up, trend common issues, identify opportunities for improvement, and facilitate sharing of learnings.

It is the desired outcome of any stop work intervention that the identified safety concern(s) has been addressed to the satisfaction of all involved persons prior to the resumption of work. Most issues can be adequately resolved in a timely manner at the job site. Occasionally additional investigation and corrective actions may be required to identify and address root causes.

62. SUBCONTRACTOR MANAGEMENT PROGRAM

1.635 Purpose and Scope

The purpose of this program is to ensure that subcontractors meet our safety standards, adhere to regulatory requirements, and contribute to a safe and productive work environment.

This program applies to all T&T Industrial, Inc. employees using subcontractors.

1.636 Resources

Number	Title
CMS-FM-0049	Subcontractor Safety Pre-Qualification Form

1.637 Subcontractor Management

The subcontractor management plan shall contain key components to consider ensuring that existing issues, vulnerabilities, and risks are adequately addressed. All subcontractors utilized to perform work on behalf of the company must adhere to the appropriate health, safety, and environmental standards required by the governing jurisdiction.

All subcontractors, suppliers, and/or vendors must meet or exceed the Company's own safety program requirements in addition to compliance with minimum jurisdictional regulatory requirements.

Subcontractors must be prequalified prior to use. The qualification process reviews the subcontractor's safety metrics, policies and procedures, safety training, and safety programs.

Supervision and direction shall be provided by the Company to subcontractors.

1.638 Training

Written health, safety and environmental programs, and training documentation applicable to the type of work the subcontractor will perform shall be obtained and reviewed to assist with the hiring of safe subcontractors.

1.639 Statistics

Past performance is a key indicator of future performance. Incident statistics shall be obtained and analyzed to ensure that only safe subcontractors are hired. A copy of the subcontractor's OSHA logs, Experience Modifier Report (EMR), TRIR, DART, etc. shall be obtained to compare their performance to others in the industry’s minimum safety metric. Those who outperform the industry shall be selected whenever practicable.

1.640 Orientation

Subcontractors shall be provided a site orientation that addresses health, safety, security, and environmental concerns.

1.641 Drug and Alcohol Policy

The company is responsible for ensuring that subcontractors are aware of the hiring client's Drug and Alcohol policy. Subcontractors must adhere to the requirements of the Drug and Alcohol policy at all times while at the work site.

1.642 Assessments

Subcontractors shall be included in pre-job meetings, job safety analysis (JSA) development, hazard and risk assessments, tailgate meetings, and job site audits.

Post-job performance reviews for safety performance shall be conducted for subcontractors in post job evaluations. A combination of factors may be considered including, but not limited to, housekeeping, cost, safety, and quality of work.

1.643 Incidents

The company must ensure that subcontractors are aware of incident reporting requirements.

All subcontractors must report to the company all incidents and accidents in a prompt and timely manner.

If a subcontractor is involved in an incident, the company is responsible for reporting the incident to the hiring client. The company must ensure the incident is investigated, and must participate in the investigation.

Subcontractors involved in any incident or accident must perform an incident investigation, identify root causes and causal factors, develop corrective actions, and have a methodology for determining the effectiveness of the corrective actions.

Appendix 39 **Subcontractor Safety Pre-Qualification Form**

General Information			
1. Subcontractor Information:			
Subcontractor Name:		Telephone No.:	
Address:		Email:	
		Website:	
NAICS Code:		Date:	
2. Officers			
President:			
Vice President:			
Treasurer:			
3. How many years has your organization been in business under the present name?			
4. Parent Company Information:			
Name and Address:			
Subsidiaries:			
5. Under current management since (date):			
6. Contact for Insurance Information:			
Name and Title:	Phone No.:	Email:	

7.	Insurance Carrier(s):	
Name	Type of Coverage	Phone No.

8. Workers' Compensation Account Information *(include a copy of your workers' comp insurance certificate)*	
Account No.:	Industry Code:

9. Contact for Requesting Bids:		
Name and Title:	Phone No.:	Email:

10. Contractor Evaluation Form Completed By:		
Name and Title:	Phone No.:	Email:

Health, Safety, and Environmental Performance

Provide the following data for your company for the past 3 years.

Safety Performance Definitions and Guidance:

- **a)** **Hours Worked**: Employee hours worked last 3 years. Report actual scheduled total hours worked, and total overtime worked. If actual hours worked are not available for certain individuals, it may be estimated at 2000 hours per individual per year.
- **b)** **Recordable Incidents**: Cases that involve any work-related injury or illness, including death but excluding first-aid cases.
- **c)** **Lost Workday Cases**: Medical cases that involve fatalities, days away, or restricted work cases.
- **d)** **Motor Vehicle Incidents**: Any incident event involving a motor vehicle that is owned, leased, or rented that results in death, injury, or property damage unless the vehicle is properly parked.

T&T Industrial, Inc.

Subcontractor Management Program

Health and Safety Incidents	2021	2022	2023
Total Hours Worked			
Total Recordable Incidents # Fatalities # Medical Aids # Days Away from Work Cases # Restricted Work Cases			
Total Recordable Incident Rate (TRIR) <u>Total # recordable incidents x 200,000</u> Total # hours worked			
Lost Workday Cases (LWC) # Fatalities # Days Away from Work Cases # Restricted Work Cases			
Lost Workday Incident Rate (LWDR) <u>Total # lost workday incidents x 200,000</u> Total # hour worked			
Motor Vehicle Incidents (MV) # Motor Vehicle Incidents # Miles Driven			
Motor Vehicle Incident Frequency Rate <u>Total # motor vehicle incidents x 1,000,000</u> Total # miles driven			
Environmental Incidents	**2021**	**2022**	**2023**
# Petroleum Spills to Water # Chemical Spills to Water # Petroleum Spills to Land # Chemical Spills to Land			
Enforcement Action	**2021**	**2022**	**2023**
# Health and Safety Citations # Environmental Citations Provide details:			

OSHA Citations	2021	2022	2023
Number and type of violations. Penalties assessed by OSHA. Were the citations contested / vacated? What specific corrective actions were taken to prevent further penalties / injuries?			

Worker's Compensation (WC) Experience Modification Rate (EMR)	
Year	**Rate**
1.	
2.	
3.	
4. Provide a letter from your WC insurance carrier certifying the above EMRs.	
5. If your WC carrier has not issued your company an EMR because you have not accrued enough WC costs, provide a copy of your WC Loss Run (available from your carrier).	
6. If your current EMR is greater than 1.0, provide a written explanation of the safety methods that are being implemented by your company to reduce this rate.	

Health, Safety, and Environmental Management		
Highest ranking HSE professional:		
Name and Title:	Phone No.:	Email:
Do you have a written HSE program?	Yes ☐	No ☐
Does your HSE program include the following?		
1. HSE policy statement signed by management	Yes ☐	No ☐
2. Management involvement and commitment	Yes ☐	No ☐
3. Hazard identification and risk control	Yes ☐	No ☐
4. Rules and work procedures	Yes ☐	No ☐
5. Training	Yes ☐	No ☐
6. Communications	Yes ☐	No ☐

7. Incident reporting and investigation	Yes ☐	No ☐
Does the program include the below work practices and procedures?		
1. Permit to Work including isolation of energy	Yes ☐	No ☐
2. Confined space entry	Yes ☐	No ☐
3. Injury and illness recording	Yes ☐	No ☐
4. Fall protection	Yes ☐	No ☐
5. Personal protective equipment	Yes ☐	No ☐
6. Portable electrical / power tools	Yes ☐	No ☐
7. Motor vehicle / driving safety	Yes ☐	No ☐
8. Compressed gas cylinders	Yes ☐	No ☐
9. Electrical equipment grounding assurance	Yes ☐	No ☐
10. Powered industrial trucks (cranes, forklifts, etc.)	Yes ☐	No ☐
11. Housekeeping	Yes ☐	No ☐
12. Incident reporting and investigation	Yes ☐	No ☐
13. Hazard reporting	Yes ☐	No ☐
14. Emergency preparedness and evacuation plan	Yes ☐	No ☐
15. Waste disposal and pollution prevention	Yes ☐	No ☐
16. Regular workplace inspection / audit	Yes ☐	No ☐
Do you have a drug and alcohol program?		
1. Pre-employment testing	Yes ☐	No ☐
2. Reasonable cause testing	Yes ☐	No ☐
3. Post-rehabilitation / return to work testing	Yes ☐	No ☐
Do you have a Job Safety Analysis (JSA) process?	Yes ☐	No ☐
Is there a root cause analysis process used for investigations, near misses, environmental spills?	Yes ☐	No ☐
Is there a Management of Change (MOC) process?	Yes ☐	No ☐

Do you have programs for the following?		
1. Respiratory protection	Yes ☐	No ☐
2. Hazard communication	Yes ☐	No ☐
3. High hazard work such as highly hazardous chemicals, explosives, blasting agents	Yes ☐	No ☐
Do you have a corrective action process for addressing employee safety and health performance deficiencies?	Yes ☐	No ☐
Do you conduct medical examinations?	Yes ☐	No ☐
Do you have personnel trained to perform first aid and CPR?	Yes ☐	No ☐
Is applicable PPE provided to employees?	Yes ☐	No ☐
Do you have a program to ensure that PPS is inspected and maintained?	Yes ☐	No ☐

Do you hold HSE meetings for?			**Frequency**
1. Field Supervisors	Yes ☐	No ☐	
2. Employees	Yes ☐	No ☐	
3. New Hires	Yes ☐	No ☐	
4. Subcontractors	Yes ☐	No ☐	
Inspections and audits:			**Frequency**
1. Do you conduct internal HSE inspections?	Yes ☐	No ☐	
2. Do you conduct internal HSE program audits?	Yes ☐	No ☐	
3. Are corrections or deficiencies to internal HSE program or equipment communicated and documented until closure?	Yes ☐	No ☐	

Equipment and Materials:		
1. Do you own or lease equipment and materials?	Yes ☐	No ☐
2. Do you have a system for establishing applicable health, safety, and environmental specifications for acquisition of materials and equipment?	Yes ☐	No ☐
3. Do you conduct inspections on operating equipment in compliance with regulatory requirements?	Yes ☐	No ☐
4. Do you maintain operating equipment in compliance with regulatory requirements?	Yes ☐	No ☐
5. Do you maintain the applicable inspection and maintenance certification records for operating equipment?	Yes ☐	No ☐
6. Do you conduct corrections or deficiencies from equipment inspections and maintenance?	Yes ☐	No ☐
Subcontractor Management:		
1. Do you subcontract any work?	Yes ☐	No ☐
2. Do you have a written contractor safety management process?	Yes ☐	No ☐
3. Do you use HSE performance criteria in selection of subcontractors?	Yes ☐	No ☐
4. Do you evaluate the ability of subcontractors to comply with applicable HSE requirements as part of the selection process?	Yes ☐	No ☐
5. Do your subcontractors have a written HSE program?	Yes ☐	No ☐
6. Do you include your subcontractors in the following:		
• HSE orientation	Yes ☐	No ☐
• HSE meetings	Yes ☐	No ☐
• HSE equipment inspections	Yes ☐	No ☐
• HSE program audits	Yes ☐	No ☐
• Are corrections or deficiencies documented	Yes ☐	No ☐
Employee and Job Training:		
1. Have employees been trained in appropriate job skills?	Yes ☐	No ☐

2. Are employees' job skills certified where required by regulatory or industry standards?	Yes ☐	No ☐

T&T Industrial, Inc.
Subcontractor Management Program

Health, Safety, and Environmental Orientation:	**New Hires**	**Supervisors**
1. Do you have an HSE orientation program for new hires and newly hired or promoted supervisors?	Yes ☐	No ☐
2. Does the program provide instruction on the following:		
• New employee orientation	Yes ☐	No ☐
• Safe work practices	Yes ☐	No ☐
• Safety supervision	Yes ☐	No ☐
• Toolbox meetings	Yes ☐	No ☐
• Emergency procedures	Yes ☐	No ☐
• First aid procedures	Yes ☐	No ☐
• Fire protection and prevention	Yes ☐	No ☐
• Safety Intervention	Yes ☐	No ☐
• Hazard communication	Yes ☐	No ☐
Health, Safety, and Environmental Training:		
1. Do you know the regulatory HSE training requirements for your employees?	Yes ☐	No ☐
2. Have your employees received the required HSE training and re-training?	Yes ☐	No ☐
3. Do you have a specific HSE training program for supervisors?	Yes ☐	No ☐
Training Records:		
1. Do you have HSE training records for your employees?	Yes ☐	No ☐
2. Do the training records include the following:		
• Employee identification	Yes ☐	No ☐
• Date of training	Yes ☐	No ☐
• Name of trainer	Yes ☐	No ☐
• Method used to verify understanding	Yes ☐	No ☐

Signatures			
Completed By:		Title:	
Signature:		Date:	

63. Traffic Control Program

1.644 Purpose and Scope

The purpose of this program is to minimize the risk of accidents, injuries, and traffic congestion by providing guidelines and safety protocols for the control and regulation of vehicular and pedestrian traffic.

This program applies to all T&T Industrial, Inc. employees working near vehicular and pedestrian traffic through and adjacent to the project area.

1.645 Traffic Control Program

This section includes identifying safety hazards and then furnishing all necessary labor, materials, tools, and equipment including, but not limited, to signs, barricades, traffic drums, cones, flashers, construction fencing, flag persons, variable message boards, uniformed police officers, warning devices, temporary pavement markings, temporary sidewalk, delineators, etc., to maintain vehicular and pedestrian traffic through and adjacent to the project area.

These measures and actions must be taken to safely maintain the accessibility of public and construction traffic by preventing potential construction hazards.

All materials, work, and incidental costs related to maintenance of traffic will be paid for at the contract lump sum price.

Individuals must be certified or properly trained to create traffic control plans, to place channeling devices, and or warning or traffic control signs and devices.

The program must adhere to all the requirements set forth in the Manual on Uniform Traffic Control Devices, in addition to localized traffic control requirements, including appropriate traffic control devices and traffic control signs.

1.646 Requirements

Pre-work site assessments shall be conducted to identify potential hazards in and around the work zone.

Training shall be provided to workers involved in the planning, setup, operation, maintenance, or removal of traffic control to the level of their responsibility.

All affected employees will receive training on this program upon hire, and regular refresher training as required, and corresponding training documentation will be retained.

The Traffic Control Plan must conform to the following standards:

- Sequence the work in a manner that will minimize disruption of vehicular and pedestrian access through and around the construction area.
- Traffic planning and control for the maintenance and protection of pedestrian and vehicular traffic affected by this Company's work includes, but is not limited to:
  - Construction and maintenance of any necessary detour equipment and facilities.
  - Providing necessary facilities for access to residences and businesses.
  - Furnishing, installing, and maintenance of traffic control and safety devices (e.g., signage, barricades, barriers, message boards, etc.), and flag persons as appropriate during construction.
  - Control of water runoff, dust, and any other special requirements for safe and expeditious movement of traffic.
- Planning, maintenance, and control of traffic must be provided at the Company's expense. The Company will bear all expenses of maintaining the vehicle and pedestrian traffic throughout the work area.
- The Company will remove temporary equipment and facilities when no longer required, restore grounds to original, or to specified conditions.
- Before employees begin work in the vicinity of vehicular or pedestrian traffic that may endanger them, warning signs or flags and other traffic-control devices must be placed in conspicuous locations to alert and channel approaching traffic.

1.647 Submittals

Submit at Contractor's own expense a Traffic Control Plan for approval by the controlling roadway agency (DOT, County Public Works, or other local government) having jurisdiction over the road for approval.

The Traffic Control Plan will detail procedures and protective measures proposed by the Company to provide for protection and control of traffic affected by the work consistent with the following applicable standards:

- Standard Specifications for Road and Bridge Construction, latest edition including all subsequent supplements issued by the Department of Transportation (DOT Spec.).
- Manual of Traffic Control and Safe Practices for Street and Highway Construction, Maintenance and Utility Operations, DOT.
- Right-of-Way Utilization Regulations, latest edition.

All references to the respective agencies in the above referenced standards must be construed to also include the municipality as applicable for this work.

The Traffic Control Plan will be signed and sealed by a Professional Engineer registered and must include proposed locations and time durations of the following, as applicable:

- Pedestrian and public vehicular traffic routing.
- In all instances on the work site, pedestrian traffic has the right-of-way.
- Lane and sidewalk closures, other traffic blockage, and lane restrictions and reductions anticipated to be caused by construction operations. Show and describe the proposed location, dates, hours and duration of closure, vehicular and pedestrian traffic routing and management, traffic control devices for implementing pedestrian and vehicular movement around the closures, and details of barricades.
- Location, type, and method of shoring to provide lateral support to the side of an excavation or embankment parallel to an open travel-way.
- Allowable on-street parking within the immediate vicinity of worksite.
- Access to buildings immediately adjacent to worksite.
- Driveways blocked by construction operations.
- Temporary traffic control devices, temporary pavement striping and marking of streets and sidewalks affected by construction
- Temporary commercial and industrial loading and unloading zones.
- Construction vehicle reroutes, travel times, staging locations, and number and size of vehicles involved.

Obtain and submit prior to erection, or otherwise impacting traffic, all required permits from all authorities having jurisdiction, including County Public Works, if applicable required for traffic control and the use of barricades.

1.648 Materials and Equipment

The Company must furnish, erect, and maintain all necessary traffic control devices, including flag person, in accordance with the Manual of Uniform Traffic Control Devices for Streets and Highways published by the U.S. Department of Transportation, Federal Highway Administration.

1.648.1 Flag Persons

All flag persons used will adhere to the following requirements:

- Any person acting as a flag person will have attended a training session taught by a Contractor's qualified trainer before the start date.
- The Company's qualified trainer will have completed a "Flag Person Train the Trainer Session" in the 5-years previous or before the start date and will be on file as a qualified flag person trainer.
- The flag person trainer's name and qualification number will be furnished by the Company at the pre-construction meeting. The Company will provide all flag persons with the Flag Person Handbook and will observe the rules and regulations contained therein. This handbook will be in the possession of all flag persons while flagging.
- Flag persons will not be assigned other duties while working as authorized flag persons.
- Any person replacing flag person for break must have the same training.

1.649 Execution Notifications

The Company will notify individual owners, owner's agents, and tenants of buildings affected by the construction, with copies to the county, 72-hours in advance of any construction activities.

The Company must notify residents and pedestrians via variable message boards no later than 10 days prior to the closure of any road, lane, or pedestrian thoroughfare.

The Company must notify Emergency Management Services agencies no less than 7 days prior to such closures or whenever roads are impassable.

Implement closing of vehicle or pedestrian thoroughfare in accordance with the construction drawings and the approved Traffic Control Plan.

The Company will immediately notify the County of any vehicular or pedestrian safety or efficiency problems incurred as a result of the construction.

1.650 General Traffic Control

The Company will sequence and plan construction operations and will generally conduct work in such a manner as not to unduly or unnecessarily restrict or impede normal traffic.

Unless otherwise provided, all roads within the limits of the work will be kept open to all traffic. The Company will keep the portion of the project being used by public traffic, whether it is through or local traffic, in such condition that traffic will be adequately accommodated.

The Company will be responsible for installation and maintenance of all traffic control devices and requirements for the duration of the construction period. Necessary precautions for traffic control will include, but not be limited to, warning signs, signals, lighting devices, markings, barricades, canalizations, and hand signaling devices.

The Company will provide and maintain in a safe condition temporary approaches or crossings and intersections with trails, roads, streets, businesses, parking lots, residences, garages, and farms.

The Company will provide emergency access to all residences and businesses at all times. Residential and business access will be restored and maintained at all times outside of normal working hours.

Traffic is to be maintained on one section of existing pavement, proposed pavement, or a combination thereof. Alternating one-way traffic may be utilized and limited to a maximum length of 500-feet during construction hours. Lane width for alternating one-way traffic will be kept to a minimum width of 10-feet, or as directed by the County.

Travel lanes and pedestrian access will be kept reasonably smooth, dry, and in a suitable condition at all times.

The Company will make provisions at all "open cut" street crossings to allow for free passage of vehicles and pedestrians, either by bridging or other temporary crossing structures. Such structures will be of adequate strength and proper construction and will be maintained in such a manner as not to constitute an undue traffic hazard.

The Company will keep all signs in proper position, clean, and legible at all times. Care will be taken so that weeds, shrubbery, construction materials, equipment, and soil are not allowed to obscure any sign, light, or barricade. Signs that do not apply to construction conditions should be removed or adjusted so that the legend is not visible to approaching traffic.

The County may determine the need for, and extent of, additional striping removal and restriping.

Excavated material, spoil banks, construction materials, equipment, and supplies will not be located in such a manner as to obstruct traffic, as practicable. The Company will immediately remove from the site all demolition material, exercising such precaution as may be directed by the County. All material excavated must be disposed of so as to minimize traffic and pedestrian inconvenience and to prevent damage to adjacent property.

During any suspension, the Company will make passable and open to traffic such portions of the project and/or temporally roadways as directed by the County for accommodation of traffic during the anticipated period of suspension. Passable conditions will be maintained until issuance of an order for the resumption of construction operations. When work is resumed, the Company will replace or renew any work or materials lost or damaged because of such temporary use in every respect as though its prosecution had been continuous and without interferences.

1.651 Temporary Shoring

Use shoring to maintain traffic when it is necessary to provide lateral support to the side of an excavation or embankment parallel to an open travel-way.

Provide shoring when a theoretical 2:1 or steeper slope from the bottom of the excavation or embankment intersects the existing ground line closer than 5-feet (1.5 m) from the edge of pavement of the open travel-way.

The Company will furnish, install, and remove sheeting, shoring, and bracing necessary to maintain traffic at locations shown on the Traffic Control Plan and other locations determined during construction.

Barricades shall be used where additional employee protection is necessary, Excavated areas shall be protected with barricades and warning lights prominently displayed at night.

1.652 Qualifications for Flaggers

Because flaggers are responsible for public safety and make the greatest number of contacts with the public of all highway workers, they should be trained in safe traffic control practices and public contact techniques.

When work activity occurs on or adjacent to a surface being used by the public, the Company shall provide flagger(s) to direct traffic.

Flaggers should be able to satisfactorily demonstrate the following abilities:

- Ability to receive and communicate specific instructions clearly, firmly, and courteously
- Ability to move and maneuver quickly in order to avoid danger from errant vehicles
- Ability to control signaling devices (such as paddles and flags) in order to provide clear and positive guidance to drivers approaching a project zone in frequently changing situations
- Ability to understand and apply safe traffic control practices, sometimes in stressful or emergency situations

- Ability to recognize dangerous traffic situations and warn workers in sufficient time to avoid injury

1.653 High-Visibility Safety Apparel

For daytime and nighttime activity, flaggers shall wear high-visibility safety apparel that meets the Performance Class 2 or 3 requirements of the ANSI/ISEA 107–2004/107-1999 publication entitled “American National Standard for High-Visibility Apparel and Headwear” (see Section 1A.11) and labeled as meeting the ANSI 107-2004 standard performance for Class 2 or 3 risk exposure.

The apparel background (outer) material color shall be fluorescent orange-red, fluorescent yellow-green, or a combination of the two as defined in the ANSI standard.

The retroreflective material shall be orange, yellow, white, silver, yellow-green, or a fluorescent version of these colors, and shall be visible at a minimum distance of 1,000 feet in any direction.

The retroreflective safety apparel shall be designed to clearly identify the wearer as a person.

For nighttime activity, high-visibility safety apparel that meets the Performance Class 3 requirements of the ANSI/ISEA 107–2004\107-1999 publication entitled “American National Standard for High-Visibility Apparel and Headwear” and labeled as meeting the ANSI 107-2004 standard performance for Class 3 risk exposure should be considered for flagger wear.

When uniformed law enforcement officers are used to direct traffic within a zone, they shall wear high-visibility safety apparel as described in this section.

In lieu of ANSI/ISEA 107-2004 apparel, law enforcement personnel within the zone may wear high-visibility safety apparel that meets the performance requirements of the ANSI/ISEA 207-2006 publication entitled “American National Standard for High-Visibility Public Safety Vests” and labeled as ANSI 207-2006.

1.654 Hand Signaling Devices

The STOP/SLOW paddle should be the primary and preferred hand-signaling device because the STOP/SLOW paddle gives road users more positive guidance than red flags. Use of flags should be limited to emergency situations.

The STOP/SLOW paddle shall have an octagonal shape on a rigid handle. STOP/SLOW paddles shall be at least 18 inches wide with letters at least 6 inches high. The STOP face shall have white letters and a white border on a red background. The SLOW face shall have black letters and a black border on an orange background. When used at night, the STOP/SLOW paddle shall be retro-reflectorized.

The STOP/SLOW paddle should be fabricated from light semi-rigid material.

The optimum method of displaying a STOP or SLOW message is to place the STOP/SLOW paddle on a rigid staff that is tall enough that when the end of the staff is resting on the ground, the message is high enough to be seen by approaching or stopped traffic.

The STOP/SLOW paddle may be modified to improve conspicuity by incorporating either white or red flashing lights on the STOP face, and either white or yellow flashing lights on the SLOW face. The flashing lights may be arranged in any of the following patterns:

- Two white or red lights, one centered vertically above and one centered vertically below the STOP legend; and/or two white or yellow lights, one centered vertically above and one centered vertically below the SLOW legend
- Two white or red lights, one centered horizontally on each side of the STOP legend; and/or two white or yellow lights, one centered horizontally on each side of the SLOW legend
- One white or red light centered below the STOP legend; and/or one white or yellow light centered below the SLOW legend
- A series of eight or more small white or red lights no larger than 1/4 inch in diameter along the outer edge of the paddle, arranged in an octagonal pattern at the eight corners of the border of the STOP face; and/or a series of eight or more small white or yellow lights no larger than 1/4 inch in diameter along the outer edge of the paddle, arranged in a diamond pattern along the border of the SLOW face
- A series of white lights forming the shapes of the letters in the legend

If flashing lights are used on the STOP face of the paddle, their colors shall be all white or all red. If flashing lights are used on the SLOW face of the paddle, their colors shall be all white or all yellow.

If more than eight flashing lights are used, the lights shall be arranged such that they clearly convey the octagonal shape of the STOP face of the paddle and/or the diamond shape of the SLOW face of the paddle.

If flashing lights are used on the STOP/SLOW paddle, the flash rate shall be at least 50, but not more than 60, flashes per minute.

Flags, when used, shall be red or fluorescent orange/red in color, shall be a minimum of 24 inches square, and shall be securely fastened to a staff that is approximately 36 inches in length.

The free edge of a flag should be weighted so the flag will hang vertically, even in heavy winds.

When used at nighttime, flags shall be retro reflectorized red.

When flagging in an emergency situation at night in a non-illuminated flagger station, a flagger may use a flashlight with a red glow cone to supplement the STOP/SLOW paddle or flag.

When a flashlight is used for flagging in an emergency situation at night in a non-illuminated flagger station, the flagger shall hold the flashlight in the left hand, shall hold the paddle or flag in the right hand as shown in Figure 1, and shall use the flashlight in the following manner to control approaching road users:

- To inform road users to stop, the flagger shall hold the flashlight with the left arm extended and pointed down toward the ground, and then shall slowly wave the flashlight in front of the body in a slow arc from left to right such that the arc reaches no farther than 45 degrees from vertical.
- To inform road users to proceed, the flagger shall point the flashlight at the vehicle's bumper, slowly aim the flashlight toward the open lane, then hold the flashlight in that position. The flagger shall not wave the flashlight.
- To alert or slow traffic, the flagger shall point the flashlight toward oncoming traffic and quickly wave the flashlight in a figure eight motion.

1.655 Flagger Procedures

The use of paddles and flags by flaggers is illustrated in Figure 1.

Flaggers shall use a STOP/SLOW paddle, a flag, or an Automated Flagger Assistance Device (AFAD) to control road users approaching a zone. The use of hand movements alone without a paddle, flag, or AFAD to control road users shall be prohibited except for law enforcement personnel or emergency responders at incident scenes.

The following methods of signaling with paddles shall be used:

- To stop road users, the flagger shall face road users and aim the STOP paddle face toward road users in a stationary position with the arm extended horizontally away from the body. The free arm shall be held with the palm of the hand above shoulder level toward approaching traffic.
- To direct stopped road users to proceed, the flagger shall face road users with the SLOW paddle face aimed toward road users in a stationary position with the arm extended horizontally away from the body. The flagger shall motion with the free hand for road users to proceed.
- To alert or slow traffic, the flagger shall face road users with the SLOW paddle face aimed toward road users in a stationary position with the arm extended horizontally away from the body.

To further alert or slow traffic, the flagger holding the SLOW paddle face toward road users may motion up and down with the free hand, palm down.

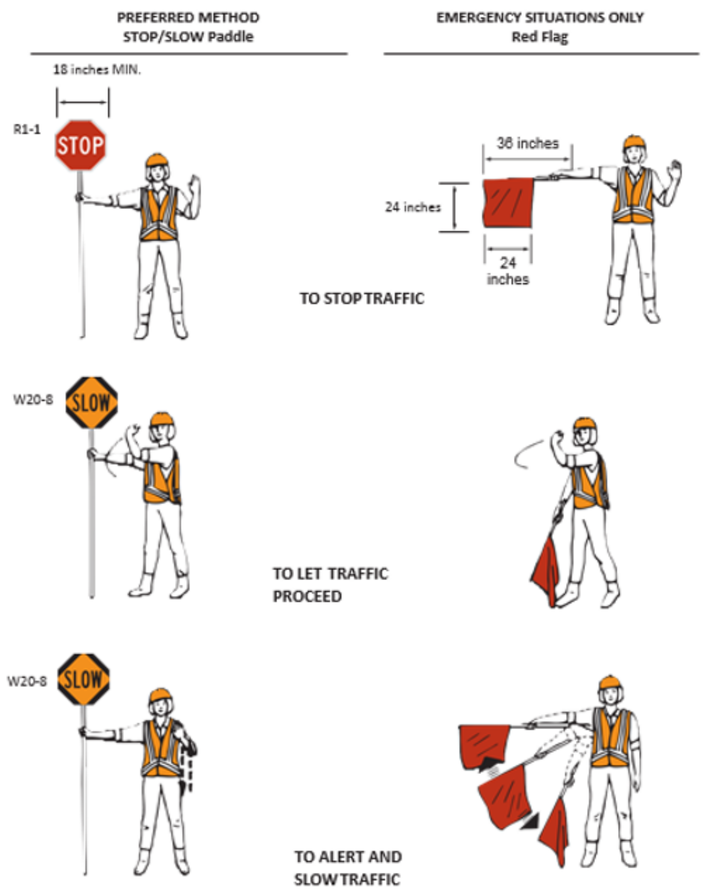
The following methods of signaling with a flag shall be used:

- To stop road users, the flagger shall face road users and extend the flag staff horizontally across the road users' lane in a stationary position so that the full area of the flag is visibly hanging below the staff. The free arm shall be held with the palm of the hand above shoulder level toward approaching traffic.
- To direct stopped road users to proceed, the flagger shall face road users with the flag and arm lowered from the view of the road users and shall motion with the free hand for road users to proceed. Flags shall not be used to signal road users to proceed.
- To alert or slow traffic, the flagger shall face road users and slowly wave the flag in a sweeping motion of the extended arm from shoulder level to straight down without raising the arm above a horizontal position. The flagger shall keep the free hand down.

The flagger should stand either on the shoulder adjacent to the road user being controlled or in the closed lane prior to stopping road users. A flagger should only stand in the lane being used by moving road users after road users have stopped. The flagger should be clearly visible to the first approaching road user at all times. The flagger also should be visible to other road users. The flagger should be stationed sufficiently in advance of the workers to warn them (for example, with audible warning devices such as horns or whistles) of approaching danger by out-of-control vehicles. The flagger should stand alone, away from other workers, work vehicles, or equipment.

At spot lane closures where adequate sight distance is available for the reasonably safe handling of traffic, the use of one flagger may be sufficient.

Figure 1
Signaling Devices by Use of Hand-Flaggers



When a single flagger is used, the flagger should be stationed on the shoulder opposite the spot lane closure or workspace, or in a position where good visibility and traffic control can be maintained at all times.

1.656 Flagger Stations

Flagger stations shall be located such that approaching road users will have sufficient distance to stop at an intended stopping point.

The distances shown in Table 1, which provides information regarding the stopping sight distance as a function of speed, may be used for the location of a flagger station. These distances may be increased for downgrades and other conditions that affect stopping distance.

Flagger stations should be located such that an errant vehicle has additional space to stop without entering the workspace. The flagger should identify an escape route that can be used to avoid being struck by an errant vehicle.

Except in emergency situations, flagger stations shall be preceded by an advance warning sign or signs. Except in emergency situations, flagger stations shall be illuminated at night.

Table 1 Stopping Sight Distance as a Function of Speed

Speed	Distance
20 mph	115 feet
25 mph	155 feet
30 mph	200 feet
35 mph	250 feet
40 mph	305 feet
45 mph	360 feet
50 mph	425 feet
55 mph	495 feet
60 mph	570 feet
65 mph	645 feet
70 mph	730 feet
75 mph	820 feet

*Posted speed, off-peak 85^{th} percentile speed prior to work starting, or the anticipated operating speed.

1.657 Temporary Traffic Control (TTC) Zone Devices

The design and application of TTC devices used in TTC zones should consider the needs of all road users (motorists, bicyclists, and pedestrians), including those with disabilities.

All roadside appurtenances such as traffic barriers, barrier terminals and crash cushions, bridge railings, sign and light pole supports, and work zone hardware used on the National Highway System meet the crashworthy performance criteria contained in the National Cooperative Highway Research Program (NCHRP) Report 350, "Recommended Procedures for the Safety Performance Evaluation of Highway Features." The FHWA website at "http://safety.fhwa.dot.gov/programs/roadside_hardware.htm" identifies all such hardware and includes copies of FHWA acceptance letters for each of them. In the case of proprietary items, links are provided to manufacturers' websites as a source of detailed information on specific devices. The website also contains an "Ask the Experts" section where questions on roadside design issues can be addressed.

Various sections of the MUTCD require certain traffic control devices, their supports, and/or related appurtenances to be crashworthy. Such MUTCD crashworthiness provisions apply to all streets, highways, and private roads open to public travel. Also, State Departments of Transportation and local agencies might have expanded the NCHRP Report 350 crashworthy criteria to apply to certain other roadside appurtenances.

Crashworthiness and crash testing information on devices described are found in AASHTO's "Roadside Design Guide".

"Crashworthy" is a characteristic of a roadside appurtenance that has been successfully crash tested in accordance with a national standard such as the NCHRP Report 350, "Recommended Procedures for the Safety Performance Evaluation of Highway Features".

Traffic control devices shall be defined as all signs, signals, markings, and other devices used to regulate, warn, or guide road users, placed on, over, or adjacent to a street, highway, private roads open to public travel, pedestrian facility, or bikeway by authority of a public body or official having jurisdiction.

All traffic control devices used for construction, maintenance, utility, or incident management operations on a street, highway, or private road open to public travel shall comply with the applicable provisions of this Manual.

1.658 General Characteristics of Signs

Employees struck by vehicles or mobile equipment account for many work zone injuries or fatalities. Work zones should be marked by traffic control devices such as signals, message boards, cones, barrels, barricades, and delineator posts.

TTC zone signs convey both general and specific messages by means of words, symbols, and/or arrows and have the same three categories as all road user signs regulatory, warning, and guide.

The colors for regulatory signs shall follow the Standards for regulatory signs. Warning signs in TTC zones shall have a black legend and border on an orange background, except for the Grade Crossing Advance Warning (W10-1) sign which shall have a black legend and border on a yellow background, and except for signs that are required or recommended to have fluorescent yellow-green backgrounds. Colors for guide signs shall follow the Standards, except for guide signs as otherwise provided.

Where the color orange is required, the fluorescent orange color may also be used.

1.658.1 Support:

The fluorescent version of orange provides higher conspicuity than standard orange, especially during twilight.

Existing warning signs that are still applicable may remain in place.

In order to maintain the systematic use of yellow or fluorescent yellow-green backgrounds for pedestrian, bicycle, and school warning signs in a jurisdiction, the yellow or fluorescent yellow-green background for pedestrian, bicycle, and school warning signs may be used in TTC zones.

Standard orange flags or flashing warning lights may be used in conjunction with signs.

When standard orange flags or flashing warning lights are used in conjunction with signs, they shall not block the sign face.

The sizes for TTC signs and plaques shall be as shown in the Standard. The sizes in the minimum column shall only be used on local streets or roadways where the 85th-percentile speed or posted speed limit is less than 35 mph.

The dimensions of signs and plaques may be increased wherever necessary for greater legibility or emphasis.

Deviations from standard sizes as prescribed in this Manual shall be in 6-inch increments.

Sign design details are contained in the "Standard Highway Signs and Markings" book.

The Standard contains additional information regarding the design of signs, including an option allowing the development of special word message signs if a standard word message or symbol sign is not available to convey the necessary regulatory, warning, or guidance information.

All signs used at night shall be either retroreflective with a material that has a smooth, sealed outer surface or illuminated to show the same shape and similar color both day and night.

Warning lights shall be displayed prominently at night.

The requirement for sign illumination shall not be considered to be satisfied by street, highway, or strobe lighting.

Sign illumination may be either internal or external.

Signs may be made of rigid or flexible material.

1.659 Sign Placement

Signs should be located on the right-hand side of the roadway unless otherwise provided in this Manual.

Where special emphasis is needed, signs may be placed on both the left-hand and right-hand sides of the roadway. Signs mounted on portable supports may be placed within the roadway itself. Signs may also be mounted on or above barricades.

The provisions of this Section regarding mounting height apply unless otherwise provided for a particular sign elsewhere in this Manual.

The minimum height measured vertically from the bottom of the sign to the elevation of the near edge of the pavement, of signs installed at the side of the road in rural areas shall be 5 feet.

The minimum height, measured vertically from the bottom of the sign to the top of the curb, or in the absence of curb, measured vertically from the bottom of the sign to the elevation of the near edge of the traveled way, of signs installed at the side of the road in business, commercial, or residential areas where parking or pedestrian movements are likely to occur, or where the view of the sign might be obstructed, shall be 7 feet.

The minimum height measured vertically from the bottom of the sign to the sidewalk, of signs installed above sidewalks shall be 7 feet.

The height to the bottom of a secondary sign mounted below another sign may be 1 foot less than the height provided previously.

Neither portable nor permanent sign supports should be located on sidewalks, bicycle facilities, or areas designated for pedestrian or bicycle traffic. If the bottom of a secondary sign that is mounted below another sign is mounted lower than 7 feet above a pedestrian sidewalk or pathway, the secondary sign should not project more than 4 inches into the pedestrian facility.

1.660 Temporary Traffic Control Plans

A TTC plan describes TTC measures to be used for facilitating road users through a work zone or an incident area. TTC plans play a vital role in providing continuity of effective road user flow when a work zone, incident, or other event temporarily disrupts normal road user flow. Important auxiliary provisions that cannot conveniently be specified on project plans can easily be incorporated into Special Provisions within the TTC plan.

TTC plans range in scope from being very detailed to simply referencing typical drawings contained in this Manual, standard approved highway agency drawings and manuals, or specific drawings contained in the contract documents. The degree of detail in the TTC plan depends entirely on the nature and complexity of the situation.

TTC plans should be prepared by persons knowledgeable (for example, trained and/or certified) about the fundamental principles of TTC and work activities to be performed. The design, selection, and placement of TTC devices for a TTC plan should be based on engineering judgment.

Coordination should be made between adjacent or overlapping projects to check that duplicate signing is not used and to check compatibility of traffic control between adjacent or overlapping projects.

Provisions for effective continuity of accessible circulation paths for pedestrians should be incorporated into the TTC process. Where existing pedestrian routes are blocked or detoured, information should be provided about alternative routes that are usable by pedestrians with disabilities, particularly those who have visual disabilities. Access to temporary bus stops, travel across intersections with accessible pedestrian signals, and other routing issues should be considered where temporary pedestrian routes are channelized. Barriers and channelizing devices that are detectable by people with visual disabilities should be provided.

Reduced speed limits should be used only in the specific portion of the TTC zone where conditions or restrictive features are present. However, frequent changes in the speed limit should be avoided. A TTC plan should be designed so that vehicles can travel through the TTC zone with a speed limit reduction of no more than 10 mph.

A reduction of more than 10 mph in the speed limit should be used only when required by restrictive features in the TTC zone. Where restrictive features justify a speed reduction of more than 10 mph, additional driver notification should be provided. The speed limit should be stepped down in advance of the location requiring the lowest speed, and additional TTC warning devices should be used.

Reduced speed zoning (lowering the regulatory speed limit) should be avoided as much as practical because drivers will reduce their speeds only if they clearly perceive a need to do so.

1.661 Temporary Traffic Control Zones

A TTC zone is an area of a highway, urban street, or rural street where road user conditions are changed because of a work zone, an incident zone, or a planned special event through the use of TTC devices, uniformed law enforcement officers, or other authorized personnel.

A work zone is an area of a highway, urban street, or rural street with construction, maintenance, or utility work activities. A work zone is typically marked by signs, channelizing devices, barriers, pavement markings, and/or work vehicles. It extends from the first warning sign or high intensity rotating, flashing, oscillating, or strobe lights on a vehicle to the END ROAD WORK sign or the last TTC device.

An incident zone is an area of a highway, urban street, or rural street where temporary traffic controls are imposed by authorized officials in response to a traffic incident It extends from the first warning device (such as a sign, light, or cone) to the last TTC device or to a point where road users return to the original lane alignment and are clear of the incident.

A planned special event often creates the need to establish altered traffic patterns to handle the increased traffic volumes generated by the event. The size of the TTC zone associated with a planned special event can be small, such as closing a street for a festival, or can extend throughout a municipality for larger events. The duration of the TTC zone is determined by the duration of the planned special event.

1.662 Components of Temporary Traffic Control Zones

Most TTC zones are divided into four areas: the advance warning area, the transition area, the activity area, and the termination area. Figure 2 illustrates these four areas.

1.663 Advance Warning Area

The advance warning area is the section of highway where road users are informed about the upcoming work zone or incident area.

The advance warning area may vary from a single sign or high intensity rotating, flashing, oscillating, or strobe lights on a vehicle to a series of signs in advance of the TTC zone activity area.

Typical distances for placement of advance warning signs on freeways and expressways should be longer because drivers are conditioned to uninterrupted flow. Therefore, the advance warning sign placement should extend on these facilities as far as 1/2 mile or more.

On urban streets, the effective placement of the first warning sign in feet should range from 4 to 8 times the speed limit in mph, with the high end of the range being used when speeds are relatively high. When a single advance warning sign is used (in cases such as low-speed residential streets), the advance warning area can be as short as 100 feet. When two or more advance warning signs are used on higher-speed streets, such as major arterials, the advance warning area should extend a greater distance (see Table 2).

Since rural highways are normally characterized by higher speeds, the effective placement of the first warning sign in feet should be substantially longer—from 8 to 12 times the speed limit in mph. Since two or more advance warning signs are normally used for these conditions, the advance warning area should extend 1,500 feet or more for open highway conditions (see Table 2).

The distances contained in Table 2 are approximate, are intended for guidance purposes only, and should be applied with engineering judgment. These distances should be adjusted for field conditions, if necessary, by increasing or decreasing the recommended distances.

Figure 2 Component Parts of a Temporary Traffic Control Zone

Figure 6C-1. Component Parts of a Temporary Traffic Control Zone

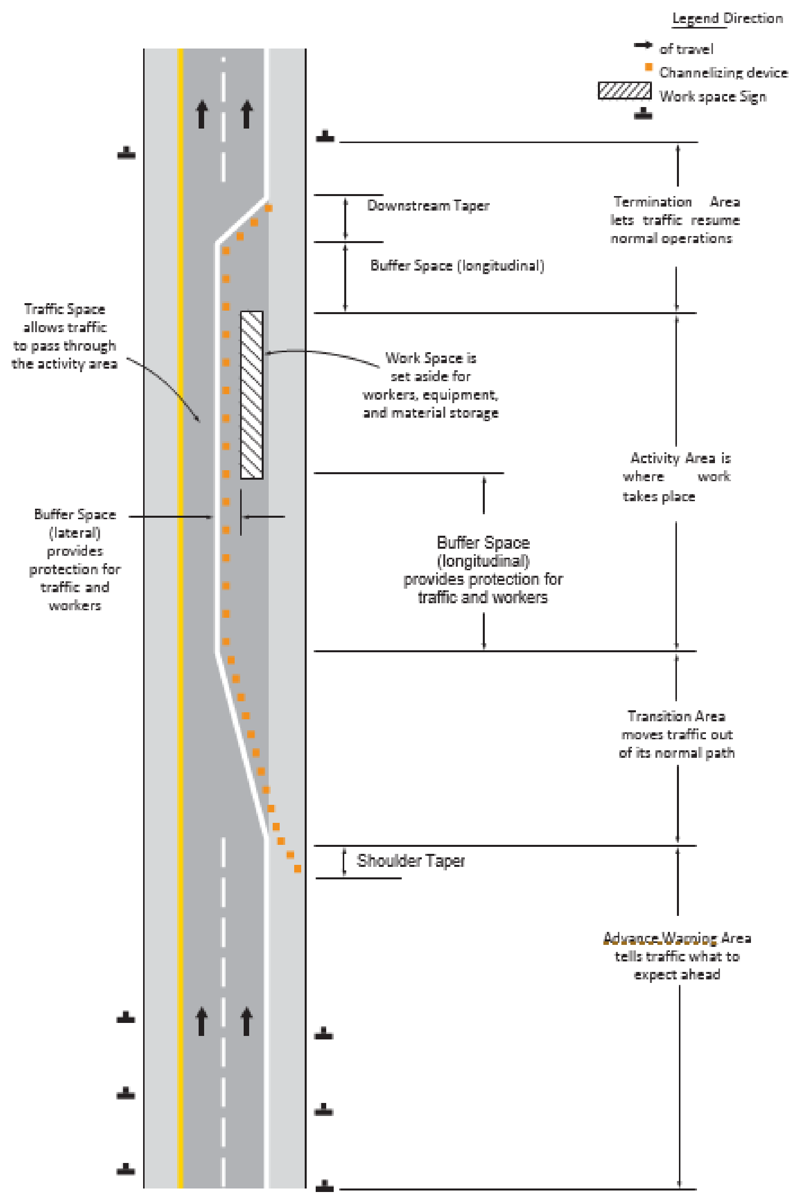


Table 2 Recommended Advance Warning Sign Minimum Spacing

Road Type	**Distance Between Signs****		
	A	B	C
Urban (low speed)*	100 feet	100 feet	100 feet
Urban (high speed)*	350 feet	350 feet	350 feet
Rural	500 feet	500 feet	500 feet
Expressway / Freeway	1,000 feet	1,500 feet	2,640 feet

*Speed category to be determined by the highway agency.

**The column headings A, B, and C are the dimensions shown in the Standard. The A dimension is the distance from the transition or point of restriction to the first sign. The B dimension is the distance between the first and second signs. The C dimension is the distance between the second and third signs. (The "first sign" is the sign in a three-sign series that is closest to the TTC zone. The "third sign" is the sign that is furthest upstream from the TTC zone.)

The need to provide additional reaction time for a condition is one example of justification for increasing the sign spacing. Conversely, decreasing the sign spacing might be justified in order to place a sign immediately downstream of an intersection or major driveway such that traffic turning onto the roadway in the direction of the TTC zone will be warned of the upcoming condition.

Advance warning may be eliminated when the activity area is sufficiently removed from the road users' path so that it does not interfere with the normal flow.

1.664 Transition Area

The transition area is that section of highway where road users are redirected out of their normal path. Transition areas usually involve strategic use of tapers, which because of their importance are discussed separately in detail.

When redirection of the road users' normal path is required, they shall be directed from the normal path to a new path.

Because it is impractical in mobile operations to redirect the road user's normal path with stationary channelization, more dominant vehicle-mounted traffic control devices, such as arrow boards, portable changeable message signs, and high-intensity rotating, flashing, oscillating, or strobe lights, may be used instead of channelizing devices to establish a transition area.

1.665 Activity Area

The activity area is the section of the highway where the work activity takes place. It is comprised of the workspace, the traffic space, and the buffer space.

The workspace is that portion of the highway closed to road users and set aside for workers, equipment, and material, and a shadow vehicle if one is used upstream. Workspaces are usually delineated for road users by channelizing devices or, to exclude vehicles and pedestrians, by temporary barriers.

The workspace may be stationary or may move as work progresses.

Since there might be several workspaces (some even separated by several miles) within the project limits, each workspace should be adequately signed to inform road users and reduce confusion.

The traffic space is the portion of the highway in which road users are routed through the activity area.

The buffer space is a lateral and/or longitudinal area that separates road user flow from the workspace or an unsafe area and might provide some recovery space for an errant vehicle.

Neither work activity nor storage of equipment, vehicles, or material should occur within a buffer space.

Buffer spaces may be positioned either longitudinally or laterally with respect to the direction of road user flow. The activity area may contain one or more lateral or longitudinal buffer spaces.

A longitudinal buffer space may be placed in advance of a workspace.

The longitudinal buffer space may also be used to separate opposing road user flows that use portions of the same traffic lane, as shown in Figure 3.

If a longitudinal buffer space is used, the values shown in Table 1 may be used to determine the length of the longitudinal buffer space.

Typically, the buffer space is formed as a traffic island and defined by channelizing devices.

When a shadow vehicle, arrow board, or changeable message sign is placed in a closed lane in advance of a workspace, only the area upstream of the vehicle, arrow board, or changeable message sign constitutes the buffer space.

The lateral buffer space may be used to separate the traffic space from the workspace, as shown in Figures 2 and 3, or such areas as excavations or pavement edge drop-offs. A lateral buffer space also may be used between two travel lanes, especially those carrying opposing flows.

The width of a lateral buffer space should be determined by engineering judgment.

When work occurs on a high-volume, highly congested facility, a vehicle storage or staging space may be provided for incident response and emergency vehicles (for example, tow trucks and fire apparatus) so that these vehicles can respond quickly to road user incidents.

1.666 Termination Area

The termination area is the section of the highway where road users are returned to their normal driving path. The termination area extends from the downstream end of the work area to the last TTC device such as END ROAD WORK signs, if posted.

An END ROAD WORK sign, a Speed Limit sign, or other signs may be used to inform road users that they can resume normal operations.

A longitudinal buffer space may be used between the workspace and the beginning of the downstream taper.

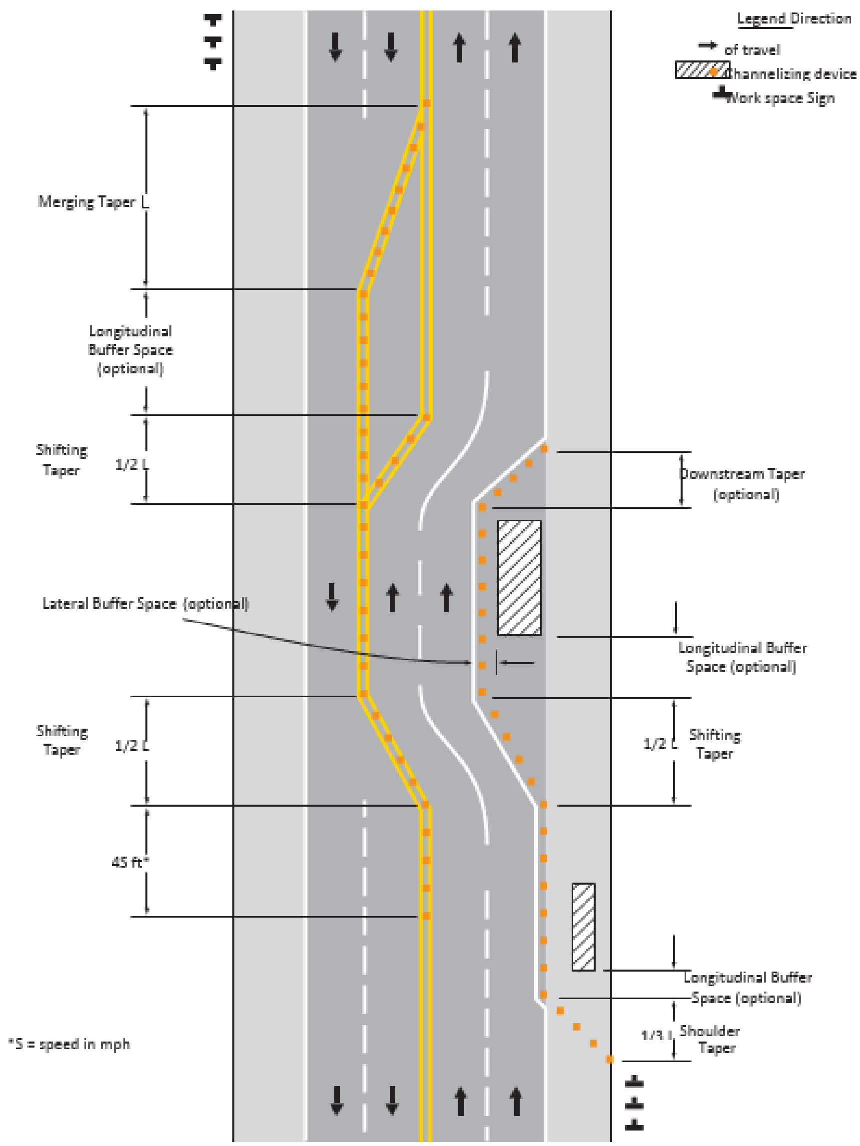
1.667 Tapers

Tapers may be used in both the transition and termination areas. Whenever tapers are to be used in close proximity to an interchange ramp, crossroads, curves, or other influencing factors, the length of the tapers may be adjusted.

Tapers are created by using a series of channelizing devices and/or pavement markings to move traffic out of or into the normal path. Types of tapers are shown in Figure 3.

Figure 3 Types of Tapers and Buffer Spaces

Figure C-2. Types of Tapers and Buffer Spaces



Longer tapers are not necessarily better than shorter tapers (particularly in urban areas with characteristics such as short block lengths or driveways) because extended tapers tend to encourage sluggish operation and to encourage drivers to delay lane changes unnecessarily. The test concerning adequate lengths of tapers involves observation of driver performance after TTC plans are put into effect.

The appropriate taper length (L) should be determined using the criteria shown in Tables 3 and 4.

The maximum distance in feet between devices in a taper should not exceed 1.0 times the speed limit in mph.

A merging taper requires the longest distance because drivers are required to merge into common road space.

A merging taper should be long enough to enable merging drivers to have adequate advance warning and sufficient length to adjust their speeds and merge into an adjacent lane before the downstream end of the transition.

A shifting taper is used when a lateral shift is needed. When more space is available, a longer than minimum taper distance can be beneficial. Changes in alignment can also be accomplished by using horizontal curves designed for normal highway speeds.

A shifting taper should have a length of approximately 1/2 L (see Tables 3 and 4).

A shoulder taper might be beneficial on a high-speed roadway where shoulders are part of the activity area and are closed, or when improved shoulders might be mistaken as a driving lane. In these instances, the same type, but abbreviated, closure procedures used on a normal portion of the roadway can be used.

If used, shoulder tapers should have a length of approximately 1/3 L (see Tables 3 and 4). If a shoulder is used as a travel lane, either through practice or during a TTC activity, a normal merging or shifting taper should be used.

A downstream taper might be useful in termination areas to provide a visual cue to the driver that access is available back into the original lane or path that was closed.

If used, a downstream taper should have a minimum length of 50 feet and a maximum length of 100 feet with devices placed at a spacing of approximately 20 feet.

The one-lane, two-way taper is used in advance of an activity area that occupies part of a two-way roadway in such a way that a portion of the road is used alternately by traffic in each direction.

Traffic should be controlled by a flagger or temporary traffic control signal (if sight distance is limited), or a STOP or YIELD sign. A short taper having a minimum length of 50 feet and a maximum length of 100 feet with channelizing devices at approximately 20-foot spacing should be used to guide traffic into the one-lane section, and a downstream taper should be used to guide traffic back into their original lane.

An example of a one-lane, two-way traffic taper is shown in Figure 4.

Table 3 Taper Length Criteria for Temporary Traffic Control Zones

Type of Taper	Taper Length
Merging Taper	At least L
Shifting Taper	At least 0.5 L
Shoulder Taper	At least 0.33 L
One-Lane, Two-Way Traffic Taper	50 feet minimum, 100 feet maximum
Downstream Taper	50 feet minimum, 100 feet maximum

Table 4 Formulas for Determining Taper Length

Speed (S)	Taper Length (L) in feet
40 mph or less	$L = WS^2 / 60$
45 mph or more	$L = WS$

Where:

- L = taper length in feet
- W = width of offset in feet
- S = posted speed limit, or off-peak 85th percentile speed prior to work starting, or the anticipated operating speed in mph.

1.668 Detours and Diversions

A detour is a temporary rerouting of road users onto an existing highway in order to avoid a TTC zone.

Detours should be clearly signed over their entire length so that road users can easily use existing highways to return to the original highway.

A diversion is a temporary rerouting of road users onto a temporary highway or alignment placed around the work area.

1.669 One Lane, Two Way Traffic Control

Except as provided previously, when traffic in both directions must use a single lane for a limited distance, movements from each end shall be coordinated.

Provisions should be made for alternate one-way movement through the constricted section via methods such as flagger control, a flag transfer, a pilot car, traffic control signals, or stop or yield control.

Control points at each end should be chosen to permit easy passing of opposing lanes of vehicles.

If traffic on the affected one-lane roadway is not visible from one end to the other, then flagging procedures, a pilot car with a flagger used as described in the Standard, or a traffic control signal should be used to control opposing traffic flows.

If the workspace on a low-volume street or road is short and road users from both directions are able to see the traffic approaching from the opposite direction through and beyond the worksite, the movement of traffic through a one-lane, two-way constriction may be self-regulating.

1.670 Flagger Method of One Way, Two Way Traffic Control

Except as provided previously, traffic should be controlled by a flagger at each end of a constricted section of roadway. One of the flaggers should be designated as the coordinator. To provide coordination of the control of the traffic, the flaggers should be able to communicate with each other orally, electronically, or with manual signals. These manual signals should not be mistaken for flagging signals.

When a one-lane, two-way TTC zone is short enough to allow a flagger to see from one end of the zone to the other, traffic may be controlled by either a single flagger or by a flagger at each end of the section.

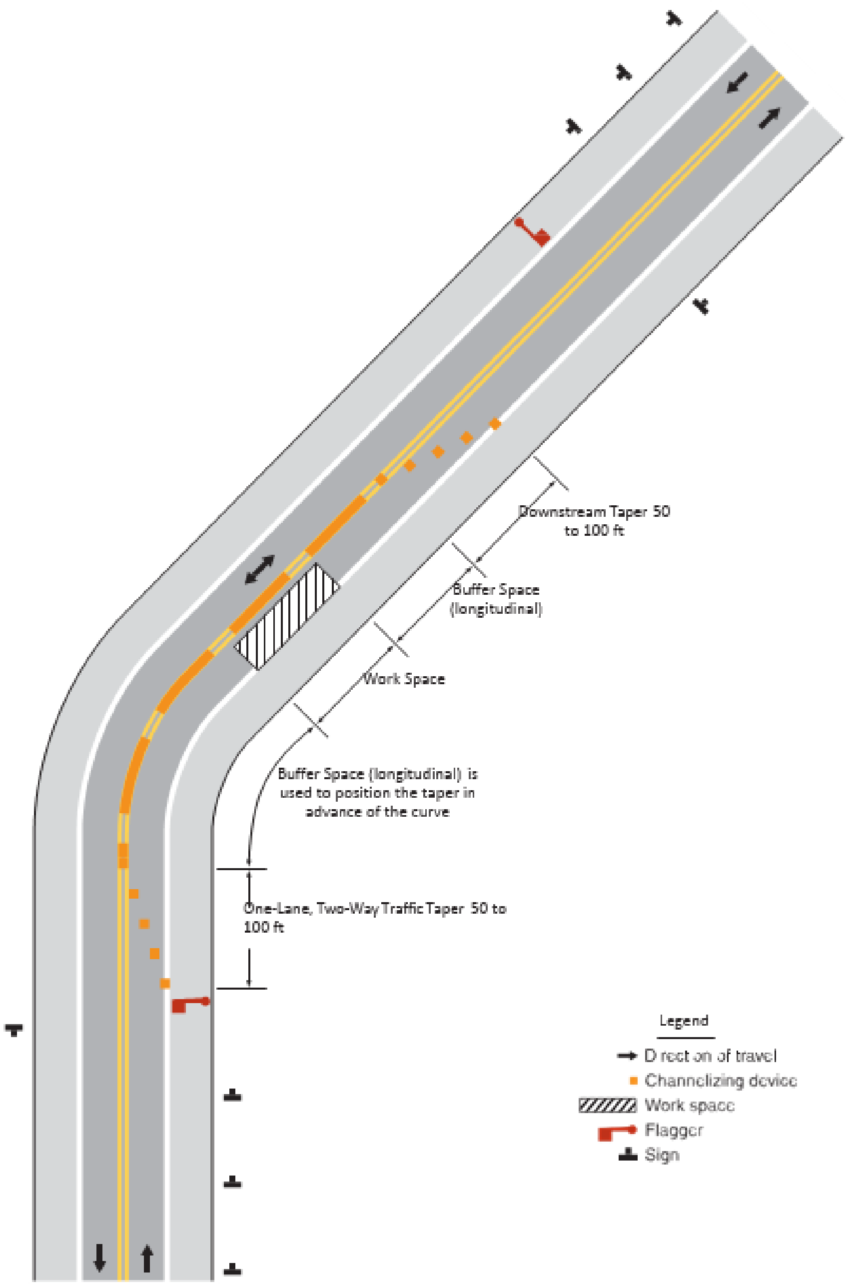
When a single flagger is used, the flagger should be stationed on the shoulder opposite the constriction or workspace, or in a position where good visibility and traffic control can be maintained at all times. When good visibility and traffic control cannot be maintained by one flagger station, traffic should be controlled by a flagger at each end of the section.

1.671 Flag Transfer Method of One Lane, Two Way Traffic Control

The driver of the last vehicle proceeding into the one-lane section is given a red flag (or other token) and instructed to deliver it to the flagger at the other end. The opposite flagger, upon receipt of the flag, then knows that traffic can be permitted to move in the other direction. A variation of this method is to replace the use of a flag with an official pilot car that follows the last road user vehicle proceeding through the section.

The flag transfer method should be employed only where the one-way traffic is confined to a relatively short length of a road, usually no more than 1 mile in length.

Figure 4 Example of a One Lane, Two Way Traffic Taper



1.672 Pilot Car Method of One Lane, Two Way Traffic Control

A pilot car may be used to guide a queue of vehicles through the TTC zone or detour.

The pilot car should have the name of the contractor or contracting authority prominently displayed.

The PILOT CAR FOLLOW ME (G20-4) sign shall be mounted on the rear of the pilot vehicle.

A flagger shall be stationed on the approach to the activity area to control vehicular traffic until the pilot vehicle is available.

1.673 Temporary Traffic Control Signal Method of One Lane, Two Way Traffic Control

Traffic control signals may be used to control vehicular traffic movements in one-lane, two-way TTC zones.

1.674 Stop or Yield Control Method of One Lane, Two Way Traffic Control

STOP or YIELD signs may be used to control traffic on low-volume roads at a one-lane, two-way TTC zone when drivers are able to see the other end of the one-lane, two-way operation and have sufficient visibility of approaching vehicles.

If the STOP or YIELD sign is installed for only one direction, then the STOP or YIELD sign should face road users who are driving on the side of the roadway that is closed for the work activity area.

1.675 DEAD END or NO OUTLET Signs

The DEAD END and NO OUTLET signs may be used to warn road users of a road that has no outlet or that terminates in a dead end or cul-de-sac.

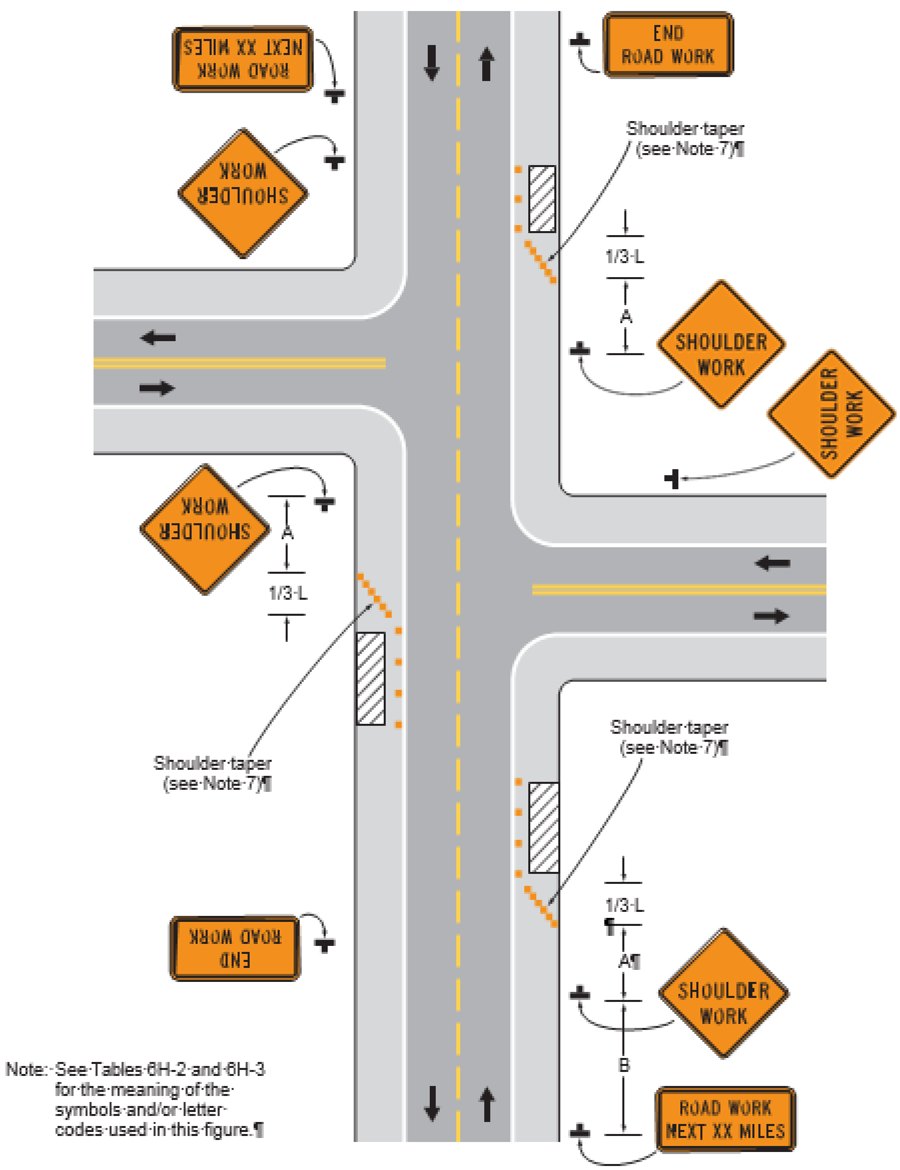
If used, these signs should be placed at a location that gives drivers of large commercial or recreational vehicles an opportunity to select a different route or turn around.

T&T Industrial, Inc.

Traffic Control Program

Figure 5 Work on the Shoulder

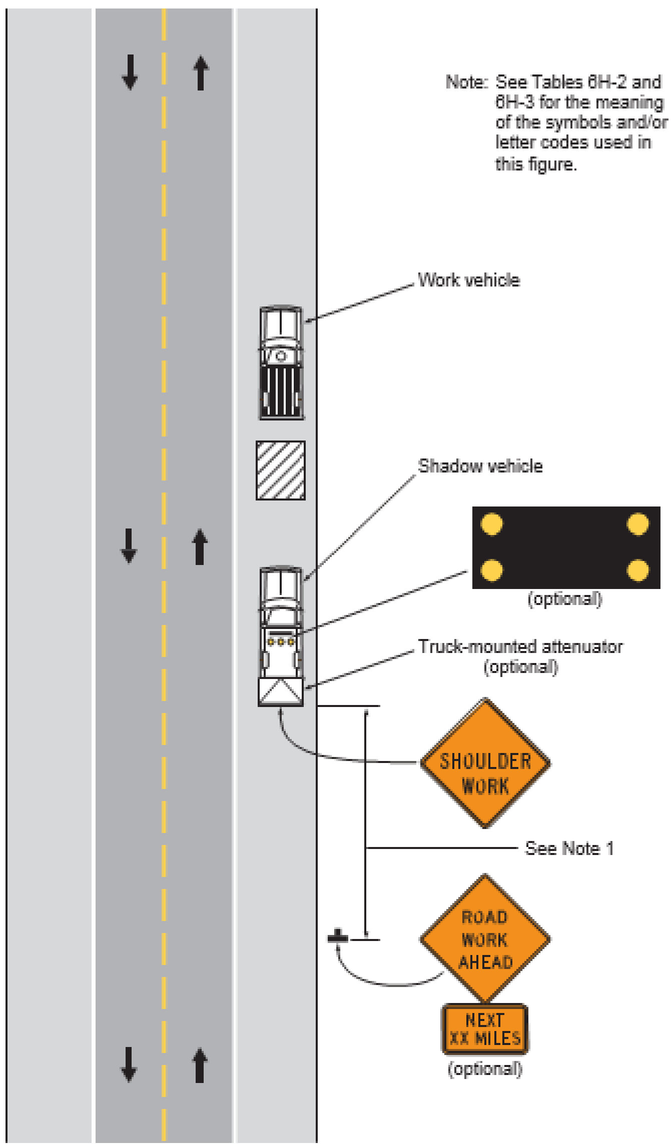
Figure 6H-3. Work on the Shoulders (TA-3)



Typical Application 3

Figure 6 Short Duration or Mobile Operation on a Shoulder

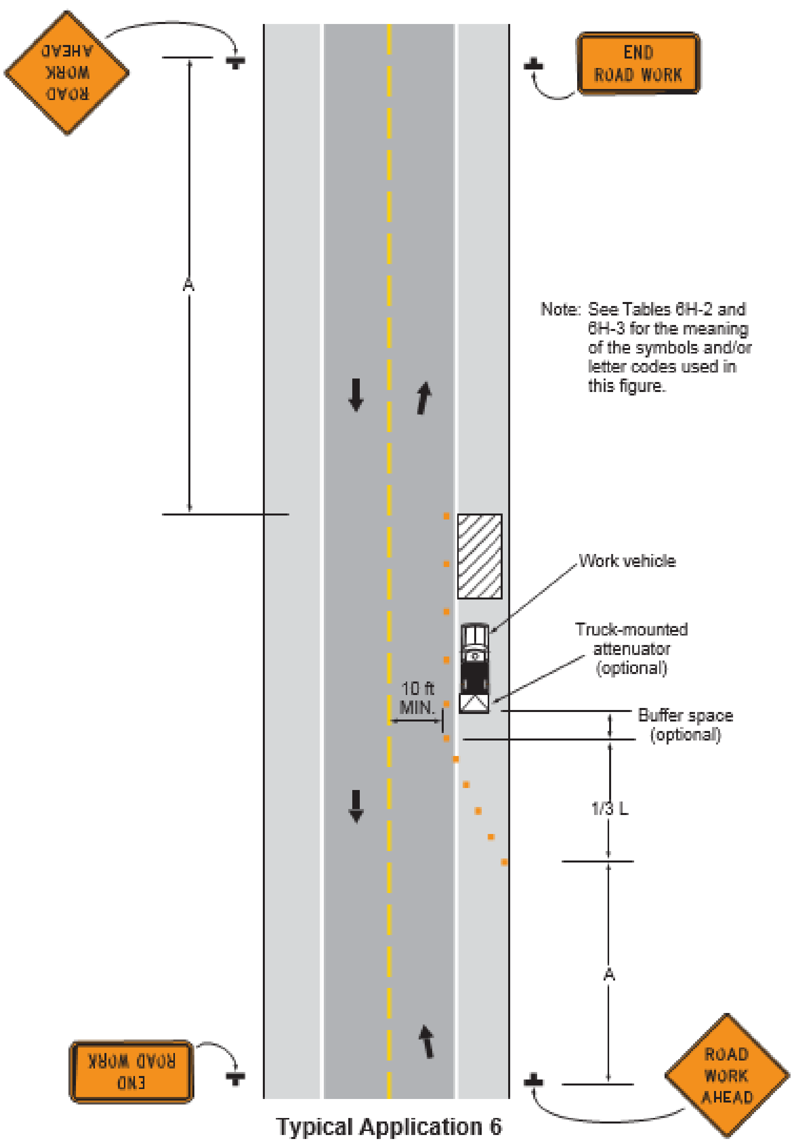
Figure 6H-4. Short-Duration or Mobile Operation on a Shoulder (TA-4)



Typical Application 4

Figure 7 Shoulder Work with Minor Encroachment

Figure 6H-6. Shoulder Work with Minor Encroachment (TA-6)



Typical Application 6

64. Welding, Cutting, Hot Work Program

1.676 Purpose and Scope

The purpose of this program is to establish hot work requirements to ensure all hazards are evaluated and the appropriate safety measures and controls are administered prior to and during any process that involved welding and cutting or any other hot work.

This program applies to all T&T Industrial, Inc. employees involved with hot work.

1.677 Resources

Number	Title
29 CFR 1910 Subpart Q	Welding, Cutting, and Brazing
Cal/OSHA T8CCR Subchapter 7	General Industry Safety Orders – Gas Systems for Welding and Cutting
NFPA 51B	Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
CMS-FM-0050	Hot Work Permit

1.678 Definitions

Acronym/Term	Definition
Welder and Welding Operator	Any operator of electric or gas welding and cutting equipment.
Approved	Listed or approved by a nationally recognized resting laboratory.
Confined Space	A relatively small or restricted space such as a tank, boiler, pressure vessel, or small compartment of a ship.

1.679 Welding, Cutting, and Hot Work Program

Welding, cutting, and hot work such as brazing or grinding present a significant opportunity for fire and injury. All precautions of this program shall be applied prior to commencing any hot work.

1.680 Responsibilities

Management shall recognize its responsibility for the safe usage of cutting and welding equipment on the property and:

- Based on fire potentials of plant facilities, establish areas for cutting and welding, and establish procedures for cutting and welding, in other areas.
- Designate an individual responsible for authorizing cutting and welding operations in areas not specifically designed for such processes.
- Insist that cutters or welders and their supervisors are suitably trained in the safe operation of their equipment and the safe use of the process.
- Advise all contractors about flammable materials or hazardous conditions of which they may not be aware.

The **Supervisor**:

- Shall be responsible for the safe handling of the cutting or welding equipment and the safe use of the cutting or welding process.
- Shall determine the combustible materials and hazardous areas present or likely to be present in the work location.
- Shall protect combustibles from ignition by the following:
  - Have the work moved to a location free from dangerous combustibles.
  - If the work cannot be moved, have the combustibles moved to a safe distance from the work or have the combustibles properly shielded against ignition.
  - See that cutting and welding are so scheduled that operations that might expose combustibles to ignition are not started during cutting or welding.
- Shall secure authorization for the cutting or welding operations from the designated management representative.
- Shall determine that the cutter or welder secures approval that conditions are safe before going ahead.
- Shall determine that fire protection and extinguishing equipment are properly located at the site.
- Where fire watches are required, see that they are available at the site.

1.681 Training

Cutters, welders, and their supervisors shall be suitably trained and qualified in the safe operation of hot work equipment and safe use of the process.

Assigned fire watchers shall be trained in the use of fire extinguishing equipment and familiar with the facilities for sounding an alarm in the event of a fire.

Affected employees assigned to maintain or operate welding or cutting equipment must be familiar with 29 CFR 1910.254 and 29 CFR 1910.252(a-c).

Affected employees whose duties require arc welding or cutting must be trained in a manner such that those employees are qualified to operate such equipment.

1.682 Special Precautions

Employees must cease hot work operations if welding and cutting cannot be performed in a safe manner.

The Qualified Hot Work Supervisor shall verify that all fire suppression systems present in the area are fully operational. If fire suppression systems cannot be operational, additional protective measures shall be put in place.

If the object to be welded or cut cannot readily be moved, all movable fire hazards in the vicinity shall be taken to a safe place.

If the object to be welded or cut cannot be moved and if all the fire hazards cannot be removed, then guards shall be used to confine the heat, sparks, and slag, and to protect the immovable fire hazards.

Where practicable, all combustibles shall be relocated at least 35 feet from the work site. Where relocation is impracticable, combustibles shall be protected with flameproof covers or otherwise shielded with metal or asbestos guards or curtains.

If the requirements for fire hazards and guarding cannot be followed, then welding and cutting shall not be performed.

Wherever there are floor openings or cracks in the flooring that cannot be closed, precautions shall be taken so that no readily combustible materials on the floor below will be exposed to sparks which might drop through the floor. The same precautions shall be observed regarding cracks or holes in walls, open doorways, and open or broken windows.

The frame or case of welding machines, except engine-driven machines, shall be grounded.

Before starting operations all connections to welding machines shall be checked to make certain they are properly made.

Ducts and conveyor systems that might carry sparks to distant combustibles shall be suitably protected or shut down.

Where cutting or welding is done near walls, partitions, ceiling, or roof of combustible construction, fire-resistant shields or guards shall be provided to prevent ignition.

If welding is to be done on a metal wall, partition, ceiling or roof, precautions shall be taken to prevent ignition of combustibles on the other side, due to conduction or radiation, preferably by relocating combustibles. Where combustibles are not relocated, a fire watch on the opposite side from the work shall be provided.

Welding shall not be attempted on a metal partition, wall, ceiling, or roof having a combustible covering nor on walls or partitions of combustible sandwich-type panel construction.

Cutting or welding on pipes or other metal in contact with combustible walls, partitions, ceilings, or roofs shall not be undertaken if the work is close enough to cause ignition by conduction.

1.683 Fire Watch

Fire watchers shall be required whenever welding or cutting is performed in the following situations:

- Locations where other than a minor fire might develop.
- Appreciable combustible material is closer than 35 feet to the point of operation.
- Appreciable combustibles are more than 35 feet away but are easily ignited by sparks.
- Wall or floor openings within a 35-foot radius expose combustible material in adjacent areas including concealed spaces in walls or floors.
- Combustible materials are adjacent to the opposite side of metal partitions, walls, ceilings, or roofs and are likely to be ignited by conduction or radiation.

Fire watchers shall have fire extinguishers readily available and shall be trained in its use. They shall be familiar with facilities for sounding an alarm in the event of a fire. They shall watch for fires in all exposed areas, try to extinguish them only when obviously within the capacity of the equipment available, or otherwise sound the alarm. A fire watch shall be maintained at least half an hour after the welding or cutting operation is completed. Depending on location and local regulations, a fire watch shall be maintained at a minimum of 60 minutes after the welding or cutting operation is complete. A minimum of 2 hours of surveillance must be maintained after completing hot work.

1.684 Hot Work Permit

Before cutting or welding is permitted, the area shall be inspected by the individual responsible for authorizing cutting and welding operations. Precautions shall be designated to be followed in granting authorization to proceed, preferably in the form of a written permit.

1.685 Prohibited Areas

Cutting or welding shall not be permitted in areas not authorized by management, in sprinkled buildings while such protection is impaired, in the presence of explosive atmospheres, areas near the storage of large quantities of exposed, readily ignitable materials.

1.686 Personal Protective Equipment (PPE)

Employees exposed to the hazards created by welding, cutting, or brazing operations shall be protected by personal protective equipment (PPE).

Goggles or other suitable eye protection shall be used during all gas welding or oxygen cutting operations. Spectacles without side shields, with suitable filter lenses are permitted for use during gas welding operations on light work, for torch brazing, or for inspection.

Helmets or hand shields shall be used during all arc welding or arc cutting operations.

1.687 Ventilation

Local exhaust or general ventilating systems shall be provided and arranged to keep the amount of toxic fumes, gases, or dusts below the maximum allowable concentration.

1.688 Confined Spaces

When hot work is to be performed in confined spaces, ventilation, securing cylinders, lifelines, electrode removal, gas cylinders shutoff, and warning signs shall be addressed.

When welding or cutting is being performed in any confined spaces the gas cylinders and welding machines shall be left on the outside. Before operations are started, heavy portable equipment mounted on wheels shall be securely blocked to prevent accidental movement.

Where a welder must enter a confined space through a manhole or other small opening, means shall be provided for quick removal in case of emergency. When safety belts and lifelines are used for this purpose, they shall be so attached to the welder's body that the body cannot be jammed in a small exit opening. An attendant with a preplanned rescue procedure shall be stationed outside to observe the welder at all times and be capable of putting rescue operations into effect.

When arc welding is to be suspended for any substantial period of time, such as during lunch or overnight, all electrodes shall be removed from the holders and the holders carefully located so that accidental contact cannot occur, and the machine shall be disconnected from the power source.

In order to eliminate the possibility of gas escaping through leaks of improperly closed valves, when gas welding or cutting, the torch valves shall be closed and the fuel-gas and oxygen supply to the torch positively shut off at some point outside the confined area whenever the torch is not to be used for a substantial period of time, such as during lunch hour or overnight. Where practicable, the torch and hose shall also be removed from the confined space.

After welding operations are completed, the welder shall mark the hot metal or provide some other means of warning other employees.

1.689 Equipment

The operator shall report any equipment defect or safety hazard to the supervisor and the use of the equipment shall be discontinued until safety has been assured. Repairs shall be made only by qualified personnel.

First aid equipment shall be available at all times.

Only approved apparatus such as torches, regulators or pressure-reducing valves, acetylene generators, and manifolds shall be used.

Employees in charge of the oxygen or fuel-gas supply equipment, including generators, and oxygen or fuel-gas distribution piping systems shall be instructed and judged competent by the Company for this important work before being left in charge.

1.690 Identification

Compressed gas cylinders shall be legibly marked, for the purpose of identifying the gas content, with either the chemical or the trade name of the gas. Such marking shall be by means of stenciling, stamping, or labeling, and shall not be readily removable. Whenever practical, the marking shall be located on the shoulder of the cylinder.

1.691 Transportation and Storage

Cylinders shall be kept away from radiators and other sources of heat. Inside of buildings, cylinders shall be stored in a well-protected, well-ventilated, dry location, at least 20 feet from highly combustible materials and any flammable or petroleum products.

Cylinders shall be stored in assigned places away from elevators, stairs, or gangways. Assigned storage spaces shall be located where cylinders will not be knocked over or damaged by passing or falling objects, or subject to tampering by unauthorized persons.

Cylinders shall not be kept in unventilated enclosures such as lockers and cupboards.

Empty cylinders shall have their valves closed. Valve protection caps, where cylinder is designed to accept a cap, shall always be in place except when cylinders are in use or connected for use.

Appendix 40 **Hot Work Permit**

General Information			
Hot Work Operator:		Date:	
Fire Watch Name:		Job No.:	
Location of Work:			
Description of Work:			
Date Work to Begin:		Date Work to End:	
Time Work to Begin:		Time Work to End:	

Emergency Notification	
Contact	Phone No.

T&T Industrial, Inc.

Welding, Cutting, Hot Work Program

Building Information			
Building Name:		Building Number:	
Floor:		Specific Location:	
Do building fire alarm system devices have to be deactivated?		Yes ☐	No ☐
Device number(s) requiring deactivation:			
Date(s) device(s) will be deactivated:			

Special Precautions			
1.	Proper training verified by hot work operator.	Yes ☐	No ☐
2.	Can hot work be moved outside or to a safe location?	Yes ☐	No ☐
3.	Hot work area swept and clear of combustible materials within a 35-foot radius.	Yes ☐	No ☐
4.	Combustible construction has been shielded.	Yes ☐	No ☐
5.	All tools, equipment, and PPE inspected and in good working condition.	Yes ☐	No ☐
6.	Adequate ventilation.	Yes ☐	No ☐
7.	Fire extinguishers are in service / operable.	Yes ☐	No ☐
8.	Fire watch has fire extinguisher and has completed fire extinguisher training.	Yes ☐	No ☐
9.	Management has evaluated the facility.	Yes ☐	No ☐
10.	Operator understands that work must stop if unsafe.	Yes ☐	No ☐
11.	Confined space entry permit required.	Yes ☐	No ☐

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Welding, Cutting, Hot Work Program

12.	Lockout / tagout required.	Yes ☐	No ☐
13.	Hot work area will be monitored for 30 minutes after work.	Yes ☐	No ☐
Additional Precautions:			

Signatures			
I verify that the above location has been examined and the required precautions have been taken.			
Hot Work Operator Name:		Date:	
Hot Work Operator Signature:			
I verify that the above location has been examined and the required precautions have been taken.			
Post-Work Fire Watch Finish Time:			
Fire Watch Name:		Date:	
Fire Watch Signature:			
I verify that the above location has been examined, the required precautions have been taken, and permission is authorized for this work.			
Permit Authorizer Name:		Date:	
Permit Authorizer Signature:			

65. Working Alone Program

1.692 Purpose and Scope

The purpose of this program is to encourage awareness and promote safe work practices to minimize the risk of injury for employees who work alone.

This program applies to all T&T Industrial, Inc. employees who work alone.

1.693 Resources

Number	Title
CMS-FM-0051	Working Alone Checklist

1.694 Definitions

Acronym/Term	Definition
High Risk Work	Work that exposes employees to hazards that, should an incident occur, may result in a fatality or permanent disability.
Working Alone	Any work performed not within normal sight or hearing distance of another employee.

1.695 Working Alone Program

Employees who work alone tend to be more vulnerable than those who have co-workers present. Even though emergency incidents are not that common, when they do occur, the consequences can be serious.

1.696 Training

Employee training shall include how to:

- Perform tasks safely
- Operate machines and equipment safely
- Use and maintain any required personal protective equipment (PPE)
- Identify and report hazards
- Follow the written person check procedures

1.697 Hazard Assessment

A hazard assessment shall be conducted to evaluate the risk of working alone. Hazards shall be addressed, and control measures identified to minimize the risk associated with working alone.

1.697.1 High Risk Areas

Employees shall not work alone in high-risk areas such as work involved with:

- Live electrical
- Diving operations
- Extreme temperatures
- Acutely toxic chemicals
- Confined spaces
- Dangerous machinery
- Working at heights

1.697.2 Low Risk Areas

Examples of low-risk areas include:

- Maintenance, other than on active equipment
- Security, except in hazardous locations
- Janitorial or custodial positions, except when location is hazardous
- Transportation under ideal conditions
- Offices or while in transit on a business trip

1.698 Communication Plan

Employees shall carry a mobile phone or electronic monitoring device at all times while working alone. This means of communication shall be established between the lone employee and the designated check person.

A designated check person shall be established for check-in with the employee working alone.

Employees shall be monitored at regular intervals, or the employee shall contact the designated check person at pre-determined intervals based on determinations made in the risk assessment.

A backup form of communication shall be established in the event that communication is unavailable.

Employee status at check-in shall be documented.

Emergency response procedures, including search procedures, shall be followed if the employee working alone cannot be contacted or fails to check in.

If effective electronic communication is not practicable at the work site, ensure that: (a) the employer or designate visits the worker, or (b) the worker contacts the employer or designate at intervals appropriate to the nature of the hazard associated with the worker's work.

T&T Industrial, Inc.

Working Alone Program

Appendix 41 **Working Alone Checklist**

General Information			
Employee Name:		Date:	
Phone No.:		Supervisor:	
Location:			

Communication Details	
Emergency Phone No.:	
Name of Check Person:	
Contact for Check Person:	
Method to be used for checking:	
Intervals for contact:	

Checklist		
Has a risk assessment been conducted for this task (attach)?	Yes ☐	No ☐
Is the employee trained in working alone procedures?	Yes ☐	No ☐
Is the employee aware of all risks associated with the task?	Yes ☐	No ☐
Has a safety inspection been completed prior to operating equipment?	Yes ☐	No ☐
Have emergency plans been discussed?	Yes ☐	No ☐
Is PPE and training on proper use available?	Yes ☐	No ☐
Have standard operating procedures been provided and discussed?	Yes ☐	No ☐
Have additional controls been put in place for the task?	Yes ☐	No ☐

T&T Industrial, Inc.

Working Alone Program

Communication Arrangements
Describe the communications arrangements:
Action to be taken if contact is not made as described above:
Signatures
I, ______________________________________, am aware and agree to abide by all Company procedures when working alone. I also agree to abide by any additional requirements as listed in the attached risk assessment for this task.

Employee Signature:		Date:	
Supervisor Signature:		Date:	
Approval Begins:		Approval Ends:	

66. SILICA EXPOSURE PROGRAM

1.699 Purpose and Scope

The purpose of this program is to provide information about silica, the potential health effects associated with exposure, and safety procedures that should be followed to reduce exposure and protect the health of employees.

This program applies to all T&T Industrial, Inc. employees.

1.700 Resources

Number	Title
29 CFR 1910 Subpart Z	Toxic and Hazardous Substances – Respirable Crystalline Silica
29 CFR 1926 Subpart Z	Toxic and Hazardous Substances – Respirable Crystalline Silica

1.701 Definitions

Acronym/Term	Definition
Action Level	A concentration of airborne respirable crystalline silica of 25 μg/m^3, calculated as an 8-hour (time weighted average) TWA.
Competent Employee / Person	A person who can identify existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.
High-Efficiency Particulate Air (HEPA) Filter	A filter that is at least 99.97 percent efficient in removing mono-dispersed particles of 0.3 micrometers in diameter.
Objective Data	Information, such as air monitoring data from industry-wide surveys or calculations based on the composition of a substance, demonstrating employee exposure to respirable crystalline silica associated with a particular product or material or a specific process, task, or activity. The data must reflect workplace conditions closely resembling or with a higher exposure potential than the processes, types of material, control methods, work practices, and environmental conditions in the employer's current operations.
Respirable Crystalline Silica	Quartz, cristobalite, and/or tridymite contained in airborne particles that are determined to be respirable by a sampling device designed to meet the characteristics for respirable-particle-size-selective samplers specified in the International Organization for Standardization (ISO) 7708:1995: Air Quality—Particle Size Fraction Definitions for Health-Related Sampling.

1.702 Silica Exposure

Silica is the compound formed from the elements silicon (Si) and oxygen (O) and has a molecular form of SiO_2. Silica is the second most common mineral on earth, found in the common form as sand and rock. The three main forms or polymorphs of silica are alpha quartz, cristobalite, and tridymite. The polymer most abundant and most hazardous to human health is alpha quartz and is commonly referred to as crystalline silica.

The health hazards of silica come from breathing in the dust. If crystalline silica becomes airborne through industrial activities, exposures to fine crystalline silica dust (specifically exposure to the size fraction that is considered to be respirable) can lead to disabling, sometimes fatal disease called silicosis.

No employee shall be exposed to an airborne concentration of respirable crystalline silica in excess of 50µg /m^3, calculated as an 8-hour time weighted average (TWA).

1.703 Training

All employees who are exposed to action level respirable crystalline silica at or above the action level (8-hour TWA of 25µg/m^3) shall be provided with training initially and refresher training periodically.

The training shall ensure that employees covered by the written exposure control plan can demonstrate knowledge and understanding of at least the following:

- The health hazards associated with respirable crystalline silica.
- The specific tasks in the workplace that could result in exposure to respirable crystalline silica.
- The specific measures taken to protect employees from exposure to crystalline silica.
- The contents of the respirable crystalline silica rule.
- The designated competent person.
- The purpose of the medical surveillance program.
- The appropriate use of respiratory protection related to respirable crystalline silica. Appropriate respirators, approved by the National Institute for Occupational Safety and Health (NIOSH) and based upon the measured exposure levels of workers and the assigned protection factors (APF) of the respirators must be selected.

1.704 Assessments

The exposure of each employee who is or is expected to be exposed to respirable crystalline silica at or above the action level (8-hour TWA of 25 µg/m³) shall be assessed. This assessment can be performed by monitoring employees individually or taking a representative sample from employees.

Exposures shall be reassessed whenever there is a change in the production, process, control equipment, personnel, or practices that may reasonably be expected to result in new or additional exposures at or above the action level, or when there is any reason to believe that new or additional exposures at or above the action level have occurred.

Each affected employee shall be individually notified in writing of the results of that assessment or post the results in an appropriate location accessible to all affected employees. Whenever an exposure assessment indicates that employee exposure is above the permissible exposure limit (PEL), the corrective action being taken to reduce employee exposure to or below the PEL shall be described in the written notification.

1.705 Personal Protective Equipment (PPE)

Employees with exposure to silica must be issued and must wear appropriate personal protective equipment (PPE) to include but is not limited to proper eye and respiratory protection.

1.706 Respiratory Protection

Appropriate respirators shall be provided to employees who are or will be exposed to actionable levels of respirable crystalline silica. Table 1 of 1926.1153 (c) shall be referenced for specified exposure control methods. If an employee is performing a task listed in Table 1 that does not require the use of a respirator, then they are not required. All other tasks not covered by Table 1 shall be accounted for by providing respirators if necessary.

Respiratory protection is required:

- Where exposures exceed the PEL when installing or implementing engineering and work practice controls.
- During tasks for which engineering and work practice controls are not feasible.
- When all feasible engineering and work practice controls are not sufficient to reduce exposures to or below the PEL.
- During periods when the employee is in a regulated area.

1.707 Regulated Area

A regulated area shall be established wherever an employee's exposure to airborne concentrations of respirable crystalline silica is, or can reasonably be expected to be, in excess of the PEL.

1.708 Controls

Engineering and work practice controls shall be used to reduce and maintain employee exposure to respirable crystalline silica to or below the PEL. Wherever such feasible engineering and work practice controls are not sufficient, the Company shall use them to reduce employee exposure to the lowest feasible level and shall supplement them with the use of respiratory protection.

1.709 Exposure Control Plan

A written exposure control plan that contains at least the following elements shall be established and implemented:

- Description of the tasks in the workplace that involve exposure to respirable crystalline silica.
- Description of the engineering controls, work practices, and respiratory protection used.
- Description of the housekeeping measures.

The effectiveness of the written exposure control plan shall be reviewed and evaluated at least annually. The written exposure control plan shall be readily available for examination and copying upon request. Copies may be available electronically or physically, depending on location needs and requirements.

A competent person shall be designated to make frequent and regular inspections of job sites, materials, and equipment to implement the written exposure control plan.

Exposure tasks may include but are not limited to activities such as:

- Sawing
- Drilling
- Jackhammering
- Grinding

Any changes in task shall be communicated to affected employees.

Situations where reevaluation may be necessary include regulatory updates, changes in equipment, and exposure incidents. Any changes resulting from this process shall be communicated to affected employees.

1.710 Housekeeping

A description of housekeeping measures used to limit exposure to respirable crystalline silica shall be in place. Some examples of these measure include:

- HEPA-filtered vacuuming
- Wet sweeping
- Wetting
- Any other techniques used to limit the amount of respirable crystalline silica exposure during housekeeping activities.

Compressed air shall be used to clean clothing or surfaces where doing so could contribute to employee exposure to respirable crystalline silica.

Dry sweeping or dry brushing shall not be allowed where such activity could contribute to employee exposure to respirable crystalline silica unless wet sweeping.

1.711 Medical Surveillance

A medical surveillance program shall be established for employees who are exposed to the action level of 8-hour TWA of 25μg/m³ of respirable crystalline silica at no cost to the employee, at a reasonable time and place.

A baseline medical assessment shall be available to exposed employees within 30 days of initial assignment unless they have previously received a suitable medical examination in the past 3 years. This applies to employees who would be required to wear a respirator more than 30 days per year or who are exposed to action level respirable crystalline silica for more than 30 days per year. A suitable prescreen that meets the same requirements is also acceptable.

1.712 Recordkeeping

An accurate record shall be made available and maintained of all exposure measurements taken to assess employee exposure to respirable crystalline silica, all objective data relied upon to comply with the requirements, and accurate records for each employee covered by medical surveillance.

This record of exposure measurements shall include at least the following information:

- The date of measurement for each sample taken.
- The task monitored.
- Sampling and analytical methods used.
- Number, duration, and results of samples taken.
- Identity of the laboratory that performed the analysis.
- Type of personal protective equipment, such as respirators, worn by the employees monitored.
- Name and job classification of all employees represented by the monitoring, indicating which employees were actually monitored.

This record of objective data shall include at least the following information:

- The crystalline silica-containing material in question.
- The source of the objective data.
- The testing protocol and results of testing.
- A description of the process, task, or activity on which the objective data were based.
- Other data relevant to the process, task, activity, material, or exposures on which the objective data were based.

The record of medical surveillance shall include the following information about the employee:

- Name and social security number.
- A copy of the physician or other licensed health care professionals' and specialists' written medical opinions.
- A copy of the information provided to the physician or other licensed health care professionals and specialists.

67. Commercial Fleet Safety Program

1.713 Purpose and Scope

The purpose of this program is to strive to reduce or eliminate motor vehicle accidents and associated injuries by following the safe practices established in this program.

Compliance with this program is mandatory for all company commercial drivers.

Violations of this program may result in disciplinary action up to and including suspension of driving privileges or termination.

Any deviations from this program must be immediately brought to the attention of the employee's supervisor or the Program Administrator.

This program applies to all T&T Industrial, Inc. employees.

1.714 Program Administrator

The Program Administrator is responsible for the following:

- Evaluating prospective company drivers
- Maintaining an accurate qualified drivers list
- Maintaining accurate qualification records
- Maintaining accurate substance abuse testing records
- Ensuring company vehicles are maintained mechanically
- Selection / procurement of all company vehicles
- Ensuring all qualified drivers are trained in the safe operation of company's vehicles
- Monitoring drivers to ensure compliance with all elements of this program

1.715 Driver Trainers

Driver Trainers of the Company are responsible for the following:

- Conducting on-road driving tests for new employees and existing employees at least annually
- Making recommendations to the Program Administrator regarding the retention or release of employees based on driving tests

1.716 Commercial Drivers

Drivers of the Company are responsible for conducting themselves in accordance with this program. All drivers will:

- Meet all minimum qualification criteria
- Be medically qualified to drive a commercial motor vehicle
- Maintain satisfactory evaluations from the company's Driver Trainer.
- Receive negative drug / alcohol tests
- Maintain an acceptable motor vehicle record (MVR)

A driver is anyone who may, in the course of their employment, operate a company-owned commercial motor vehicle, a rented / leased commercial motor vehicle, and/or a personal commercial motor vehicle on company business.

1.717 Commercial Driver Qualification Criteria

Commercial driver applicants will not be considered for employment unless they meet the minimum requirements listed below.

- Be at least 25 years old to operate a commercial motor vehicle intrastate
- Be able to read and speak English sufficiently to converse with the general public, to understand highway traffic signs and signals, to respond to official inquiries, and to make entries on reports and records
- Be physically and mentally qualified to drive a company vehicle and possess a valid medical certificate as defined in 49 CFR Part 391.
- Possess a current and valid commercial driver's license or chauffer's license and proper endorsements for the type of commercial vehicle to be driven.
- Must not be disqualified to drive a commercial motor vehicle under the rules and regulations set forth in 49 CFR Part 391.15.
- Meet all of the requirements and be able to perform all of the tasks and essential duties of the job description
- Have at least 2 years of verifiable driving experience with like type vehicles
- Have at least 5 years verifiable driving experience, if required to transport hazardous materials

- Have not been convicted of any of the following violations within the previous 5 years:
  - Driving under the influence of alcohol and/or drugs,
  - Reckless driving / speed contests
  - Hit and run accidents
  - Vehicular manslaughter / homicide
  - Leaving the scene of an accident
  - Failure to report an accident
  - Improper or erratic lane changing
  - Following too closely
  - Distracted driving (including cell phone use [texting or talking] while driving),
  - Flee / elude police officer
  - Passing a stopped school bus
  - Speeding 15 miles per hour over the posted speed limit
  - Refusal to submit to an alcohol and/or drug test
  - Operating with a suspended or revoked license
- Have not experienced any of the following within the previous 3 years:
  - Two “at fault” accidents
  - Three moving violations
  - Two moving violations and one “at fault” accident
  - Tested positive for drugs or alcohol

1.718 Hiring Process

The Company employment hiring process is designed to ensure that the safest individuals are hired to operate our motor vehicles.

This multi-step process shall be used for all applicants and will be administered uniformly without bias toward race, color, religion, gender, age, national origin, disability, sexual orientation, or any other criteria deemed unlawful by state, federal, or local law.

1.718.1 Application

All commercial driver applicants must submit a completed, accurate, signed, and dated application for employment.

The hiring / screening process will not continue until all information on the application has been verified.

1.718.2 Previous Employment

The employment history will be collected and verified for every commercial driver applicant.

All commercial driver applicants must provide the following employer information on all driving positions they have held for the previous 10 years:

- Names, addresses, and phone numbers or other contact information of previous employers
- Names, titles, and phone numbers or other contact information of previous supervisors

1.718.3 Motor Vehicle Records

The driving record from the previous 5 years will be examined for all commercial driver applicants from the appropriate agency of every state in which the applicant held a motor vehicle license or permit.

The driver qualification and hiring process will not continue until all driving record information has been verified and no disqualifying items have been found (See qualification requirements above).

1.718.4 Drug and Alcohol History

All applicants will be asked if they have tested positive, or refused to test, on any preemployment drug and alcohol test administered by a previous employer.

If the employee admits to any of the above, without documented successful completion of DOT return-to-duty requirements, the applicant will not be considered for employment.

All applicants who indicate no drug or alcohol violations must provide written consent for a drug and alcohol history to be obtained for the preceding 2 years from all DOT-regulated employers.

If the applicant fails to provide this consent, they will not be considered for employment. Any positive indication of drug or alcohol use at the following levels will immediately disqualify the applicant.

- Alcohol test with a result of 0.04 or higher
- Verified positive drug test
- Verified adulterated or substituted drug test results
- Violations of DOT agency drug and alcohol testing regulations

Individuals who have successfully completed DOT return-to-duty requirements after a drug or alcohol regulation violation will continue through the hiring process.

1.719 Pre-Employment Screening Program

All applicants must provide written approval for the Company to request a copy of the applicant's commercial driving record from the FMCSA's Pre-Employment Screening Program.

Non-compliance with hours of service, cargo securement, vehicle inspections, etc. will be evaluated in the hiring process. Significant or repeated violations may disqualify the applicant.

1.720 Background and Fair Credit Reporting Act Investigations

All applicants must provide written approval for the Company to perform a Criminal Background Check and a Credit Report Check.

These checks will be made on all commercial driver applicants and other applicants that may be required to operate a motor vehicle while conducting company business.

1.721 Proof of Citizenship and Right to Work

All commercial driver applicants shall be required to provide either proof of US citizenship or proof of their legal right to work in the United States.

1.722 Personal Interviews

All applicants will be given an in-person interview by the Program Administrator.

1.723 Drug / Alcohol Screening

All commercial driver applicants will submit to a drug / alcohol screening after an initial offer of employment is extended.

Only the designated Company drug / alcohol testing facility will be used.

Drug / alcohol test results from the commercial driver applicant's previous employer will not be accepted.

A negative test result is a condition of employment.

No driver applicant will perform any work or activity for the Company until a negative test result has been obtained for the driver applicant.

Be advised that marijuana remains a drug listed in Schedule I of the Controlled Substances Act. It is unacceptable for any employee subject to drug testing under the DOT's drug testing regulation to use marijuana medicinally or recreationally.

1.724 Medical Qualification

All applicants shall be medically examined and certified as physically qualified to operate a commercial motor vehicle by a licensed, DOT-certified medical examiner designated by the Company. All drivers must ensure they are physically able to drive as deemed by a licensed physician at least every 24 months.

1.725 Driving Evaluation

All applicants will be required to submit to a driving test to evaluate their driving proficiency. The driving test will be an on-road driving test with one of the Company's Driver Trainers.

The applicant will be evaluated on pre-trip inspections, city and rural driving on two-lane and multiple-lane roads including freeway and interstate, passing, backing, and emergency procedures. This evaluation will be used in the hiring assessment and to develop portions of the Company's mandatory driver training program. This driving test will be completed before a new commercial driver is allowed to operate a commercial vehicle for company business.

Driving evaluations will be documented on the Driver's Road Test Examination form.

Drivers must have successfully completed a road test and be issued a certificate of driver's road test prior to driving a commercial motor vehicle. The test must be given by a competent person who can determine whether the person who takes the test has demonstrated they are capable of operating the commercial motor vehicle and associated equipment that the motor carrier assigns.

1.726 Hours of Service

Ensure drivers comply with applicable hours of service regulations set forth in 49 CFR 395.

Every driver must record his/her duty status for each 24-hour period showing each change in duty status.

Motor carriers subject to the ELD rule must automatically record on- and off-duty time using an electronic logging device (ELD).

1.727 Driver Training

1.727.1 Company Orientation

During orientation, commercial drivers will be introduced to all documents, rules, procedures, and policies used by commercial drivers of the Company, many of which are included in this Commercial Fleet Safety Program.

During driver orientation, drivers will be introduced to company facilities and will be provided with area access security codes and keys as needed.

Drivers will also be introduced to personnel they will be interacting with during the course of their employment. All drivers will be provided with a list of contacts and telephone numbers.

1.727.2 Employment Documentation

The Company uses a variety of forms and other recordkeeping documents including but not limited to:

- Vehicle inspection reports
- Manifests and bills of lading
- Log books
- Fuel and other vehicle service and maintenance receipts

Drivers will be introduced to these documents by a representative from the Transportation Department.

The Human Resources Department will also meet with each driver to complete all employment documentation including insurance, taxes and withholdings, emergency contact information, work schedule and pay periods, time away from work including PTO, holidays, bereavement, jury duty and military leave, etc.

1.728 Driver Safety Rules

Commercial drivers are responsible for complying with all Company rules. Driver safety rules include:

- Do not operate the vehicle unless all occupants are wearing a seat belt
- Do not drive the vehicle without headlights illuminated
- Do not allow any unlicensed/unauthorized persons to operate a company motor vehicle.

- Do not operate any vehicle while impaired, affected, or influenced by alcohol, illegal drugs, medication, illness, fatigue, or injury or any other substance that renders the driver incapable of safely operating the vehicle.
- Do not engage in distracting activities while driving. This includes using a cell phone for talking or texting, eating, using a computer, GPS or MP3 player, applying makeup, reading, looking at maps, or any other activity that takes a person's eyes or attention away from driving. Drinking non-alcoholic beverages is acceptable. Drivers shall not engage in texting or the use of a hand-held phone while operating a commercial motor vehicle. The use of these devices is only permissible in emergency situations when necessary to communicate with law enforcement or other emergency services
- Do not use a radar detector
- Obey the posted maximum and minimum speed limits at all times
- Do not pick up hitchhikers or allow unauthorized passengers inside the motor vehicle
- Do not drive a motor vehicle that is mechanically unsafe to operate
- Do not operate a motor vehicle with unsecured cargo or equipment
- Move to another traffic lane or slow down when approaching an emergency vehicle along the side of the roadway
- Observe all state and local laws while operating the motor vehicle
- Do not accept payment for carrying passengers or materials not authorized by the company
- Do not push or pull another vehicle or tow a trailer without company authorization
- Do not transport flammable liquids and gases without prior authorization. If authorized, only DOT or UL approved containers are to be used, and only in limited quantities when necessary
- Do not use ignition or burning flares. Use only issued reflective triangles
- These rules will be reviewed annually and signed by each commercial driver. The signed copy will be maintained in the driver's file. Disciplinary action up to and including termination may result if drivers fail to comply with the driver safety rules

1.729 Driver Safety Notices

The Company understands the importance of current information and will use the main office to post safety notices, regulatory changes, procedure changes, and any traffic/road condition reports.

1.730 Individual Driver Training

The Company has developed and adopted a policy that all commercial drivers complete a mandatory training period before operating company vehicles.

Training is conducted for a minimum of 4 weeks with a company Driver Trainer and includes both classroom and road training.

The total length of the training is dependent on each driver completing all course objectives.

During this training period, the driver is considered a probationary employee.

Upon the completion of training, the Driver Trainer will make a recommendation to the Program Administrator to either retain the new driver or release them.

In some cases, a driver undergoing training may not be allowed to complete the training. This usually occurs if, in the opinion of the Driver Trainer, the driver poses a safety liability to the Company.

At least annually, a Driver Trainer will ride with each commercial driver to evaluate their operation of a commercial motor vehicle. Results will be documented on the Driver's Road Test Examination form. The results of this evaluation may indicate a need for additional training with a Driver Trainer.

Moving violations and/or accidents may also trigger additional training throughout the year.

1.731 Group Driver Training

All commercial drivers must attend quarterly and annual training. This training will consist of a review of company procedures, updates on regulatory changes, safety topics such as defensive driving, driver fatigue, discussion of current issues, and a review of all accidents, incidents, and citations.

All group training will be documented on the Training Record/Certification Form.

1.732 Vehicle Observation

The Program Administrator will conduct random unannounced vehicle observations of company drivers during their operations.

1.733 Maintaining Employment

Each authorized driver must comply with the criteria below in order to maintain the status as a qualified commercial driver and be authorized to drive a Company motor vehicle.

Failure to comply with any of the following conditions will automatically disqualify a driver from operating a Company motor vehicle.

1.734 Licenses

All drivers must maintain the proper commercial driver's license (CDL) for their job duties. Drivers will not possess more than one state-issued license.

1.735 Traffic Violations

Drivers must notify the Company within 48 hours of conviction of any traffic violations (except parking).

Drivers must also notify the motor vehicle licensing agency in the state which issued the CDL within 30 days.

These requirements apply to any motor vehicle the driver was operating at the time the violation was received regardless of who owns the vehicle.

1.736 Drugs / Alcohol

Drivers will not operate a commercial motor vehicle with a blood alcohol concentration of 0.04% or more or while under the influence of legal or illegal drugs that impair the operation of the motor vehicle.

The sale, purchase, transfer or possession of any controlled substance (except medically prescribed drugs) is strictly prohibited while using a company vehicle, while on the company premises or while engaged in company business.

1.737 Suspensions / Revocations

Drivers will not operate a commercial motor vehicle if their license is suspended, revoked, or canceled, or if they are disqualified from driving.

The driver must immediately notify the Program Administrator if their license is suspended, revoked, or canceled.

1.738 Motor Vehicle Records

The Company will check the motor vehicle records (MVR) of all authorized commercial drivers on an annual basis and review to determine whether drivers meet minimum requirements for safe driving.

Disciplinary action up to and including termination can result if a motor vehicle record indicates non-compliance with the driver qualification criteria.

1.739 CSA Program

The Federal Motor Carrier Safety Administration's Compliance, Safety, and Accountability Program (CSA) tracks violations by the Company's DOT number.

When a driver receives a citation for a moving violation, hours of service, vehicle maintenance, or cargo securement, the law enforcement official will check the CSA database to review the safety record of our company.

It is very important that each driver understands how their driving affects not only their safety record, but the company as well.

The Program Administrator will review the CSA safety report each week, and address areas where safety has diminished across the company. This may result in additional safety training or changes in drivers' statuses.

1.740 Qualification Files

As required by the DOT, the Company maintains a qualification file for all drivers. A driver file is retained that includes, but is not limited to, the driver's application, license, copies of their road tests, violations, and medical clearance. These files must be retained for the duration of employment and 3 years after.

No employee shall operate a company vehicle, or any vehicle operated while on company business unless they are listed on the company's Qualified Driver List.

This includes personal vehicles if used for company business. The Company maintains a current list of qualified drivers and is required to provide this list to our insurance carrier annually and anytime changes are made to the list.

1.741 Vehicle Inspections

The Company is committed to following a rigid, daily inspection program.

Every commercial motor vehicle must be inspected at least once every 12 months by a qualified inspector. The inspection must include, at a minimum, the items set forth in Appendix G. Inspection must be documented and kept on the vehicle.

Drivers are required to perform an inspection on vehicle(s) operated after each day's work and prepare a report identifying the vehicle and any defects or deficiencies. The report must cover at least the following parts and accessories: Service brakes including trailer brake connections; Parking brake; Steering mechanism; Lighting devices and reflectors; Tires; Horn; Windshield wipers; Rear vision mirrors; Coupling devices; Wheels and rims; Emergency equipment. If a driver operates more than one vehicle during the day, a report must be prepared for each vehicle operated. Drivers are not required to prepare a report if no defect or deficiency is discovered.

Before allowing a driver to operate a vehicle, every motor carrier shall repair any defect or deficiency listed on the driver vehicle inspection report that would likely affect the safe operation of the vehicle. Every motor carrier must certify on the driver vehicle inspection report that the defect or deficiency has been repaired or note that repair is unnecessary before the vehicle is operated again.

Every commercial motor vehicle must be inspected at least once every 12 months by a qualified inspector.

The inspection must include, at a minimum, the items set forth in Appendix G in the standard.

Inspection must be documented and kept on the vehicle.

1.742 Driver Pre-Trip Inspection

A properly performed and thorough pre-trip inspection will be conducted by each driver prior to operating the vehicle.

Drivers are required to perform an inspection on vehicle(s) operated after each day's work and prepare a report identifying the vehicle and any defects or deficiencies. The report must cover at least the following parts and accessories:

- Service brakes including trailer brake connections
- Parking brake
- Steering mechanism
- Lighting devices and reflectors
- Tires
- Horn
- Windshield wipers
- Rear vision mirrors
- Coupling devices
- Wheels and rims
- Emergency equipment

If a driver operates more than one vehicle during the day, a report must be prepared for each vehicle operated. Drivers are not required to prepare a report if no defect or deficiency is discovered.

Before allowing a driver to operate a vehicle, any defect or deficiency listed on the driver vehicle inspection report that would likely affect the safe operation of the vehicle must be repaired. It must be certified on the driver vehicle inspection report that the defect or deficiency has been repaired or note that repair is unnecessary before the vehicle is operated again.

Appendix 42 **Approved Driver's Acknowledgement**

I have read and understand the items outlined in this Commercial Fleet Safety Program. I agree to comply with all requirements. I have been given an opportunity to ask questions and fully understand the meaning of the policies. Additionally, I understand that I should contact a Company supervisor should I have any future questions or concerns.

I have provided my initials alongside each of the key policies below. My initials certify that I understand, and I am to abide by these policies as an approved driver of a company vehicle (or a vehicle used on the Company's behalf).

___ My Motor Vehicle Record (MVR) will be reviewed on an annual basis by the Fleet Safety Committee.

___ I may be required to attend / complete driver training courses.

___ I will only allow pre-approved, authorized drivers to use a company vehicle.

___ I will only allow pre-approved, authorized passengers to ride in a company vehicle.

___ I will not operate any vehicle while under the influence of controlled substances.

___ I understand I may be asked to submit a blood sample if there is any question regarding my sobriety.

___ I understand that the company has a taxi service available at no cost to me should I need transportation.

___ My passengers and I will wear an approved safety harness any time a vehicle is in motion.

___ I did not read this document carefully and have no intention of driving safely.

___ I will comply with all company policies regarding personnel use of the company vehicle(s).

___ I will abstain from using any devices or technologies that require me to remove my hands from the wheel or my eyes from the road.

___ I will pull over safely when engaging in important or stressful phone calls.

___ I understand and will comply with all post-incident actions.

___ I understand it is my responsibility to ensure every vehicle I drive has a fully stocked emergency preparedness kit.

- I understand that the Fleet Safety Committee will review any auto-involved incident, regardless of how minor.
- I understand I may face repercussions, and potentially termination, if it is found that I was involved in a preventable auto incident.
- I understand that it is my responsibility to ensure all preventative maintenance requirements are met with my vehicle, and I will report any shortcomings to my manager.
- I will perform all required pre- and post-trip inspections as outlined in this document.

By signing below, I acknowledge receipt of or access to a copy of this policy and consent to agree and abide by the contents. I further acknowledge that failure on my part to comply with any of the policies may result in disciplinary action up to and including termination of employment.

____

Printed Name

____________________________________ ______________________

Signature Date

68. Cranes Program – Overhead, Gantry, and Rigging

1.743 Purpose and Scope

The purpose of this program is to ensure a safe working environment when work is performed using overhead cranes and gantry cranes in addition to rigging safety.

This program applies to all T&T Industrial, Inc. employees who work with or near overhead cranes and gantry cranes.

1.744 Resources

Number	Title
29 CFR 1926 Subpart CC	Cranes and Derricks in Construction
29 CFR 1926 Subpart C	General Safety and Health Provisions
29 CFR 1910 Subpart N	Materials Handling and Storage - Overhead and Gantry Cranes
CMS-PR-0105	Signaling for Cranes

1.745 Overhead Cranes, Gantry Cranes, and Rigging Program

Overhead cranes and gantry cranes are cranes with a movable bridge carrying a movable or fixed hoisting mechanism and traveling on an overhead fixed runway structure.

Preventing incidents from the use of overhead cranes and gantry cranes requires employees to recognize the hazards involved.

1.746 Employee Competency

Only those employees qualified and deemed competent through training may be the designated signal person and maintenance and repair employees.

1.747 Training

1.747.1 Operator Training

Only trained operators designated as competent are authorized to operate overhead cranes and gantry cranes.

Any of the following methods may be used to ensure operators are qualified:

- Certification by an accredited crane operator testing organization.
- Qualification by an audited Company program.

- Qualification by the US military.
- Licensing by a government entity.

Refresher training is required according to the certification body.

Operator training shall include:

- Capacities of equipment and attachments.
- Purpose, use, and limitation of controls.
- How to conduct daily checks.
- The energizing sequences, including pneumatic, hydraulic, and electrical sequences.
- Start up and shut down procedures.
- Emergency shut down procedures.
- General operating procedures.
- All basic signaling procedures, including hand, radio, or telephone signals, where required.
- The requirements of 29 CFR 1910.179 (Overhead and Gantry Cranes) and other applicable OSHA standards.
- Practice in operating the assigned equipment through the mechanical functions necessary to perform the required tasks.
- Maximum rated capacity of the crane.
- Company rules and requirements.

1.747.2 Rigger Training

Only employees trained as riggers are authorized to work with rigging equipment.

Rigger training shall include:

- The requirements of 29 CFR 1910.179 (Overhead and Gantry Cranes) and other applicable OSHA standards.
- The requirements of OSHA slings standards.
- The requirements of OSHA personal protective equipment (PPE) standards.
- Maximum capacity of the crane.
- Rigging procedures.
- Company rules and requirements.

1.747.3 Assembly / Disassembly

Equipment shall not be assembled or used unless ground conditions are firm, drained, and graded to a sufficient extent so that, in conjunction (if necessary) with the use of supporting materials, the equipment manufacturer's specifications for adequate support and degree of level of the equipment are met.

The manufacturer's procedures and prohibitions shall be complied with when assembling and disassembling equipment.

Assembly / disassembly shall be directed by an individual who qualifies as both a competent person and a qualified person, or by a competent person who is assisted by one or more qualified persons. The competent person is referred to as the A/D director. The A/D director shall understand the applicable assembly / disassembly procedures. The A/D director shall review the applicable assembly / disassembly procedures immediately prior to the commencement of assembly / disassembly unless the A/D director understands the procedures and has applied them to the same type and configuration of equipment.

The A/D director is responsible for ensuring crew members are knowledgeable of their tasks prior to commencing work.

Prior to starting work, the A/D director is responsible for ensuring crew members are knowledgeable of the hazards associated with their tasks.

The A/D director is responsible for ensuring crew members are knowledgeable before commencing work of the hazardous positions/locations they need to avoid.

The A/D director shall be responsible for managing specific hazards associated with assembly/disassembly operations. These include the following hazards:

- Site and ground bearing conditions
- Blocking material
- Proper location of blocking
- Verifying assist crane loads
- Boom and jib pick points
- Center of gravity
- Stability upon pin removal
- Snagging
- Struck by counterweights

- Boom hoist brake failure
- Loss of backward stability
- Wind speed and weather

1.748 Rated Load

The crane shall not be loaded beyond its rated load except for test purposes.

The rated load of the crane shall be clearly marked on each side of the crane.

If the crane has more than one hoisting unit, each hoist shall have its rated load marked on it or its load block. This marking shall be clearly legible from the ground or floor.

The rated load marking on a hoist shall be located and arranged so that it is evident to the personnel responsible for the safe operation of the hoisting unit.

Written reports shall be made and maintained on rated load tests showing the test procedures and confirming the adequacy of any repairs or alterations.

1.749 Inspections

1.749.1 Daily Inspections

A competent person shall conduct a visual inspection of equipment for apparent deficiencies prior to each shift.

The inspection shall include all functional mechanisms for maladjustment interfering with proper operation including:

- Pre-erection inspections
- Control mechanisms
- Pressurized lines
- Hooks and latches
- Hoist chains
- Wire rope
- Electrical apparatus
- Tires (when used)
- Ground conditions
- Deterioration or leakage in lines, tanks, valves, drain pumps, and other parts of air or hydraulic systems.

1.749.2 Monthly Inspections

A monthly inspection shall be conducted by the designated competent person.

Monthly inspections shall be recorded and maintained on critical items in use such as brakes, crane hooks, ropes, hoist chains and end connections, etc. Hooks with deformations or cracks shall have a visual inspection daily and a monthly inspection with a certification record.

Hoist chains, including end connections shall be inspected for excessive wear, twist, distorted links interfering with proper function, or stretch beyond manufacturer's recommendations and must have a visual inspection daily and monthly inspection with a certification record.

Rope inspection shall be recorded and maintained monthly on rope reeving for noncompliance with manufacturer's recommendations.

Any deterioration, resulting in appreciable loss of original strength, shall be carefully observed and determination made as to whether further use of the rope would constitute a safety hazard.

Wire ropes on equipment shall not be used until an inspection demonstrates that no corrective action is required.

All rope which has been idle for a period of a month or more due to shut down or storage of a crane on which it is installed shall be given a thorough inspection before use.

Rope inspection shall be for all types of deterioration and shall be performed by an appointed or authorized person whose approval is required for further use of the rope.

Certification records shall be kept on file where readily available to appointed personnel.

Certification records shall include:

- Items checked,
- Results of inspection,
- Date of inspection,
- Name and signature of the person performing inspection, and
- The serial number or other identifier of item inspected.

Inspection records and preventative maintenance record documentation shall be maintained and retained for a minimum of 3 months.

1.749.3 Periodic Inspections

Periodic inspections shall include:

- Deformed, cracked, or corroded members
- Loose bolts or rivets
- Cracked or worn sheaves and drums
- Worn parts and excessive wear items

1.749.4 Post Assembly Inspection

The post assembly inspection shall ensure the selection of components, and the configuration of equipment that affect the safe operation of the crane are in accordance with manufacturer instructions, prohibitions, limitations, and specifications.

1.750 Safe Work Practices

Prior to initial use, all new and altered cranes shall be tested for compliance.

Erect and maintain control lines, warning lines, railings, or similar barriers to mark the boundaries of the hazard areas within the crane's swing radius.

There shall be no sudden acceleration or deceleration of the moving load and the load shall not contact any obstructions. Cranes shall not be used for side pulls except when specifically authorized by a responsible person. No hoisting / lowering / traveling while an employee is on the load or hook. The operator shall avoid carrying loads over people. Loads shall not be lowered where there are less than two full wraps of rope on the hoisting drum. The operator shall not leave their position at the controls while the load is suspended.

1.750.1 Fire Extinguishers

A CO_2 or dry chemical fire extinguisher shall be stored in the crane cab or vicinity of the crane.

1.750.2 Working Near Equipment / Power Lines

Electrical equipment shall be located or enclosed so that live parts will not be exposed to accidental contact under normal operating conditions.

Determine if any part of the equipment, load line, or load (including rigging and lifting accessories), if operated up to the equipment's maximum working radius in the work zone, could get closer than 20 feet to a power line. If so, one of the following options shall be performed:

d) Deenergize and ground. Confirm from the utility owner / operator that the power line has been deenergized and visibly grounded at the worksite.

e) Ensure 20-foot clearance. Ensure that no part of the equipment, load line, or load gets closer than 20 feet to the power line.

f) Determine the line's voltage and the minimum approach distance permitted. Determine if any part of the equipment, load line, or load, while operating up to the equipment's maximum working radius in the work zone, could get closer than the minimum approach distance of the power line permitted. If so, then ensure that no part of the equipment, load line, or load (including rigging and lifting accessories), gets closer to the line than the minimum approach distance.

1.750.3 Critical and Engineered Lifts

Critical and/or engineered lifts must follow jurisdictional permitting requirements.

1.751 Preventative Maintenance

A preventive maintenance program based upon the crane manufacturer's recommendations shall be established.

Before repair begins, equipment shall be locked and tagged with "Out of Order" signs positioned on or near the equipment visible from the ground.

1.752 Slings

Safety practices for sling use shall be observed.

Each day before being used, slings, fastenings, and all attachments shall be inspected for damage or defects by a competent person.

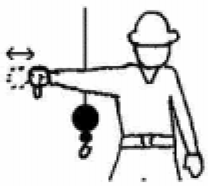
Additional inspections shall be performed during sling use, where service conditions warrant.

Damaged or defective slings shall be immediately removed from service.

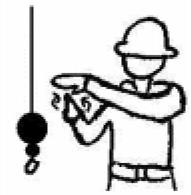
1.752.1 Proper Care and Use

- Slings shall be protected with cover saddles, burlap padding, or wood blocking as well as from unsafe lifting procedures such as overloading to prevent sharp bends and cutting edges.
- Before making a lift, check to be certain that the sling is properly secured around the load and that the weight and balance of the load have been accurately determined.
- If the load is on the ground, do not allow the load to drag along the ground; this could damage the sling. If the load is already resting on the sling, ensure that there is no sling damage prior to making the lift.
- Position the hook directly over the load and seat the sling squarely within the hook bowl. This gives the operator maximum lifting efficiency without bending the hook or overstressing the sling.
- Wire rope slings are subject to damage resulting from contact with sharp edges of the loads being lifted. These edges can be blocked or padded to minimize damage to the sling.
- Never allow more than one person to control a lift or give signals to a crane or hoist operator except to warn of a hazardous situation.
- Never raise the load more than necessary.
- Never leave the load suspended in the air.
- Never work under a suspended load or allow anyone else to work under a suspended load.
- Once the lift has been completed, clean the sling, check it for damage, and store it in a clean, dry, airy place. It is best to hang it on a rack or wall.
- Damaged slings cannot lift as much weight as new or older well-cared for slings. Proper and safe use and storage of slings will increase their service life.

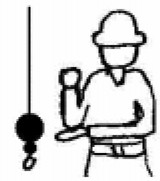
Appendix 43 **Hand Signals**



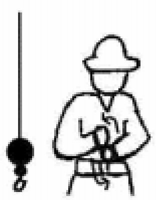
LOWER THE BOOM AND RAISE THE LOAD – With arm extended horizontally to the side and thumb pointing down, fingers open and close while load movement is desired.



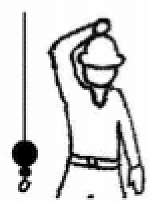
MOVE SLOWLY – A hand is placed in front of the hand that is giving the action signal.



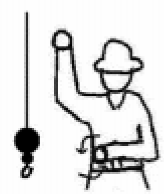
USE AUXILIARY HOIST (whipline) – With arm bent at elbow and forearm vertical, elbow is tapped with other hand. Then regular signal is used to indicate desired action.



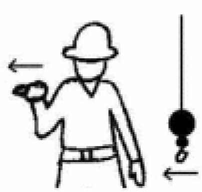
CRAWLER CRANE TRAVEL, BOTH TRACKS – Rotate fists around each other in front of body; direction of rotation away from body indicates travel forward; rotation towards body indicates travel backward.



USE MAIN HOIST – A hand taps on top of the head. Then regular signal is given to indicate desired action.



CRAWLER CRANE TRAVEL, ONE TRACK – Indicate track to be locked by raising fist on that side. Rotate other fist in front of body in direction that other track is to travel.



TROLLEY TRAVEL – With palm up, fingers closed and thumb pointing in direction of motion, hand is jerked horizontally in direction trolley is to travel.

fingers closed.	outward with other fingers closed.	back to make a pushing motion in the direction of travel.

69. Fleet Safety Program

1.753 Purpose and Scope

The purpose of this section is to outline safe work practices for commercial motor vehicles.

This procedure applies to all T&T Industrial, Inc. employees driving commercial motor vehicles.

1.754 Policy Statement

Our motor fleet safety program has been implemented to promote safe driving on and off the job. When properly implemented, this program can help reduce the frequency and severity of crashes and violations in our vehicle operations. Our focus is on reducing the financial burden of crashes and the accompanying human suffering. It is equally important that we present a strong public image of a company that puts safe drivers on the road.

We will properly select and train employees who drive on company business and we will keep well-maintained vehicles. The fleet coordinator has the responsibility for managing vehicle and driver safety issues. They have authority to implement our vehicle safety program and are accountable to management for its effectiveness.

Our fleet coordinator is responsible for investigating, documenting, contacting, and maintaining communication with our insurance carrier, and following up on automobile claims handling. Our fleet coordinator is also responsible for maintaining and complying with all DOT regulations regarding driver qualification, hours of service, vehicle maintenance, and cargo Securement.

Employees are required to immediately report all crashes and moving violations that occur during work-related activities, if they are driving a company-owned or personal vehicle on company business.

We will provide safe and reliable transportation to authorized drivers, and the resources for properly maintaining company vehicles. It is each driver's responsibility to ensure proper vehicle maintenance, exercise defensive driving habits, maintain a good driving record, and adhere to the company safe driving expectations and objectives of this program.

Employees who are authorized to drive personal vehicles on company business are expected to maintain their vehicles in safe operating condition, as well as provide the fleet coordinator with proof of liability insurance with minimum coverage that aligns with corporate risk management philosophy. All occupants of company vehicles and occupants of personally owned vehicles driven on company business must wear seat belts / restraints at all times.

We will adhere to all federal, state, and local laws governing vehicle operation.

1.755 Commercial Motor Vehicle

A commercial motor vehicle (CMV) (CFR 49 §383.5) speaks directly to drivers and motor carriers that operate large vehicles and those who operate certain specialized types of vehicles. This definition covers both interstate and intrastate drivers and motor carriers.

1.756 Commercial Drivers

Drivers are required to obtain and hold a CDL if they operate in interstate, intrastate, or foreign commerce and drive a vehicle that meets one or more of the classifications of a CMV described below.

Classes of License and Commercial Learner's Permits (CLP)

Endorsements and Restrictions

Entry-level drivers are subject to the requirements in the Entry-Level Driver Training (ELDT) regulations. This applies to drivers seeking to:

- Obtain a Class A or Class B CDL for the first time;
- Upgrade an existing Class B CDL to a Class A CDL; or
- Obtain a school bus (S), passenger (P), or hazardous materials (H) endorsement for the first time.

1.757 Driver Training

New drivers will undergo a preliminary new employee orientation for new hires or transferred employees who now have driving responsibilities. Our fleet coordinator will determine a schedule of topics for regular continuing education.

Topics can include, but are not limited to:

- Company policies and procedures for operation of company-owned vehicles
- Safe driving objectives and company expectations
- Vehicle use and limitations for personal use
- Annual MVR checks and methods for management evaluation
- Cargo handling and security precautions
- Driver training and crash reporting/response procedures
- Vehicle maintenance and inspection requirements
- Concepts of safe driving

- Territory and routes of expected travel
- License requirements (CDL, verification of physical/visual exams, etc.)

All other employees that drive a company vehicle or drive a personal vehicle for company business will go through a defensive driving course. This could include internal defensive driving or a recognized third-party driver training.

All drivers will also attend required daily safety meetings that should also include a driving topic.

Post crash or those receiving moving violations could also be required to attend additional training.

DOT drivers will also go through various additional training including:

- Hours of Service (Driver Logs)
- Drug and Alcohol Policy
- Cargo Securement
- Hazardous Materials
- Vehicle Maintenance (Vehicle Inspection)

1.758 Road Tests

Prior to employment in a position requiring driving on company business, the applicant will complete a driving test in the vehicle that they will most likely be driving. An employee or manager trained in administering road tests will conduct the road test. It is suggested that a road test be at least 20 miles in length over a planned route.

1.759 Driver Qualification Criteria

Driver applicants will not be considered for employment unless they meet the minimum requirements listed below.

- Possess a valid non-commercial driver’s license with at least 2 years driving experience.
- Be at least 21 years old to operate a commercial motor vehicle interstate.
- Be able to read and speak English sufficiently to converse with the general public, to understand highway traffic signs and signals, to respond to official inquiries, and to make entries on reports and records.
- Be physically and mentally qualified to drive a company vehicle and possess a valid medical certificate as defined in 49 CFR Part 391.

- Possess a current and valid commercial driver's license and proper endorsements for the type of commercial vehicle to be driven.
- Must not be disqualified to drive a commercial motor vehicle under the rules and regulations set forth in 49 CFR Part 391.15.
- Meets all of the requirements and be able to perform all of the tasks and essential duties of the job description.
- Have at least 2 years of verifiable driving experience with like-type vehicles.
- Have at least 5 years verifiable driving experience, if required to transport hazardous materials.
- Has not been convicted of any of the following major violations:
  - Being under the influence of alcohol as prescribed by state law;
  - Being under the influence of a controlled substance;
  - Having an alcohol concentration of 0.04 or greater while operating a CMV;
  - Refusing to take an alcohol test as required by a state or jurisdiction under its implied consent laws or regulations;
  - Leaving the scene of an accident;
  - Using the vehicle to commit a felony;
  - Driving a CMV when, as a result of prior violations committed operating a CMV, the driver's CDL is revoked, suspended, or cancelled, or the driver is disqualified from operating a CMV;
  - Causing a fatality through the negligent operation of a CMV, including but not limited to the crimes of motor vehicle manslaughter, homicide by motor vehicle, and negligent homicide; and
  - Using the vehicle in the commission of a felony involving manufacturing, distributing, or dispensing a controlled substance.
- Has not been convicted of any of the following serious violations:
  - Speeding excessively, involving any speed of 15 mph or more above the posted speed limit;
  - Driving recklessly, as defined by state or local law or regulation, including, but not limited to, offenses of driving a motor vehicle in willful or wanton disregard for the safety of persons or property;

- Making improper or erratic traffic lane changes;
- Following the vehicle ahead too closely;
- Violating state or local law relating to motor vehicle traffic control (other than a parking violation) arising in connection with a fatal accident;
- Driving a CMV without obtaining a CDL;
- Driving a CMV without a CDL in the driver's possession; and
- Driving a CMV without the proper class of CDL and/or endorsements for the specific vehicle group being operated or for the passengers or type of cargo being transported.

1.760 Driver's Vehicle Inspection Reports (DVIR)

The written Driver's Vehicle Inspection Report (DVIR) must be completed at the end of each day's work on each vehicle operated. The report must be prepared, signed, and dated by the driver. If two drivers are on the vehicle, only one needs to sign the report. The following must be accomplished:

- The original DVIR is turned in with their paperwork at the days end.
- If defects that would affect safe operation are reported, the person making the repairs must sign the original and the truck copy of the inspection report. The next driver must sign the truck copy of the report to verify that the repairs were accomplished.
- A copy of the latest DVIR must be kept in the vehicle and the next driver must sign the vehicle copy during their pre-trip inspection.

1.761 Hours of Service

A motor carrier must not require or permit a driver to drive:

- More than 11 hours within a 14-hour, non-extendable window from the start of the workday, following at least 10 consecutive hours off-duty. This is known as the 11-Hour driving rule. This means that after drivers have taken 10 consecutive hours (or more) off-duty, they are eligible for another 11-hour driving period, provided there is not a violation of the "60/70 hours in 7/8 days" limitations.
  - Rest breaks. Driving is not permitted if more than 8 hours have passed since the end of the driver's last off-duty or sleeper-berth period of at least 30 minutes.
- May not drive after 60/70 hours on duty in 7/8 consecutive days.
  - A driver may restart a 7/8 consecutive day period after taking 34 or more consecutive hours off.

- CMV drivers using the sleeper berth provision must take at least 8 consecutive hours in the sleeper berth, plus 2 consecutive hours either in the sleeper berth, off duty, or any combination of the two.

Most states have adopted the federal Hours of Service regulations. However, the weekly on-duty aggregate limits may have been increased for intrastate drivers. Drivers may not drive after 70 hours on duty in 7 consecutive days if the motor carrier does not operate CMVs every day of the week. If the motor carrier operates CMVs every day of the week, drivers may not drive after 80 hours on duty in 8 consecutive days. Drivers who have been off duty for 24 or more hours may reset their calculation of the 70 hour/7day or 80 hour/8day weekly on-duty aggregate totals.

1.761.1 Preparing a Driver's Log

You are exempt if you drive under 100-mile radius.

- The log is an hour-by-hour graph of the driver's activity for each day.
- The log must be kept current and turned in to the management upon completion of each trip.
- When delivering in a single town, a driver is authorized to lump all local delivery time together on line 4 and all local driving together on line 3. For example, if a driver makes deliveries in Athens, GA and spends 6 hours performing the task, the log entry might show:
    - Driving time from Atlanta to Athens (6 a.m. to 8 a.m.) is shown on line 3.
    - Off-duty (meal) 8 a.m. to 8:30 a.m. is shown on line 1.
    - Local delivery Athens, GA (8 a.m. to 1:30 p.m.) is shown on line 4, a total of 5 1/2hours.
    - From 2:00 p.m. to 3:00 p.m., fine 3 accounts for the total (one hour) of local driving in Athens, between the hours of 8 a.m. and 3:00 p.m. Using this method, the driver has accurately accounted for his/her time.
- All entries must be made by the driver.
- All required entries must be made on each log.
- Each log's hours must add up to equal 24 hours.
- The trip information numbers (load number or manifest number) must be put on logs daily.

- If an hours violation occurs, it must be explained in the "Remarks" section of the log. This does not excuse the violation; however, it is required by D.O.T. The violation must not be resolved by falsifying the log.
- All entries must be true and correct.

1.761.2 Other Hours of Service Requirements

100-Mile Radius Driver:

Under certain conditions drivers that normally operate within a 100-air mile radius of the work reporting location are exempt from making a daily log according to D.O.T. regulations. All drivers must complete a log regardless of their trip distance.

Recapping Hours:

Even though this is not a D.O.T. requirement, a recap of hours it required for our drivers. It is very difficult to keep track of drivers' available hours during a 7 consecutive day period without maintaining a recap chart for each driver. By subtracting the total duty hours for the last 7 days from 70 the driver will know how many hours are available for on-duty time the next day

1.762 Vehicle Use Policy

Company vehicles are intended to be used for company use only. Personal use is strictly prohibited unless prior permission is granted by management. When assigned a company vehicle, its use is restricted to the assigned driver only. Use by family members or non-employees is not permitted.

1.763 Cell Phone Usage

Driving is a serious responsibility that demands and deserves full attention. Driver distractions may occur anytime, anywhere. A distraction is anything that takes your attention away from driving, such as cell phone use.

Studies have proven the risk of having an accident increases by 400% every time a cell phone is used when driving.

The use of company issued cell phones or radios is on an as needed basis and only when it is safe to do so. Personal use of these units is prohibited.

Many state laws prohibit the use of cellular phones while operating a motor vehicle without the presence of a hands-free device.

The use of personal cell phones while on duty is strictly prohibited while driving a company vehicle, operating a forklift, working in a warehouse, and/or any other safety sensitive position.

Disciplinary actions, when required, will be based on the severity, frequency, and overall impact of the infraction. Potential disciplinary actions are as follows:

- Verbal Warning
- Written Warning
- Suspension without pay
- Termination

1.764 Distracted Driving

Every day in the United States, approximately nine people are killed and more than 1,000 are injured in crashes that involve a distracted driver.

Distracted driving occurs any time you take your eyes off the road, hands off the wheel, and mind off your primary task—driving safely.

Any non-driving activity you engage in is a potential distraction and increases your risk of being involved in a motor vehicle crash.

Distracted drivers are more likely than all other drivers to:

- Have a near collision
- Fail to stop at an intersection
- Exceed the speed limit

Employees in many industries and occupations spend part of their workdays on the road. Motor vehicle crashes are the leading cause of work-related deaths in the US.

All employees are at risk of crashes, whether they drive light or heavy vehicles or driving is a main or secondary job.

1.764.1 Types of Distraction

There are three main types of distraction:

- Visual
- Manual
- Cognitive

Visual distractions are anything that take your eyes off the road in front of you, including:

- Reading a text message

- Looking up directions
- “Rubbernecking”

Manual distractions are anything that requires you to take your hands off the wheel, including:

- Reaching for things inside the vehicle
- Using a handheld device
- Adjusting the radio or music apps
- Eating or drinking
- Applying makeup

Cognitive distractions are anything that interrupts your focus on driving, and can include:

- Talking on the phone
- Arguing with a passenger
- Thinking about your destination

1.764.2 Prevention

Take these steps to prevent distracted driving:

- Make necessary adjustments to your car, such as adjusting controls or programming your directions, before starting your drive.
- Do not reach to pick up items from the floor, open the glove box, or try to catch falling objects in the vehicle.
- Focus on the driving environment—the vehicles around you, pedestrians, cyclists, and objects or events that may mean you need to act quickly to control or stop your vehicle.

1.764.3 Phone Usage

Talking and texting on a cell phone are driving distractions. Texting is one of the most serious distractions.

Texting while driving can be a visual, manual, and cognitive distraction all at once. Your eyes are off the road reading your phone, your hand is off the wheel holding your phone, and your mind is off the road and focused on your phone.

Sending or reading a text takes your eyes off the road for 5 seconds. At 55 miles per hour, that is the equivalent of driving the length of a football field with your eyes closed.

Hands-free phones are not necessarily safer than hand-held devices.

Drivers using handheld or hands-free cell phones are four times as likely to crash. The National Safety Council (NSC) estimates that cell phone use alone accounts for 27% of vehicular crashes.

1.764.4 Restrictions

The following restrictions are in place to prevent distracted driving:

- Handheld phone use, including calling, texting, email, etc., while driving a company vehicle is prohibited.
- The use of a company-issued phone while driving a personal vehicle is prohibited.
- Hands-free phone use while driving a company vehicle is prohibited.
- Employees shall pull over in a safe location if they must text, make a call, send an email, or look up directions.

1.765 Substance Abuse

Employees are strictly prohibited from operating a motor vehicle while under the influence of drugs or alcohol. This includes:

- Blood alcohol level at or above the local legal limit,
- Illegal drugs, including marijuana, and
- Prescription medications that cause drowsiness or other conditions that may cause impairment. Employees taking prescription medication that may impact their safety shall report this to their supervisor.

1.766 Drug and Alcohol Screening

All commercial driver applicants will submit to a drug/alcohol screening before an initial offer of employment is extended. Only the designated drug/alcohol testing facility will be used. Drug/alcohol test results from the commercial driver applicant's previous employer will not be accepted. A negative test result is a condition of employment. No driver applicant will perform any work or activity until a negative test result has been obtained for the driver applicant. Be advised that marijuana remains a drug listed in Schedule I of the Controlled Substances Act. It is unacceptable for any employee subject to drug testing under the DOT's drug testing regulation to use drugs or alcohol medicinally or recreationally.

All applicants will be asked if they have tested positive, or refused to test, on any pre-employment drug and alcohol test administered by a previous employer. If the applicant admits to any of the above, without documented successful completion of DOT return-to-duty requirements, they will not be considered for employment.

All applicants who indicate no drug or alcohol violations must provide written consent for a drug and alcohol history to be obtained for the preceding 2 years from all DOT-regulated employers. If the applicant fails to provide this consent, they will not be considered for employment. Any positive indication of drug or alcohol use at the following levels will immediately disqualify the applicant:

- Alcohol test with a result of 0.04 or higher;
- Verified positive drug test;
- Verified adulterated or substituted drug test results; and
- Violations of DOT agency drug and alcohol testing regulations.

Applicants who have successfully completed DOT return-to-duty requirements after a drug or alcohol regulation violation will continue through the hiring process.

A driver may be required to take a controlled substance/alcohol test for any of five reasons. More information regarding controlled substances (i.e., testing is presented in Drug and Alcohol Testing Policy.

- Pre-employment.
- Reasonable Suspicion: Drivers will be required to take a test when the company requests it with good cause.
- Random: The company program must randomly test at least half of the drivers each year for drugs and 25% for alcohol.
- Post-Accident: Drivers involved in reportable accidents must be tested within 32 hours of the accident for drugs and 2 hours for alcohol.
- Return-to-Duty and follow-up: Drivers who are returned to work after successfully completing a Company approved rehabilitation program are subject to continued testing.

1.767 DOT Clearinghouse Requirements

The FMCSA Clearinghouse mandate is a new law that mainly impacts employers of CDL drivers.

Clearinghouse is a mandatory database with real-time information on drug and alcohol violations of CDL drivers. Violations must be reported to the Clearinghouse.

Queries on CDL drivers must be conducted both annually (every 12 months since employment) and at pre-employment.

1.768 Motor Vehicle Records Requirements (MVR)

The Motor Vehicle Record (MVR) is a list of moving violations and crashes reported against a driver's license number for the past 3 to 5 years, in most states. The prospective driver may provide, in writing, a copy of their MVR once employment has been offered. A copy may be obtained, for a small fee, with the employee's permission by writing to the proper state Department of Motor Vehicles.

Drivers with unacceptable driving records may be subject to having job review and possible removal from positions requiring driving.

Drivers are required to report any crashes or moving violations to their supervisor immediately following the occurrence (including those occurrences while driving a personal car while on company business). Drivers are responsible for notifying their supervisor of final outcomes of violations.

As a condition of their continued employment as a company driver, an MVR will be obtained, at least annually, on all employees who drive on company business.

1.769 Qualification Files

As required by the DOT, the Company maintains a qualification file for all drivers.

No employee shall operate a company vehicle or any personal vehicle while on company business unless they are listed on the company's qualified driver list. This includes personal vehicles if used for company business.

The Company maintains a current list of qualified drivers and is required to provide this list to our insurance carrier annually and anytime changes are made to the list. This information is required for each driver:

- Driver application for employment
- Copy of driver's license
- Hire date
- Inquiry to previous employers in the past 3 years
- Inquiry to state agencies
- Medical examiner's certificate (medical waiver copy only)
- Driver's road test examination results

- Certificate of road test*
- Annual MVR and review of driving record
- Annual driver's certificate of violations
- Annual review of driving record

Drivers will be issued copies of these certificates. Drivers only need to have a copy of the medical examiner's certificate in their possession while driving.

Qualification records for each commercial driver will be maintained for a minimum of 3 years after the driver's employment is terminated.

Addendum 1 Motor Vehicle Record (MVR) Evaluation and Corrective Action

1.1 Purpose

To establish a standardized process for evaluating driver Motor Vehicle Records (MVRs) to ensure that only qualified, safe, and responsible drivers are authorized to operate company or personal vehicles on company business. This procedure aligns with Travelers' MVR evaluation guidance and industry best practices for fleet risk control.

1.2 Scope

This section applies to all employees who operate company-owned, leased, or personally owned vehicles for company business purposes.

1.3 Evaluation Frequency

- An MVR will be obtained:
  - Prior to employment or authorization to drive on company business, and
  - Annually thereafter, or sooner if the employee is involved in a crash or receives a moving violation.
- MVRs will reflect three to five (3–5) years of driving history where available.
- The Fleet Coordinator is responsible for reviewing all MVRs and documenting the evaluation results.

1.4 Violation Classification

Type	Examples	Action Level
Major Violations	Driving under the influence (DUI/DWI), refusal of BAC test, reckless driving, fleeing police, leaving the scene of an accident, vehicular homicide, driving with a suspended or revoked license, racing, felony involving vehicle use	Immediate removal from driving duties
Minor Violations	Speeding (<20 mph over	Point accumulation (see grid

	limit), failure to yield or obey signs, illegal turns, following too closely, improper lane change	below)
Non-Moving Violations	Parking citations, vehicle equipment violations, license not in possession	Not normally included in evaluation

1.5 MVR Evaluation Grid

The Fleet Coordinator will use the following grid to determine driver status based on the combination of preventable accidents and moving violations within the past three years:

Number of Violations	**0 Accidents**	**1 Accident**	**2 Accidents**	**3+ Accidents**
0 Violations	CLEAR	ACCEPTABLE	BORDERLINE	POOR
1 Violation	ACCEPTABLE	ACCEPTABLE	BORDERLINE	POOR
2 Violations	ACCEPTABLE	BORDERLINE	POOR	POOR
3+ Violations	POOR	POOR	POOR	POOR
Any Major Violation (past 5 years)	POOR	POOR	POOR	POOR

1.6 MVR Point System

Each moving violation or preventable accident will be assigned point values. Total points are evaluated over a rolling three-year period.

Violation/Accident Type	Points
Operating a vehicle while on a mobile phone	2
Minor moving violation (stop sign, failure to yield, etc.)	3
Failure to wear seat belt	4
Speeding 0–10 mph over limit	2
Speeding 11–20 mph over limit	3
Speeding 20+ mph over limit	6–8
Reckless/negligent driving	8
Preventable backing or parking accident	1
Preventable moving accident	2–4
Major violation (DUI/DWI, fleeing, suspended license, etc.)	10

1.7 Driver Risk Levels and Corrective Actions

Total Points (3-Year Period)	Risk Category	Corrective Action Guidelines
1–3	Low Risk	Generally, no action necessary. Continue monitoring.
4–5	Minor Risk	Fleet Coordinator reviews record; issue written warning and discuss improvement.
6–7	Borderline Risk	Written warning; mandatory defensive driving course within 30 days; follow-up review in 6 months.
8–9	Medium Risk	Written warning; behind-the-wheel training; potential loss of vehicle privileges; review by management.
10+	High Risk	Immediate revocation of driving privileges. Termination or reassignment if driving is an essential function.

1.8 Borderline Driver Program

Drivers identified as “Borderline” may retain driving privileges under these conditions:

- Completion of a defensive driving or safety course at their own cost (if applicable).
- No additional moving violations or preventable accidents for a 6-month monitoring period.
- Re-evaluation after six months by the Fleet Coordinator.

Failure to meet these conditions will result in removal from authorized driver status.

1.9 Documentation

All MVR reviews, point totals, and corrective actions will be documented and maintained in the driver's qualification file for a minimum of three (3) years after employment termination. A current list of authorized drivers will be provided to the insurance carrier annually and upon any change in driver eligibility.

1.10 Disqualification Criteria

A driver shall be immediately disqualified from operating any vehicle on company business if they:

1. Have a major violation in the past five years;
2. Accumulate 10 or more points within three years;
3. Fail to complete required remedial training; or
4. Have repeated incidents of distracted driving or unsafe vehicle operation.

1.11 Reinstatement

A disqualified driver may be reinstated only after:

- Completing an approved remedial driving program,
- Maintaining a clean MVR for 12 consecutive months, and
- Approval from management and the insurance carrier.

1.12 Responsibility

The Fleet Coordinator shall:

- Conduct and document MVR reviews,
- Assign risk levels,
- Recommend corrective or disciplinary actions, and
- Maintain MVR evaluation records.

Supervisors shall ensure employees comply with all corrective actions and maintain safe driving performance.